United States Air Force
Independent Duty Medical Technician

Medical and Dental Treatment Protocols

30 July 2013
OPR: 383 Training Squadron
Edition 1.0: Lt Col Samuel B. Munro, DO

MEMORANDUM FOR ALL USAF 4N0X1C INDEPENDENT DUTY MEDICAL TECHNICIANS

FROM: HEADQUARTERS UNITED STATES AIR FORCE/SG1E

SUBJECT: USAF INDEPENDENT DUTY MEDICAL TECHNICAN (IDMT) PROTOCOLS

1. The approved USAF IDMT Protocols are attached and replace AFMAN 44-158, *The Air Force Independent Duty Medical Technician Medical and Dental Treatment Protocols* dated 1 Dec 99. The revised protocols are to be used by all IDMTs deployed, assigned or attached to Squadron Medical Elements (SMEs), Geographically Separated Units (GSUs), Mobile Medical Units (MMUs), or when performing IDMT duties.

2. The source document for these treatment protocols is the October 2007 4N0XX Career Field Education and Training Plan (CFETP). Further support provided by the 19th Edition of the Merck Manual, Up To Date physician's online resource, and the 2010 American Heart Association ECC guidelines. All medical practices listed were peer reviewed by clinical specialists in medicine, optometry, and dentistry.

3. Implementation is effective immediately. Any deviations from the protocols must be coordinated through the applicable MAJCOM SG or designated representative IAW AFI 44-103, *The Air Force Independent Duty Medical Technician Program*, 1 May 2005.

4. AFMAN 44-158 is rescinded 30 July 2013.

5. Questions concerning medical procedures or practical application of the protocols should be directed to the IDMT preceptor or IDMT Course Medical Director via the IDMT Course at DSN 420-3940/3945 or Comm. 210-808-3940/3945. Requests for changes and updates to the protocols should be directed to the IDMT Consultant, SMSgt Kevin Nies, DSN 969-9206 or Comm. 210-395-9206 or kevin.nies@us.af.mil.

//Signed/mjs/26 Jul 13//
MICHAEL J. STEINER, CMSgt, USAF
Aerospace Medical Service Career Field Manager
HAF/SG1N

FOR ALL AEROSPACE MEDICAL TECHNICIAN 4N0X1C AIR FORCE INDEPENDENT DUTY MEDICAL TECHNICIANS

PREFACE

This is the second compilation of medical instructions for use by Independent Duty Medical Technicians throughout the Air Force. In addition to fundamental changes in patient management since the 1999 version, this format is decidedly more user friendly and able to be amended on a regular basis. Nearly 200 disorders were added along with expanded appendices and references. Although the format has changed, the foundation for the protocols still rests on the career field's core documents, the 4N0X1 Career Field Education and Training Plan and AFI 44-103.

These protocols are derived from the section 10 of the 4N0X1 CFETP and contain up-to-date medical practices from the most current resources available. Three years of clinical experience with IDMTs, IDMT students, and adult active duty populations provided the margins of inclusion for over 350 listed disorders. Arranged by system, each disorder contains a brief description of the disease followed by signs, symptoms and treatment options. Recommended drugs are listed by type and name only. Since preceptor preferences vary and treatments change with the progress of the medical arts, dosing regimens are not included in this text. The intentional omission of this "technical data" provides greater flexibility of medical care and encourages the IDMT to research each medication immediately prior to prescribing it.

With research and portability in mind, the attachments to these protocols were also expanded. In an era of biological, chemical, and nuclear threats, these protocols address both common illnesses and potential weapons exposures. The attachments now include several tools and best practices taken from the field. They serve as reminders and quick references for materials once learned, but not easily committed to long-term memory. The Table of Contents of the online version of this manual is hyperlinked to the appropriate area of the text by keeping the "Control" key depressed while selecting a disorder.

Our attempt was to create a comprehensive, but practical guide for medical treatment. If you have questions regarding any of the content within these protocols, please contact the IDMT Program at DSN 420-3945/3940 or email the IDMT Superintendent on the IDMT Knowledge Junction (https://kx.afms.mil/kxweb/dotmil/kj.do?functionalArea=IDMT&iPlanetDirectoryPro=AQIC5wM2LY4SfcxB7sCjYgcpH35n3DUKDjqwWGf1UG0hGlc%3D%40AAJTSQACMDE%3D%23)

Our mission was to provide the AFMS with a flexible, updatable, useable resource that is well suited to meet the diverse and dynamic missions found throughout the AFMS. It's my belief that this revision of the USAF IDMT Medical and Dental Treatment Protocols contains the information and direction necessary to protect and preserve our nation's most sacred resource: our Soldiers, Sailors, Airmen and Marines.

//Signed/sbm/26 Jul 13//
SAMUEL B. MUNRO, LT COL, USAF, DO
Medical Director, IDMT Program

How to Use This Book

For IDMTs Deploying to Remote or Austere Locations
Print a hard-copy of this manual prior to departure in the event you do not have access to internet services. This handbook serves as a guide for you to provide medical care and treatment in a wide variety of Stateside and deployed locations. The formatting has been designed so that you can reduce and print the manual and all of the content will still be legible.

For Medical, Dental, and Ophthalmologic Preceptors
Independent Duty Medical Technicians have many strengths- they have excellent history and examination skills, as well as verbal and written skills to convey information accurately to you. They are well versed in acute illnesses and injuries, and have outstanding emergency skills and are capable of triaging, stabilizing and managing critical patients until the patient can be evacuated to higher level of care.

There are many things, however, that IDMT's do not know and should not be called upon to do. IDMT's have little training in the management of chronic conditions. For example: an IDMT knows well that acute dehydration is the most common cause of acute kidney failure in the healthy, deployed individual and therefore will aggressively hydrate oliguric patients with a mechanism for dehydration. They have minimal knowledge of chronic renal failure, and are unlikely to know that aggressive hydration in chronic renal failure can be dangerous. They have only basic knowledge of emergency obstetrics and have little knowledge of care of the pregnant patient. They have little experience with labs, x-rays, or other imaging since they rarely have access to these services when serving remotely.

Keep this guide handy when precepting. Use the Table of Contents as a quick-reference by organ system so you know what the IDMT will use for their differential diagnosis (and what they will omit). Review the treatment plans outlined in this manual so that you are familiar with IDMT treatment plans. You may ask your IDMT to deviate from these protocols anytime based on your clinical experience and preferences, however all deviations must be annotated in the patient's medical record.

For IDMT Students
You are the eyes of your preceptor. You must master your history and physical exam skills. Use these protocols as you begin to learn disorders; they will teach you what questions to ask and what exams to perform. Your treatment plans should come from these pages. For most disorders, several options for medicines are given as well as non-pharmacological treatments. Do not get trapped believing you need to prescribe a drug for every patient. These protocols are the basis for discussion with your preceptor, but your preceptor may change any or all of your diagnoses and plans. If that happens, remember that they are complementing the information you gathered and conveyed.

Acknowledgements

Author	Co-Author	Project Lead
Maj Sarah N. Vick, MD Former Medical Director, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	SMSgt Claude L. Weidow Former Superintendent, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	CMSgt Cleveland Wiltz Former 4N0X1 Career Field Manager Defense Health Headquartes Falls Church, Virginia
Contributor	**Contributor**	**Contributor**
Capt Roy D. Blount, MPAS, PA-C Course Supervisor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	MSgt Christopher (Shawn) Couch Instructor Supervisor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	Capt Breanne Kormendy, MPAS, PA-C Instructor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas
Contributor	**Contributor**	**Contributor**
TSgt Jennifer Cutlip Instructor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	TSgt Monti D. Cutlip Jr. Instructor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas	TSgt Michele S. Edmond Instructor, IDMT Program Medical Education and Training Campus Fort Sam Houston, Texas
Contributor	**Content Review- Medicine**	**Content Review-Dentistry**
LtCol Elizabeth A. Decker, FNP-BC, RNC Commander, 383 Training Squadron Medical Education and Training Campus Fort Sam Houston, Texas	Christopher A. Dyer, MD, FACP Assistant Professor, University of Texas Health Science Center, San Antonio Chief, Renal Section Audie L Murphy VA Hospital	LtCol James M. Kutner, DDS Program Director, Dental Assistant Program 381 Training Squadron Medical Education and Training Campus Fort Sam Houston, Texas
Content Review-Dentistry	**Content Review- Mental Health**	**Content Review- Optometry**
LtCol Elizabeth N. Kutner, DDS Flight Commander, Clinical Dentistry 359 Dental Squadron Randolph AFB, Texas	Maj Roseanne Visco, PhD Flight Commander, Mental Health 383 Training Squadron Medical Education and Training Campus Fort Sam Houston, Texas	LtCol Benjamin Franklin, OD Program Director, Ophthalmic Technicians Medical Education and Training Campus Fort Sam Houston, Texas

Table of Contents

CHAPTER 1: FEBRILE ILLNESS PROTOCOLS 19

ACUTE ALCOHOL WITHDRAWAL SYNDROME (DELERIUM TREMENS) 19
ANTHRAX 19
APPENDICITIS, RUPTURED APPENDIX 19
COMMUNITY ACQUIRED PNEUMONIA, BACTERIAL 20
FUNGAL PNEUMONIAS, MUCORMYCOSIS 20
ENCEPHALITIS 20
GASTROENTERITIS, YERSINIA ENTEROCOLITICA 20
GASTROINTESTINAL PARASITES, AMEBIASIS ("AMOEBIC DYSENTERY") 20
GASTROINTESTINAL PARASITES, SCHISTOSOMIASIS ("KATAYAMA FEVER") 20
GASTROINTESTINAL PARASITES, TRICHINOSIS 21
HEAT STROKE 21
HEMORRHAGIC FEVERS (EBOLA AND MARBURG) 21
HEPATITIS (VIRAL) 22
HIGH ALTITUDE PULMONARY EDEMA (HAPE) 24
INFLUENZA 24
HUMAN IMMUNODEFICIENCY VIRUS (HIV) SEROCONVERSION 24
KALA AZAR 25
LYME DISEASE 25
MALARIA 26
MENINGITIS 27
PELVIC INFLAMMATORY DISEASE (PID) 27
PLAGUE 28
PNEUMONIA, MYCOPLASMA (ATYPICAL PNEUMONIA) 29
MONONUCLEOSIS (EPSTEIN-BARR VIRUS) 29
PROSTATITIS (ACUTE) 29
PYELONEPHRITIS, ACUTE 30
RABIES 31
ROCKY MOUNTAIN SPOTTED FEVER 32
SECONDARY SYPHILIS 33
TOXIC SHOCK SYNDROME 33

CHAPTER 2: EMERGENCY MEDICINE PROTOCOLS 34

ANAPHYLAXIS 34
UNCONSCIOUS PATIENT UNKNOWN CAUSE 35
CARDIAC EMERGENCIES 37
STROKE 38
TRAUMA 38
MULTIPLE ORGAN SYSTEM TRAUMAS 38
SHOCK 40
CERVICAL SPINE TRAUMA 40
HEAD TRAUMA 41
TENSION PNEUMOTHORAX 42
OPEN PNEUMOTHORAX ("SUCKING CHEST WOUND") 42
FLAIL CHEST 43
ABDOMINAL TRAUMA 43
PREGNANT TRAUMA PATIENT 44

CHAPTER 3: EYE PROTOCOLS 46

CENTRAL RETINAL ARTERY OCCLUSION (CRAO) 46
CHALAZION 46
OCULAR CHEMICAL BURNS 47
OCULAR RADIATION BURNS 48
BACTERIAL CONJUNCTIVITIS 48
VIRAL CONJUNCTIVITIS 49
ALLERGIC CONJUNCTIVITIS 49
CORNEAL ABRASION 50
CORNEAL ULCER 51
DACROCYSTITIS 52
DETACHED RETINA 52
FOREIGN BODY 52
GLAUCOMA, PRIMARY OPEN-ANGLE 53
GLAUCOMA, ACUTE ANGLE 54
HORDEOLUM 54
HYPHEMA (ANTERIOR CHAMBER HEMORRHAGE) 55
ORBITAL CELLULITIS 55
PINGUECULA/PTERYGIUM 56
TRAUMA AND CONTUSIONS INVOLVING THE EYE 56
GLOBE TRAUMA 57
TRAUMA, PENETRATING WOUND OF THE EYE 58
SUBCONJUNCTIVAL HEMORRHAGE 59
UVEITIS 59

CHAPTER 4: EAR, NOSE AND THROAT PROTOCOLS 60

EAR 60
OTITIS EXTERNA 60
ACUTE PURULENT OTITIS MEDIA 60
BAROTRAUMA, EAR- SEE "ENVIRONMENTAL INJURY PROTOCOLS, BAROTRAUMA, EAR" 61
FOREIGN BODY IN EXTERNAL AUDITORY CANAL 61
RUPTURED OR PERFORATED EARDRUM 62
SEROUS OTITIS MEDIA 62
VERTIGO 63
NOSE 63
ACUTE VIRAL RHINOSINUSITIS 63
ALLERGIC RHINITIS 64
EPISTAXIS 66
THROAT 67
STREP THROAT 67
VIRAL PHARYNGITIS 68
MONONUCLEOSIS (EPSTEIN-BARR VIRUS) 68
MUCOCUTANEOUS LEISHMANIASIS 68
PERITONSILLAR ABSCESS 68
LARYNGITIS 69

CHAPTER 5: DENTAL PROTOCOLS 70

ACUTE NECROTIZING ULCERATIVE GINGIVITIS ("ANUG") 70
ALVEOLAR OSTEITIS ("DRY SOCKET") 70
APHTHOUS ULCER 71
AVULSED TOOTH 72
DENTAL CARIES 72
DISPLACED TOOTH 73
FRACTURED TOOTH OR CROWN 73
GINGIVITIS 74
ODONTOGENIC SPACE INFECTION 74
PERIAPICAL ABSCESS 74
PERICORNITIS 75
PERIODONTAL ABSCESS 76
REVERSIBLE PULPITIS 76
IRREVERSIBLE PULPITIS 76

CHAPTER 6: NEUROLOGIC PROTOCOLS 70

ACUTE ALCOHOL INTOXICATION 78
ACUTE ALCOHOL WITHDRAWAL SYNDROME (DELERIUM TREMENS) 78
ACUTE SPINAL CORD TRAUMA 79
CEREBROVASCULAR ACCIDENT 80
GULLAIN-BARRÉ SYNDROME 80
MIGRAINE HEADACHE 80
CLUSTER HEADACHE 81
TENSION HEADACHE 82
HEAD INJURIES 82
HERNIATED NUCLEUS PULPOSIS (HNP) WITH RADICULOPATHY 82
ENCEPHALITIS 83
MENINGITIS 83
SEIZURES 84
SLEEP DISORDERS, SHIFT WORKERS, AND "JET LAG"
TRAUMATIC BRAIN INJURY (“CONCUSSION”) 85

CHAPTER 7: CARDIOPULMONARY PROTOCOLS 87

ASTHMA, ACUTE EXACERBATION/ASTHMA “ATTACK 87
ASTHMA, MILD INTERMITTENT 87
ANTHRAX, INHALATIONAL 88
ACUTE BRONCHITIS 88
ACUTE RESPRATORY DISTRESS SYNDROME (ARDS) IN THE ADULT 88
BAROTRAUMA, PULMONARY- SEE “ENVORNMENTAL INJURY PROTOCOLS, BAROTRAUMA, PULMONARY” 89
COSTOCONDRITIS 89
DIPHERHERIA 89
FLAIL CHEST 89
HUMAN IMMUNODEFICIENCY VIRUS (HIV) SEROCONVERSION 89
HYPERTENSION 89
HYPERTROPHIC OBSTRUCTIVE CARDIOMYOPATHY 90
PERICARDITIS 91
PERTUSSIS (“WHOOPING COUGH”) 91
COMMUNITY ACQUIRED PNEUMONIA, BACTERIAL 92
PNEUMONIA, VIRAL 92
PNEUMONIA, MYCOPLASMA (ATYPICAL PNEUMONIA) 93
PNEUMONIAS, FUNGAL- ASPERGILLOSIS 94
PNEUMONIAS, FUNGAL- HISTOPLASMOSIS 95
PNEUMONIAS, FUNGAL- COCCIDIOMYCOSIS (“VALLEY FEVER”) 95
PNEUMONIAS, FUNGAL- MUCORMYCOSIS 95
OTHER PNEUMONIAS 95
SPONTANEOUS PNEUMOTHORAX 96
TRAUMATIC PNEUMOTHORAX/TENSION PNEUMOTHORAX 96
PULMONARY EDEMA 97
PULMONARY EMBOLI 97
STABLE ANGINA 98
TUBERCULOSIS 98

CHAPTER 8: GASTROINTESTINAL PROTOCOLS 100
ABDOMINAL TRAUMA- "SEE EMERGENCY MEDICINE PROTOCOLS, ABDOMINAL TRAUMA" 100
ANAL FISSURES 100
ANTHRAX, GASTROINTESTINAL 100
APPENDICITIS 100
BAROTRAUMA, GI 101
CHOLECYSTITIS 101
CONSTIPATION 101
GASTROENTERITIS, GENERAL 102
GASTROENTERITIS, VIRAL 103
GASTROENTERITIS, VIBRIO CHOLERAE (CHOLERA) 103
GASTROENTERITIS, CLOSTRIDIUM BOTULINUM (BOTULISM) 103
GASTROENTERITIS, ESCHERICHIA COLI ("TRAVELLER'S DIARRHEA") 104
GASTROENTERITIS, CLOSTRIDIUM DIFFICILE ("C-DIFF") 104
GASTROENTERITIS, STAPHYLOCOCCAL ENTERITIS 105
GASTROENTERITIS, CLOSTRIDIUM PERFRINGENS 105
GASTROENTERITIS, SHIGELLA ("BACCILLARY DYSENTERY" OR "SHIGELLOSIS") 105
SALMONELLA 106
GASTROENTERITIS, CAMPYLOBACTER JEJUNI 106
GASTROENTERITIS, YERSINIA ENTEROCOLITICA 107
GASTROINTESTINAL PARASITES, AMEBIASIS ("AMOEBIC DYSENTERY") 107
GASTROINTESTINAL PARASITES, SCHISTOSOMIASIS ("KATAYAMA FEVER") 108
GASTROINTESTINAL PARASITES, GIARDIASIS 108
GASTROINTESTINAL PARASITES, TAPE WORMS 108
GASTROINTESTINAL PARASITES, PINWORMS (ENTEROBIASIS) 109
GASTROINTESTINAL PARASITES, TRICHINOSIS 109
GASTROINTESTINAL PARASITES, TRICHURIASIS (WHIPWORMS) 109
GASTROINTESTINAL PARASITES, ASCARIASIS 110
GASTROINTESTINAL PARASITES, HOOKWORM 110
GASTRITIS 110
DYSPEPSIA ("HEARTBURN") 111
HEMORRHOIDS 112
HEPATITIS (VIRAL) 112
INGUINAL AND FEMORAL HERNIAS 112
MOTION SICKNESS 113
PANCREATITIS 114
PEPTIC ULCER DISEASE ("PUD") 114
PERITONITIS 115
ULCERATIVE COLITIS AND CROHN'S DISEASE (INFLAMMATORY BOWEL DISEASES) 115

CHAPTER 9: GENITOURINARY/GYNECOLOGICAL PROTOCOLS....117

CANDIDIASIS (CANDIDAL VULVOVAGINITIS).... 117
CHANCROID 117
CHLAMYDIA (NONGONOCCOCAL URETHRITIS OR "NGU") 118
ECTOPIC PREGNANCY.... 118
EPIDIDYMITIS 119
GENITAL HERPES.... 119
GENITAL WARTS (CONDYLOMA ACUMINATA) 120
GLOMERULONEPHRITIS (ACUTE).... 120
GONORRHEA.... 121
HIV INFECTION 122
HYDROCELE 122
MENSTRUAL IRREGULARITY 122
NEPHROTIC SYNDROME.... 123
ORCHITIS.... 123
PELVIC INFLAMMATORY DISEASE (PID).... 124
PREGNANCY.... 124
PROSTATITIS (ACUTE) 124
PYELONEPHRITIS, ACUTE.... 124
RENAL FAILURE, (ACUTE) 124
RHABDOMYOLYSIS 125
SYPHILIS 126
TESTICULAR MASS.... 127
TESTICULAR TORSION 127
TOXIC SHOCK SYNDROME.... 128
TRICHOMONIASIS.... 128
URINARY TRACT CALCULI ("KIDNEY STONES").... 129
URINARY TRACT INFECTION
BACTERIAL VAGINOSIS.... 130
VARICOCELE 130

CHAPTER 10: ENDOCRINE PROTOCOLS....131

DIABETES MELLITUS 131
DIABETIC KETOACIDIOSIS (DKA).... 131
HYPOGLYCEMIA 131
HYPOTHYROIDISM.... 132
HYPERTHYROIDISM 132

CHAPTER 11: INTEGUMENTARY PROTOCOLS 133
ACNE 133
ECZEMA (ATOPIC DERMATITIS) 134
CELLULITIS 135
CHICKEN POX
CONTACT DERMATITIS 136
CORNS
DERMATITIS MEDICAMENTOSA 137
FOLLICULITIS 137
HERPES SIMPLEX VIRUS 138
HERPES ZOSTER 138
IMPETIGO 139
INGROWN TOENAIL 140
LEISHMANIASIS 140
LYMPHANGITIS 141
LYME DISEASE 141
MOLLUSCUM CONTAGEOSUM 142
PEDICULOSIS (LICE) INCLUDING CAPITIS/CORPORIS/PUBIS 142
PSEUDOFOLLICULITIS BARBAE 143
PSORIASIS 144
ROCKY MOUNTAIN SPOTTED FEVER 144
SCABIES 144
SEBORRHEIC DERMATITIS 145
SKIN ABSCESSES AND FURUNCLES 146
SMALLPOX 147
SUBUNGUAL HEMATOMA 148
TINEA CAPITIS/CORPORIS/PUBIS/PEDIS 149
TINEA VERSICOLOR 151
URTICARIA 151
WARTS 152

CHAPTER 12: MUSCULOSKELETAL PROTOCOLS 154
INTRODUCTION TO MUSCULOSKELETAL INJURIES 154
LYME DISEASE 154
GOUT 154
SUSPECTED FRACTURES 154
DISLOCATIONS 155
LIGAMENT SPRAINS AND TEARS 155
TENDONITIS/MUSCLE STRAIN/TENDON RUPTURE 156
BURSITIS 156
NECK, BACK, AND TORSO DISORDERS 157
ACUTE NECK PAIN 157
COSTOCHONDRITIS 158
SUSPECTED RIB FRACTURE 158
LOW BACK PAIN 158
SHOULDER DISORDERS 160
ROTATOR CUFF STRAIN/TEAR 160
BRACHIAL PLEXUS INJURY 161
CLAVICULAR FRACTURE 161
ACROMIOCLAVICULAR (AC) SEPARATION 162
GLENOHUMERAL DISLOCATION 162
ADHESIVE CAPSULITIS 163
BICEPS TENDONITIS AND BICEPS TENDON RUPTURE 163
ELBOW DISORDERS 164
EPICONDYLITIS (MEDIAL OR LATERAL) 164
ELBOW DISLOCATION AND ELBOW FRACTURE 164
WRIST AND HAND DISORDERS 165
CARPAL TUNNEL SYNDROME 165
SCAPHOID "SNUFFBOX" FRACTURE 166
DORSAL GANGLION 166
DEQUERVAIN'S TENOSYNOVITIS 166
BOXER'S FRACTURE 167
HIP/PELVIS/UPPER LEG DISORDERS 167
ILIOTIBIAL BAND SYNDROME 167
TROCHANTERIC BURSITIS 168
PELVIS FRACTURE 168
FEMUR FRACTURE 169
KNEE/LOWER LEG DISORDERS 170
CALF STRAIN/RUPTURE 170
DEEP VENOUS THROMBOSIS (DVT) 170
PREPATELLAR BURSITIS 170
PATELLOFEMORAL SYNDROME 172
PATELLAR TENDONITIS "JUMPER'S KNEE" 172
SHIN SPLINTS/TIBIAL STRESS FRACTURE 173
EXERCISE INDUCED COMPARTMENT SYNDROME 174
MENISCAL TEAR 174
ACL/PCL/MCL/LCL SPRAINS AND TEARS 175
ANSERINE BURSITIS 176
ANKLE AND FOOT DISORDERS 176
ACHILLES TENDONITIS AND ACHILLES TENDON RUPTURE 176
ANKLE SPRAIN/FRACTURE 177
PLANTAR FASCIITIS 178
STRESS FRACTURE OF THE FOOT 179
PES PLANUS ("FLAT FEET") 179

TARSOMETATARSAL FRACTURE ("LISFRANC" FRACTURE) 180
LEG LENGTH DISCREPANCY 180

CHAPTER 13: MENTAL HEALTH/SUBSTANCE ABUSE PROTOCOLS 181
MOOD DISORDERS 181
DEPRESSION 181
ADJUSTMENT DISORDER WITH DEPRESSED MOOD 182
BIPOLAR DISORDER ("MANIC DEPRESSION") 183
HOMESICKNESS 184
ANXIETY AND DISSOCIATIVE DISORDERS 185
GENERALIZED ANXIETY DISORDER 185
PANIC DISORDER 186
OBSESSIVE-COMPULSIVE DISORDER ("OCD") 186
ANOREXIA NERVOSA AND BULIMIA NERVOSA 187
PHOBIC DISORDER 188
DISSOCIATIVE DISORDER 188
SOMATAFORM DISORDERS 189
CONVERSION DISORDER 189
SOMATIZATION DISORDER 190
PERSONALITY DISORDERS 190
PERSONALITY DISORDERS 190
ACUTE STRESS CONDITIONS 192
POST TRAUMATIC STRESS DISORDER ("PTSD") 192
COMBAT OPERATIONAL STRESS REACTION ("ACUTE STRESS DISORDER") 193
SUICIDAL OR HOMICIDAL GESTURES/ATTEMPTS 193
SUBSTANCE ABUSE 194
OPIATE (NARCOTIC) INTOXICATION/OVERDOSE 194
OPIATE WITHDRAWAL SYNDROME 195
BARBITURATES AND TRANQUILIZER INTOXICATION/OVERDOSE 196
BARBITURATE/BENZODIAZEPINE WITHDRAWAL SYNDROME 197
STIMULANT DRUG INTOXICATION/OVERDOSE 197
HALLUCINOGEN INTOXICATION/OVERDOSE/FLASHBACK 198
MARIJUANA INTOXICATION 198
ALCOHOL INTOXICATION AND ALCOHOL WITHDRAWAL 199

CHAPTER 14: POISONINGS, BITES, AND STINGS PROTOCOLS 200

MARINE LIFE 200
GENERAL CONSIDERATIONS 200
COELENTERATES/CNIDARIA 200
CONE SHELLS, SNAILS, OCTOPUS, AND NUDIBRANCH (SEA SLUGS) 202
VENOMOUS FISH STINGS 203
SEA SNAKES 204
SEA URCHINS, CORALS, AND STINGRAYS 205
ARACHNIDS AND INSECTS 206
BLACK WIDOW SPIDER BITE 206
BROWN RECLUSE SPIDER BITE 207
BEES, WASPS, HORNETS, AND YELLOW JACKETS 208
FIRE ANT STINGS 208
CENTIPEDES, MILLIPEDES, AND CATERPILLARS 209
SCORPIONS 210
LIZARD BITES 211
GILA MONSTER AND BEADED LIZARD 211
SNAKE BITES 212
VIPERS 213
CORAL SNAKES, SEA SNAKES, AND BOOMSLANGS 213
MISCELLANEOUS EXPOSURES 214
POISONOUS PLANTS 214
ANIMAL BITES AND RABIES 214

CHAPTER 15: ENVIRONMENTAL INJURY PROTOCOLS 216

HEAT STRESS INJURIES 216
HEAT SYNCOPE 216
HEAT CRAMPS 216
HEAT EXHAUSTION 217
HEAT STROKE 218
COLD STRESS INJURIES 219
TRENCH FOOT AND IMMERSION FOOT/SYNDROME 219
FROSTBITE 220
HYPOTHERMIA 221
BURNS 222
THERMAL AND RADIATION BURNS 223
CHEMICAL BURNS 225
ELECTRICAL INJURIES 226
ALTITUDE ILLNESSES 228
ACUTE MOUNTAIN SICKNESS (AMS) 228
HIGH ALTITUDE CEREBRAL EDEMA (HACE) 229
HIGH ALTITUDE PULMONARY EDEMA (HAPE) 230
BAROTRAUMA 230
BAROTRAOUMA, SINUS 231
BAROTRAUMA, DENTAL 231
BAROTRAUMA, GASTROINTESTINAL 231
BAROTRAUMA, PULMONARY 232

CHAPTER 16: OCCUPATIONAL EXPOSURE PROTOCOLS 233

LEAD POISONING 233
CARBON MONOXIDE POISONING (CO POISONING) 233
BENZENE 234
HYDROCARBONS (PETROLEUM DISTILLATES) 235
ORGANOPHOSPHATE POISONING 236
RADIATION SYNDROME 237

CHAPTER 17: BASIC LABORATORY INFORMATION 239

MACROSCOPIC URINE ANALYSIS (URINE DIPSTICK) 239
BASIC METABOLIC PANEL ("BMP", "CHEM 7") 240
COMPLETE BLOOD COUNT ("CBC") 242
TESTING FOR SEXUALLY TRANSMITTED INFECTIONS 243

ATTACHMENT 1: GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 247

WEBSITES 247
REFERENCES 247

ATTACHMENT 2: DRUG FORMULARY 251

ANTIHISTAMINE / DECONGESTANT / EXPECTORANT/ANTITUSSIVE MEDICATIONS 251
SYSTEMIC ANTI-INFECTIVE AGENTS - ALL REQUIRE PHYSICIAN PRESCRIPTION/CONSULTATION 251
TOPICAL ANTI-INFECTIVE AGENTS 252
ANALGESIC DRUGS 253
OVERDOSE / POISON ANTIDOTES 253
TOPICAL PREPARATIONS 254
OPHTHALMIC OINTMENTS AND SOLUTIONS 254
OTIC AND NASAL PREPARATIONS 255
BRONCHODILATORS / ASTHMA MEDICATIONS 255
SYSTEMIC STEROID MEDICATIONS 255
GASTROINTESTINAL DRUGS 256
CARDIAC DRUGS 256
PSYCHOLOGICAL/NEUROLOGICAL DRUGS 257
I.V. SOLUTIONS 257
IMMUNIZATIONS/MISCELLANEOUS 257

ATTACHMENT 4: PATIENT ENCOUNTER NOTETAKER 263

ATTACHMENT 5: HISTORY AND PHYSICAL EXAM GUIDE 265
SUBJECTIVE 265
PATIENT IDENTIFICATION 265
CHIEF COMPLAINT (CC) 265
HPI 265
ASSOCIATED SYMPTOMS 265
OTHER PROBLEMS 265
PMH 265
CURRENT MEDICATIONS 265
IMMUNIZATIONS 265
ALLERGIES 265
HABITS 265
HOBBIES 266
RELIGIOUS PREFERENCE 266
LIVING CONDITIONS 266
EXPOSURES 266
FAMILY HISTORY 266
ROS 266
OBJECTIVE 266
VITAL SIGNS 266
GENERAL IMPRESSION 266
HEAD 266
TMJ 266
EYES 266
EARS 266
NOSE 267
THROAT 267
NECK 267
RESPIRATORY 267
CARDIOVASCULAR 267
BREASTS 268
ABDOMEN 268
GENITALIA (MALE) 268
GENITALIA (FEMALE) 268
RECTAL 268
NEUROLOGICAL 268
SHOULDERS 269
ELBOWS 269
WRISTS 269
HANDS/FINGERS 269
SPINE 269
HIPS 270
KNEES 270
ANKLES 270
FEET/TOES 270
ASSESSMENT 270
PLAN 270
ADDITIONAL INFORMATION 271
COLORS 271
MASSES AND LESIONS DESCRIPTION GUIDE 271
GUIDE TO LOCATION AND DIRECTION WORDS 271

ATTACHMENT 6: MATERIALS NEEDED FOR PELVIC EXAM 272

ATTACHMENT 7: BIOLOGICAL AGENTS WITH POTENTIAL FOR WEAPONIZATION 273

CATEGORY A AGENTS 273
CATEGORY B AGENTS 273
CATEGORY C AGENTS 273

ATTACHMENT 8: CHEMICALS WITH POTENTIAL FOR WEAPONIZATION 274

NERVE AGENTS 274
TOXIC ASPHYXIANTS 274
PULMONARY IRRITANTS 274
BLISTER AGENTS (VESICANTS) 274

ATTACHMENT 9: HEAD-TO-TOE PHYSICAL EXAM 275

ATTACHMENT 10: QUICK REFERENCE GUIDES 281

EYE DISORDERS QUICK REFERENCE 281
EAR DISORDERS QUICK REFERENCE 282
NOSE AND THROAT DISORDERS QUICK REFERENCE 283
NEUROLOGIC QUICK REFERENCE 284
CHEST PAIN QUICK REFERENCE 285
SHORTNESS OF BREATH QUICK REFERENCE 286
ABDOMINAL PAIN QUICK REFERENCE 287
ABDOMINAL PAIN QUICK REFERENCE, CONTINUED 288
GENITOURINARY QUICK REFERENCE 289
PSYCHIATRIC QUICK REFERENCE GUIDE 290

HEALTH RECORD SF 600

Chapter 1: Febrile Illness Protocols

I. **ACUTE ALCOHOL WITHDRAWAL SYNDROME (DELERIUM TREMENS)-** See "Neurologic Protocols, Acute Alcohol Withdrawal Syndrome"

II. **ANTHRAX-** *A bacterial infection caused by Bacillus anthracis (gram positive bacilli) that releases damaging spores and toxins. This bacterium is very stable in the environment and has the potential for weaponization.*

A. Signs and Symptoms

1. Incubation period of 5-7 days
2. Three different clinical syndromes
   a. Cutaneous anthrax
      i. Evolving lesions that start as pruritic papules, enlarge to develop central vesicle that ruptures, leaving a painless, necrotic black eschar
      ii. Sores will have extensive surrounding erythema and lymphadenopathy
   b. Inhalation anthrax
      i. May come from weaponization or from breathing spores from contaminated wool or hides
      ii. Evolving respiratory syndrome that begins with 4-5 day prodrome of myalgia, fever, malaise
      iii. Rapidly progresses to hemoptysis, severe dyspnea/hypoxia and shock
   c. Gastrointestinal anthrax
      i. Comes from ingestion of undercooked meat contaminated with spores
      ii. Patient develops ulcers in gut with extensive edema
      iii. Illness starts with prodrome of fever, flushing and conjunctivitis followed by nausea, vomiting and abdominal pain a few days later
      iv. Rapidly progresses to ascites, hypovolemia and shock

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Patient must receive diagnostic testing and antibiotics
   b. Arrange hospital admission
   c. Discuss antibiotic therapy, such as oral or IV ciprofloxacin (Cipro®) or doxycycline (Vibramycin®)
2. **INITIATE OUTBREAK INVESTIGATION**
   a. Anthrax is primarily spread through exposure to the bacteria or spores, not person-to-person
   b. Identify source of outbreak
   c. Consider antibiotic prophylaxis for all potentially exposed personnel such as ciprofloxacin (Cipro®) or doxycycline (Vibramycin®)

III. **APPENDICITIS, RUPTURED APPENDIX-** See Gastrointestinal Protocols, Appendicitis"

IV. **COMMUNITY ACQUIRED PNEUMONIA, BACTERIAL**- See "Cardiopulmonary Protocols, Community Acquired Pneumonia, Bacterial"

V. **FUNGAL PNEUMONIAS, MUCORMYCOSIS**- *A devastatingly invasive fungal infection in patients with immunocompromise, trauma, or malnutrition.*

A. Signs and Symptoms

1. Inhaled spores cause fever, sinusitis, nasal congestion, purulent nasal discharge and sinus pain
2. Symptoms quickly progress to orbital areas, palate and brain. Symptoms include periorbital edema, proptosis, and blindness
3. Black eschar of necrotic tissue may be visible on nasal or palatine mucosa

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. This disorder is rapidly progressive
   b. Administer Amphotericin B (Amphocin®) if available
   c. Monitor vitals hourly until evacuation
   d. Administer oxygen and IV as required

VI. **ENCEPHALITIS**- See "Neurologic Protocols, Encephalitis"

VII. **GASTROENTERITIS, YERSINIA ENTEROCOLITICA**- See "Gastrointestinal Protocols, Gastroenteritis, Yersinia Enterocolitica"

VIII. **GASTROINTESTINAL PARASITES, AMEBIASIS ("AMOEBIC DYSENTERY")**- See "Gastrointestinal Parasites, Amebiasis"

IX. **GASTROINTESTINAL PARASITES, SCHISTOSOMIASIS ("KATAYAMA FEVER")**- *A blood fluke (worm) that inhabits the blood vessels and digestive tract of the host. Schistosomes invade directly through the skin when the host exposes skin to freshwater that is also inhabited by snails (the other host of the schistosome life cycle). Schistosomiasis occurs around the world in tropical and semi-arid areas including Iran, Iraq, the Nile River Valley, Japan, Philippines, Brazil, Puerto Rico, and much of Africa.*

A. Signs and Symptoms

1. Erythemic, pruritic papular rash, usually on feet or lower legs at the site of schistosome entry that occurs within one day of exposure
2. "Katayama fever" develops 2-8 weeks later with fever, malaise, dry cough, diarrhea, abdominal pain, weight loss and headache
3. Primary findings on exam are lymphadenopathy and hepatosplenomegaly
4. Urinary frequency, dysuria, and hematuria if schistosomes invade the urinary system

B. Treatment

1. Maintain fluid and electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Definitive treatment should be administered only after positive identification of the parasite
   b. Consider evacuation for further evaluation and management

C. Prevention

1. Do not allow personnel to wade in fresh water
2. Counsel personnel to avoid washing or splashing in fresh water

CLINICAL NOTE: Consider schistosomiasis in all GI disorders refractory to treatment if the disease is endemic. Schistosomiasis presents much like bacillary or amoebic dysentery.

X. **GASTROINTESTINAL PARASITES, TRICHINOSIS-** *Parasitic roundworm infection affecting humans. Infection occurs by ingesting contaminated raw or undercooked pork.*

A. Signs and Symptoms

1. History of eating raw or undercooked, smoked, or dried meat
2. Usually comes from pig, bear, walrus, boar, horse or wild carnivore meat
3. Incubation 5-15 days and lasts 5 weeks
4. Diarrhea, abdominal cramps, malaise, nausea and vomiting
5. Fever (low to high grade)
6. Muscle tenderness, especially with breathing, chewing or using large muscle groups
7. Edema and spasms
8. Periorbital edema, sweating, photophobia, fatigue
9. Dysphagia, dyspnea, coughing and mucous membrane hemorrhages
10. May display signs of meningitis, encephalitis, myocarditis, pneumonitis, nephritis and CN disorders (depending on where the worm travels)

B. Treatment

1. Establish IV access and correct fluid/electrolyte balance if indicated
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. If evacuation will be delayed and history includes eating undercooked meat in the past 2 weeks, consider antihelminthic medication and steroids
   b. Bed rest with bathroom privileges
3. Identify contaminated meat source and destroy/eliminate it

C. Prevention

1. Ensure all pork and wild meat is always cooked "well done"
2. Smoking, salting, or drying wild meat **WILL NOT** kill the organism

XI. **HEAT STROKE-** See "Environmental Injury Protocols, Heat Stroke"

XII. **HEMORRHAGIC FEVERS (EBOLA AND MARBURG)-** *Filoviruses are the most deadly pathogens in humans. Most infections come from unintentional exposure to an infected animal or physical contact with the body fluids of a sick patient. A major public health concern is that these viruses have the potential to be weaponized and the disease has the potential to be transmitted before diagnosis is made.*

A. Signs and Symptoms

1. Abrupt onset of illness with high fever, bradycardia, severe malaise
2. Muscle and joint pains
3. Petechiae in the conjunctivae, easy bruising
4. Confusion rapidly progressing to coma
5. Skin flushing, nonpruritic maculopapular rash on the upper body is very characteristic filoviruses

6. Spontaneous hemorrhaging with rapid progression to hypotension, shock, multiorgan failure, and death

B. Treatment

1. Airway, breathing, O_2, monitor, intubate if necessary
2. Initiate two large-bore IV's with Lactated Ringer's and bolus as necessary
3. "Spacesuit" isolation is not necessary; however strict respiratory, droplet, and contact precautions should be followed
    a. Private room, keep stethoscope with patient
    b. Restrict access to patient to as few people as possible and also keep them isolated as much as possible
    c. Gloves, gowns, face shields, surgical masks, eye protection, N-95 masks; remove all disposable gear while exiting the patient room and decontaminate shoes outside of room
    d. Dispose of sharps with extreme care
4. Investigate source of contamination/outbreak

XIII. HEPATITIS (VIRAL)- *Liver inflammation caused by specific viruses. On exam, liver can be tender, be enlarged, or be small with nodules on it. Careful physical exam is critical.*

CLINICAL NOTE: Many viruses are implicated in the production of acute hepatitis. The more common of these are hepatitis A, B, C (non-A, non-B), D (delta agent) and E (enteric hepatitis).
Hepatitis A is primarily spread through fecal-oral contamination, but may be food borne.
Hepatitis B/C are typically transmitted through contact with blood, semen, or vaginal fluids.
Hepatitis C (non-A, non-B) is responsible for most transfusion related cases of hepatitis.
Hepatitis D is seen in conjunction with hepatitis B (never alone) and frequently exacerbates chronic hepatitis B infections.
Hepatitis E (enteric hepatitis) is has been identified in waterborne outbreaks in many countries—including India, Burma, Afghanistan, Algeria and Mexico.

ACTION ALERT: Hepatitis A is extremely contagious. Strict isolation is not necessary; however standard contact precautions (private room and bathroom, gown, gloves) are required to prevent its spread. Thorough hand washing by caregivers handling contaminated utensils bedding or clothing is essential. Extreme caution must be exercised when using and disposing of sharps and other biohazards.

A. Signs and Symptoms

1. <u>Incubation period varies between viruses</u>
    a. Hepatitis A: 15-15 days
    b. Hepatitis B: 40-180 days (second most common)
    c. Hepatitis C: 20-120 days (transmitted by blood)
    d. Hepatitis D: 30-180 days
    e. Hepatitis E: 14-60 days
2. <u>The prodromal (pre-icteric) Phase</u>

    a. Anorexia, malaise, nausea, vomiting and fever or right upper quadrant abdominal pain
    b. Dark urine
3. Icteric or Jaundice Phase (Acute)
    a. Patient appears yellow but other symptoms begin to resolve
    b. Enlarged and tender liver
    c. Jaundice (yellow skin) and icterus (yellowing of the sclera) peaks in 1-2 weeks then fades
4. Recovery (convalescent) Phase
    a. Appetite returns
    b. Acute viral hepatitis usually resolves spontaneously after 4-8 weeks

B. Treatment

1. Wear gloves when handling utensils, soiled facial tissue (Kleenex®), feces, urine or any of the seven specified fluids requiring precautions
2. Use only single-use disposable syringes and needles
3. Good personal hygiene and controlled sanitation methods are a must
4. Patient should be assigned separate bathroom facilities
5. Provide tissue and paper bags for patient to use-dispose of used tissue in sealed plastic bags
6. Bed rest during the acute phase
7. No strenuous physical exertion, alcohol or hepatotoxic agents [i.e. sedatives, morphine, or acetaminophen (Tylenol®)] during the acute phase
8. All medications must be cleared by Preceptor
9. Provide liquid, soft, or bland nutritional diet, as tolerated
10. Patients or those in close contact should not be permitted to prepare food for others for a period of at least 2 weeks after onset of jaundice to preclude spreading the disease
11. IV normal saline bolus and/or maintenance if nausea and vomiting are pronounced or if oral intake is substantially reduced
12. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
    a. Although the IDMT would have the skills necessary to manage an uncomplicated case of acute viral hepatitis, it is unrealistic for the IDMT to do so given the lack of private facilities when deployed and the risk of transmission to the rest of the unit
    b. Patient should also receive lab workup and have lab available during recovery
13. If evacuation is delayed or if patient has any mental status changes, consider restricting protein in diet until clinical improvement

CLINICAL NOTE: IMMUNIZATIONS FOR HEPATITIS PROPHYLAXIS
Hepatitis A immunization for all active duty personnel. Hepatitis A Immune globulin (IG) should be routinely given to all close personal contacts of patients with hepatitis A. Recommended dose: 0.2 ml/kg IM. Hepatitis B immunization for health care personnel (including IDMT's) is also required. Hepatitis B immune globulin is administered for accidental needle stick exposure to Hep B infected blood, and for sexual partners of infected patients (within two weeks of sexual contact). Document in PHS, Immunization Record. Hepatitis C: No prophylactic treatment is available.

XIV. **HIGH ALTITUDE PULMONARY EDEMA (HAPE)-** See "Environmental Injury Protocols, High Altitude Pulmonary Edema"

XV. **INFLUENZA-** *Rapidly mutating virus that causes acute respiratory illness in epidemics worldwide, primarily during the winter and early spring.*

A. Signs and Symptoms

1. Fevers to 102°, malaise, weakness/fatigue, myalgias, night sweats, headache
2. Sneezing, nonproductive cough, sore throat, nasal discharge
3. Nausea/vomiting/diarrhea; may lose up to 5 lbs
4. In rare cases can progress to encephalitis, pericarditis or Guillain-Barré
5. Flushing of the face, throat, ear canals
6. Consider nasal washing for influenza and adenovirus, if available

CLINICAL NOTE: Acute HIV seroconversion mimics influenza and other viral syndromes and is often missed due to not considering it a possible diagnosis for viral syndrome.

B. Treatment

1. Supportive therapy
2. Increase fluid intake enough to urinate every 2 hours
3. Administer IV NS until able to tolerate PO or if patient is orthostatic
4. Control nausea with IV or PO diphenhydramine (Benadryl®) or ondansetron (Zofran®)
5. Use Acetaminophen (Tylenol®) PO or PR PRN for control of fever and myalgias
6. Avoid use of aspirin due to risk of potentially deadly Reye's Syndrome
7. Consider zanamivir (Relenza®) or oseltamivir (Tamiflu®) if available within first 36 hours of symptoms

C. Prevention

1. Ensure annual vaccination in active duty troops and high-risk populations
2. Respiratory isolation for anyone who is ill with above symptoms to prevent spread to healthy individuals
3. Avoid markets where live birds are present
4. Do not eat or handle birds or bird meat in countries where avian flu is endemic
5. Regular briefings on handwashing, coughing into sleeve, etc.

XVI. **HUMAN IMMUNODEFICIENCY VIRUS (HIV) SEROCONVERSION-** *A world pandemic, Human immunodeficiency virus is the causative agent of Acquired Immune Deficiency Syndrome (AIDS). Highest risk populations in the US include IV drug users, anal sex participants, individuals who have acquired other sexually transmitted infections, and healthcare workers with needle stick injury. According to the Center for Disease Control and Prevention, several African countries now have more than 25% of their population affected.*

A. Signs and Symptoms

1. Incubation period usually 2-4 weeks, but can be much longer
2. Fever- usually low grade but may reach 104°F

3. Nontender lymphadenopathy, pharyngitis, myalgias, arthralgias
4. Headache which can be severe with photophobia and stiff neck, or milder and diffuse, usually located retro-orbitally and worse with eye movements
5. Shallow oral ulcers and/or genital and/or anal ulcers that indicate route of exposure
6. Nausea/vomiting/diarrhea
7. May have maculopapular rash on upper chest, collar area, and face that can also include scalp, palms and soles
8. Often missed or confused with "The Flu"

B. Treatment

1. **IMMEDIATE ACTION**
   a. Supportive therapy
   b. Increase fluid intake enough to urinate every 2 hours
   c. Administer IV NS until able to tolerate PO or if patient is orthostatic
   d. Control nausea with IV or PO diphenhydramine (Benadryl®) or ondansetron (Zofran®)
   e. Use Acetaminophen (Tylenol®) PO or PR PRN for control of fever and myalgias
2. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
   a. Evacuation for further evaluation and testing is indicated if HIV seroconversion is suspected
   b. Use proper personal protective equipment when handling blood and body fluids
   c. Saliva, tears, and urine are at low risk for transmission
   d. Having a high index of suspicion is key. Collect a thorough history including sexual contacts and sexual trauma

C. Prevention

1. Many individuals will come to clinic seeking sexually transmitted infection screening after a high-risk sexual encounter; screening HIV blood test commonly performed in the military will not turn positive until after seroconversion
2. Patients seeking sexually transmitted infection screening need to return to clinic for HIV test 6 months after exposure
3. Counsel patients to avoid high risk sexual exposure
4. Encourage use of condoms every time, including with anal and oral sex
5. As a healthcare worker, avoid needle stick injury by proper use and disposal of sharps and by wearing all appropriate personal protective equipment

XVII. **KALA AZAR-** See "Integumentary Protocols, Leishmaniasis"

XVIII. **LYME DISEASE**- *Spirochete infection caused by Borrelia burgdorferi transmitted by Ixodes ticks. Disease has three clinical phases: early localized, early disseminated, and late persistent.*

A. Signs and Symptoms

1. <u>Stage 1 (early localized)</u>
   a. Flat or slightly raised red lesion at site that starts about 1 week after tick bite and expands over several days

b. Red lesion may have a typical "bull's eye" ("target") appearance called "Erythema Migrans" due to central clearing, but may also have central intensification or a homogeneous
c. "Target" lesion commonly seen in the groin, legs, or axillae
d. Skin lesion may be associated with viral-like illness (myalgias, arthralgias, headache, fever, fatigue)
e. Skin lesions resolve in 3-4 weeks

2. Stage 2 (early disseminated)
   a. Occurs within weeks of original infection and is marked by multiple skin lesions, malaise, fatigue, fever, headache, neck pain, and generalized aching
   b. May develop cardiac abnormalities such as myocarditis or heart blocks
   c. May develop neurologic abnormalities such as signs of meningitis or facial nerve palsy
3. Stage 3 (late persistent)
   a. Months to years after initial infection marked by single or multiple large joint pains and encephalopathy
   b. If disease wasn't diagnosed in stage 1 or 2 it is frequently missed in stage 3; have a high index of suspicion for any chronic large joint pains

B. Treatment

1. If stage I, CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Treat Stage 1 (erythema migrans) with doxycycline (Vibramycin®)
   b. No labs required if erythema migrans is seen
2. If Stage 2 or 3, CONTACT PHYSICIAN PRECEPTOR- EVACUATION
   a. Will require further evaluation and possible specialist consultation
   b. Consider starting doxycycline if evacuation will be delayed

CLINICAL NOTE: Tick must feed for at least 36 hours to transmit Borrelia burgdorferi. Prevent Lyme Disease by checking for and removing ticks nightly.

- Emphasize prevention-cover exposed skin with long sleeve shirts, pants tucked into socks, light colored clothing with repellant. Inspect for ticks nightly in regions where exposure is possible.
- Remove tick as soon as possible. Use fine tipped tweezers at its mouthparts or gloved fingers until it releases its hold (do not pull or squeeze on body). Pull straight up and firmly. Do not jerk or twist. If mouthparts remain in the skin, leave them alone and they will be expelled naturally.
- Do not use petroleum jelly, hot match, fingernail polish, or isopropyl alcohol. These will irritate the tick and cause further regurgitation of tick contents into host.
- Disinfect the skin with soap and water after tick has been removed.

XIX. **MALARIA-** *Relapsing, febrile illness caused by multiple different species of Plasmodium parasites that are transmitted through the bite of an infected Anopheles mosquito. The parasites infect the liver and blood cells. Malaria is highly infective and should be considered in the differentials of anyone who has a fever and has been in or to a malaria-endemic region within the past year.*

A. Signs and Symptoms

1. Feverish paroxysms occur at irregular intervals
2. Tachycardia, tachypnea, chills, malaise, fatigue, diaphoresis, cough
3. Nausea, anorexia, vomiting, abdominal pain, diarrhea
4. Headache, arthralgias, myalgias
5. Palpable, tender, enlarged spleen
6. May have mild jaundice
7. May progress to "complicated malaria" where blood cells get stuck causing extensive circulatory, respiratory, and neurologic destruction leading to death
8. Diagnosis made based on peripheral blood smear or rapid diagnostic testing

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
   a. Anyone having relapsing fevers and the symptoms above should be evacuated for further evaluation
   b. Supportive therapy until evacuation including IV fluids if necessary
2. Medications are tailored to the specific region where the illness was acquired due to the high incidence of resistant strains
3. Treatment often requires two or more medications

C. Prevention

1. Wear long sleeves and pants tucked into socks/boots to reduce amount of skin exposure
2. Avoid outdoor activity between dusk and dawn (when mosquitoes feed)
3. Wear insect repellent; 50% DEET is effective for 3-6 hours
4. Sleep under bed nets treated with permethrin
5. Follow local guidance regarding prophylactic medications
6. Eliminate standing water areas in camp

XX. **MENINGITIS**- See "Neurologic Protocols, Meningitis"

XXI. **PELVIC INFLAMMATORY DISEASE (PID)**- *Acute infection of the uterus and/or ovaries caused by an untreated or incompletely treated sexually transmitted infection.*

A. Signs and Symptoms

1. Bilateral lower abdominal pain worse during intercourse or after menses; pain with jarring movements, may have abnormal spotting or bleeding
2. Risk factors include prior history of STIs, intercourse without condoms and douching
3. Diffuse tenderness worse in lower quadrants, rebound tenderness and decreased bowel sounds
4. Severe cervical motion tenderness and/or tenderness around the ovaries; may have purulent endocervical discharge (due to untreated STI)
5. Fever
6. Lab: Urine HCG for pregnancy

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access and give fluid bolus if indicated
   b. Administer analgesic such as aspirin or acetaminophen (Tylenol®)
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

a. PID will present with a clinical syndrome similar to acute appendicitis or other surgical abdomen and therefore immediate further evaluation is required
b. If patient is febrile and evacuation is delayed, consider cefoxetin (Mefoxin®) and doxycycline (Vibromyin®) until patient is afebrile or transferred
c. If patient is afebrile initially, or has been afebrile for 24 hours and evacuation is delayed, consider PO therapy of levofloxacin (Levaquin®) and metronidazole (Flagyl®)

ACTION ALERT: It is essential to rule out ectopic pregnancy (and other causes of a surgical abdomen) on all female patients with severe abdominal pain and elevated temperature.

XXII. **PLAGUE**- *A potentially fatal infection caused by the gram negative coccobacillus, Yersinia pestis. The infection has been responsible for many pandemics through the centuries, including the "Black Death" of the 14th century, during which 1 out of every 3 citizens in England and some countries in mainland Europe died of the infection. It is transmitted through the bite of infected fleas. In this country deaths from plague still sometimes occur in the Southwestern United States, where some prariedog populations still carry infected fleas. This organism has the potential for weaponization.*

A. Signs and Symptoms

1. Bubonic Plague- sudden onset fever, chills, weakness, headache, followed by intense pain and swelling of the lymph nodes and redness of the overlying skin (called buboes, most commonly in the inguinal lymph nodes; over several days buboes progress to septic shock and death
2. Septicemic plague- very rare; same initial presentation but progresses directly to septic shock and death without bubo formation
3. Pneumonic plague- Acquired by inhalation of respiratory secretions or aerosolized droplets from infected animals; can also come from contact with blood from an infected bubo; presents with sudden onset dyspnea, high fever, pleuritic chest pain, cough with grossly bloody sputum that progresses rapidly to death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, circulation, O_2 monitor
   b. Strict isolation and droplet precautions
   c. Limit number of individuals having contact with patient
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. If known contact with rodent population or recent flea bites and the above symptoms occur, do not delay treatment with antibiotics
   b. Consider gentamicin (Garamycin®) if available; may use doxycline (Vibramycin®)

C. Prevention

1. Avoid contact with animal carcasses or skins
2. Maintain rodent control and flea control in worksites and living quarters

3. Promote appropriate insect repellant measures such as proper washing of ABU uniform and application of topical insect repellants

XXIII. **PNEUMONIA, MYCOPLASMA (ATYPICAL PNEUMONIA)- See** "Cardiopulmonary Protocols, Pneumonia, Mycoplasma"

XXIV. **MONONUCLEOSIS (EPSTEIN-BARR VIRUS)-** *Self limiting viral illness characterized by a triad of fever, sore throat, and lymphadenopathy. It is often difficult to distinguish from strep throat in the absence of throat culture or blood tests.*

A. Signs and Symptoms

1. Malaise, headache, sore throat
2. Low grade fever for 1-3 days prior to onset of sore throat, and lymphadenopathy
3. Profound fatigue suggestive of diagnosis
4. May have white, gray-green or necrotic exudates in posterior oropharynx
5. Splenomegaly is not always present, but if present, is suggestive of diagnosis

CLINICAL NOTE: Splenomegaly is present in 50-60% of cases, but not diagnostic of Epstein-Barr Virus. Spleen must be checked in every patient that has EBV. Enlarged spleens have high incidence of life-threatening splenic rupture. Due to the overlap of symptoms with strep throat, antibiotics are often mistakenly given to patients with EBV, which results in the development of an urticarial, maculopapular rash in 70-90% of cases.

B. Treatment

1. Quarters (up to several weeks) may be necessary when severe fatigue is present
2. If splenomegaly is present or if blood test is positive, limit physical activities that will impact the spleen for a minimum of 1 month. Activities include, but not limited to, contact sports, bumpy rides, load bearing, heavy lifting, etc.
3. Increase fluid intake
4. Bland or liquid diet. Advance as tolerated
5. Acetaminophen (Tylenol®) or NSAIDs for fever, malaise and body aches

C. Prevention

1. Hand washing education
2. Frequent decontamination of shared spaces, doorknobs, equipment, etc.
3. Avoid shared eating/drinking utensils

CLINICAL NOTE: EBV is often shed from asymptomatic patients for several months after symptoms resolve.

XXV. **PROSTATITIS (ACUTE)-** *An inflammatory condition of the prostate that may be acute or chronic. Often occurs with gonorrhea or chlamydial infection.*

A. Signs and Symptoms

1. Urinary retention/hesitation, dysuria, perineal or suprapubic pain
2. Fever, chills and myalgias
3. Nausea, vomiting, anorexia

4. Warm, tender prostate
5. Lab: Macroscopic UA positive for WBCs, nitrites and RBCs

CLINICAL NOTE: Symptoms of prostatitis resemble UTI's. UTI's are uncommon in men and symptoms are almost always caused by prostatitis.

ACTION ALERT: Avoid excessive manipulation of the prostate during exam. Septicemia may result.

B. Treatment

1. Control pain with aspirin (Bayer®)
2. Sitz baths 2-3 times/day (for 1 hour) may relieve pain and spasm (See "Gastrointestinal Protocols, Hemorrhoids")
3. Ensure adequacy of hydration, but do not force fluids
4. Observe for urinary retention due to swelling of prostatic tissue
5. Suspect obstructed urethra if patient does not void for 6 hrs or if they complain of increasing bladder discomfort with inability to void

ACTION ALERT: If urinary obstruction is suspected, **do not catheterize!** Refer the patient to a higher level of care immediately.

6. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. If over 35 years, consider Trimethroprim-Sulfamethoxazole (Septra®), Ciprofloxacin (Cipro®), or tetracycline (Tetra®)
   b. Treat with antibiotics for 1 month if symptoms have lasted longer than one month
   c. If under 35 years old, consider Ceftriaxone (Rocephin®) and Doxycycline (Vibramycin®)
   d. If allergic, call preceptor for alternative regimen
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if symptoms persist

XXVI. PYELONEPHRITIS, ACUTE- *An infectious inflammatory disease of the kidney usually caused by the spread of a urinary tract infection. Usually caused by Escherichia coli (gram negative bacilli).*

A. Signs and Symptoms

1. Dysuria, urgency and frequency
2. Fever, flank pain, shaking chills
3. Pronounced CVA tenderness
4. Associated with nausea, vomiting and diarrhea
5. Lab: Macroscopic UA positive for WBCs, leukocyte esterase, bacteria and possibly RBCs

B. Treatment

1. Maintain hydration: 1-2 Liters Lactated Ringer's followed by normal saline if not able to tolerate adequate PO intake
2. Administer acetaminophen (Tylenol®) PRN for fever

3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION
    a. May treat as outpatient with preceptor approval
    b. Consider levofloxacin (Levaquin®), ciprofloxacin (Cipro®) or trimethoprim-sulfamethoxazole (Septra®)
    c. Administer analgesics for relief of severe pain, meperidine (Demerol®), as needed
    d. If not tolerating PO, consider immediate transfer
4. Reevaluate patient daily to ensure clinical improvement
5. Evacuate and arrange for hospital admission if febrile or fever not controlled with medications
6. Monitor and record urine output
7. Consider promethazine (Phenergan®) if needed for nausea and vomiting
8. If patient appears unstable and immediate transfer is not possible, consider administering ceftriaxone (Rocephin®) + anti-pseudomonal penicillin such as ticarcillin (Ticar®) or pipercillin (Pipracil®) as directed by preceptor

ACTION ALERT: Clinical improvement does not necessarily indicate cure of infection. Continue to monitor closely.

XXVII. RABIES- *Rabies is a potentially fatal virus that is transmitted when the patient is bitten by an infected dog, raccoon, bat, skunk, fox, or coyote. The virus travels along nerves to the brain and spinal cord; once clinical manifestations occur it will universally progress to neurologic deficits, encephalitis, and death. Assume any animal that bites has rabies unless it is a pet and its vaccination status is well documented.*

A. Signs and Symptoms of Rabies

1. Incubation period- lasts days to months after bite or exposure; bites close to the head will have shorter incubation periods than bites on the extremities
2. Prodromal period- flu-like symptoms lasting up to a week that include nausea, malaise, anorexia, sore throat, headache, irritability, and low grade fever; percussion of muscular areas will cause mounding up of tissue at the percussion site
3. Encephalitic rabies- fear of water, (hydrophobia) painful pharyngeal spasms upon encountering a draft of air, excessive salivation, fluctuating consciousness, cardiac arrhythmias lasting 2-7 days
4. Paralytic rabies- less common than encephalitic rabies, it resembles Guillain-Barré syndrome with weakness progressing to flaccid paralysis and respiratory failure; it is different than Guillain-Barré in that it is accompanied by persistent fever, intact sensation, and bladder dysfunction
5. Coma- both encephalitic and paralytic rabies progress to coma, flaccid paralysis, and respiratory and vascular collapse

B. Treatment

1. Apply direct pressure to control bleeding
2. Cleanse superficial wounds with soap and water; if tissue damage is extensive inject local anesthesia into wound prior to cleansing and clean with gently pressurized sterile water; remove any devitalized tissue prior to closure

3. The IDMT should not close bite wounds from cats or humans, puncture wounds, bites > 12 hours old, bites associated with crush injury, or bites on the hands or feet
4. Inject Rabies Immune Globulin (RIG) into wound if available
5. Survival from rabies is dependent on being vaccinated prior to the appearance of symptoms; therefore it is critical that patient and animal be evaluated for rabies whenever an animal bite from a potential carrier occurs
6. Every effort should be made to capture/kill the animal so that it can be evaluated for presence of rabies virus
7. In the absence of the animal, the patient may be required to undergo lengthy and painful treatments such as HDCV (Human Diploid Cell Vaccine)
8. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION for further evaluation and management

XXVIII. **ROCKY MOUNTAIN SPOTTED FEVER**- *Potentially lethal infection occurring 2-14 days after bite from an infected tick transmitting the gram-negative bacteria Rickettsia Rickettsii. Infected ticks are found throughout the United States, Canada, Mexico, and Central and South America.*

A. Signs and Symptoms

1. About 1 week after tick bite patient will develop fever, chills, headache, nausea, vomiting, myalgias, restlessness, insomnia and irritability
2. Faint macules that progress to maculopapules and then petechiae appear between 2nd and 6th days after bite
3. This characteristic rash appears first on wrists and ankles and spreads centrally to arms/legs/trunk over 2-3 days following onset of fever and malaise
4. Rash includes palms and soles
5. Be alert in dark-skinned individuals because rash may not be visible or may be very difficult to see
6. Up to 10% of patients do not develop rash or petichiae
7. Advanced disease includes abdominal pain, cough, bleeding, confusion, seizures, and death

CLINICAL NOTE: Rocky Mountain Spotted Fever has a high likelihood of successful treatment if it is diagnosed in the first week after tick bite. According to the Center for Disease Control data, approximately 25% of patients die when treatment is delayed until the second week. Don't forget to include infectious diseases in your differentials!

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish large-bore IV and rehydrate with normal saline if indicated
   b. Treat nausea with promethazine (Phenergan®) IM or ondansetron (Zofran®) IV
2. **CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND IMMEDIATE EVACUATION**
   a. Doxycycline (Vibramycin®) orally or IV is the mainstay of treatment

b. Do not use any other antibiotics unless directed by preceptor; sulfa drugs may worsen infection and most other antibiotics increase mortality compared to doxycylcine (Vibramycin®)

C. Prevention

1. Wear protective clothing and insect repellants
2. For successful removal of ticks, see "Lyme Disease" clinical note on tick removal procedures

XXIX. SECONDARY SYPHILIS- See "Genitourinary Protocols, Syphilis"

XXX. TOXIC SHOCK SYNDROME- *Shock caused by infection with Staphylococcus aureus (gram positive coccus) that has entered the bloodstream and released toxins. Often caused by use of super-absorbent tampons or retained vaginal foreign bodies. In males, other staphylococcal infections may be the cause. This toxin has the potential to be weaponized.*

A. Signs and Symptoms

1. Rapidly progressive syndrome occurring in 1-3 days
2. Fever, orthostatic hypotension, and characteristic diffuse, red rash resembling sunburn that includes palms and soles of feet
3. Profuse, watery diarrhea, shock, decreased urine output, cyanosis, edema of the extremities
4. Neurologic symptoms that include confusion, agitation, irritability, hallucinations, decreased level of consciousness
5. Skin desquamation (blistering and sloughing skin) and seizures are late and ominous findings
6. Lab: Urine HCG

B. Treatment

1. **IMMEDIATE ACTION**
   a. Remove retained foreign object (nasal packing can also cause TSS)
   b. Initiate IV with Lactated Ringer's and bolus as indicated
   c. Patient may require 10-20 Liters of fluid per day-discuss with preceptor
   d. Monitor vital signs
2. **CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND IMMEDIATE EVACUATION**
   a. Consider antibiotic treatment with clindamycin (Cleocin®)
   b. Seizure precautions

Chapter 2: Emergency Medicine Protocols

I. **ANAPHYLAXIS-** *Potentially fatal reaction to bite/sting, medication, or food due to massive immune response. Occurs within minutes to several hours following exposure.*

A. Signs and Symptoms

1. Skin/mucosa- hives, itching, flushing, swelling of lips, tongue, uvula, periorbital edema (swelling around eyes)
2. Respiratory- wheeze, dyspnea, tachypnea, hypoxemia, reduced peak flow, sensation of throat closure, tightness, or itching
3. Cardiovascular- hypotension, syncope, dizziness, tachycardia, may have incontinence
4. Gastrointestinal- nausea, vomiting, cramping abdominal pain

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, circulation, O_2, monitor, expose patient
   b. Place patient supine with legs elevated
   c. Epinephrine 1:1000 aqueous solution (which is the same as epinephrine 1 mg/mL) give 0.5 mL IM; repeat every 5 minutes until symptoms improve
   d. Obtain 2 large-bore IVs with normal saline at maintenance rate if no hypotension; if hypotension is present administer 1-2 L normal saline bolus over 1 hour
   e. If no improvement in anaphylactic symptoms after 3 doses of IM epinephrine, prepare an epinephrine infusion
      i. Using 1:10,000 epinephrine solution give 2 mL per minute IV/IO
      ii. Titrate to response

ACTION ALERT: The intramuscular administration of epinephrine is the single most important action in the treatment of an anaphylactic reaction. IT MUST BE PERFORMED IMMEDIATELY. If epinephrine 1:10,000 is not available, may dilute epinephrine 1:1000 for use in IV.

1. Put 0.1 mL of 1:1000 epinephrine into 10 mL of normal saline; this solution contains 0.1 mg of epinephrine
2. Give 2 mL per minute IV or IO, titrated to response

   f. If patient develops stridor or complains of difficulty getting air or sensation of throat closure, consider immediate intubation prior to the development of laryngeal edema
   g. If unable to insert endotracheal tube, perform emergency cricothyroidotomy (may also use ET tube for epinephrine if no other route available)
   h. If wheezing is present but airway is still intact, administer albuterol 5 mg diluted in 3 mL normal saline inhaled PO via continuous nebulizer; may repeat every 10 minutes as needed for maximum of 3 doses (if

nebulizer is not available, administer 2 puffs albuterol (Ventolin®) PO via metered dose inhaler or HFA inhaler with spacer; repeat every 5 minutes as needed for maximum of 3 doses [six puffs total])

i. If itching, flushing, or rash is present, consider diphenhydramine (Benadryl®) 25-50 mg IV single dose
j. For gastrointestinal symptoms, consider ranitidine (Zantac®) 50 mg IV or famotidine (Pepcid®) 20 mg IV single dose
k. Monitor for hypothermia and keep patient warm

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Reassess vital, temperature, and breath sounds every 5 minutes
   b. If BP < 90/50 initiate rapid volume expansion with normal saline or preceptor preference; titrate to keep BP > 90/50 (rapid expansion of intravascular fluid volume is necessary due to large amounts of fluid escaping to the extra-vascular spaces)
   c. If not responding to epinephrine, consider dopamine (Intropin®) continuous infusion IV per Preceptor
   d. Consider methylprednisolone (Medrol®) 125 mg IV or dexamethasone (Decadron®) 20 mg IV per Preceptor
3. If evacuation is delayed, most cases of anaphylaxis will resolve within 12 hours with appropriate treatment
   a. Once acute symptoms have resolved, administer diphenhydramine hydrochloride (Benadryl®) 25 mg PO every six hours for one week
   b. Monitor for "biphasic reaction", or new onset anaphylaxis within 72 hours of initial episode
   c. Even if symptoms have fully resolved, patient should still be evaluated by physician

II. **UNCONSCIOUS PATIENT UNKNOWN CAUSE-** *In general, consciousness requires three main components: intact neural pathways, oxygen reaching the brain, and sugar reaching the brain.* Neural pathways *can be disrupted by head trauma, electrical injury, swelling of brain or meninges, infection, toxins, metabolic abnormalities, environmental temperature extremes, fever, seizures, or drugs/medications.* Oxygen delivery *to the brain can be disrupted by dehydration/hypotension, shock, cardiac abnormalities, inadequate oxygen intake (bronchospasm, decreased respiratory drive due to sedation or chest wall pain, reduced oxygen levels at altitude, pneumothorax, etc), inadequate perfusion of the lungs (chest trauma, pulmonary embolism), or disruption in the oxygen exchange barrier (pneumonia, pulmonary edema, acute respiratory distress syndrome, etc). Inadequate* glucose delivery *to the brain can be caused by dehydration/hypotension, shock, inadequate glucose intake (nausea/vomiting, not eating enough to meet body's energy demands, etc) or inadequate absorption (diabetes, gastroenteritis, diarrhea, weight loss supplements, laxatives, etc). These are just a few examples and not meant to be an exhaustive list. Use clues in patient's environment to help determine cause.*

A. Signs and Symptoms

1. Bystander discovers unresponsive patient
2. Be alert to interventions bystanders may have performed

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish unresponsiveness

b. Initiate C-spine precautions
c. Airway, breathing, circulation, O_2, monitor; initiate CPR per ACLS guidelines if indicated
d. Establish baseline vitals signs and treat any abnormalities found
e. Estimate Glasgow Coma score; consider intubation if score <8

f. Initiate two large-bore IV lines, bolus 1 Liter normal saline; if IV access can not be obtained may use IO access

CLINICAL NOTE: The Glasgow Coma Scale (GCS) is used to assess level of neurologic disability. Consider intubation for GCS < 8. A score of 15 (max score) indicates patient is awake, interactive, and oriented. The minimum score is 3.

I. Eye opening (max score = 4)
   a. Spontaneous eye opening = 4
   b. Eyes open when someone speaks = 3
   c. Eyes open to pain = 2
   d. No response = 1

II. Verbal response (max score = 5)
   a. Oriented = 5
   b. Confused = 4
   c. Inappropriate = 3
   d. Incomprehensible = 2
   e. No response = 1

III. Motor response (max score = 6)
   a. Obeys commands = 6
   b. Localizes to pain = 5
   c. Withdraws from pain = 4
   d. Flexion to pain = 3
   e. Extension to pain = 2
   f. No response = 1

g. Obtain baseline glucose reading if possible; administer 50 mL of 50% glucose solution in IV if glucose reading not available or if glucose < 200 mg/dL
h. If no response to above measures, administer naloxone hydrochloride (Narcan®) 2 mg IV/IM/subcutaneous
i. Obtain venous blood sample for portable blood analyzer (iSTAT®) to evaluate for electrolyte abnormalities (CHEM 7) if available
j. Obtain arterial blood gas sample for portable blood analyzer (iSTAT®) to check for respiratory and metabolic abnormalities (ABG) if available
k. Expose patient; perform a thorough secondary survey
l. When evaluating patient's eyes, if resistance or response to passive eye opening occurs, patient is not unconscious

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

C. Initial Investigation to Determine Cause (complete once patient is stabilized)

1. Gather information from patient's peers
    a. Anything abnormal in behavior, movements, or mood?
    b. Any complaints in past 24 hours?
    c. How long ago since was patient last observed to be normal?
    d. Any recent medical visits?
    e. Any chronic medical conditions?
    f. Does the patient normally take any medications, herbs or supplements?
    g. What were they doing before they became unconscious?
2. Gather information from patient's location
    a. Any environmental hazards? (electricity, chemicals, fall risks, etc)
    b. Any fumes in environment?
    c. Anyone else sick in the unit?
3. Gather information from patient's living area (requires Commander's approval)
    a. Any herbs, supplements, or medication bottles?
    b. Any chance of accidental or intentional overdose?
    c. Any illegal drugs or paraphernalia?
    d. Any evidence of severe depression or discontent?

III. CARDIAC EMERGENCIES- *Includes myocardial infarction, symptomatic bradycardia, tachycardias, acute coronary syndrome, and arrhythmias.*

A. Signs and Symptoms

1. Chest pain, palpitations, "fluttering" in chest, uncomfortable chest pressure, fullness, squeezing sensations in center of chest
2. Pressure or pain may radiate to shoulders, neck, arms, jaw, or back
3. May or may not have pain; may also manifest as lightheadedness, dizziness, fainting, sweating, nausea, or vomiting
4. Unexplained sudden shortness of breath
5. Diaphoresis (sweating), cyanosis

ACTION ALERT: A conscious cardiac patient can rapidly become an unconscious cardiac patient. It is critical to have quick access to the following equipment throughout patient care and evacuation: AED or monitor that can deliver shocks, advanced airway equipment and supplies, and suction equipment and supplies.

B. Treatment

1. Follow American Heart Association's current Advanced Cardiac Life Support (ACLS) treatment algorithms
2. Healthy active duty patients usually have a stressor that triggered their cardiac abnormality
3. Consider "H's and T's"- hypovolemia, hypothermia, hypoxia, hydrogen ion (acidosis), hypoglycemia, hyperkalemia, hypokalemia, toxins, trauma, cardiac tamponade, tension pneumothorax, thrombosis (cardiac or pulmonary)

C. CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION

ACTION ALERT: The Automated External Defibrillator (AED) should never be applied to a conscious patient unless the AED has cardiac monitoring capability. An AED is only indicated for the pulseless unconscious patient.

IV. **STROKE-** *Blood clot or hemorrhage causing sudden loss of focal brain functioning while other brain functions remain intact. Generally, focal neurologic deficits WITH headache or nausea/vomiting are more likely to be hemorrhage, trauma, or increased intracranial pressure. Focal neurologic deficits without headache or nausea/vomiting are more likely to be ischemic stroke.*

A. Signs and Symptoms

1. Asymmetrical facial, arm, or leg weakness
2. Speech difficulties such as word finding or slurring
3. Visual field defects
4. Listen for carotid bruits (avoid palpation of the neck since this may dislodge further

B. Treatment

1. Follow current ACLS treatment guidelines
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Complete baseline full neurologic exam
4. Repeat neurologic exam every 30 minutes to identify improvement or deterioration in functioning

Trauma

V. **MULTIPLE ORGAN SYSTEM TRAUMAS-** *The vast majority of deaths and disabilities in active duty military personnel are caused by accidental or battlefield traumas. Trauma is the leading cause of death in the United States for all people under the age of 45. It is critical for the IDMT to have a systematic approach to handling trauma. Initial response to multiple organ system traumas is addressed here, following by specific considerations for individual organ systems. The treatment described below is best accomplished in controlled, protected environments.*

A. Indicators of Severe Trauma

1. Fall from greater than twice the patient's height
2. Extrication of patient took longer than 20 minutes
3. A death occurred in the same vehicle
4. Patient was ejected from vehicle

B. Initial Assessment

1. Airway- assess airway; secure airway (oropharyngeal, nasopharyngeal) provide simultaneous manual protection of C-Spine
2. Breathing- Check rate, efficiency, effort, administer high flow oxygen using nonrebreather mask if needed, assist ventilation if needed; check for symmetrical vs paradoxical chest movement

3. Circulation- Check pulse, capillary refill, and blood pressure; start CPR if indicated, control life-threatening bleeding
4. Disability- assess level of consciousness using AVPU (A = alert/awake, V= responds to voice, P = responds to pain, U = unresponsive) or Glasgow Coma scale
5. Expose patient- Examine whole patient, including patient's back and buttocks; complete rectal exam for occult blood, rectal wall tenderness/bony fragments, and decreased sphincter tone; check genital region for blood, incontinence, and spontaneous ejaculation; complete full set of vital signs including temperature

C. Additional Treatments

1. Airway- reassess airway; secure permanent airway (Combitube®, endotracheal intubation, cricothyroidotomy, etc), secure protection of C-Spine
2. Breathing- Reassess rate, efficiency, effort, assist ventilation if needed, auscultate lung sounds; if diminished on one side perform needle decompression for presumed pneumothorax (See "Tension Pneumothorax" below); palpate for crepitus; percuss lungs for tympany, dullness, or hyper-resonance
3. Circulation- Check pulse, capillary refill, and blood pressure; control obvious bleeding; initiate two large bore IV/IO lines and administer up to 2 L Normal Saline bolused
4. Disability- reassess patient's level of disability; perform rapid inspection, auscultation, percussion, and palpation of abdomen; consider urinary catheter insertion; DO NOT insert urinary catheter in male patients with signs of lower abdominal/genitourinary trauma or if there is blood at the urethra; strictly monitor fluid intake and output
5. Cover patient- repeat vitals signs including temperature, cover above and below patient to help prevent hypothermia

D. CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION

1. Discuss fluid resuscitation; preceptor may advise switch to Lactated Ringers, ½ normal saline, D5 ½ normal saline, or hypertonic saline depending on injuries; after first two Liters of Normal Saline reduce rate per preceptor's guidelines
2. Reassess critical patient's vital signs every 5 minutes; reassess stable trauma patients every 15 minutes
3. The two most common causes of preventable death in multi-organ system trauma patients are failure to maintain patent airway and failure to recognize life-threatening internal hemorrhage
4. The most common cause of poor response to fluid therapy is continuing hemorrhage
5. If evacuation is delayed longer than 24 hours, maintain urine output > 30 ml/hour

E. Treat non-life threatening injuries

1. SAMPLE history (signs/symptoms, allergies, medications, past medical history, last meal/medications, events leading up to emergency)
2. Immobilize fractures
3. Dress and bandage all wounds

4. Package patient appropriately for weather conditions

VI. **SHOCK-** *Inadequate tissue perfusion, leading to inadequate oxygen delivery to tissues, which results in end-organ damage. Causes include trauma/hemorrhage, diarrhea/vomiting, severe infection, heat stroke, burns, toxins, anaphylaxis, spinal cord injury, and cardiac failure. Trauma-induced hypovolemia is discussed here.*

A. Signs and Symptoms

1. Pre-shock- patient is still able to compensate; may already have lost up to 10% blood volume; may have mild increase or decrease in blood pressure and mildly elevated pulse
2. Shock- compensatory system is overwhelmed and symptoms of organ dysfunction appear including dyspnea, tachypnea, tachycardia, restlessness, cool clammy skin/diaphoresis, and decreased urine output; may have lost up to 25% of blood volume in this stage; may have respiratory rate less than 10 or more than 30, capillary refill longer than 2 seconds, thready pulse
3. End organ damage/failure- absent urine output, agitation, lethargy, coma, death

B. Treatment

1. Pre-shock- Establish IV access with Lactated Ringer's and provide high-flow oxygen as needed
2. Shock- Rapid volume resuscitation with Lactated Ringer's; high flow oxygen to offset blood loss, may require blood transfusion
3. End-organ damage- loss of life imminent, continue Lactated Ringer's, plan for transfusion and surgical intervention

VII. **CERVICAL SPINE TRAUMA-** *Always suspect C-Spine trauma in patients with decreased mental status, with history or evidence of blunt or penetrating trauma above the clavicle, or with mechanisms of severe trauma (see "Multiple Organ System Trauma" above)*

A. Signs and Symptoms

1. Mechanism suggests neck trauma
2. Neck pain or spasm
3. Changes in sensory or motor function
4. Decreased respiratory drive ("C3, C4, C5 keep the diaphragm alive")

B. Treatment

1. Protect cervical spine from flexion-extension-rotation-bending with semi-rigid collar backboard, and head immobilization devices
2. Use jaw thrust/chin lift maneuver to open the airway
3. Insert airway adjunct (oropharyngeal airway or nasopharyngeal airway) if indicated
4. Consider advanced airway placement- endotracheal intubation is extremely difficult to perform while maintaining cervical spine in neutral position; consider other appropriate airway such as Combitube®, LMA®, or cricothyroidotomy

C. Field Clearance of Cervical Spine

1. Cervical spine precautions do not need to be maintained if all of the following conditions are met
    a. Mechanism was not a "severe trauma" as defined above
    b. No midline cervical spine tenderness

c. Patient has normal level of alertness
d. The patient is not intoxicated
e. There are no abnormal neurologic findings
f. There are no other injuries severe enough (painful enough) to distract patient from identifying additional injuries
g. Patient is able to communicate (no foreign language barrier, not a child)

2. If patient meets all of the above criteria, remove cervical collar and ask patient to actively rotate neck left and right
   a. Patient should be able to rotate at least 45 degrees in each direction; if so, patient does not require C-Spine immobilization or x-rays
   b. Patient may have pain with this maneuver; clearance is based on ability to actively rotate neck; can still be cleared even if pain is present
   c. If unable to rotate neck at least 45 degrees in each direction, replace C-collar and continue patient care with patient immobilized

VIII. HEAD TRAUMA- *Approximately 50% of trauma deaths are associated with head injury.*

A. Signs and Symptoms (Severe Head Trauma)

1. Altered mental status
2. Glasgow Coma Scale Score less than 8 or change of 2 points or more while under observation
3. Persistence of changes in levels of consciousness lasting longer than 2 minutes
4. Decreased heart rate/respirations
5. Bradycardia suggests increased intracranial pressure
6. Unequal pupils suggests unilateral head trauma
7. Fixed, dilated pupils may be due to brain ischemia, pain, drug overdose, or drug withdrawal
8. Unilateral sensory or motor changes
9. Drainage of clear, bubbly, or bloody discharge from ears or nose may indicate basilar skull fracture
10. Open injuries, especially with presence of suspected brain tissue or bony fragments

B. Treatment

1. **IMMEDIATE ACTION**
   a. Protect cervical spine
   b. Follow "Multiple Organ System Trauma" protocols above
   c. Patient requires continuous observation; do not leave patient unattended
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consult physician preceptor regarding type and amount of IV fluids for resuscitation
   b. Do not administer any medications unless directed by a preceptor
   c. Continually reassess for changes in neurologic status

CLINICAL NOTE: Narcotics and sedatives should not be administered to a patient with a head injury because these medications will alter future neurologic evaluations and may worsen neurologic complications (decreased respiratory drive, hypothermia, etc).

IX. **TENSION PNEUMOTHORAX**- *Collection of air between the lung and the chest wall prevents the lung from expanding properly during inhalation. As the collection of air increases, lung volume decreases. Common causes include rib fracture and over-ventilation during resuscitation efforts.*

A. Signs and Symptoms

1. Initial presentation includes dyspnea, pain with deep inspiration or inability to take a deep breath, guarding or splinting the chest wall on affected side
2. Progresses to respiratory distress/tachypnea, hypotension, cyanosis
3. Unilaterally absent breath sounds, hyper-resonance with chest wall percussion on affected side, distended neck veins

B. Treatment

1. **IMMEDIATE ACTION**
   a. Perform immediate needle decompression on affected side
   b. If unsure whether or not pneumothorax may be present, perform needle decompression anyway! Rush of air will be diagnostic as well as therapeutic. If no rush of air, consider alternative diagnoses

ACTION ALERT: Performing a needle decompression:

1. Select insertion site- 2nd intercostal space at mid-clavicular line on same side as hyper-resonance and diminished breath sounds
2. Cleanse site with povidone-iodine (Betadine®) or alcohol
3. Select 10-14 gauge needle over the catheter device at least 2 inches in length; use 10 gauge and longer needles for patients with increased bulk in chest
4. Insert needle in a "Z" pattern going just over top of lower rib; nerves and blood vessels run along the inferior surface of each rib and so these areas should be avoided
5. Aspirate during insertion until air breaks vacuum
6. Remove needle and syringe and leave catheter attached to flutter valve or 3-way stopcock

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Needle decompression may need to be repeated multiple times while awaiting transport
   b. Definitive management is insertion of a chest tube with suction device

X. **OPEN PNEUMOTHORAX ("SUCKING CHEST WOUND")**- *Chest wall penetrating injury causes air to collect between lung and chest wall, preventing adequate lung expansion. As patient attempts to breathe, air is pulled straight through opening in chest wall, rapidly causing lung on affected side to collapse.*

A. Signs and Symptoms

1. Open chest wall wound
2. May have audible gurgling or sucking sounds from wound
3. May have frothy blood or bubbling at puncture sites

B. Treatment

1. Follow "Multiple Organ System Trauma" protocols, above
2. Apply occlusive dressing to wound and attach securely
3. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

CLINICAL NOTE: Taping an occlusive dressing on 3 sides vs 4 sides for open pneumothorax remains controversial. Advantages of taping on 3 sides is that air has a way to escape from the pleural cavity but not to get in, where it can cause further damage. Disadvantages of taping on three sides are that it is more difficult to secure the occlusive dressing and that it can increase the risk of infection. Four-sided occlusive dressings reduce the risk of infection, but increase the risk of tension pneumothorax from a buildup of air in the pleural space. This will require needle decompression. Four-sided taping is most common for transport times under 30 minutes, because these wounds are unlikely to have time to develop a tension pneumothorax. For the IDMT, transport times will usually be longer than 30 minutes, and therefore 3-sided taping is recommended although both are permitted. In either case, continuously reassess patient and "burp" dressing or perform needle decompression at any sign of tension pneumothorax.

XI. **FLAIL CHEST-** *Blunt trauma to the chest wall causes a segment of ribs to break free from the rest of the chest wall. Usually 3 or more ribs are broken in two places each, creating the flail segment. Flail chest causes decreased respiratory ability on its own, but due to the force of the trauma there is also usually underlying organ damage.*

A. Signs and Symptoms

1. Crepitus of the chest wall, asymmetrical chest wall motion, paradoxical respirations (chest wall seems to contract with inspiration and expand with exhalation)
2. Decreased respiratory rate, splinting or guarding on affected side, shallow breaths, chest wall pain
3. Contusions on chest wall

B. Treatment

1. **IMMEDIATE ACTION**
   a. Follow "Multiple Organ System Trauma" protocols, above
   b. Stabilize flail segment by applying direct pressure over injury site; this will relieve pain and improve ventilation
   c. Assume spinal trauma; transport with C-collar, backboard, and head immobilization
   d. Transport with affected side slightly elevated, if possible
   e. Shock is likely, but hydrate cautiously; use Lactated Ringer's solution and administer just enough to keep systolic blood pressure greater than 90 mmHg
   f. Aggressive fluid resuscitation can lead to pulmonary edema and death
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Do not administer narcotics unless directed by preceptor (narcotics may be necessary for pain control but will worsen respiratory depression)
   b. Discuss fluid resuscitation to include type and rate of administration

XII. **ABDOMINAL TRAUMA-** *Abdominal traumas are at increased risk of multiple life-threatening injuries including aortic dissection, spleen or liver laceration; shearing forces on kidney, liver, or spleen causing blood vessel rupture, or open wounds such as*

evisceration. Due to the high concentration of bacteria and debris inside the GI tract, abdominal wounds are at high risk for chronic infection.

A. Signs and Symptoms

1. Abdominal pain, guarding, nausea/vomiting, rebound tenderness, pelvic instability
2. Abdominal distention, abdominal wall ecchymosis, decreased bowel sounds

CLINICAL NOTE: Diminished bowel sounds will be one of the first physical exam findings of severe abdominal injury. Take the time to listen to each quadrant for one full minute; as soon as bowel sounds are heard, continue with the remainder of exam and reassess again after 15 minutes. If no bowel sounds are heard following major trauma, assume patient is bleeding into abdominal cavity. Prepare for rapid decline in patient's status progressing to shock.

B. Treatment

1. **IMMEDIATE ACTION**
   a. Follow "Multiple Organ System Trauma" protocols, above
   b. Assume spinal trauma; transport with C-collar, backboard, and head immobilization
   c. Cover evisceration with moist (saline) dressing; however exposed abdominal organs can rapidly lower the body's core temperature. Avoid moist dressings unless core body temperature can be maintained
   d. Reassess bowel sounds every 15 minutes in stable patients, every 5 minutes in unstable patients
   e. FAST (Focused Assessment with Sonography for Trauma) if available and trained
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XIII. PREGNANT TRAUMA PATIENT- *Ask every female of childbearing age if there is a possibility of pregnancy. Special consideration must be made for the pregnant trauma patient.*

A. Anatomical Changes in the Pregnant Patient

1. Cardiovascular- increased heart rate, changes in blood pressure, increased vascularity, increased plasma volume, EKG changes may resemble myocardial infarction, decreased red blood cell count, increased clotting in 3rd trimester
2. Pulmonary- Decreased lung volume, increased respiratory rate
3. Gastrointestinal- Decreased stomach volume, increased risk of aspiration; much greater risk of abdominal injury but more difficult to detect because guarding, rebound tenderness, and decreased bowel sounds are difficult to assess or not present despite severity of injury
4. Renal- Normal reduced renal function of pregnancy increases risk of kidney hypo-perfusion and injury during trauma

B. Treatment

1. **IMMEDIATE ACTION**
   a. Follow "Multiple Organ System Trauma" protocols, above unless otherwise specified

b. Pregnant patient will require more oxygen than non-pregnant patient with same injuries
c. CPR is more difficult in pregnant patients due to decreased chest wall flexibility
d. Ask patient to report all fetal movements; check fetal heart tones with Doppler device if available (auscultation of fetal heart tones with stethoscope difficult until after 20 weeks)
e. Keep patient in left lateral recumbent position to reduce compression of vena cava; this is very difficult if performing CPR
f. Pregnant patients will have increased fluid requirements compared to non-pregnant patients with same injuries, however due to their decreased quantity of red blood cells, they will have more difficulty maintaining tissue oxygenation. Stop all bleeding as soon as possible to avoid loss of red blood cells!

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

Chapter 3: Eye Protocols

CLINICAL NOTE: Differential Diagnosis of "RED EYE"—Ruling out vision-threatening causes of red eye is critical in the management of all cases of red eye. Most "red eyes" that are seen in the clinical setting will be caused by allergic, bacterial or viral conjunctivitis. Prior to evaluating these etiologies, the following vision-threatening etiologies should be considered. Red eye accompanied by headache, nausea and vomiting is suggestive of glaucoma. Acute onset of red eye suggests trauma or irritation of the conjunctiva. Bilateral itching and watery discharge suggests allergic conjunctivitis. Scratchiness or foreign body sensation suggests corneal abrasion. A deep ocular ache accompanies scleritis. Photophobia suggests corneal abrasion, corneal ulcer or uveitis. Decreased visual acuity suggests severe pathology including glaucoma, corneal ulcer, hyphema and uveitis. Discharge corresponds to allergic, viral or bacterial infection. Unilateral exophthalmos suggests retro-bulbar pathology such as an abscess or tumor. Irregular corneal light reflex occurs with corneal edema. Redness around the cornea suggests uveitis, glaucoma or keratitis. Increased intraocular pressure is diagnostic for glaucoma. Pupil irregularity may be due to trauma, uveitis, or glaucoma.

I. **CENTRAL RETINAL ARTERY OCCLUSION (CRAO)-** *Blockages of the central retinal artery producing sudden, unilateral blindness.*

A. Signs and Symptoms

1. Poor pupillary response with direct light, but brisk response with consensual light
2. Pale retina with "cherry red" macula
3. Affected field of vision corresponds with size of emboli and affected arterial branches

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Immediate treatment by an ophthalmologist is essential
3. If evacuation is delayed, consider digital massage over the closed eye
4. Perform cardiac exam for arrhythmias, murmurs, and bruit

CLINICAL NOTE: digital massage can help dislodge the clot and allow it to enter a smaller branch of the artery and therefore limit the area of retinal ischemia.

II. **CHALAZION-** *a chronic enlargement of a meibomian gland followed by inflammation of the gland or surrounding tissue.*

A. Signs and Symptoms

1. Difficult to distinguish from hordeolum
2. Lid edema, swelling and irritation are common
3. History of a slowly-growing, mobile, indurated mass in the lid
4. Red or grey mass visible on conjunctival side of eye lid

B. Treatment

1. Test and record visual acuity as a baseline for later comparisons
2. Apply warm compresses to eyes for 15 minutes QID x 2 weeks then recheck

3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Typically most chalazion clear spontaneously in weeks to months
    b. Symptomatic patients with recalcitrant lesions can be referred to an ophthalmologist for incision and curettage or direct steroid injection
4. If no resolution after six weeks, refer to physician preceptor for further evaluation

CLINICAL NOTE: A stye appearing near the lower inner canthus must be differentiated between dacryrostenosis (blockage of the nasolacrimal duct causing excessive tearing).

III. **OCULAR BURNS, CHEMICAL-** *Can be caused by either acidic or basic solutions. Use pH paper to determine when the eye has returned to its normal pH. Use high volume but low pressure when irrigating.*

A. Signs and Symptoms

1. History of splash or powder exposure to the face
2. Severe pain and sensation of burning
3. May have photophobia, sensation of foreign body, vision changes
4. Copious lacrimation and conjunctival injection are the prominent symptoms
5. Coexisting lid edema and burns are common

B. Treatment

1. **IMMEDIATE ACTION**
    a. IRRIGATE! Continuous irrigation is best chance to preserve vision

ACTION ALERT: The most important treatment of chemical burns is extensive immediate irrigation. Sterile higher osmotic solutions such as amphoteric solution (Diphoterine) or buffered solutions (BSS or Lactated Ringers) are ideal. If not available, sterile isotonic saline is an appropriate irrigant. DO NOT USE hypotonic solutions, such as water, because these result in deeper penetration of corrosive material into the corneal structures due to the cornea's higher osmotic gradient.

The duration and amount of irrigation is determined by the eye pH. Continue irrigation until the pH remains at normal level for 30 minutes. Use of a Morgan lens or other eye irrigation system can minimize interference from blepharospasm, which can often be severe. If these are unavailable, the lid can be retracted manually with a Desmarres retractor, lid speculum, or bent paperclip. The end of intravenous tubing can direct the stream of sterile fluid across the eye. In addition, use a cotton swab to remove any particulate matter that may be retained in the fornices (corners).

    b. Instill proparacaine hydrochloride (Alcaine®) ophthalmic solution, 1-2 gtts. Prior to irrigation if blepharospasm or pain prevents irrigation
    c. Direct lavage of cornea with continuous sterile solution to dislodge particles
    d. Swab superior and inferior cul-de-sac to remove particles. Dilate with a cycloplegic mydriatics drug (i.e. 2% homatropine 1 gtt BID)
    e. Administer NSAID and/or narcotic analgesic PRN such as

i. Morphine sulfate 10-30 mg IM/Sub-Q every 3-4 hrs
ii. Morphine Sulfate 1-4 mg IV/IO every 4hrs PRN for pain
iii. Meperidine hydrochloride (Demerol®) 50 mg IM every 3-4 hours PRN for pain
iv. Meperidine hydrochloride (Demerol®) 25 mg IV/IO every 3 -4 hrs PRN for pain
v. Be prepared to manage respiratory depression any time IV/IO/IM narcotics are used

f. Evaluate and record visual acuity as a baseline for later comparisons
g. **CONSULT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   i. Needs urgent referral to ophthalmologist
   ii. Administer ocular antibiotic such as Neomycin or Gentamicin
h. Reassess vital signs and visual acuity every 30 minutes until evacuation
i. All ocular burns require topical antibiotics, tetanus immunization, cycloplegics and mydriatics

IV. **OCULAR BURNS, RADIATION-** *Mild cases often go unrecognized in military population. Commonly caused by intense light exposure such as snow-blindness, prolonged exposure to direct sunlight without sunglasses, or use of tanning booths.*

A. Signs and Symptoms
1. Tearing, sensation of foreign body, photophobia
2. May have coexisting "sunburn" on eyelids and orbits

B. Treatment
1. Mirrors treatment for corneal abrasion
2. Evaluate and record visual acuity as a baseline for later comparisons
3. Symptoms should resolve within 24 hours with conservative treatment

V. **CONJUNCTIVITIS, BACTERIAL-** *inflammation of the bulbar and palpebral conjunctivae commonly caused by Staphylococcus or Streptococcus species, but may also be caused by exposure to genital secretions containing gonorrhea or chlamydia.*

A. Signs and Symptoms
1. Injection of the conjunctivae
2. Itching
3. White/yellow/green mucoid discharge best seen along lid margins and in corners
4. Must ask thorough sexual history for possible STI exposure
5. Regardless of history, suspect gonorrhea or chlamydia if discharge is thick, purulent, or any vision changes are present (floaters, halos, etc)

B. Treatment
1. Evaluate and record visual acuity as a baseline for later comparisons
2. Apply warm compresses to eyes for 15 minutes every 3-4 hours
3. Remove crusts and debris
4. Usually self-limiting lasting 10 - 14 days
5. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Instill antibiotic erythromycin (E-mycin®) ophthalmic ointment TID until asymptomatic

b. Alternatives are polymixin/ trimethroprim (Polytrim®) gtts or bacitracin (Eacticin®) ointment
c. If patient wears contact lenses, stop lens wear and consider ciprofloxacin (Cipro®) gtts.
d. If gonorrhea or chlamydia are suspected, patient requires further evaluation by a physician and oral anitibiotics
    i. If evacuation is delayed, begin systemic antibiotics
    ii. Treat patient for both gonorrhea and chlamydia until culture is available
    iii. Refer to Genitourinary protocols for further information

ACTION ALERT: Gonoccocal conjunctivitis is a vision-threatening infection and characterized by copious, mucoid discharge, pain, periorbital swelling/tenderness to palpation**. IMMEDIATE EAVCUATION** to an ophthalmologist is required.

VI. **CONJUNCTIVITIS, VIRAL-** *Highly contagious infection of the conjunctivae caused by multiple different viruses. Usually presents without systemic symptoms.*

A. Signs and Symptoms

1. Self-limited course lasting 2-4 weeks (typically lasts longer than bacterial conjunctivitis)
2. Pre-auricular lymphadenopathy suggests viral etiology
3. Watery discharge with scant mucous
4. May have history of recent URI or coexisting URI
5. May complain of burning, sandy or gritty feeling with "eye matted closed" especially upon waking
6. Second eye typically displays symptoms 24-48 hours after the first eye

B. Treatment

1. Test and record visual acuity as a baseline for later comparisons
2. Apply warm compresses to eyes for 15 minutes every 3 or 4 hours
3. Remove crusts and debris
4. Usually self limiting and resolves over 2-3 weeks
5. No medications indicated; discontinue eye makeup and throw away current contact lenses
6. Use glasses until 2 days after symptoms have completely resolve

ACTION ALERT: Herpetic infections of the eye can be serious and characterized by grouped vesicles over an erythematous base with serosanguinous crusts. **IMMEDIATE EVACUATION** to an ophthalmologist is required. Do not use steroids.

VII. **CONJUNCTIVITIS, ALLERGIC-** *Injection of the conjunctivae associated with immune response to an irritant. Improves or resolves when changing environments or if irritant can be identified and removed.*

A. Signs and Symptoms

1. Itchiness is the primary distinguishing symptom
2. Copious, bilateral, watery discharge

3. Periorbital edema
4. Additional allergic symptoms such as runny nose or sneezing

B. Treatment

1. Visine AC® eye drops (for a short period of time) AND
2. Systemic antihistamines such as diphenhydramine (Benadryl®) or loratadine (Claritin®)
3. See "Ear, Nose, and Throat Protocols- Allergic Rhinitis" for more information

VIII. CORNEAL ABRASION- *Conjunctival and corneal abrasions are common and occur as the result of foreign bodies in the eye, dirt, debris, or contact use. Differential diagnosis should include foreign body, corneal ulcer and penetrating trauma.*

A. Signs and Symptoms

1. Foreign body sensation
2. Irritation and increased lacrimation
3. May be accompanied by photophobia
4. Flourescein stain highlights abrasion

ACTION ALERT: Warning signs of a more serious problem that should prompt evaluation by an ophthalmologist include:

1. Reduction or change in visual acuity
2. Ciliary flush: A pattern of injection in which the redness is most pronounced in a ring at the limbus (the limbus is the transition zone between the cornea and the sclera)
3. Photophobia
4. Severe foreign body sensation that prevents the patient from keeping the eye open
5. Corneal opacity
6. Fixed pupil
7. Severe headache with nausea

B. Treatment

1. **IMMEDIATE ACTION**
   a. Test and record visual acuity as a baseline for later comparisons
   b. If patient's pain prevents an adequate exam, contact physician preceptor and instill topical ophthalmic anesthetic such as 1 gtt of proparacaine hydrochloride (Ophthaine®)
   c. Check cornea and conjunctiva with the Wood's lamp after instilling fluorescein stain into the eye to rule out foreign body and diagnose abrasion
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Instill antibiotic, polymyxin B-bacitracin-neomycin (Neosporin®) ophthalmic ointment, into involved eye TID to QID
   b. For significant abrasion, instill 1-2 gtts of 5% homatropine (Isopto Homatropine®) to prevent a secondary iritis and help with discomfort
   c. Administer PO analgesics such as ibuprofen (Motrin®) or acetaminophen with codeine (Tylenol#3®)
3. Remove patient from duties requiring eyestrain or intense visual focus

4. Recheck eye after 24 hours to be certain cornea has healed and to rule out the development of a corneal ulcer
5. Repeat treatment if not completely re-epithelized and contact physician preceptor to consider referral to ophthalmologist

ACTION ALERT: Do not use topical ophthalmic corticosteroids. They will make the condition worse because they tend to promote fungal growth. Be sure to use only medications designed for ophthalmologic use.

C. Prevention

1. Ensure all work areas have eye protection sufficient for everyone
2. Ensure workout facilities have appropriate eye protection for activities at high risk of eye injury (Squash, racquetball, etc)
3. Check hobby locations to make sure eye protection is available (wood shop, framing shop, etc)

IX. **CORNEAL ULCER-** *Necrosis of corneal tissue due to infection by bacteria, fungus, virus or other microorganism is often due to extended contact lens wear.*

A. Signs and Symptoms

1. Ache and foreign body sensation
2. Photophobia and increased lacrimation
3. Dull, grayish, circumscribed superficial opacity with an excavated center
4. Lesion will stain green with fluorescein
5. Erythema and corneal neovascularization occurs with long-standing ulcer
6. Pus may appear in the anterior chamber

CLINICAL NOTE: Corneal ulcers are frequently the result of prolonged contact lens wear or inadequate cleansing of lenses between uses. Additionally, individuals will frequently use disposable lenses past their functional use- for example wearing 2-week lenses for 5 or 6 weeks. Even with proper cleansing and care, breakdown of the disposable lens material can cause infections to begin.

B. Treatment

1. **IMMEDIATE ACTION**
   a. Test and record visual acuity as a baseline for later comparisons
   b. Check cornea and conjunctiva using Wood's lamp or ophthalmoscope to rule out foreign body
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Administer optic antibiotics: ciprofloxacin (Cipro®), tobramycin (Tobradex®) or ofloxacin (Ocuflox®) gtts every 1-2 hours until seen by an ophthalmologist
   b. Administer PO analgesics ibuprophen (Motrin®) or acetaminophen with codeine (Tylenol #3®) PRN for pain

C. Prevention

1. Educate all contact lens wearers of the importance of proper contact lens cleansing and the need to dispose of used lenses according to package instructions
2. Ensure that all personnel follow local guidance regarding use of contacts while wearing additional eye protection
3. Do not allow personnel to wear contacts while in MOPP gear

X. **DACROCYSTITIS-** *An acute inflammation of the nasolacrimal duct, causing excessive tearing. Dacryocystitis can be complicated by orbital cellulitis, sepsis, or meningitis.*

A. Signs and Symptoms

1. Tearing and swelling along the nasolacrimal fold or at the puncta
2. Warmth, tenderness of the lacrimal sac
3. Regurgitation of tears or mucous with palpation of the lacrimal apparatus
4. Conjunctival irritation

B. Treatment

1. Massage lacrimal duct 3-4 times per day by applying gentle, downward pressure on the lacrimal duct, pressing towards the nasal fold
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. If infection is mild, prescribe clindamycin (covers MRSA), ciprofloxacin or levofloxacin (Levaquin®)
    b. If severe, consider evacuation, IV vancomycin hydrochloride (Vancocin®) and inpatient admission

XI. **DETACHED RETINA-** *Separation of the retina from the underlying epithelium. It can be caused by trauma, or as a result of chronic disease such as diabetes. Nearsightedness is a risk factor for detached retina.*

A. Signs and Symptoms

1. Retinal detachment is painless
2. Vitreous floaters
3. Patient reports Hx of seeing flashes or blurred/darkened vision
4. If the macula is involved, central visual acuity fails
5. Fundoscopic exam may show irregularities of the retina such as loss of red reflex and darkened blood vessels

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Must be treated immediately by an ophthalmologist

XII. **FOREIGN BODY-** *Conjunctival and corneal foreign bodies are the most common eye injuries. Differential diagnosis of ocular penetration should be ruled out first.*

A. Signs and Symptoms

1. Foreign body sensation
2. Irritation/increased lacrimation
3. May be accompanied by photophobia
4. Fluorescein stain highlights abrasions and foreign bodies
5. Multiple vertical lacerations indicate foreign body is trapped under the upper lid; lacerations are caused when the patient blinks and drags the foreign body across the conjunctiva

B. Treatment

1. **IMMEDIATE ACTION**
    a. Test and record visual acuity as a baseline for later comparisons
    b. Locate the foreign body by examining the eye, using a Wood's lamp ophthalmoscope
    c. May install fluorescein stain into the conjunctival sac if globe penetration is not suspected
    d. Attempt to wash out the foreign body with copious amounts of sterile saline
        i. If this fails, contact physician preceptor and anesthetize eye with 1 drop of proparacaine hydrochloride (Ophthaine®) ophthalmic solution into the eye
        ii. Use a wet cotton swab to remove the foreign body

CLINICAL NOTE: If unable to remove foreign body or if there is any suspicion the foreign body has penetrated the globe, patch both eyes and **CONSULT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION.**

2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. For larger foreign bodies and corneal abrasions, consult preceptor prior to applying cycloplegic medications such as homatropine
    b. Instill antibiotic, polymyxin B-bacitracin-neomycin (Neosporin®) ophthalmic ointment, or erythromycin (E-mycin®) ophthalmic ointment into involved eye
    c. Consult physician preceptor regarding whether or not to patch the involved eye.
3. Administer PO analgesics ibuprofen (Motrin®) or acetaminophen with codeine (Tylenol#3®)
4. Recheck patient every 24 hours until signs and symptoms resolve
    a. Check for secondary infection of conjunctiva with symptoms such as redness or discharge
    b. Injury has high likelihood of developing corneal ulcer
5. Follow up should be performed by an ophthalmologist within 1-2 days after injury if possible

XIII. GLAUCOMA, PRIMARY OPEN-ANGLE- *Increased intraocular pressure (IOP) associated with an open anterior chamber angle. This is the most common form of glaucoma. Both eyes are usually affected, but not equally.*

A. Signs and Symptoms

1. No early warning symptoms
2. Peripheral vision loss occurs first and usually not noticed by the patient
3. Patients may complain of difficulty reading, driving or performing coordinated activities
4. Intraocular pressure may be elevated (>21mmHg)
5. Risk factors include age, heritage (African American), family history, myopia (nearsightedness) and diabetes
6. Fundoscopic exam will show abnormalities of the cup and cup will be more than half the size of the disc

B. Treatment

1. **CONTACT PHYSICAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Therapy is aimed at either causing miosis, increasing aqueous outflow, decreasing aqueous production or drawing fluid out of the anterior chamber
    b. Apply topical β-blocker, or acetazolamide (Diamox®) or osmotic drug per preceptor
    c. Physician must clear prior to return to duty
2. Keep patient in position of comfort

XIV. **GLAUCOMA, ACUTE ANGLE**- *Increased intraocular pressure (IOP) associated with a closed anterior chamber angle.*

A. Signs and Symptoms

1. May be asymptomatic
2. Redness, eye pain, blurred vision, and headaches improving with rest
3. Decreased vision, colored halos around lights
4. Nausea and vomiting
5. Increased lacrimation, lid swelling, steamy cornea, fixed/dilated pupil
6. If the contralateral eye is affected, consider other diagnosis

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Apply combination of topical β-blocker, acetazolamide (Diamox®) and/or osmotic drug per preceptor
    b. Preceptor may recommend systemic medications such as acetazolamide (Diamox®) or mannitol (Osmitrol®) to control intraocular pressures
2. Keep patient in position of comfort

XV. **HORDEOLUM**- *An acute, localized infection of the glands of Zeis or Moll (external hordeolum) or occasionally of the Meibomian glands (internal hordeolum or stye)*

A. Signs and Symptoms

1. <u>External hordeolum</u>
    a. Pain, redness and tenderness of the lid margin initially, followed by a point of induration or pustule
    b. Photophobia, lacrimation and foreign body sensation
    c. Localized edema, but may be diffuse
    d. Symptoms are relieved if the abscess ruptures, releasing purulent discharge
2. <u>Internal hordeolum</u>
    a. More severe and occur with less frequency than external styes
    b. Small, elevated, yellow punctal lesion at the site of the infected gland
    c. Abscess may come to a point through the conjunctiva or occasionally through the anterior side of the eye lid
    d. Spontaneous rupture does not usually occur

B. Treatment

1. Test and record visual acuity as a baseline for later comparisons
2. Apply warm, moist compresses to eyes for 10 minutes 3-4 times per day until spontaneous rupture

3. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Topical antibiotics may be needed to treat internal hordeola
    b. Consider erythromycin (E-Mycin®) ophthalmic
4. Reevaluate patient in 24 hours
5. If stable, continue treatment for 5 days
6. CONTACT PHYSICIAN PRECEPTOR- EVACUATION if symptoms last longer than 1 week or if at any point vision is affected

XVI. **HYPHEMA (ANTERIOR CHAMBER HEMORRHAGE)**- *Bleeding into the anterior chamber of the eye may follow blunt trauma. Hyphemas are an emergency requiring immediate attention from an ophthalmologist.*

A. Signs and Symptoms

1. Blood pooling in the anterior chamber of the eye with obvious blood-fluid level
2. Caused by blunt trauma to the globe or orbit
3. May have distorted pupil
4. May be followed by recurrent bleeding, glaucoma, and staining of the cornea

ACTION ALERT: Trauma affecting the eye should be treated by an ophthalmologist. Primary concerns are increased IOP (intraocular pressure) and globe rupture. Aggressive pain/nausea management is essential to avoid increases in IOP.

B. Treatment

1. **IMMEDIATE ACTION**
    a. Test and record visual acuity as a baseline for later comparisons
    b. Place patient at bed rest
    c. Patient will remain at bed rest – (Fowler's position) 45° - for 6 to 7 days
    d. Do not allow patient to lie flat. Keep the patient's head elevated at 45° or greater
    e. Patch affected eye without causing additional pressure on the affected eye
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. If evacuation is delayed, consider a carbonic anhydrase inhibitor such as one acetazolamide (Diamox®) tablet every 4 hours
    b. Do not administer NSAIDs or aspirin- may increase bleeding
    c. Give acetaminophen (Tylenol®) for pain relief
    d. Consider acetaminophen with oxycodone (Percocet®) or acetaminophen with hydrocodone (Vicodin®) for severe pain
    e. For nausea, consider ondansetron (Zofran®)
    f. Do not give mydriatic drugs or stain the eye unless directed by an ophthalmologist

XVII. **ORBITAL CELLULITIS**- *Inflammation of the tissues surrounding the orbit caused by an infection that extends from any tissue in the face, mouth or sinuses. Cellulitis can also be caused by bacteria introduced by direct trauma.*

A. Signs and Symptoms

1. Trauma to the eyelid and infections of the sinus cavity are the most common causes
2. Pain, eyelid redness, swelling, hyperemic mucous membranes
3. Exophthalmos, decreased mobility of the eye
4. Fever
5. Double vision
6. Complications include blindness, meningitis and encephalitis

B. Treatment

1. Find the primary cause of the infection
2. Test and record visual acuity as a baseline for later comparisons
3. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Start antibiotic therapy
   b. Consider cefuroxime (Ceftin®) or cefoxitin (Mefoxin®) PO/IV
4. Reevaluate daily
5. If antibiotics are not effective, refer to an ophthalmologist

ACTION ALERT: Any decreased mobility of the eye or history of double vision accompanied by fever is highly suggestive of meningeal spread or sepsis. Emergency evacuation is required.

XVIII. PINGUECULA/PTERYGIUM- *A pinguecula is a benign neoplasm of the conjunctiva. A pterygium is a corneal degeneration. Both are yellow, bumpy surface changes on the eye related to increased UV/sun exposure.*

A. Signs and Symptoms

1. Pinguecula- raised, yellow-white mass on the bulbar conjunctiva adjacent to the cornea at the 3 or 9 o'clock position
2. Pterygium- wedge-shaped growth extending onto the cornea with visible intrusion over the iris
3. Both pinguecula and pterygium are painless but may be accompanied by foreign body sensation, irritation or localized erythema

B. Treatment

1. Does not directly effect eye's ability to see, but can eventually limit field of vision by direct extension across the cornea
2. No treatment required in the acute setting
3. May use OTC ophthalmologic lubricants (Artificial Tears® or Clear Eyes CLR®) to reduce irritation if needed

XIX. TRAUMA AND CONTUSIONS INVOLVING THE EYE- *Direct trauma to the face or nose can manifest as "black eye", even if the orbit did not sustain trauma itself. Swelling around the orbits can make it difficult to examine the structures of the globe; if globe is unable to be assessed due to surrounding edema, CONTACT PHYSICIAN PRECEPTOR- EVACUATION is required to ensure orbital fracture, hyphema, or vision-threatening problems have not occurred.*

A. Signs and Symptoms

1. Lid Contusions
   a. "Black Eye"
   b. Periorbital ecchymosis and edema

2. Lid Lacerations- minor
    a. Horizontal
    b. Not full thickness
3. Lid Lacerations- extensive (any of the following)
    a. Vertical
    b. Involving nasal or tear duct areas
    c. Full thickness

B. Treatment

1. Lid contusions
    a. Apply ice packs to the region for the first 48 hours
    b. Apply warm compresses to the region for the next 48 hours
    c. Ensure trauma is limited to lid and does not involve the globe
    d. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
        i. **IMMEDIATE EVACUATION** is required if any of the extraoccular movements or pupillary reflexes are abnormal
        ii. Other treatments at discretion of preceptor
2. Lid Lacerations- minor
    a. Cleanse and suture with 6-0 silk
    b. Many minor upper lid horizontal lacerations do not require sutures
    c. Administer tetanus toxoid 0.5 ml or immunoglobulin as indicated
    d. Apply sterile dressing and keep dry
3. Lid Lacerations- extensive
    a. **CONSULT PRECEPTOR- IMMEDIATE EVACUATION**
        i. Cleanse
        ii. Apply sterile dressing with wound margins approximated if possible
        iii. Administer tetanus toxoid 0.5 ml or immunoglobulin as indicated
        iv. IDMT's should not suture due to the wound's increased complexity

ACTION ALERT: NEVER SHAVE THE EYEBROW. It may not grow back. Do not use ophthalmic ointments when a penetrating wound of the eye is present. They will cause permanent eye floaters. Never attempt to suture a vertical laceration of the eyelid or laceration involving the lid margin as an associated penetrating injury to the eye must be ruled out.

XX. GLOBE TRAUMA- *Blunt trauma affecting the continuity of the conjunctiva or sclera, causing an opening of either of the chambers of the eye and distorting the globe. Includes laceration of the iris, dislocated lens, hemorrhage into the globe, and hemorrhage or rupture of the eyeball.*

A. Signs and Symptoms

1. History of trauma
2. May have irregular, teardrop-shaped or nonreactive pupil
3. Hyphema
4. Extruded vitreous humor
5. Unable to open eye

6. Do not use fluorescein in patients with suspected globe trauma; however if fluorescein staining was done and "streaming" fluorescein is seen, this is diagnostic of an open chamber
    a. "Streaming" is caused by leaking aqueous humor
    b. Avoid fluorescein backwashing into wound

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Administer meperidine HCl (Demerol®) IV for pain
    b. Apply sterile protective dressing to both eyes
    c. Penicillin PO/IM or other systemic antibiotic as directed
    d. If patient **IS** allergic to penicillin, give erythromycin (E-mycin®) PO until evacuation
    e. Administer tetanus toxoid, 0.5 ml, IM, or immunoglobulin as indicated
2. Evacuate to a medical facility with ophthalmologic and maxillofacial surgery as soon as possible

XXI. TRAUMA, PENETRATING WOUND OF THE EYE- *Occurs when a foreign body has passed through the thickness of the conjunctiva and cornea or sclera causing an opening into one of the chambers of the eye.*

A. Signs and Symptoms

1. May or may not be able to visualize foreign body depending on the location
2. History of foreign object sensation or trauma
3. May have irregular, teardrop-shaped or nonreactive pupil
4. Hyphema
5. Extruded vitreous humor
6. Unable to open eye
7. Do not use fluorescein in patients with suspected penetration of the globe; however if fluorescein staining was done and "streaming" flourescein is seen, this is diagnostic of an open chamber
    a. "Streaming" is caused by leaking aqueous humor
    b. Avoid fluorescein backwashing into wound

B. Treatment

1. **IMMEDIATE ACTION**
    a. Place patient on his/her back
    b. Keep manipulation of the eye to an absolute minimum
    c. Do not clean wounds of the globe
    d. Reduce visual stimulation
        i. Turn off lights
        ii. Apply sterile, protective dressing to both eyes
    e. Examine head and other areas of the body for additional injuries
2. **CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND IMMEDIATE EVACUATION**
    a. Administer meperidine hydrochloride (Demerol®), 50 mg, IM, every 3 to 4 hours, as necessary, for pain
    b. Evacuate to a medical facility with ophthalmologic and maxillofacial surgery as soon as possible

XXII. SUBCONJUNCTIVAL HEMORRHAGE- *A wedge-shaped flat collection of blood between the sclera and conjunctiva, occurring suddenly after a transient increase in intraocular pressure such as crouching, sneezing or vomiting, but can also occur secondary to blunt trauma.*

A. Signs and Symptoms

1. Usually painless and asymptomatic
2. Focal collection of bright red blood between sclera and conjunctiva

ACTION ALERT: If size or shape of the blood spot is changing, or if symptoms are accompanied by pain or foreign body sensation, consider trauma or foreign body.

B. Treatment

1. None
2. Defect will resolve spontaneously over 1-3 weeks

XXIII. UVEITIS- *Inflammation of the iris, cilliary body, or chorioid. Distinguished from conjunctivitis because redness usually at or crossing the iris border, whereas in conjunctivitis the injection usually spares the iris.*

A. Signs and Symptoms

1. Pain, redness, photophobia, and decreased vision
2. Patient may complain of "floaters"
3. Redness worst adjacent to the cornea radiating from the iris

CLINICAL NOTE: This pattern of redness is different from that seen with viral, allergic, and bacterial conjunctivitis because in those cases, the injection usually spares the area around the cornea. In the case of uveitis, the injection usually appears like a "sunburst" radiating from the iris or cornea.

B. Treatment

1. **IMMEDIATE ACTION**
   a. Test and record visual acuity as a baseline for later comparisons
   b. Instill fluorescein stain into conjunctival sac and examine eye to rule out a corneal lesion
   c. Warm compresses will decrease pain
2. **CONTACT PHYISICAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Antimicrobials are used to treat infectious uveitis; consult preceptor
   b. Consider instilling 1 drop of scopolamine ophthalmic solution (Isopto Hyoscine®) (0.25%) into involved eye every 12 hours, until asymptomatic if directed by preceptor
   c. Administer non-steroidal anti-inflammatory drug or narcotic/NSAID combination PO as needed for pain relief

Chapter 4: Ear, Nose and Throat Protocols

Ear

I. **OTITIS EXTERNA, ACUTE-** *An inflammation of the external auditory canal. May be precipitated by other ENT infections, allergies or flaking of the skin. Trauma from cotton-tipped applicator use is the most common source of infection.*

A. Signs and Symptoms

1. Itching, pain, and drainage of the affected ear
2. History of Q-tip® use or other manipulation of the auditory canal
3. History of wet environment (i.e. swimming, headphones, ear plugs etc.)
4. Inflammation of external auditory canal
5. Tenderness to tragus/pinna with manipulation
6. Visible debris or discharge in externl auditory canal

B. Treatment

1. Remove visible debris from canal
2. Administer NSAIDs or acetaminophen for pain
3. Apply heat to the affected ear for comfort
4. Advise patient to keep the ear dry and avoid earplugs, headphones, etc. until the infection has completely cleared
5. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Apply ciprofloxacin otic (Cipro HC®), Cortisporin Otic® or other acidic combination of steroid + antiseptic + antibiotic drops into the ear canal
    b. If otic antibiotics are not available, apply undiluted acetic acid (vinegar) or alcohol TID
    c. Use ear wick placed in canal to administer drops if canal is too swollen to apply drops directly
    d. Culture ear discharge for pseudomonas if greenish appearance or failure to respond to above measures
    e. Administer appropriate systemic antimicrobial therapy if indicated by fever, or involvement of surrounding tissues
    f. Patient requires emergency evacuation if systemic symptoms occur

II. **OTITIS MEDIA, ACUTE PURULENT-** *"Ear infection" marked by ear pain and bulging red tympanic membrane. The EAC can also be erythematous in severe cases with erythema extending away from the tympanic membrane. If the canal is inflamed and inflammation is not worse adjacent to the TM, then a diagnosis of external otitis should be considered.*

A. Signs and Symptoms

1. Ear pain
2. Decreased hearing
3. Possible fever
4. May be accompanied by history of recent upper respiratory infection
5. Red, bulging TM
6. Purulent discharge in external auditory canal if TM has ruptured
7. Conductive hearing loss (Weber/Rinné test with 512 tuning fork: sound lateralizes to affected ear and BC>AC)

8. Anterior and posterior auricular lymphadenopathy

B. Treatment

1. Administer analgesics as needed for pain; acetaminophen (Tylenol®) or ibuprofen (Motrin®)
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Administer appropriate antimicrobial therapy
   b. If patient **IS NOT** allergic to penicillin-give amoxicillin (Trimox®), OR amoxicillin clavulanate (Augmentin®)
   c. Alternatives are clarithromycin (Cleocin®) or azithromycin (Zithromax®)

CLINICAL NOTE: Mastoid bone tenderness and uncontrolled fevers are emergencies due to the potential spread of the infection to the deep structures of the brain and sinuses. Contact physician preceptor for evacuation consideration.

III. **BAROTRAUMA, EAR**- See "Environmental Injury Protocols, Barotrauma, Ear"

IV. **FOREIGN BODY IN EXTERNAL AUDITORY CANAL-** *Ear foreign body in adults is typically caused by crawling insects, but can also be related to environmental hazards, foreign objects used to clean or scratch the ear canal, or ear protection/headphone malfunction. Removal strategies are based on cause.*

A. Signs and Symptoms

1. Pain to the affected ear
2. Bleeding
3. Change in hearing or sensation of movement in the ear canal (insect)
4. Visible obstruction or trauma
5. Hearing changes

B. Treatment

1. If the object is easily grasped (the exterior ½ of the canal) remove slowly with hemostats or alligator forceps
2. For plastic objects, irrigate with warm water using 30-50 cc syringe and butterfly catheter
3. For removal of insects that are deeper than external ½ of canal, attempt to entice insect out with at light source at external meatus ; large insects will not be able to turn and light will not help them
4. If the TM is verified as intact, and if the insect cannot or will not be enticed out, immobilize the insect before manually removing it by filling the external auditory canal with lidocaine, mineral oil, or alcohol
5. If the TM can be verified to be intact, you may flush the external auditory canal with mineral oil and then remove the foreign object manually as it gets closer to the auditory meatus

ACTION ALERT: Do not use irrigation for batteries or chemicals. If unable to remove the object, CONTACT PHYSICIAN PRECEPTOR- EVACUATION.

6. If foreign body and signs of infection are present at the same time, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
7. Reassess hearing after foreign body is removed
8. Reevaluate patient in 24 hours for infection and to recheck hearing

V. **RUPTURED OR PERFORATED EARDRUM-** *Ruptured tympanic membrane may be found incidentally on exam or may be related to external otitis, otitis media, or trauma.*

A. Signs and Symptoms

1. Severe escalating ear pain followed by abrupt improvement
2. Drainage from the ear
3. History of ear surgeries/tubes/frequent ear infections, hearing loss
4. May complain of sensation of air in the ear or "rushing" air sounds
5. Visible hole in TM (may enable clear view of ossicles)
6. Drainage may be evident in canal
7. Potentially abnormal Weber, Rinné or Whisper tests

B. Treatment

1. Keep ear absolutely dry- use cotton balls in shower, etc.
2. Treat underlying condition if rupture is due to otitis media or other infection
3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Do not irrigate or use curette in ear to remove debris unless instructed to do so
   b. Follow up daily for reexamination and reevaluation of hearing
   c. Administer analgesics for pain, as needed
      i. Ibuprofen (Motrin®) PRN for pain, or
      ii. Acetaminophen with codeine sulfate (Tylenol #3®) with preceptor approval
   d. Surgical intervention may be required

VI. **OTITIS MEDIA, SEROUS -** *Blockage of the eustacean tube causing fluid buildup behind the tympanic membrane and inability to clear ears.*

A. Signs and Symptoms

1. Feeling of fullness
2. Hearing change
3. Often associated with recent URI, allergies or air travel
4. Retracted TM
5. Thick or amber colored TM
6. Bubbles or fluid line visible behind the TM
7. Conductive hearing loss (Weber/Rinné test with 512 tuning fork: sound lateralizes to affected ear and BC>AC)

B. Treatment

1. Treat underlying cause
2. Administer oral decongestants such as pseudoephedrine hydrochloride (Sudafed®)
3. Administer oxymetazoline hydrochloride (Afrin®) nose drops for treatment of associated URI. Do not use for more than 3 days
4. Perform self-insuflation of the ear using Valsalva maneuver after each instillation of nose drops

ACTION ALERT: Patients must be able to clear their ears (perform Valsalva maneuver) before being allowed to fly. Flying with serous otitis media predisposes patients to tympanic membrane rupture since they are unable to equalize pressure on both sides of the membrane.

VII. **VERTIGO -** *Symptom related to several disorders affecting the inner ear. A false sensation of movement causing deficits in proprioception, balance and gait.*

A. Signs and Symptoms

1. "Dizziness", imbalance, or sensation of spinning or tilting made worse by moving or tilting the head
2. May be accompanied by severe nausea or vomiting
3. Patient may appear to be having a "panic attack"- often the vertigo is so scary they get sensation of panic with tachycardia and dyspnea
4. Nystagmus, gait instability, and falling to side while attempting Romberg test can all be present

B. Treatment

1. Attempt to determine the underlying cause of the vertigo
2. Treat the underlying cause when applicable
3. Administer diphenhydramine (Benadryl®) PO, or if symptoms are severe, administer diazepam (Valium®) PO
4. If patient is still nauseous 30 minutes after diphenhydramine administration, consider ondansetron (Zofran®)
5. Check hearing
6. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. If tinnitus or hearing changes are present, evacuate for further evaluation
    b. If vertigo follows otitis media, meningitis, or signs of other infectious process, antibiotics are indicated. Discuss with preceptor.
7. Place patient at bedrest, preferably in a darkened room, until severe symptoms subside
8. For mild symptoms or motion sickness prophylaxis or treatment, give dimenhydrinate (Dramamine®), or meclizine (Antivert®) PO

Nose

VIII. **RHINOSINUSITIS, ACUTE VIRAL/"UPPER RESPIRATORY INFECTION"-** *A "cold". An inflammation and infection of the paranasal sinuses caused by a variety of communicable viruses.*

A. Signs and Symptoms

1. Nasal congestion/obstruction
2. Loss of sense of smell
3. Rhinorrhea (may be purulent)
4. Ear pressure/pressure when leaning forward
5. Symptoms are typically less than 10 days
6. Swollen, erythematous turbinates
7. Maxillary pressure with palpation and percussion

B. Treatment

1. Symptoms should improve or resolve within 10 days with/without therapy
2. If cough is present, check pulse oximetry and peak flows
3. Consider nasal washes for influenza and adenovirus

4. Mechanical flushing of the nasal cavity with salt solution reduces symptoms and prevents bacterial superinfection
    a. Use syringe or netipot to irrigate nasal passageways
    b. Mix 1 Tbsp Kosher salt (no iodine) with 8 oz warm water
    c. Encourage patient that although initially unpleasant, once mastered this technique will dramatically improve symptoms and pressure
5. If ear pressure is also a symptom, administer oxymetazoline hydrochloride (Afrin ®) for no more than 3 days
6. Provide supportive therapy and encourage PO fluid intake >3 L per day or enough for patient to urinate every 2 hours while awake

CLINICAL NOTE: Saline spray, zinc, and pseudoephedrine (Sudafed®) are all widely used but lack clinical evidence sufficient to recommend their use.

ACTION ALERT: Avoid using oxymetazoline hydrochloride (Afrin ®) nose drops for more than 3 consecutive days. Rebound congestion may occur with prolonged or excessive use. The above drugs must be used with caution with patients having a cardiac history. Consult with preceptor prior to administering if such history is present.

IX. **RHINITIS, ALLERGIC**- *Differentiated from viral and bacterial rhinosinusitis due to multiple factors. Symptoms of allergic rhinitis are intermittent and change based on season or environment. Usually symptoms all start at the same time and do not evolve. Nasal symptoms tend to occur before sore throat. This is in contrast to a "cold", which often starts as malaise and sore throat and progresses to cough.*

A. Signs and Symptoms

1. Thin watery, "dripping faucet" nasal secretions; watery eyes and tearing
2. Absence of pain
3. Itching of the nose, throat, eyes, or face
4. Sneezing
5. All symptoms start at once and leave at once without progression of symptoms
6. Pale grey, white or blue swollen turbinates which may appear to have dangling masses or polyps
7. Visible post nasal drip in posterior oropharynx or pharyngeal erythema
8. Associated symptoms include: increased lacrimation, conjunctival injection/cobblestoning, and eyelid edema

B. Treatment

1. If cough is present, check pulse oximetry and peak flows
2. Avoid allergens if known
3. 1st line treatment- nasal glucocorticoids such as fluticasone (Flonase®) or mometasone (Nasonex®)
4. If symptoms are severe, consider adding 1st /2nd generation antihistamine such as hydroxyzine (Atarax®) or diphenhydramine (Benadryl®)

CLINICAL NOTE: 1st line treatment, or if planning to treat with a single agent, use a nasal glucocorticoid such as fluticasone (Flonase®) or mometasone (Nasonex®). Additionally, may add 1st generation antihistamine as above. Loratidine (Claritin®), cetirizine (Zyrtec®) and fexofenadine (Allegra®) may be used instead of 1st generation antihistamine due to their decreased side effect of drowsiness, but they have been shown to be less effective than the first generation antihistamines. Decongestants such as pseudoephedrine (Sudafed®) do not demonstrate significant improvement when compared to nasal glucocorticoids.

X. **BACTERIAL SINUSITIS**- *An acute inflammation of the paranasal sinuses. Caused by multiple different strains of bacteria.*

A. Signs and Symptoms

1. Nasal congestion/obstruction, purulent nasal discharge, fever, loss of sense of smell, ear pressure, halitosis, decreased appetite, and headache all possible
2. Symptoms >10 days duration
3. Positive head tilt, positive tooth rake
4. Swollen, erythematous turbinates
5. Maxillary sinus tenderness (especially unilateral) with palpation/percussion
6. Initial improvement followed by worsening symptoms

B. Treatment

1. If fever < 101° and pain is mild, preferred treatment is close observation and nasal saline washes (See "Acute Viral Sinusitis")
2. If cough is present, check pulse oximetry and peak flows
3. Consider nasal washings for influenza and adenovirus
4. If mucous is thick, may use mucolytics such as guaifenesin (Mucinex®)
5. If nasal swelling, congestion, or sinus pressure is a predominant symptom, give nasal glucocoricoid such as Flonase® or Nasonex®

ACTION ALERT: If patient has fever> 103°F, visual changes, periorbital edema, or severe headache, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION. Consider evacuation for further evaluation and management.

6. Administer analgesics/antipyretics such as acetaminophen (Tylenol®) if required for relief of pain and fever
7. If ear pressure is also a symptom, administer oxymetazoline hydrochloride (Afrin®) nasally for no more than 3 days
8. Provide supportive therapy and encourage PO fluid intake >3 L per day or enough for patient to urinate every 2 hours while awake
9. Zinc preparations, systemic decongestants such as pseudoephedrine hydrochloride (Sudafed®), and antihistamines have little efficacy compared to side effect profile and should be avoided, although patient may be adamant that they help
10. If fever >101° or severe pain, CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. 1st line- amoxicillin, trimethoprim-sulfamethaxazole (Septra®), or erythromcyin
    b. Do not use amoxicillin in penicillin-allergic patients

CLINICAL NOTE: Antibiotics are not recommended if pain is mild and temperature is less than 101° even if symptoms have lasted >10 days. Spontaneous resolution is likely. Follow viral rhinosinusitis guidelines.

XI. **BAROTRAUMA, SINUS**- See "Environmental Injury Protocols, Barotrauma, Sinus"

XII. **EPISTAXIS**- *Bloody nose" incidence is increased in winter weather due to decreased humidity in the air and drying of the nasal passages. May result from facial trauma or arise spontaneously. Other factors that predispose patients to epistaxis include upper respiratory infection, excessive alcohol intake, allergic rhinitis, digital trauma and ingesting medications that interfere with clotting (i.e. NSAIDS and aspirin), and nasal steroid use* (*Flonase*®).

A. Signs and Symptoms

1. Bleeding from the nose, usually unilateral, often brisk
2. History of one or more of the above predisposing factors
3. Evaluate for possible foreign body during history and physical
4. Determine site of bleed (have patient blow their nose and attempt examination)

B. Treatment

1. **IMMEDIATE ACTION**
   a. Monitor the airway
   b. Keep patient in a sitting position; do not lean patient forward or backward; keep patient's head in a neutral position
   c. Instruct patient to blow their nose and spray 2 sprays of oxymetazoline (Afrin®) in affected nare
   d. Instruct patient to apply pressure by pinching the soft lobular portion of the nose (pressure should be maintained for 5 to 10 minutes)
   e. This is often sufficient; however, if bleeding continues:
      i. Pack the bleeding nostril with a cotton ball or tampon that has been saturated with a vasoconstrictor such as oxymetazoline hydrochloride (Afrin®) or lidocaine with epinephrine (place a small amount of Bacitracin® on the tip of the tampon or cotton ball for ease of insertion)
      ii. After insertion, inject approximately 10 cc of saline into the tampon to cause tampon to expand
      iii. Leave tampon in place for >15 minutes
      iv. If bleeding continues and originates from a visible, bare blood vessel in the anterior portion of the nostril, use a silver nitrate stick to cauterize the bleeding site
      v. If the above measures are insufficient, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION and possible evacuation

ACTION ALERT: If bleeding is severe or in the posterior area and uncontrollable by the above treatments, use the postnasal pack procedure to control bleeding. The balloon of a Foley catheter may be used to obstruct the choana (After consulting with preceptor).

SINUSITIS, MUCORMYCOSIS (FUNGAL)- See "Febrile Illness Protocols, Fungal Pneumonias, Mucormycosis"

Throat

CLINICAL NOTE: For all sore throats, consider possible life threats of peritonsillar abscess and rheumatic fever (a complication of strep throat). Sore throats can be broken down into categories of bacterial vs viral vs other. This section reviews major causes of sore throat and diagnosis and treatment for each category. Subjective data should include the Centor criteria, if rash is present, and whether or not the patient is able to swallow saliva.

XIII. **STREP THROAT-** *For empirical treatment with antibiotics, patients should meet 3 of 4 Centor criteria. If patient does not meet Centor criteria but with severe sore throat, close followup is recommended.*

A. Signs and Symptoms

1. Centor criteria
    a. Erythematous, exudative tonsils and posterior oropharynx
    b. Tender anterior cervical lymphadenopathy
    c. Fever by history or exam
    d. No cough
2. May also have halitosis, hematuria, rashes, vision changes, joint pain and orthostatic hypotension

CLINICAL NOTE: For rheumatic fever, check cardio for S3 sounds, murmur, tachycardia, and rubs; check skin for rash and check joints for pain/inflammation. Check lungs for rales/crackles at bases. If splenomegaly is present, diagnosis is more-likely EBV.

B. Treatment

1. **IMMEDIATE ACTION**
    a. Monitor vital signs
    b. Initiate IV fluids if needed to correct orthostasis
    c. Advise patient to rest and eat a light diet. Encourage fluid intake
    d. Advise patient to decrease or stop smoking
    e. Administer analgesics/antipyretics as required. May give up to 1 gram of acetaminophen (Tylenol®) and/or 800mg of ibuprofen (Motrin®) if fever >101°F and no hematuria
    f. Perform rapid strep test if available
        i. If rapid test is negative, perform throat culture and sensitivity tests to identify causative organism
        ii. If rapid strep test is positive, throat culture is not necessary
    g. Consider nasal washes for influenza and adenovirus
    h. Administer warm (not hot) saline gargles every 2 hours
    i. Quarters until afebrile for 24 hours
    j. CONTACT PHYSICIAN PRECEPTOR-MEDICATION

i. Administer antibiotics if 3 or more Centor criteria are met and/or rapid strep test is positive
ii. If patient is **NOT** allergic to penicillin, give phenoxymethyl penicillin (Pen Vee K®) or IM equivalent.
iii. If patient **IS** allergic to penicillin, give clarithromycin (Biaxin®), azithromycin (Zithromax®), or erythromycin (E-mycin®)

CLINICAL NOTE: Ceftriaxone (Rocephin®) may be given for PCN sensitivity such as a rash, but not for PCN anaphylaxis).

ACTION ALERT: Strep throat can have serious or even deadly consequences such as poststreptococcal glomerulonephritis (marked by hematuria and other urinary changes) and acute rheumatic fever (fine pink sandpaper-like rash, vision changes, joint pain, and valvular heart disease). To reduce the chance of complications such as rheumatic fever or nephritis, treatment of pharyngitis should be continued for the full course.

XIV. **VIRAL PHARYNGITIS-** *Likely etiology when patient meets <3 of the Centor criteria. Etiologic agents can include Epstein-Barr virus (EBV), influenza, rhinovirus, adenovirus, respiratory syncytial virus (RSV) and HIV.*

A. Signs and Symptoms

1. Review signs and symptoms for peritonsillar abscess and strep throat
2. Will not meet Centor criteria
3. May display the following:
   a. Low grade or no fever, rhinitis, cough/hoarse voice, sinusitis, headache and body aches and decreased appetite
   b. Erythema of the posterior oropharynx, normal to slightly enlarged tonsils without exudates
   c. Mild, nontender lymphadenopathy or absence of palpable lymph nodes

B. Treatment

1. Supportive care
   a. Lozenges or numbing sprays may provide symptomatic relief
   b. Encourage fluid intake to >3 L per day
   c. Soft or liquid diet; advance as tolerated
   d. Acetaminophen (Tylenol®) for pain, body aches and fever

XV. **MONONUCLEOSIS (EPSTEIN-BARR VIRUS)-** See "Febrile Illness Protocols, Mononucleosis"

XVI. **MUCOCUTANEOUS LEISHMANIASIS-** See "Integumentary Protocols, Leishmaniasis"

XVII. **PERITONSILLAR ABSCESS -** *An acute infection/abscess of pharyngeal structures common in young adults. This disorder can progress rapidly and may pose a significant threat to the patient's airway.*

A. Signs and Symptoms

1. Difficulty swallowing, progressing to
    a. Sensation of airway closure when lying supine, progressing to
    b. Inability to swallow saliva with frequent spitting/drooling, progressing to
    c. Airway collapse requiring surgical airway management
2. Unilateral swollen tonsil
3. Deviation of uvula to unaffected side
4. Tonsillar, cervical and submandibular lymphadenopathy
5. Snoring sounds from upper airway obstruction
6. Orthostatic hypotension

B. Treatment

1. **IMMEDIATE ACTION**
    a. Evaluate and monitor airway
    b. Be prepared for intubation and crichothyroidotomy for airway management
    c. If patient has orthostatic hypotension, initiate IV access, give 1 liter fluid bolus and then titrate until orthostasis is corrected
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Administer antimicrobial therapy
        i. If patient is **NOT** allergic to penicillin (Pen Vee K®), initiate IV ampicillin-sublactam (Unasyn®) or IV clindamycin (Cleocin®).
        ii. PO antibiotics are not as effective, but may be used if IV antibiotics are not available. Consider ampicillin (Omnipen®) or clindamycin (Cleocin®)PO
    b. Administer analgesics for relief of pain, acetaminophen or ibuprofen (Motrin®) as necessary. If unable to tolerate PO pain medicine, consider low dose IV narcotic such as morphine
    c. Ensure adequate fluid intake IV or PO until evacuation- approximately 3 L per day or enough to urinate every two hours while awake

XVIII. LARYNGITIS- *Inflammation of the larynx due to a virus or overuse. May also be caused by gastroesophageal reflux or chronic bronchitis.*

A. Signs and Symptoms

1. Change of voice
2. Hoarseness and raw sensation of the pharynx
3. Constant urge to clear throat may occur

B. Treatment

1. Advise patient to rest voice (use a note pad and pencil to communicate; whispering will also stress the voice)
2. Advise patient to stop smoking
3. Administer steam inhalations
4. Administer analgesics, as needed, for relief of pain, acetaminophen (Tylenol®)
5. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Administer antimicrobial therapy if patient meets Centor criteria, see "Strep Throat" protocols

Chapter 5: Dental Protocols

I. **ACUTE NECROTIZING ULCERATIVE GINGIVITIS ("ANUG")**- *Infectious disease of the marginal gingival and attachment apparatus of the tooth. Synonyms include "trench mouth" and "Vincent's Infection."*

A. Signs and Symptoms

1. Factors associated with ANUG are: smoking, stress and poor oral hygiene
2. Acute pain
3. Halitosis
4. Interproximal necrosis resulting in characteristic, "punched out" appearance to the papillae

B. Treatment

1. Administer analgesics PO as necessary for pain. Consider acetaminophen (Tylenol®) or ibuprofen (Motrin®)
2. Manually remove as much plaque as possible with a toothbrush
3. Benzocaine 20% (Hurricane® Topical Anesthetic gel) may be placed on the gingiva before brushing (tissue will be very sensitive with or without the gel)
4. Counsel the patient on
   a. The need for good nutrition, oral hygiene and plenty of rest
   b. The need to stop smoking
   c. The use of dental floss and thorough brushing several times a day is a must!
5. Have patient swish with 1 capful of Peridex® for 30 seconds and expectorate, twice daily for 7 days
6. **CONTACT DENTAL PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider antibiotic therapy such as Penicillin (Pen VK®) if symptoms do not improve
   b. Provide further analgesia as necessary; consider acetaminophen with codeine (Tylenol #3®)

II. **ALVEOLAR OSTEITIS ("DRY SOCKET")**- *Post-extraction alveolitis is characterized by pain emanating from the mandible or maxilla. It occurs when the clot that had formed in the removed tooth's socket becomes dislodged.*

A. Signs and Symptoms

1. More common in smokers and oral contraceptive users
2. Associated with post-extraction of the 3rd molars
3. Pain begins 2-3 days after tooth extraction and radiates to the ear

B. Treatment

1. Use sterile H_2O or saline to gently irrigate the socket and remove necrotic debris
2. Apply a palliative medication: Gel Foam® slightly moistened with Eugenol® placed in the socket for 24 hours (should relieve the intense ache within 30 - 40 minutes)
3. Continue changing dressing every 24 hours for 3 days, gently irrigating extraction site with sterile saline before replacing dressing

4. Administer analgesics, PO as necessary for pain. Consider acetaminophen (Tylenol®), ibuprofen (Motrin®) or acetaminophen with codeine (Tylenol #3®)
5. CONTACT DENTAL PRECEPTOR– MEDICATION
    a. Antibiotics are not generally warranted, but each case should be discussed with a dental preceptor
        i. Penicillin (Pen-Vee K®) if patient is not penicillin allergic
        ii. Erythromycin (E-Mycin®) if patient is penicillin is allergic
    b. Notify dental clinic of any persistent symptoms and arrange for patient to be seen as soon as possible

III. **APHTHOUS ULCER-** *Recurrent, round, painful lesions of the oral mucosa with an unknown etiology. Other causes of oral ulcers must be considered and ruled out including herpes, ANUG, candidiasis, hand-foot-mouth disease, gonorrhea and syphilis.*

A. Signs and Symptoms

1. History of recurrence since childhood
2. Stress
3. History of similar lesions 1-2 times per year
4. Pain or burning for 1-2 days preceding ulcers
5. Pain disproportional to the size of the lesion
6. Lesions are non-grouped, well defined, shallow, and round with a necrotic center over an erythematous base

B. Treatment

1. Administer topical anesthetic, such as viscous lidocaine, 1 tablespoon, 4 times a day (before meals and at bedtime) or apply topical anesthetic gel Benzocaine 20% (Hurricane Gel®) to provide short-term relief and to facilitate eating if patient has multiple ulcers. Have patient swish medication in mouth for one to two minutes and expectorate
2. Apply a protective dental paste (Orabase®) to individual ulcers 4 times a day (after meals and at bedtime) to prevent irritation by the teeth and oral fluids
3. Have patient remain on soft bland diet until resolution
4. Notify dental preceptor if condition worsens or does not resolve in 7-10 days

IV. **TRAUMATIC ULCER-** *Ulcerations associated with chemical burns, cheek or tongue biting secondary to mastication, abrasions from fractured teeth or a thermal burn from hot food.*

A. Signs and Symptoms

1. Round Painful lesion in close proximity to a fractured tooth, near the biting plane on buccal mucosa, near the palate as a result of hot food entry into the oral cavity, or at the placement site for improper use of aspirin or other analgesics as patient's self-treatment for a tooth ache

B. Treatment

1. Administer topical anesthetic, such as viscous lidocaine, 1 tablespoon, 4 times a day (before meals and at bedtime) or apply topical anesthetic gel Benzocaine 20% (Hurricane Gel®) to provide short-term relief and to facilitate eating if patient has multiple ulcers. Have patient swish medication in mouth for one to two minutes and expectorate
2. Apply a protective dental paste (Orabase®) to individual ulcers 4 times a day (after meals and at bedtime) to prevent irritation by the teeth and oral fluids

3. Have Patient remain on soft bland diet until resolution
4. Notify dental preceptor if condition worsens or does not resolve in 7-10 days

V. **AVULSED TOOTH**- *Missing tooth following trauma to the face.*

A. Signs and Symptoms

1. Missing tooth
2. Gum trauma

B. Treatment

1. **IMMEDIATE ACTION**
   a. Examine socket area and gums for any obvious bone fragment or deformity (remove any deformity)
   b. Place tooth in Save A Tooth® solution (Hank's Balanced Salt Solution®) for 20 min
   c. Reimplant tooth in socket site; if unable to reimplant leave it in solution

> **CLINICAL NOTE:** Do not scrape tooth. If Save A Tooth® solution is not available, other storage solution options include the following (in order of preference): milk, saline, saliva, or sterile water.

   d. Place a small amount of wax on the avulsed tooth and adjacent teeth to help stabilize tooth
   e. Administer analgesics- PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®), acetaminophen with codeine (Tylenol #3®) for severe pain
1. CONTACT DENTAL PRECEPTOR- EVACUATION AND MEDICATION
   a. If patient is NOT allergic to penicillin, administer penicillin (Pen-VK®)
   b. If patient IS allergic to penicillin, administer clindamycin (Cleocin®)
   c. Administer tetanus toxoid 0.5 ml or immunoglobulin as indicated
   d.

VI. **BAROTRAUMA, DENTAL- See "Environmental Injury Protocols, Barotrauma, Dental"**

VII. **DENTAL CARIES**- *Tooth decay known as "cavities."*

A. Signs and Symptoms

1. Tender, painful teeth
2. Initially produces no symptoms
3. Pain with exposure to hot, cold or sweet foods/beverages
4. May have pain with chewing or percussion (tooth rake/tooth sleuth)

B. Treatment

1. Remove any gross debris if visible with saline irrigation or floss
2. Administer analgesics--PO for pain as required. Consider acetylsalicylic acid (Aspirin®), acetaminophen (Tylenol®), or ibuprofen (Motrin®). acetaminophen with codeine (Tylenol #3®) for severe pain
3. CONTACT DENTAL PRECEPTOR- CONSULTATION
   a. If symptoms are relieved, make a routine scheduled appointment
   b. If symptoms are not relieved with analgesics, notify dental preceptor for evacuation

c. If evacuation is delayed, place a temporary filling using IRM® or other restorative material

VIII. **DISPLACED TOOTH**- *Misaligned tooth following trauma to the face.*

A. Signs and Symptoms

1. Visible malalignment of the tooth
2. Trauma to the gums and mouth
3. External facial trauma and edema often visible

B. Treatment

1. **IMMEDIATE ACTION**
   a. Attempt to reposition tooth in socket with finger pressure and stabilize with wax if tooth is very loose
   b. If unable to move tooth into original position, place gauze between posterior teeth as a jaw rest
   c. Administer analgesics--PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol #3®) for severe pain
2. CONTACT DENTAL PRECEPTOR- EVACUATION

IX. **FRACTURED TOOTH OR CROWN**- *A fracture of a tooth resulting from blunt-force trauma or overuse.*

A. Signs and Symptoms

1. Visible fracture or missing portion of a tooth
2. Pain, temperature sensitivity
3. May be asymptomatic

B. Treatment

1. **IMMEDIATE ACTION**
   a. Mix glass ionomer restorative material such as Fuji 9®, IRM® or Vitrabond®
   b. Cover exposed area with restorative material
   c. Smooth surfaces of material applied
   d. Have patient bite down gently to check occlusion of teeth
   e. Remove any excess material
   f. Administer analgesics--PO for pain as required. Consider acetylsalicylic acid (Aspirin®), acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol #3®) for severe pain
   g. CONTACT DENTAL PRECEPTOR- EVACUATION

CLINICAL NOTE: If tooth is asymptomatic, immediate treatment is not required. Schedule a routine appointment.

CLINICAL NOTE: If symptomatic after treatment, notify dental clinic and arrange for patient to be seen as soon as possible (ASAP). If asymptomatic after treatment, make a routine scheduled appointment.

X. GINGIVITIS- *Inflammation of the gingiva usually caused by poor oral hygiene.*

A. Signs and Symptoms

1. Non-painful inflammation of the marginal gingiva
2. Swelling and easy bleeding

B. Treatment

1. Advise patient to maintain good oral hygiene
2. Have patient swish with 1 capful of Peridex® for 30 sec and expectorate, twice daily for 7 days
3. Administer analgesics--PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®); may use acetaminophen with codeine (Tylenol #3®) for severe pain
4. CONTACT DENTAL PRECEPTOR- CONSULTATION
    a. Arrange for patient to make a routine scheduled appointment
    b. Recheck patient in 1 week to make sure symptoms are improving

XI. ODONTOGENIC SPACE INFECTION- *Abscesses or cellulitis involving the buccal or vestibular spaces progressing to bilateral submandibular and sublingual space infections known as Ludwig's Angina. This is an instance in which a toothache may develop into a life-threatening condition.*

A. Signs and Symptoms

1. Firm or fluctuant, painful soft tissue swelling associated with necrotic tooth
2. Systemic signs of an infection include lymphadenopathy, fever and malaise
3. Trismus (can't open jaw all the way)

B. Treatment

1. Monitor patient's airway and vital signs
2. **CONTACT DENTAL PRECEPTOR- IMMEDIATE EVACUATION**
    a. <u>Abscesses</u>
        i. With the concurrence of the Dental Preceptor: incise and drain the abscess at the point of fluctuance either intra- or extra-orally
        ii. Administer analgesics--PO for pain as required. Consider acetylsalicylic acid (Aspirin®), acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol #3®) for severe pain
    b. <u>Cellulitis</u>
        i. Discuss case with Dental Preceptor
            a) If patient is NOT allergic to penicillin, administer penicillin (Pen Vee K®)
            b) If patient IS allergic to penicillin, administer clindamycin (Cleocin®)
    c. Definitive treatment involves draining the abscess and may involve a root canal or extraction

XII. PERIAPICAL ABSCESS- Inflammatory changes of the apex of the tooth root where the pulp canal contents connect with the periodontium. Usually occurs after the death of the dental pulp due to bacterial infection, traumatic ischemia or pulpal necrosis.

A. Signs and Symptoms

1. Sensitivity to percussion (tooth rake/tooth sleuth)

2. Pain may be acute, spontaneous, continuous or intermittent and often positional
3. Affected teeth usually unresponsive to cold, but may respond to heat
4. May develop a granuloma or cyst at the buccal side of the tooth apex
5. May develop a sinus tract with purulent discharge (gum boil)

B. Treatment

1. **IMMEDIATE ACTION**
   a. If obvious superficial fluctuant swelling is present, induce drainage with #11 blade or explorer
   b. Administer analgesics--PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol #3®) for severe pain
2. **CONTACT DENTAL PRECEPTOR- IMMEDIATE EVACUATION**
   a. Definitive treatment may involve a root canal or extraction
   b. If patient is NOT allergic to penicillin, administer penicillin (Pen Vee K®)
   c. If patient IS allergic to penicillin, administer clindamycin (Cleocin®)

XIII. **PERICORNITIS**- *Inflammation or infection of tissues covering or adjacent to non-erupted or partially erupted teeth—usually the 3rd molars.*

A. Signs and Symptoms

1. Trauma to the covering tissues of the erupting tooth caused by the opposing tooth. Usually, it's an upper third molar (upper wisdom tooth) opposing an erupting mandibular third molar (lower wisdom tooth)
2. Foreign body sensation under the covering tissues (operculum)
3. May develop trismus in masticatory muscles (unable to open mouth wide)

B. Treatment

1. If possible, remove obvious plaque buildup by irrigation of the area using large amounts of saline and an irrigation syringe (be gentle because this area will be very tender)
2. Stress to the patient the need for good oral hygiene to improve the condition of the gum in spite of the pain or bleeding
3. Have patient swish with 1 capful of Peridex® for 30 sec and expectorate, BID x 7 days
4. Dispense an irrigation syringe to patient and show them how to irrigate area four times a day with saline solution
5. Administer analgesics, PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol # 3®), for severe pain
6. If available, consider giving the patient a maxillary mouthguard to prevent further irritation to the lower tooth/gum tissue
7. CONTACT DENTAL PRECEPTOR- CONSULTATION AND MEDICATION
8. If relief is not evident in 4 to 8 hours, administer appropriate antimicrobial therapy and notify dental clinic of any persistent symptoms
   a. If patient is NOT allergic to penicillin, administer phenoxymethyl penicillin (Pen-Vee K®)
   b. If patient IS allergic to penicillin, administer clindamycin (Cleocin®)

9. If relief is not evident in 4 to 8 hours, administer appropriate antimicrobial therapy and notify dental clinic of any persistent symptoms

XIV. **PERIODONTAL ABSCESS**- *An abscess involving the infection of connective tissue surrounding teeth. Pre-existing loss of connective tissue and bone with periodontal pocketing are usually required for development of a periodontal abscess.*

A. Signs and Symptoms

1. Foreign body, often calculus or food debris forms a nidus for development of a localized infection with a periodontal pocket
2. Sensitivity to palpation of the tooth with tooth mobility is diagnostic

B. Treatment

1. If possible, remove obvious plaque buildup by irrigation of the area using large amounts of saline and an irrigation syringe. Be gentle because this area will be very tender
2. Stress to the patient the need for good oral hygiene to improve the condition of the gum in spite of the pain or bleeding
3. Have patient swish with 1 capful of Peridex® for 30 sec and expectorate, BID x 7 days
4. Administer analgesics, PO for pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®); acetaminophen with codeine (Tylenol #3®), for severe pain
5. CONTACT DENTAL PRECEPTOR-EVACUATION
    a. If patient is NOT allergic to penicillin, administer penicillin (PenVee K®)
    b. If patient IS allergic to penicillin, administer Clindamycin (Cleocin®)

XV. **REVERSIBLE PULPITIS**- *Pulpalgia and pulpitis describe pain and inflammatory changes possible in an affected tooth. Pulpitis can be described as reversible or irreversible. Diagnosis is based on symptoms.*

A. Signs and Symptoms

1. Hypersensitive reaction precipitated by thermal stimuli, usually cold; pain is acute and lasts less than 5 seconds after application of stimulus
2. No sensitivity to palpation/percussion (tooth rake/tooth sleuth)
3. History of defective filling or recent dental work

B. Treatment

1. Administer analgesics, PO for mild or moderate pain as required. Consider aspirin, acetaminophen (Tylenol®), or ibuprofen (Motrin®), acetaminophen with codeine (Tylenol#3®), for severe pain
2. CONTACT DENTAL PRECEPTOR

CLINICAL NOTE: if symptoms do not improve in 24 hours, notify dental clinic and arrange for patient to make a routine scheduled appointment.

XVI. **IRREVERSIBLE PULPITIS**- *Pulpalgia and pulpitis describe pain and inflammatory changes possible in an affected tooth. Pulpitis can be described as reversible or irreversible. Diagnosis is based on symptoms.*

A. Signs and Symptoms

1. May or may not be symptomatic
2. Spontaneous pain and prolonged painful response to thermal stimuli are diagnostic
3. Causes include caries, defective restorations, cracked teeth, or barodentalgia
4. May be sensitive to biting pressure

B. Treatment

1. Administer analgesics, PO for mild or moderate pain as required. Consider acetylsalicylic acid (Aspirin®), acetaminophen (Tylenol®), or ibuprofen (Motrin®), acetaminophen with codeine (Tylenol#3®), for severe pain
2. CONTACT DENTAL PRECEPTOR- EVACUATION
    a. If patient is NOT allergic to penicillin, administer penicillin (Pen Vee K®)
    b. If patient IS allergic to penicillin, administer clindamycin (Cleocin®)

Chapter 6: Neurologic Protocols

I. **ALCOHOL INTOXICATION**- *Approximately 10% of American adults abuse alcohol- it is implicated as the leading factor in traffic accidents, domestic violence, homicide, and suicide nationwide. Military personnel frequently avoid medical attention for alcohol-related trauma due to fear of career repercussions.*

A. Signs and Symptoms

1. Symptoms based on blood alcohol content percentage:
    a. 0.01-0.10%: euphoria, mild deficits in coordination, attention, and cognition
    b. 0.11-0.20%: deficits in coordination, decreased attention, slurred speech, uncoordinated walk, impaired judgment, mood volatility
    c. 0.21-0.30%: incoherence, nausea, vomiting
    d. >0.30%: stupor, loss of consciousness, respiratory depression, coma, death
2. At any of these levels, patient may be malodorous and unkempt

CLINICAL NOTE: Alcohol breakdown is linear, meaning only a certain amount can be broken down per hour regardless of how much is ingested. It is impossible to determine clinically if the patient is still absorbing ingested alcohol or if the patient has stopped absorbing and is now breaking the alcohol down. Depending on the severity of the intoxication or the reported amount ingested, be prepared for patient to worsen before getting better. Plan to observe all patients with acute intoxication brought to medical attention for several hours. Perform vitals and brief history and physical hourly until released.

B. Treatment

1. Monitor hydration status, encourage fluid intake
2. "Hangovers" typically caused by acute dehydration following diuresis from alcohol
3. Do not release patient until clinical symptoms of intoxication have passed
4. Do not release patient alone- release into care of competent adult
5. In patients with alcohol intoxication and nausea/vomiting, respiratory depression and loss of airway is a serious concern
    a. Initiate IV rehydration with Ringer's Lactate and reassess vitals every 15 minutes until stabilized
6. For patients with decreased level of consciousness, follow above measures AND administer thiamine IV
7. May place an NG tube if patient is nauseous/vomiting, but gastric lavage and/or charcoal are generally not indicated due to rapid rate of alcohol absorption

II. **ALCOHOL WITHDRAWAL SYNDROME (DELERIUM TREMENS)**- *The "DT's". Abrupt cessation of alcohol use in someone who abuses alcohol. Symptoms begin 6-12 hours after the last drink. High risk categories include individuals who consume alcohol daily and "binge drinkers" that have consumed more than five drinks per sitting with three or more binges over the past two weeks.*

A. Signs and Symptoms

1. Based on time since last drink and follow a predictable pattern
    a. Stage 1: 6-24 hours after last drink
        i. Agitation, anxiety, insomnia, tremors, diaphoresis, palpitations, mild hypertension, tachycardia less than 120 bpm, nausea
        ii. Mostly coherent at this stage
        iii. Stage 1 may resolve within 72 hours or proceed into severe withdrawal
    b. Stage 2: 24-72 hours after last drink
        i. Similar symptoms as stage 1 but more severe
        ii. Marked agitation, nausea/vomiting/ diarrhea, severe tremors, tachycardia, escalating hypertension, marked diaphoresis
        iii. May have seizures at this stage
    c. Stage 3- "Delirium Tremens": 72-96 hours after last drink
        i. Hyperactivity, fever, severe tachycardia and hypertension
        ii. Agitation, drenching sweats, hallucinations, disorientation
        iii. May have seizures at this stage
2. Patients can progress rapidly or unpredictably through the stages

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Patient likely to be combative- protect self first
3. Avoid being alone with patient
4. Monitor vitals and neurologic status every 30 minutes
5. Consider diazepam (Valium®) or lorazepam (Ativan®) for severe agitation or seizures

III. SPINAL CORD TRAUMA- *Trauma that causes direct or indirect pressure or damage to the spinal cord. Common etiologies include motor vehicle accidents, falls, gunshot wounds, and high-velocity sporting accidents (bull riding, boating accidents, etc).*

A. Signs and Symptoms

1. Transient or permanent sensation/motor function loss below level of trauma
2. Trauma to cervical spine causes quadriplegia and full-body flaccid paralysis
3. Priapism and urinary retention are common with lumbar spine injuries
4. Pain at the vertebrae or nerve roots associated with the trauma
5. Cervical spine injuries cause quadriplegia and can be fatal
    a. The diaphragm is innervated by cervical nerves at the C3-C5 spinal level
    b. Trauma at or above this area will lead to difficulty breathing in addition to flaccid paralysis of the body

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Move patient as little as possible
3. Logroll technique, C-Spine precautions, backboard
4. Do not backboard if immobilization causes deterioration in sensation or motor function
5. See "trauma" in emergency treatment protocols for further information

CLINICAL NOTE: Patients with significant mechanism for spinal cord injury who do not present with paralysis MUST BE OBSERVED for a minimum of six hours due to the possibility of swelling causing compression of the cord.

IV. **CEREBROVASCULAR ACCIDENT-** See "Emergency Medicine Protocols- Stroke" and follow current ACLS guidelines

V. **GULLAIN-BARRÉ SYNDROME**- *Progressive ascending paralysis that follows routine infection.*

A. Signs and Symptoms

1. Patient initially complains of weakness of the lower extremities which may be accompanied by back or leg pain
2. May report sensory abnormalities in hands and feet but minimal or no sensory abnormalities found on exam
3. Deep tendon reflexes are absent
4. Symptoms progress over next 2-4 weeks with ascending paralysis, respiratory depression, and potentially death from asphyxiation
5. Mental status is usually preserved but patient is unable to move or respond ("Locked in")

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Transport any patient with ascending weakness
3. Be prepared for advanced airway management and mechanical ventilation if evacuation is delayed

VI. **HEADACHE, MIGRAINE**- *Benign headache that is often confused with acute neurologic deficiency due to transient sensory or motor changes. Patients typically try to stay as still as possible.*

A. Signs and Symptoms

1. Chronic, episodic headache
2. Symptoms may last 4-72 hours
3. Character usually throbbing or pulsatile moderate to severe pain
4. Typically improves or resolves with sleep
5. Pain may be unilateral or bilateral
6. Associated symptoms include nausea/vomiting, photophobia, and phonophobia
7. Patient may have difficulty concentrating during attacks
8. Headache may be proceeded by one or more prodromal symptoms:
    a. Visual symptoms- bright zigzags, scotomata, scintillating lights
    b. Paresthesias, numbness, or tingling
    c. Speech disturbances

CLINICAL NOTE: Frequent opioid use can contribute to development of chronic daily headache and may interfere with other preventive therapies. Regular users of NSAIDs or acetaminophen are also likely to experience rebound headache.

B. Treatment

1. Place patient in dark, quiet room
2. Manage mild to moderate pain with NSAIDS or acetaminophen (Tylenol®), but not if they have had either of these in the past 24 hours
3. For incapacitating pain
    a. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
        i. Consider administration of zolmitriptan (Zomig®) or sumitriptan (Imitrex®)
        ii. Alternative therapies include acetaminophen + aspirin + caffeine (Excedrin®)
        iii. For severe pain, consider triple therapy of 1L IV NS, metoclopramide (Phenergan®) IV/IM, and either zolmitriptan (Zomig®) or sumitriptan (Imitrex®) (This therapy works best for patients with migraines and nausea, but is very effective for most migraines)

C. Prevention

1. Reduce stress, regulate sleep cycle, eat regular meals, hydrate enough to urinate every two hours
2. Discontinue caffeine, herbs, supplements, tobacco
3. Avoid triggers if possible (consider emotional stress, mensturation/hormonal irregularities, fasting or inadequate oral intake, nitrates/food additives, weather changes, sleep disturbances, aspartame, and drinking wine are all common triggers)
4. Daily preventive medication indicated if headaches are frequent, long-lasting, or if causing significant disability
5. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
    a. New repetitive headaches require further workup
    b. Physician may consider verapamil (Verelan®), nifedipine (Procardia®), amyltriptiline (Elavil®) or antidepressive therapy

VII. HEADACHE, CLUSTER- *Benign, severe, incapacitating headache that occurs in recurring patterns. Patients typically very agitated with behavior such as pacing, yelling, or thrashing.*

A. Signs and Symptoms

1. Pain typically unilateral in temporal-orbital area
2. Attacks often usually occur at the same time each day or in predictable pattern occurring in groups or "clusters"
3. Pain may wake patient from sleep
4. Pain is excruciating, but peaks within minutes and usually resolves within an hour
5. Patients usually restless and agitated
6. May experience nasal congestion, rhinorrhea, lacrimation, facial flushing and Horner's syndrome (ptosis, miosis, enophtalmos, and anhidrosis) on the ipsilateral (same) side
7. Attacks have a frequency from one every other day to eight per day

B. Treatment

1. Inhalation of 100% oxygen, 7 L per min for 15 min
2. **CONTACT PHYSICIAN PRECEPTOR- EMERGENCY EVACUATION**

VIII. **HEADACHE, TENSION-** *Benign headache associated with emotional stress and muscular fatigue. Symptoms usually recur until underlying physical and emotional stressors are addressed.*

A. Signs and Symptoms

1. Mild, generalized pain generally without the incapacity, nausea, or photophobia seen with migraine
2. Pain typically bilateral, described as "vice-like", non-throbbing
3. Often precipitated by mental stress, or physical stress of sitting still for long periods
4. Normal physical exam or muscle tenderness in the head, neck, or shoulders

B. Treatment

1. Manage pain with ibuprofen (Motrin®), naproxen (Aleve®), acetaminophen (Tylenol®) or combination analgesics such as acetaminophen, aspirin and caffeine (Excederin®) but avoid taking any of these medications regularly for headache since rebound headache may result
2. Muscle relaxants are not recommended

C. Prevention

1. Identify and treat potential triggers
   a. Sleep irregularities/disorders
   b. Poor posture
   c. Emotional stressors
   d. TMJ dysfunction
   e. Neck pain
   f. Eyestrain
2. Increase exercise, regulate sleep schedule, eat healthy diet, hydrate enough to urinate every two hours, avoid herbs, supplements, tobacco, alcohol

IX. **HEAD INJURIES-** See "Emergency Medicine Protocols, Head Trauma".

X. **HERNIATED NUCLEUS PULPOSIS (HNP) WITH RADICULOPATHY-** *"Herniated disc" or "slipped disc". A traumatic or degenerative rupture of one of the cushioning discs between the vertebrae; may be asymptomatic or present clinically due to compression of nearby nerve roots.*

A. Signs and Symptoms (if nerve roots have been compressed)

1. Unilateral sciatica, which is pain or burning sensation down posterior or lateral leg past the knee and worse with specific motions
2. Straight leg test may cause sciatica but should not be bilateral
3. Palpation of the vertebrae and discs causes pain (in musculoskeletal low back pain, palpation on the spinous processes should not be tender)
4. Extension of the spine causes pain and may cause unilateral or bilateral weakness if herniation is severe (Musculoskeletal low back pain typically has lateral back pain and spasms with flexion of spine, but extension of spine is usually not as painful)

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Do not allow patient to lift, push, or pull more than 5 lbs
3. May sit or lay in position of comfort

CLINICAL NOTE: If severe herniated disc, patient may have cauda equina syndrome, which is bowel, bladder, or penile dysfunction and may be associated with numbness in the genitals or inner thighs (ie "saddle anesthesia"). These patients may also have bilateral sciatica or weakness of the lower extremities. Any of these symptoms predict potential for paralysis and therefore require emergency evacuation. Put patient on strict bed rest in position of greatest comfort.

XI. **ENCEPHALITIS-** *Swelling of the brain itself; often associated with viral infections. Differentiated from meningitis because in meningitis, brain functionality is preserved, whereas in encephalitis, thinking, motor, sensory, and speech can be impaired. These viruses have the potential to be used as biological weapons.*

A. Signs and Symptoms

1. Altered mental status
2. Motor and sensory deficits
3. Altered behavior
4. Personality changes
5. Speech disorders
6. Movement disorders
7. Seizures
8. Often caused by viruses such as West Nile or rabies, or following viral infection such as rubella, mumps, measles, herpes, Epstein-Barr, or influenza

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Requires lumbar puncture for further evaluation
3. Give Acyclovir (Zovirax®) IV. Do not delay antiviral treatment if encephalitis is suspected, even without lumbar puncture confirmation
4. Acetaminophen (Tylenol®) for fever/pain

CLINICAL NOTE: Rabies can cause encephalitis. If you suspect rabies, start treatment for rabies with RIG and HDCV. Discuss with preceptor prior to initiating therapy. See "Rabies" in "Febrile Illnesses Protocols, Rabies".

XII. **MENINGITIS-** *Swelling of the fibrous layers (meninges) that protect the brain and spinal cord. Often caused by coexisting bacterial or viral infections.*

A. Signs and Symptoms

1. Unlike encephalitis, patients with meningitis do not generally have focal neurologic findings or altered behavior on exam until very late in course, although they may have decreased level of consciousness earlier
2. Patients may have fever and/or complain of headache or neck stiffness
   a. Bruzinski's sign- passive flexion of the neck causes flexion at the hips in the supine patient
   b. Kernig's sign- Flex hip to 90° in the supine patient and attempt to extend the knee. Kernig's sign is when patient is reluctant to allow knee extension

3. Papilledema on fundoscopic exam (blurring of cup and disc margins) with photophobia
4. Petichiae/purpura (if bacterial)

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Requires lumbar puncture for further evaluation
3. IV access, correct fluid and electrolyte disturbances
4. Do not delay initiation of antibiotics if meningitis is possible
    a. Consider IV vancomycin (Vancocin®) plus IV ceftriaxone (Rocephin®)
    b. Consider IV dexamethasone (Decadron®) as adjunctive therapy
    c. Acetaminophen (Tylenol®) for fever/pain
    d. Monitor and record vital signs every 30 minutes
    e. Monitor and maintain an adequate airway

ACTION ALERT: Meningitis is a lethal and very contagious condition. Contact preceptor immediately to begin prioritizing evacuation. Treat all meningitis as bacterial until proven otherwise.

ACTION ALERT: Close contacts should receive Rifampin 600 mg PO BID x 4 doses for meningococcal prophylaxis. Consult with preceptor.

XIII. SEIZURES- *Brain electrical activity disorder in which whole body spasms and convulses followed by a "postictal state" in which memory and cognition are impaired.*

A. Signs and Symptoms

1. Seizure typically begins as abrupt loss of consciousness, possibly with scream or shriek
2. Full body goes rigid and patient may become cyanotic for approximately a minute ("tonic phase")
3. This is followed by the violent "clonic" phase with bilateral muscle jerking lasting 1-2 minutes. Patient may bite tongue, froth at mouth, or become incontinent during this phase
4. Seizure is followed by a postictal state lasting up to several hours during which patient has altered level of consciousness, deep breathing, and confusion
5. As patient becomes more alert they generally complain of headache
6. Etiologies in healthy active duty population include acute intoxications, poisonings, alcohol intoxication or withdrawal, head trauma, infections, hypoxia, and hypoglycemia
7. Primary seizures (ie epilepsy) usually triggered by environmental or physiological stressor such as flashing lights, strong emotion, intense exercise, or loud music

CLINICAL NOTE: syncope can mimic seizures with unilateral or bilateral muscle contractions, but syncopal patients will be able to remember events preceding the syncopal episode and will not have a postictal phase.

B. Treatment

1. IMMEDIATE ACTION
    a. Establish and maintain an adequate airway using an airway adjunct (OP/NP) unless it induces gagging, vomiting or trismus is present
    b. Protect the patient from injury
    c. During seizure do not try to restrain patient
    d. Clear area so that patient's violent jerkings cause as little trauma as possible
    e. Do not attempt to force patient's mouth open
    f. May administer lorazepam (Ativan®) IV (preferred) or IM during seizure
    g. Place patient on side with head down and suction if necessary
    h. Administer 100% O_2
    i. Treat underlying cause, if known and examine for associated injuries and treat as required
    j. Be alert for possible for muscle damage from severe contractions leading to kidney damage (see "Musculoskeletal Protocols, Rhabdomyolysis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Establish IV access after initial episode
4. Treat any trauma caused by incident
5. Seizure precautions- bedrest with bathroom privileges, pad bedrails, place cot on floor

ACTION ALERT: Do not force objects between teeth as damage to tongue and/or teeth could result. DO NOT PUT YOUR FINGERS IN THE PATIENT'S MOUTH.

XIV. SLEEP DEPRIVATION, "JET LAG", AND SHIFT WORK- *"Sleep deprivation" exists when quantity or quality of sleep is insufficient to support adequate alertness, functioning, and health. Shift work frequently disrupts circadian rhythms required to maintain adequate sleep cycles. "Jet lag" refers to disruption of sleep-wake cycles due to change of multiple time zones.*

A. Signs and Symptoms

1. Daytime sleepiness, inattention/distraction, yawning
2. Mood disorders, outbursts, frustration, "short fuse", stress
3. Increased accidents or mistakes
4. Headaches, neck pain, fatigue, malaise, gastrointestinal upset

B. Treatment/Prevention

1. For chronic poor sleep
    a. Set same bedtime and wakeup routine at same time every day
    b. Use bed for sleep only- no TV, eating, reading, etc.

c. Avoid bright light or screens (TV, computer) within 1 hour of bedtime
d. Perform daily exercise with at least one period of sustained maximal exertion several hours prior to bedtime
e. Avoid caffeine, especially after lunch
f. Avoid alcohol
g. Avoid nicotine, especially after lunch
h. If worries are a contributing factor, journal with lower ambient light immediately prior to bedtime
i. Avoid big meals right before bed
j. If awake in the middle of the night, get out of bed and perform soothing activity in low light; avoid bright screens and eating

2. For jet lag
   a. Maximize time exposed to sunlight in new location
   b. Eat only at normal mealtimes (if normal routine is to eat supper at 6 pm, then eat at 6 pm at new location)
   c. Exercise (especially outdoors) will increase ability to fall asleep
   d. Avoid alcohol and stimulants to regulate sleep cycle if possible
   e. Melatonin (herbal preparation) has supporting data but is not FDA approved; advise patient of risks
3. For shift workers
   a. Avoid rotating shifts or picking up extra shifts
   b. Maintain same sleep schedule on days off
   c. For managers in charge of rotating the shift schedule
      i. Recommend keeping same schedule for longer intervals (3 months or more)
      ii. When rotating shifts, rotate each group later (afternoon shift moves to evening shift, etc)

XV. TRAUMATIC BRAIN INJURY ("CONCUSSION")- *Caused by direct head trauma or acceleration-deceleration forces on the brain. Syndromes are complicated and a significant cause of morbidity in personnel in combat zones.*

A. Signs and Symptoms

1. Constellation of symptoms that follow predictable progression
   a. Within minutes- initially patient is confused, befuddled, with amnesia of event with or without loss of consciousness
   b. Within minutes to hours- dizziness, vertigo, headache, nausea/vomiting, lack of awareness of surroundings
   c. Within hours to days- cognitive or mood disturbances, sensitivity to light or noise
2. Consider traumatic brain injury whenever any of the following are identified
   a. Slurred speech
   b. Delay in answering questions
   c. Vacant stare
   d. Emotions out of proportion to stimulus
   e. Confusion, incomprehensible statements
   f. Disorientation (not aware of person/place/time)
   g. Inability to focus attention
   h. Gross incoordination (unable to walk straight line)

i. Memory deficits or unable to remember 3 items at 5 minutes
j. Any loss of consciousness, however brief

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Every hour assess vitals, neurologic status, speech, coordination, and ability to remember 3 objects at 5 minutes
3. Seizure precautions- bed rest with bathroom privileges, pad bedrails, place cot on floor
4. Don't leave patient alone
5. Fall precautions
6. Consider Military Acute Concussion Evaluation

Chapter 7: Cardiopulmonary Protocols

I. **ASTHMA, ACUTE EXACERBATION/ASTHMA "ATTACK"-** *Acute reaction affecting the airway caused by exposure to allergens or irritants resulting in bronchoconstriction. Diagnosis is based on history, physical exam and pulmonary function testing or spirometry. Asthma is classified by degree of severity.*

A. Signs and Symptoms

1. Dyspnea, chest tightness, non-productive weak cough and wheezing
2. Respiratory distress
3. Possible cyanosis
4. Exam with prolonged expiratory phase and diminished air movement

B. Treatment

1. **IMMEDIATE ACTION**
   a. Check pulse oximetry and peak flows; normal peak flow is greater than 80% predicted value
   b. O_2 via non-rebreather mask at 15 LPM
   c. Administer albuterol via metered-dose inhaler using a spacer or use albuterol nebulizer
   d. Continue O_2 therapy until respiratory rate is <20, then slowly wean
   e. Patient must be able to maintain O_2 saturation above 97% with walking and ADLs
   f. CONTACT PHYSICIAN PRECEPTOR -CONSULTATION
2. If patient unable to maintain O_2 saturation above 97% without supplemental oxygen, CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION
   a. Consider administration of epinephrine IM (EpiPen®, Adrenalin®)
   b. Consider systemic corticosteroid such as prednisolone (Prelone®), prednisone (Deltasone®) or methylprednisolone (Solu-Medrol®)

II. **ASTHMA, MILD INTERMITTENT-** *Intermittent affecting the airway caused by exposure to allergens or irritants resulting in bronchoconstriction. Diagnosis is based on history, physical exam and pulmonary function testing or spirometry. Asthma is classified by degree of severity.*

A. Signs and Symptoms

1. Weak irritating cough especially at night and with exertion. May have reversible feelings of chest "tightness" or wheeze
2. Patient may have history of allergic rhinitis and show symptoms of allergies to include pale, boggy nasal mucosa, nasal polyps, itchy, injected, watery eyes, and clear rhinorrhea
3. Lung exam typically normal or have occasional wheeze at the end of expiration
4. Examiner may note frequent throat-clearing or weak, dry or minimally productive cough

B. Treatment

1. Check Peak flows if available. Normal is greater than 80% predicted value

2. Prescribe an albuterol MDI with spacer as needed; give first dose in clinic and repeat peak flows. Patient should demonstrate some improvement
3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider evacuation if symptoms worsen or no improvement seen
    b. If stable, patient will still eventually require further evaluation to include spirometry or pulmonary function testing when treatment facilities are available
4. Attempt to identify and control triggers
5. Treat coexisting allergic rhinitis aggressively with both intranasal corticosteroid and systemic antihistamine (see allergic rhinitis)

III. **ANTHRAX, INHALATIONAL-** See "Febrile Illness Protocols, Anthrax"

IV. **BRONCHITIS, ACUTE-** *An inflammation of the upper airway usually following an upper respiratory infection. Bronchitis is usually caused by a virus.*

A. Signs and Symptoms

1. Nonproductive cough for more than 5 days with or following URI symptoms
2. Chest pain with deep inspiration
3. No hypoxia
4. May have scattered rhonchi and wheezing; often has normal lung exam
5. Sputum is minimal compared to pneumonia, but may be clear, purulent or blood tinged

B. Treatment

1. Check pulse oximetry and peak flow
2. Acetaminophen (Tylenol®) for pain and fever
3. Avoid antitussives such as benzonate (Tessalon Perles®) or Dextromethorphan combinations (Robitussin®, Triaminic®) unless required to aid sleep
4. If wheezing, consider albuterol MDI with spacer
5. If pneumonia is suspected, CONTACT PHYSICIAN PRECEPTOR and consider antibiotic therapy only in patients with cough, fever, dyspnea and purulent sputum

V. **ACUTE RESPRATORY DISTRESS SYNDROME (ARDS) IN THE ADULT-** I*nflammatory disorder of the lung caused by sepsis, pneumonia, trauma or aspiration.*

A. Signs and Symptoms

1. Develops within 24-48 hours after initial injury or illness
2. Dyspnea, cough and chest pain
3. Tachypnea and tachycardia
4. Mottled/cyanotic skin
5. Abnormal breath sounds that includes crackles, rhonchi and wheezes
6. Respiratory failure
7. Diffuse, bilateral infiltrates on chest x-ray

B. Treatment

1. **IMMEDIATE ACTION**
    a. Apply 100% oxygen via non-rebreather at 15 LPM
    b. Establish large bore IV access with normal saline at maintenance rate
    c. Do not administer additional fluids unless directed to do so

d. Condition almost always requires endotracheal intubation and mechanical ventilation
e. If needed, administer oxygen by positive pressure (AMBU bag or ventilator)- use the lowest level of PEEP or pressure that produces adequate oxygenation

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Attempt to identify and treat underlying cause
   b. Initiate broad-spectrum antibiotics via IV treatment if infection is known or suspected

VI. **BAROTRAUMA, PULMONARY-** See "Envornmental Injury Protocols, Barotrauma, Pulmonary"

VII. **COSTOCHONDRITIS-** See "Musculoskeletal Protocols, Costochondritis"

VIII. **DIPHERHERIA-** *An acute pharyngeal infection caused by Corynebacterium diphtheria (gram positive bacilli), which produces exotoxin.*

A. Signs and Symptoms

1. Incubation 2-4 days with prodromal period of 12-24 hours
2. Respiratory diphtheria characterized by visibly swollen neck, pharynx and tonsils
3. Hoarseness, stridor and dyspnea
4. Sore throat, dysphagia, low grade fever and tachycardia
5. Nausea, emesis, chills, headache
6. Dirty grey fibrous pharyngeal membrane that bleeds easily
7. May develop cardiac dysfunction, cranial nerve abnormalities, or renal failure
8. Cutaneous diphtheria characterized by chronic, non-healing sores or shallow ulcers covered with a dirty grey membrane

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. If evacuation will be delayed, consider initiating antibiotic therapy with IV penicillin or erythromycin

IX. **FLAIL CHEST-** See "Emergency Protocols, Flail Chest"

X. **HUMAN IMMUNODEFICIENCY VIRUS (HIV) SEROCONVERSION-** See "Febrile Illness Protocols, Human Immunodeficiency Virus (HIV) Seroconversion"

XI. **HYPERTENSION-** *chronically elevated systolic or diastolic blood pressure that increases lifetime risk of heart attack or stroke.*

A. Definitions

1. Hypertension- defined as systolic blood pressure greater than or equal to 140 or diastolic blood pressure greater than or equal to 90 on 3-6 blood pressure measurements not taken on the same day
2. Prehypertension- systolic blood pressure 120-139 or diastolic blood pressure 80-89 on 3-6 measurements not taken on the same day
3. Normal blood pressure- less than 120 systolic and less than 80 diastolic

4. Isolated elevated blood pressure- any elevated blood pressure reading that does not meet the definition of "normal blood pressure" in a patient who has not undergone evaluation to determine if they meet the definitions of hypertension or prehypertension
5. Malignant hypertension- diastolic blood pressure greater than 100 accompanied by retinal hemorrhages, retinal exudates, papilledema, heart attack, or stroke (end organ damage)

B. Signs and Symptoms

1. Usually asymptomatic and elevated blood pressure is found incidentally during routine exam for other purposes
2. Blood pressure may be elevated due to pain, abnormal hydration status, nervousness, or increased activity level; blood pressure readings taken under these circumstances should not be used towards diagnosis of hypertension
3. Gather history of possible end organ damage to include questions regarding vision, cardiac, and renal function
4. Perform thorough secondary history to include diet history, exercise history, medications/supplements, and family history

C. Treatment

1. For cases of isolated elevated blood pressure, counsel the patient on any of the following, tailored to their diet and exercise history
    a. Reduce salt intake
    b. Increase aerobic exercise
    c. Stop all chronic NSAID and supplement use
    d. Lose weight if indicated. Even a 5-10 pound weight loss can make substantial improvements in a patient's blood pressure
    e. Increase fruit and vegetable intake
    f. Limit alcohol intake
    g. Stop smoking and using tobacco products
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. If chronic hypertension or prehypertension are suspected, contact preceptor for consultation regarding non-emergent blood pressure evaluation
    b. If malignant hypertension is suspected and in all cases of blood pressure greater than 160 systolic or greater than 100 diastolic, consult preceptor for further direction

XII. HYPERTROPHIC OBSTRUCTIVE CARDIOMYOPATHY- *A genetic disease that frequently goes undiagnosed even when joining the military. It often is discovered postmortem, with a typical scenario being an athlete that is performing well and suddenly dies on the field. It is caused by cardiac muscle fibers that are all going different directions; as they contract they fail to squeeze the ventricle closed. As a result, the heart increases in size, further narrowing the size of the ventricles until they are no longer effective.*

A. Signs and Symptoms

1. Usually asymptomatic, but initial presentation may include exertional dyspnea, chest pain, syncope, palpitations, lightheadedness with standing, and fatigue

2. Physical exam in an asymptomatic patient may include 4th parasternal heart sound, heaves, harsh murmur after S1 at the apex and lower left sternal border that radiates to the axilla
3. Murmur gets worse if the sitting patient suddenly stands or if the patient Valsalvas; murmur gets better if patient squats

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway breathing, O_2, monitor
   b. Keep patient in position of comfort
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XIII. PERICARDITIS- *Acute inflammation of the pericardial sac, often following viral illness.*

A. Signs and Symptoms

1. Sharp, pleuritic chest pain that is relieved by sitting up and leaning forward
2. Pericardial friction rub- a superficial scratching or squeaking sound heard with the diaphragm of the stethoscope at the left sterna border occurring with each heartbeat
3. Recent or resolving viral respiratory or gastrointestinal illness
4. May have shortness of breath
5. Chest pain may improve with sitting up and leaning forward

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, O_2, monitor
   b. Keep patient in position of comfort
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Many patients may be managed with NSAIDs as an outpatient, but additional diagnostic testing is necessary
   b. Keep patient warm

XIV. PERTUSSIS ("WHOOPING COUGH")- *A respiratory infection caused by Bordetella Pertussis (gram negative bacilli) common in children and adolescents characterized by a high-pitched, crowing inspiration.*

A. Signs and Symptoms

1. Incubation period of 7-14 days
2. Begins with URI symptoms of sneezing, lacrimation and signs similar to a common cold
3. Cough begins with paroxysms of rapid cough and characteristic "whoop"ing sound
4. Cough is productive, with thick mucous that may cause vomiting or choking

B. Treatment

1. **IMMEDIATE ACTION**
   a. Initiate respiratory isolation (gloves, gown, mask, private room)
   b. Provide supportive care with O_2 and suction as indicated
   c. Obtain IV access and administer IV fluids if indicated
2. **CONTACT PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider azitrhomycin (Zithromax®) and erythromycin (E-mycin®) depending on the stage of the disease
   b. Fevers may get very high; implement seizure precautions

XV. **COMMUNITY ACQUIRED PNEUMONIA (CAP), BACTERIAL-** *Inflammation of the lungs caused by Streptococcus pneumoniae. X-rays show a solid white consolidation in the affected lobe with air-fluid levels.*

A. Signs and Symptoms

1. Acute onset of malaise, cough, dyspnea and pleuritic chest pain
2. Cough initially dry, but becomes productive, thick, and sometimes painful
3. Dyspnea, especially with exertion
4. Fever, (100.4-105° F), tachypnea and tachycardia
5. Crackles, bronchial breath sounds, bronchophony, egophony and whispered pectoriloquy can be scattered throughout but are worse in affected lobe(s)
6. Scattered inspiratory and/or expiratory wheezes may be present
7. Tactile fremitus and dullness to percussion in the affected lobe(s)

B. Treatment

1. Place patient on bed rest and keep warm
2. Administer oxygen: 2 to 4 liters per minute via nasal cannula or 8-10 liters per minute by mask
3. Continue to increase oxygen as needed to keep SpO2 >95 % or until respiratory rate is less than 20 per minute
4. If dyspneic, consider albuterol inhaler (Ventolin®) with spacer or albuterol nebulizer if available
5. Maintain adequate hydration for patient to urinate every 2 hours using IV fluids if necessary due to sore throat
6. If fever >100.4, administer acetaminophen (Tylenol®) and/or ibuprofen (Motrin®) and observe for improvement
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider azithromycin (Zithromax®) or doxycycline (Vibramycin®) for typical presentation and levofloxacin (Levaquin®) if patient has been on antibiotics for any reason in the past 60 days
    b. Arrange for medical evacuation and hospital admission if any of the following conditions occur
        i. Patient is dyspneic>20 breaths per minute after albuterol treatment
        ii. Patient requires O_2 due to sensation of air hunger or to maintain O_2 saturation greater than 95%
        iii. Fever does not go lower than 100.4° F after first doses of acetaminophen (Tylenol®) and ibuprofen (Motrin®)
        iv. Patient is confused or excessively sleepy
        v. Patient has been on antibiotics within the past 60 days
8. If patient does not meet criteria for evacuation, patient should be placed on quarters and checked every 4 hours by nonmedical assistant
9. Mandatory medical followups every 24 hours until patient is afebrile for a full 24 hours

XVI. **PNEUMONIA, VIRAL- "***Walking pneumonia." Approximately 40% of adult cases of pneumonia are caused by viruses. Most commonly caused by influenza A/B, adenovirus, and parinfluenza. X-rays show "patchy infiltrates" or fluffy markings in one or more lobes.*

A. Signs and Symptoms

1. Gradual onset of signs and symptoms
2. Signs and symptoms are the similar to bacterial pneumonia, but the patient does not appear as ill
3. Cough producing white-yellow sputum
4. Bilateral, diffuse crackles, bronchial breath sounds, bronchophony, egophony and whispered pectoriloquy that do not localize to a specific lobe
5. Scattered wheezes may be present
6. Tactile fremitus and percussion findings, if present at all, do not localize to a specific lobe

B. Treatment

1. Maintain adequate hydration for patient to urinate every 2 hours using IV fluids if necessary; consider normal saline unless otherwise directed
2. If fever >100.4, administer acetaminophen (Tylenol®) and/or ibuprofen (Motrin®) and observe for improvement
3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION
    a. Consider antiviral therapy (if available) if viral pneumonia is suspected and it is within 48 hours of onset of symptoms
    b. Antibiotics are not indicated for suspected viral pneumonia; however if diagnosis is not clear between viral pneumonia and bacterial pneumonia and no lab or x-ray services are available, discuss possibility of antibiotics with preceptor
4. Arrange for medical evacuation and hospital admission if any of the following conditions occur
    a. Patient is dyspneic>20 breaths per minute after albuterol treatment
    b. Patient requires O_2 due to sensation of air hunger or to maintain O_2 saturation greater than 95%
    c. Fever does not go lower than 100.4° F after first doses of acetaminophen (Tylenol®) and ibuprofen (Motrin®)
    d. Patient is confused or excessively sleepy
    e. Patient has been on antibiotics within the past 60 days
5. If patient does not meet criteria for evacuation, patient should be placed on quarters and checked every 4 hours by nonmedical assistant
6. Mandatory medical followups every 24 hours until patient is afebrile for a full 24 hours

XVII. PNEUMONIA, MYCOPLASMA (ATYPICAL PNEUMONIA)- *Mycoplasma is the smallest free-living organism. It is a short bacillus that does not accept gram staining and so is invisible under the microscope. It infects approximately 1% of the population per year, predominantly in fall and winter months, and is very common in schoolchildren, colleges, and military recruits.*

A. Signs and Symptoms

1. Common in adults 20-40 years of age
2. Gradual onset of signs and symptoms that begin with malaise, sore throat and non-productive cough
3. Incessant paroxisms of cough with sputum becoming blood streaked and mucopurulant as the disease progresses
4. Unlike bacterial or viral pneumonia, mycoplasma does not usually cause pain with deep inspiration

5. Chest soreness from persistent coughing is common
6. No dullness to percussion and no increased tactile fremitus
7. Maculopapular rash occurs in 10-20% of patients

B. Treatment

1. Place patient on bed rest and keep warm
2. Administer oxygen: 2 to 4 liters per minute via nasal cannula or 8-10 liters per minute by mask
3. Continue to increase oxygen as needed to keep SpO2 >95 % or until respiratory rate is less than 20 per minute
4. If dyspneic, consider albuterol Metered Dose Inhaler with spacer or albuterol nebulizer if available
5. Maintain adequate hydration for patient to urinate every 2 hours using IV fluids if necessary due to sore throat
6. If fever >100.4°F, administer acetaminophen (Tylenol®) and/or ibuprofen (Motrin®) and observe for improvement
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider azithromycin (Zithromax®) or doxycycline (Vibramycin®) for typical presentation and levofloxacin (Levaquin®) if patient has been on antibiotics for any reason in the past 60 days
    b. Arrange for medical evacuation and hospital admission if any of the following conditions occur
        i. Patient is dyspneic >20 breaths per minute after albuterol treatment
        ii. Patient requires O_2 due to sensation of air hunger or to maintain O_2 saturation greater than 95%
        iii. Fever does not go lower than 100.4° F after first doses of acetaminophen (Tylenol®) and ibuprofen (Motrin®)
        iv. Patient is confused or excessively sleepy
        v. Patient has been on antibiotics within the past 60 days
8. If patient does not meet criteria for evacuation, patient should be placed on quarters and checked every 4 hours by nonmedical assistant
9. Mandatory medical followups every 24 hours until patient is afebrile for a full 24 hours

XVIII. PNEUMONIAS, FUNGAL- ASPERGILLOSIS- *An acute or chronic illness caused by aspergillus species of molds typically found in compost piles, burn piles, damp hay and stored grains.*

A. Signs and Symptoms

1. Typical presentation includes fever, chest pain, dyspnea, cough and hemoptysis
2. Suspect fungal infection if patient fails to respond to antibiotics
3. If fungi are ingested instead of inhaled, patients may develop abdominal pain and GI bleeding
4. Fungi can invade from gut into local tissues then the bloodstream

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
2. Attempt to determine source of infection and eliminate risk to remainder of unit

XIX. **PNEUMONIAS, FUNGAL- HISTOPLASMOSIS-** *Pulmonary fungal infection endemic to Central America, and Ohio and Mississippi River Valleys.*

A. Signs and Symptoms

1. Typical presentation similar to bacterial pneumonia, but patient does not respond to antibiotic treatment
2. Lymph nodes around the heart and lungs may get markedly enlarged
3. Difficulty swallowing due to esophageal narrowing from big lymph nodes
4. Congestive heart failure from poor venous return because big lymph nodes obstruct the vessels

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSLTATION AND MEDICATION
    a. Consider antifungal therapy such as itraconazole (Sporanox®)
    b. Most infections are self limiting and require only supportive therapy

XX. **PNEUMONIAS, FUNGAL- COCCIDIOMYCOSIS ("VALLEY FEVER")-** *a pulmonary fungal infection endemic to Southwestern US and desert climates of Central and South America.*

A. Signs and Symptoms

1. Fever, chest pain, cough and hemoptysis
2. May have skin manifestations of erythema nodosum (red or violet nodules that develop on the tibia) and erythema multiforme (target lesions often accompanied by bullae)
3. Subclinical presentation includes chronic arthralgias known as "desert rheumatism" and fatigue

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Frequently no treatment is required
    b. Evacuate and arrange for hospital admission if any of the following conditions occur
        i. Losing weight since symptoms began
        ii. Night sweats
        iii. Inability to work
        iv. Hemopysis

XXI. **PNEUMONIAS, FUNGAL- MUCORMYCOSIS-** See "Febrile Illness Protocols, Fungal Pneumonias, Mucormycosis"

XXII. OTHER PNEUMONIAS

A. Signs and Symptoms

1. Klebsiella pneumonia- acquired by immunosuppressed patients, trauma patients, or healthy individuals exposed to the papermill industry. Signs and symptoms are similar to other bacterial pneumonias, but characterized by thick, bloody sputum production that looks like "currant jelly"
2. Legionella- acquired by travelers exposed to contaminated water aerosolized in international hotels or on cruise ships; more common in smokers. Signs and symptoms are similar to other bacterial pneumonias, but frequently with lethargy and stupor

3. Acute HIV seroconversion- May resemble acute viral or bacterial pneumonia; See "Febrile Illness Protocols, HIV"

B. Treatment

1. Contact preceptor if any of the above organisms are suspected
2. Perform occupational hazard assessment if klebsiella is suspected
3. Perform water sampling if legionella is suspected
4. Locate, identify, and treat swamp coolers (evaporative air coolers that use fans blowing over static water; often get refilled without cleaning the water container leading to contamination) and window air conditioners

XXIII. SPONTANEOUS PNEUMOTHORAX- *Air in the pleural space resulting in partial or complete lung collapse caused by rupture of gaps in the subpleural space known as "blebs". Blebs are congenital (inherited).*

A. Signs and Symptoms

1. Dyspnea, pleuritic chest pain, anxiety or altered mental status
2. Sudden or gradual onset depending on severity and rate of development
3. Patients may have underlying pulmonary disease, typically tall, thin, males in teens or 20s
4. Smoking dramatically increases risk
5. Absent tactile fremitus, hyperresonance to percussion and decreased breath sounds on the affected side

B. Treatment

1. **IMMEDIATE ACTION**
   a. Administer supplemental O_2 to maintain O_2 saturation >95%
   b. Establish IV access but do not administer fluid bolus unless directed to do so
   c. If signs and symptoms are severe, to include tachycardia, hypotension, dyspnea, or cyanosis, perform a needle decompression of the involved lung
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXIV. TRAUMATIC PNEUMOTHORAX/TENSION PNEUMOTHORAX- *Air in the pleural space resulting in partial or complete lung collapse caused by penetrating or blunt trauma.*

A. Signs and Symptoms

1. History of trauma
2. Dyspnea, pleuritic chest pain and anxiety or altered mental status
3. Absent tactile femitus, hyperresonance to percussion and decreased breath sounds on the affected side
4. May have subcutaneous emphysema (feels like "Rice Krispies®" under the skin)

B. Treatment

1. **IMMEDIATE ACTION**
   a. Administer supplemental O_2 to maintain SpO_2 at >95%
   b. Establish IV access and treat for shock (See Emergency Protocols, Shock)
   c. If signs and symptoms are severe, perform a needle decompression of the involved lung; leave catheter in place and cover
   d. Cover open wounds with occlusive dressing

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Monitor for coexisting cardiac tamponade (space around the heart is reduced so heart can't beat effectively; check for diminished or absent pulses)
4. Recheck heart sounds, lung sounds, and pulses every 5 minutes
5. Repeat needle decompression as necessary
6. May require intubation and ventilation, but repeating needle decompressions will help delay need for advanced airway

XXV. PULMONARY EDEMA- *Acute left ventricular failure causing blood to back up into the lungs. Can be caused by inadequate acclimatization to high altitude and narcotic overdose in otherwise healthy adults.*

A. Signs and Symptoms

1. Dyspnea, restlessness and anxiety with sense of suffocation
2. Cough producing blood-tinged or pink frothy sputum
3. Pallor and cyanosis
4. Signs and symptoms of shock
5. Rapid/thready pulse
6. Fine crackles bilaterally both anterior/posteriorly
7. S3 characteristic, or may have S4 heart sound

B. Treatment

1. **IMMEDIATE ACTION**
   a. O_2 15 LPM via non-rebreather; be prepared for advanced airway management and mechanical ventilation
   b. Position patient upright
   c. Initiate IV with normal saline at maintenance rate
   d. Do not administer fluid bolus unless directed to do so
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider administration of furosemide (Lasix) at 1mg/kg IV push
   b. Consider administration of nitroglycerine (Nitrostat®) 0.4mg sublingually every 5 minutes
   c. Consider morphine 5mg IV push for breathlessness
   d. Maintain Systolic BP>80-90mmHg; mandatory discussion of all fluid therapy with preceptor

XXVI. PULMONARY EMBOLI- *The blockage of one or more pulmonary arteries by a thrombus, tumor, air collection, or fat collection that originates elsewhere in the body.*

A. Signs and Symptoms

1. Sudden onset of dyspnea (within a few seconds) with pleuritic chest pain
2. Tachypnea, tachycardia, hypotension
3. Amplified S2 heart sound
4. May cause patient to go into cardiac arrest
5. Risk factors include; underlying cardiovascular disease, family history, sedentary life style, smoking, history of immobilization and oral contraceptive use
6. May result from massive trauma
7. For patients with PE caused by thrombus, check for signs and symptoms of a deep venous thrombosis (See "Musculoskeletal Protocols, Deep Venous Thrombosis")

B. Treatment

1. **IMMEDIATE ACTION**
    a. O_2 15 LPM via non-rebreather
    b. Be prepared for advanced airway management and mechanical ventilation
    c. Initiate IV, NS and maintain BP>80-90 Systolic
    d. May sit or lay in position of comfort
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Monitor for life-threatening complications while awaiting evacuation
    b. Strict bed rest; minimize patient's movements
    c. Avoid jostling the patient as much as possible

XXVII. STABLE ANGINA- *Reversible chest pain or discomfort that occurs with the oxygen demand of the heart muscle exceeds its oxygen supply.*

A. Signs and Symptoms

1. Chest discomfort/pain occurs at a predictable level of exertion and is relieved with rest up to 15 minutes
2. Pain can not be reproduced with palpation or deep breathing
3. May have associated neck, jaw, shoulder, or arm discomfort or pain
4. Vitals may show increased heart rate or increased blood pressure
5. Cardiac exam may have new extra sounds (S3 or S4) or new murmur

B. Treatment

1. **IMMEDIATE ACTION**
    a. Administer oxygen: 4 to 6 liters per minute, by nasal catheter or cannula
    b. Increase to 8-10 liters per minute by mask if signs of hypoxia
    c. Keep patient calm and relaxed. Rule out emergent cardiac conditions (see "Acute Coronary Syndrome" in Emergency Protocols)
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION
    a. Consider nitroglycerin (Nitrostat®)sublingually
    b. Do not give nitroglycerin (Nitrostat®) if patient has taken sildenafil (Viagra®) or other sexual performance enhancers in the past 72 hours

XXVIII. TUBERCULOSIS- *A chronic infection of the lungs caused by a mycobacterium and is the leading infectious cause of illness and death in adults worldwide.*

A. Signs and Symptoms

1. Vague symptoms and general malaise
2. Productive cough of yellow or green sputum, usually in the morning; may have hemoptysis
3. Night sweats, weight loss, fatigue
4. Exam findings include diminished breath sounds, dullness to percussion with DECREASED tactile fremitus (compared to increased fremitus with bacterial pneumonia)
5. Rales (crackles) on inspiration or after short coughs (post-tussive rales)
6. Positive TB skin test
7. Upper lobe cavitation lesions visible on chest x-ray

B. Treatment

1. Respiratory isolation if TB is suspected (private room, gown, mask, gloves)
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION
3. Monitor other members of unit for similar symptoms

Chapter 8: Gastrointestinal Protocols

I. ABDOMINAL TRAUMA- ***"See Emergency Medicine Protocols, Abdominal Trauma"***

II. ANAL FISSURES- *A tear in the anal lining due to anal trauma, usually from the passage of a large hard stool.*

A. Signs and Symptoms

1. Sensation of tearing during passage of stool followed by small amount of blood in toilet or with wiping
2. Exam reveals a fresh laceration extending in or through anal sphincter
3. Patients generally will not tolerate digital rectal exam or anoscopy due to pain

B. Treatment

1. Treatment has three components
   a. Warm Sitz baths (see treatment for hemorrhoids) for 10-15 minutes after passage of stool PLUS
   b. Increasing dietary fiber PLUS
   c. Topical lidocaine (Xylocaine®) or witch hazel
2. Advise patient that all three are necessary for lesion to heal
3. Consider adding docusate sodium (Colace®) to soften stools
4. Do not give stimulant laxatives such as Ex-Lax®, Senecot®, or Senna® because these will increase pressure around the sphincter

III. ANTHRAX, GASTROINTESTINAL- See "Febrile Illness Protocols, Anthrax"

IV. APPENDICITIS- *Inflammation of the vermiform appendix. The most common cause for an attack of severe, acute abdominal pain that requires surgical intervention.*

A. Signs and Symptoms

1. Sudden onset epigastric or periumbilical pain followed by nausea and vomiting several hours later
2. Pain shifts to the right lower quadrant
3. Direct tenderness and rebound tenderness in the right lower quadrant at McBurney's Point.
4. Low grade fever of 100-101°F and leukocytosis of 12-15,000/μL are characteristic
5. Positive Rovsing's, Psoas and Obturator signs and rectal tenderness suggest appendicitis

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access
   b. Give fluid bolus if indicated
   c. Consider narcotics for pain
   d. Consider promethazine (Phenergan®) or ondansetron (Zofran®) for nausea
   e. Keep patient NPO, insert NG tube and monitor I&Os
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

a. Consider antibiotics if directed by a preceptor
b. Ceftriaxone (Rocephin®) IV PLUS metronidazole (Flagyl®) IV is effective

V. **BAROTRAUMA, GI**- See "Environmental Injury Protocols, Barotrauma, GI"

VI. **CHOLECYSTITIS**- *Acute inflammation of the gallbladder usually resulting from an obstruction caused by a gall stone.*

A. Signs and Symptoms

1. Recurrent colicky pain, progressing to severe and constant pain, localized right upper quadrant pain radiating to the right lower scapula
2. Pain is aggravated by ingesting a high fat meal
3. Nausea and vomiting are common
4. Gall bladder is palpable in 50% of cases
5. Patients attempt to splint the abdomen during deep inspiration
6. Positive Murphy's sign
7. Low grade fever, tachycardia

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access and bolus fluids if indicated
   b. Manage pain with meperidine (Demorol®) or IM ketoralac (Toradol®) as necessary
   c. Keep patient NPO
   d. Monitor I&Os
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. High fever, high white blood cell count, and absent bowel sounds suggest gangrenous gall bladder or rupture
   b. If signs and symptoms of rupture are present (severe pain, rebound tenderness, signs of shock), consider antibiotic therapy with ceftriaxone (Rocephin®)

VII. **CONSTIPATION**- *Difficult or infrequent passage of stool. In deployed populations, constipation is commonly caused by reduced fiber in diet.*

A. Signs and Symptoms

1. No passage of stool or inability to pass stool or infrequent stools that require straining to pass
2. Patient may see blood streaks on stool
3. Abdominal exam usually mild to moderate tenderness, areas of dullness to percussion especially left lower quadrant

CLINICAL NOTE: Abdominal exam demonstrating a tense, tympanic abdomen with diminished or absent bowel sounds suggests something is mechanically obstructing the gut (ovarian tumor pinches off a segment of gut, for example). If these symptoms are occurring, **IMMEDIATE EVACUATION IS REQUIRED.**

4. Tense, tympanic abdomen indicates mechanical obstruction

5. Causes of constipation can be numerous. Collect comprehensive history to include a life-long bowel history, changes in diet, exercise, sleep, activity, stress, hydration, herbals/supplements/medications

B. Treatment

1. Treat underlying cause if known
2. Increase dietary fiber to 35 mg/day
3. Add citrus fruits, prunes or prune juice to diet if available
4. Consider Docusate sodium (Colace®) as a daily stool regulating agent
5. Avoid stimulant laxatives unless directed by physician preceptor
6. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if recurrent, associated with a fever, or if initial treatment does not work

VIII. GASTROENTERITIS, GENERAL- *An inflammation of the stomach and intestinal tract primarily caused by an infection, but may be caused by ingested poisons and toxins.*

A. Signs and Symptoms

1. Decreased appetite
2. Nausea/vomiting
3. Diarrhea
4. Abdominal cramping

B. Treatment

1. Provide symptomatic relief
2. Maintain fluid and electrolyte balance
3. Initiate IV support if necessary and transition to oral rehydration solutions that contain water, salt, and sugar as soon as patient is able to tolerate PO
4. Start patient on dry cereal, crackers, or BRAT diet after tolerating PO fluids for 2 or more hours
5. Bed rest with bathroom privileges
6. If any of the following conditions are met, CONTACT PHYSICIAN PRECEPTOR- EVACUATION:
   a. Hypovolemia does not correct with initial treatments
   b. Stools contain blood
   c. Temp>101°F
   d. Severe abdominal pain
   e. History of antibiotic use in past month
7. Treatment of diarrhea with antibiotics may be initiated if none of the above criteria are met AND the patient is having more than four unformed stools per day, fever, and correctable hypovolemia

CLINICAL NOTE: The World Health Organization Solution

3.5 grams of sodium chloride (salt)
2.5 grams sodium bicarbonate (baking soda)
1.5 grams potassium chloride (salt substitute)
40 grams sucrose (table sugar)
1 Liter of water

OR

½ tsp salt
½ tsp baking soda
4 Tbsp table sugar
1 Liter of water

IX. **GASTROENTERITIS, VIRAL-** *The majority of cases of acute gastroenteritis are caused by viruses such as norovirus, rotavirus and adenovirus.*

A. Signs and Symptoms

1. Differentiate between bacterial causes
2. Symptoms start with vomiting
3. Incubation period of 24-48 hours
4. Illness lasts 12-60 hours

B. Treatment

1. Control nausea with IM or PO promethazine (Phenergan®) **OR** IV or PO ondansetron (Zofran®)
2. Support with IV therapy if patient is orthostatic and nauseous
3. Bed rest/quarters
4. Clear liquids for 12 hours, advance to BRAT diet and then advance as tolerated
5. No antibiotics indicated

X. **GASTROENTERITIS, VIBRIO CHOLERAE (CHOLERA)-** *(Gram negative, comma-shaped bacillus) Grows in the intestinal tract and becomes endemic when stools from infected persons contaminate water supply. This organism has the potential for weaponization.*

A. Signs and Symptoms

1. Incubation of 24-72 hours
2. Abrupt onset of watery emesis and/or liquid "rice water" diarrhea
3. Stools have "fishy" odor
4. Fever is uncommon with cholera, but fast, severe dehydration can be fatal
5. Found in developing countries in Asia, Middle East, Africa and South/Central America

B. Treatment

1. Aggressive management and replacement of fluids and electrolytes via IV
2. Switch to aggressive PO hydration with WHO rehydration solution or watered down Gatorade® as soon as PO can be tolerated
3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider antibiotic therapy with single-dose doxycycline (Vibramycin®) as soon as PO is tolerated
    b. Patient may require extreme amounts of IV fluids (20 or more liters)
4. Investigate for contaminated water supply

XI. **GASTROENTERITIS, CLOSTRIDIUM BOTULINUM (BOTULISM)-** *(Gram positive, spore-forming bacillus) Grows in anaerobic foods (canned or processed) and produces a powerful neurotoxin. The bacteria and the toxin have the potential for weaponization.*

A. Signs and Symptoms

1. Incubation period of 24-96 hours
2. Nausea, vomiting, abdominal pain, diarrhea, dry mouth and sore throat followed by
    a. Diplopia, dysphagia, dysphonia, and respiratory compromise if untreated
    b. Descending paralysis, no fever

c. Patient remains aware but unable to move or communicate ("Locked in")

B. Treatment

1. Aggressive airway management and ventilation
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Administer antitoxin if available

XII. **GASTROENTERITIS, ESCHERICHIA COLI ("TRAVELLER'S DIARRHEA")**- *(Gram negative bacillus) E. coli is normal intestinal flora that can overgrow or individuals can be exposed to a different strain. E. Coli produce toxins and invades epithelial surfaces of the GI tract.*

A. Signs and Symptoms

1. Incubation period of 24-72 hours
2. Abrupt onset of diarrhea/rare vomiting
3. History of drinking untreated, foreign water

B. Treatment

1. Usually self-limiting and resolves within 3 days
2. Maintain hydration status
3. Consider loperimide (Lomotil®, Imodium®) if no fever
4. CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION if
   a. Blood is present
   b. Symptoms are severe and incapacitating
   c. Consider antibiotic therapy with ciprofloxacin (Cipro®) or trimethoprim-sulfamethoxazole (Bactrim®)

XIII. **GASTROENTERITIS, CLOSTRIDIUM DIFFICILE ("C-DIFF")**- *(Gram positive, spore-forming rod) Normal flora in the intestinal tract. Infection occurs when patient takes antibiotics that kill off the helpful bacteria in the GI ecosystem and C-diff becomes opportunistic.*

A. Signs and Symptoms

1. Onset corresponds with administration of antibiotic therapy, but varies with individuals
2. Abrupt, foul smelling, bloody, watery diarrhea and low grade fever

CLINICAL NOTE: May develop life-threatening "toxic megacolon" with severe abdominal pain, abdominal distension, and slowing of stools due to intestinal muscles slowing. This is suggestive of impending colon rupture. **IMMEDIATE EVACUATION IS REQUIRED.**

B. Treatment

1. Stop any current antibiotic therapy unless patient already has a more dangerous infection
2. Maintain hydration and electrolyte status for patient
3. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Consider treatment with metronidazole (Flagyl®)
   b. Do not administer loperamide (Lomotil®, Imodium®) unless directed by preceptor
   c. Illness is spread fecal-orally, but should not be a risk to healthy individuals

d. Exercise good hand washing and consider contact precautions (gown, gloves) if feasible

XIV. **GASTROENTERITIS, STAPHYLOCOCCAL ENTERITIS**- *(gram positive cocci) Diarrhea caused by enterotoxin from ingested staphylococcal bacteria. Staph grows in meats, dairy, eggs, and bakery products. This bacteria has the potential for weaponization.*

A. Signs and Symptoms

1. Contaminated foods are left at room temperature and then consumed after a significant amount of toxin was produced
2. Incubation period of 1-6 hours after ingestion
3. Abrupt onset of vomiting lasting approximately 24 hours
4. Nausea and vomiting are usually severe, but diarrhea is less intense or absent
5. Occurs in persons eating the same food

B. Treatment

1. Recovery is spontaneous without antibiotics in 1-2 days
2. May give promethazine (Phenergan®) or ondansatron (Zofran®) IV or PO to reduce nausea
3. Enforce hydration
4. Initiate IV if orthostatic and nauseous

XV. **GASTROENTERITIS, CLOSTRIDIUM PERFRINGENS**- *(gram positive bacillus) Normal intestinal flora. Infection results by eating contaminated food (typically rewarmed meat-containing foods).*

A. Signs and Symptoms

1. Incubation period of 8-16 hours
2. Abrupt onset of profuse, foul-smelling diarrhea and vomiting

B. Treatment

1. Recovery spontaneous without antibiotics within 4 days
2. May give promethazine (Phenergan®) or ondansatron (Zofran®) IV or PO to reduce nausea
3. Enforce hydration
4. Initiate IV support if orthostatic and nauseous
5. If symptoms are severe, consider antibiotic therapy and CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND CONSULTATION

XVI. **GASTROENTERITIS, SHIGELLA ("BACCILLARY DYSENTERY" OR "SHIGELLOSIS")**- (*Gram negative bacillus) Common, self-limiting disease found in conjunction with poor sanitary conditions. It grows in the intestinal epithelium and produces a toxin. This bacteria has the potential for weaponization.*

A. Signs and Symptoms

1. Incubation period of 24-72 hours
2. Acute onset of bloody or mucous-containing diarrhea, lower abdominal cramps and tenesmus (rectal dry heaves)
3. Fever of 104°F, chills, malaise, decrease appetite and headache
4. Weakness and dehydration
5. Abdominal tenderness

B. Treatment

1. Maintain fluid/electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Consider oral or IV ciprofloxacin (Cipro®) or ceftriaxone (Rocephin®) IM
    b. Strict contact precautions (private room and bathroom, gown, gloves, etc)
3. Progress to liquid diet and advance as tolerated
4. Other bacteria produce a similar toxin: consider amoebic dysentery, salmonella, E. coli, and ulcerative colitis

XVII. **SALMONELLA**- *(gram negative bacillus) "Typhoid fever" is a severe systemic illness with stepwise progression caused by Salmonella typhi and is transmitted through contaminated water/food. This bacteria has the potential for weaponization.*

A. Signs and Symptoms

1. Incubation period is 5-14 days
2. Follows a predictable, sequential pattern
    a. <u>First week of illness</u>
        i. Fevers to 104°F, chills and sweats
        ii. May be misdiagnosed as malaria or dengue fever
        iii. Abdominal pain, vomiting and diarrhea
        iv. Diarrhea described as "Pea Soup" in nature
        v. Maximum temperature each day is higher than the previous day's temp and stabilizes in 1 week
    b. <u>Second week of illness</u>
        i. Fever decreases in a stair-step fashion by the 10th day
        ii. Bradycardia
        iii. Faint salmon-colored macules appear on the trunk and abdomen
        iv. Malaise, headache and cough
    c. <u>Third week of illness</u>
        i. Hepatosplenomegaly, intestinal bleeding, peritonitis, intestinal perforation and septic shock
        ii. May have changes in mentation or psychoses
    d. Observe for "typhoid state"—patient appears exhausted, motionless and unresponsive, but able to follow simple commands

B. Treatment

1. Maintain fluid/electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Consider oral or IV ciprofloxacin (Cipro®) or ceftriaxone (Rocephin®) IM
    b. Strict contact precautions (private room and bathroom, gown, gloves)
3. Progress to liquid diet and advance as tolerated
4. Other bacteria produce a similar toxin: consider amoebic dysentery, salmonella, E. coli, and ulcerative colitis

XVIII. **GASTROENTERITIS, CAMPYLOBACTER JEJUNI**- *(Gram negative spiral bacillus) Grows in the jejunum and ilium and transmitted in contaminated drinking water and unpasteurized milk.*

A. Signs and Symptoms

1. Incubation period 1-7 days (mean=3 days) after ingestion
2. Abrupt onset of periumbilical cramping and abdominal pain (mimicking acute appendicitis) and diarrhea that may be accompanied by high fever with rigors, body aches, dizziness and delirium
3. Bloody stools may develop on the second or third day of diarrhea

B. Treatment

1. Maintain fluid/electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Consider Erythromycin (E-Mycin®) or ciprofloxacin (Cipro®)
    b. Strict contact precautions (private room and bathroom, gown, gloves)
3. Progress to liquid diet and advance as tolerated
4. Other bacteria produce a similar toxin: consider amoebic dysentery, salmonella, E. coli, and ulcerative colitis

XIX. GASTROENTERITIS, YERSINIA ENTEROCOLITICA- *(gram negative coccobacillus) Yersinia Enterocolitica is transmitted from contaminated food (raw vegetables or salad ingredients contaminated with human manure) or a water supply contaminated with human feces.*

A. Signs and Symptoms

1. Incubation period 1-11 days (mean = 4 days)
2. Severe abdominal pain, diarrhea, and fever often misdiagnosed as acute appendicitis
3. Nausea and vomiting are less common
4. Blood in stool

B. Treatment

1. Maintain fluid/electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Antibiotics are not indicated unless severe disease, but symptoms are often indistinguishable from other causes of acute gastroenteritis that do require antibiotics
    b. Find source of contaminated food or water
3. Progress to liquid diet and advance as tolerated

XX. GASTROINTESTINAL PARASITES, AMEBIASIS ("AMOEBIC DYSENTERY")- *Amoebic dysentery is caused by entamoeba histolytica, which is a protozoan form of parasite.*

A. Signs and Symptoms

1. Diarrhea is the most common presentation; between 5-10 episodes per day
2. Incubation period of several weeks with insidious onset of symptoms
3. Weight loss and abdominal cramping may be accompanied by vomiting
4. Fever; may be as high as 105°F
5. Recurrent episodes of diarrhea and abdominal cramps alternating between periods of constipation is also common
6. May have blood and mucous in stool
7. Tender, enlarged liver
8. Difficult to diagnose. Requires repeated stool samples and identification of the organism or cysts in the stool. Frequently misdiagnosed as other conditions

B. Treatment

1. Maintain fluid/electrolyte balance
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Mandatory evacuation for any patient with bloody diarrhea
    b. Consider evacuation in other cases for definitive diagnosis

XXI. GASTROINTESTINAL PARASITES, SCHISTOSOMIASIS ("KATAYAMA FEVER")- See "Febrile Illness Protocols, Gastrointestinal Parasites, Schistosomiasis"

XXII. GASTROINTESTINAL PARASITES, GIARDIASIS- *Giardia lamblia (protozoan) caused by ingestion of contaminated water supply, is endemic worldwide. In the US, the disorder is common among hikers drinking inadequately filtered/treated water and in daycare centers (fecal-oral transmission).*

A. Signs and Symptoms

1. Incubation period of 7-14 days
2. 50% of infections are asymptomatic
3. Acute/chronic diarrhea alternating with constipation
4. Excessive foul flatulence
5. Diarrhea is foul and excessively fatty or greasy
6. Blood/mucous in stool and mild abdominal cramps
7. Nausea, vomiting, low grade fever and headache
8. Weight loss (up to 10% of body weight in a month) and malabsorption symptoms
9. Mimics peptic ulcer disease and cholecystitis

B. Treatment

1. Maintain fluid and electrolyte balance
2. Strict contact precautions (private room and bathroom, gown, gloves)
3. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Medication of choice is metronidazole (Flagyl®), however may not be able to differentiate giardia from other more serious conditions without lab capability
    b. If bloody stool is present, **IMMEDIATE EVACUATION** is required
4. Attempt to localize source of contamination

XXIII. GASTROINTESTINAL PARASITES, TAPE WORMS- *General classification of several worms that affect humans. Distribution is worldwide. Worms vary in size from 1 cm to 300cm.*

A. Signs and Symptoms

1. History of eating infected, undercooked or raw beef, pork, fish or seafood
2. May see worm segments in stool/toilet or may feel worm movement around anus
3. Heavy infestations cause weight loss, diarrhea, abdominal pain
4. Larvae travel throughout the body and are capable of invading all tissue types
5. CNS infections manifest with seizures, CNS depression, personality disturbances and hydrocephalus

B. Treatment

1. Diagnosis is made by finding worm segments in clothing, bedding or stool or by identification of ova on the perianal skin

2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider evacuation for diagnostic testing and treatment
    b. If illness is not severe and worm segments are identified, drug of choice is praziquantel (Biltricide®)

XXIV. GASTROINTESTINAL PARASITES, PINWORMS (ENTEROBIASIS)- *An intestinal infestation by Enterobius vermicularis that lives in tropical and temperate environments. Worms live in intestinal tract and lay eggs around the anus. Infection is by ingestion of eggs dislodged from clothing/bedding or by eating food prepared by persons with contaminated hands.*

A. Signs and Symptoms

1. Presenting complaint is often intense, perianal pruritus
2. Perianal excoriations from scratching
3. Transferred via fecal oral route or by fomites
4. Diagnosis is made by observing the female worm on the surface of the anus 1-2 hours after patient goes to bed or by identification of ova on the perianal skin

B. Treatment

1. Indicated if patient is symptomatic
2. Strict contact precautions (private room and bathroom, gown, gloves)
3. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Administer a single dose of mebendazole (Vermox®)
    b. Repeat mebendazole (Vermox®) dose in 2 weeks
4. Evaluate all close contacts
5. All bedding and clothing must be washed in hot water and hot dryer to destroy eggs
6. Clip infected patient's fingernails close to skin

XXV. GASTROINTESTINAL PARASITES, TRICHINOSIS- See "Febrile Illness Protocols, Gastrointestinal Parasites, Trichinosis"

XXVI. GASTROINTESTINAL PARASITES, TRICHURIASIS (WHIPWORMS)- *Small, skinny worms 30-50 mm in length found worldwide primarily in the tropics and warm, humid areas. Infection comes from ingestion of soil, such as eating inadequately washed raw vegetables or fruits.*

A. Signs and Symptoms

1. Usually asymptomatic, but most commonly presents with rectal prolapse
2. Abdominal pain, tenesmus (rectal dry heaves), diarrhea, abdominal distention, flatulence, nausea, vomiting and weight loss
3. May exhibit pica (eating things that aren't food such as dirt, ice, or chalk)
4. Finger clubbing also occurs
5. Barrel shaped eggs in stool visible with microscope

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Mebendazole (Vermox®) is the drug of choice
    b. If meds are unavailable, consider soapsuds enema
2. Identify source of contamination and eliminate

XXVII. GASTROINTESTINAL PARASITES, ASCARIASIS- *The most common intestinal worm. Worldwide distribution, primarily in warm, wet areas or during rainy seasons. Infection caused by fecal-oral route and contaminated water or food (especially raw fruits and vegetables). Adult worms are 20-40 cm long, mature in the intestines and migrate to the lungs.*

A. Signs and Symptoms

1. Fever, cough, hemoptysis, rales and evidence of lung involvement
2. Heavy infections present with abdominal pain and colic
3. History of coughing up a worm, or a worm passing through the nose

B. Treatment

1. Diagnosis is made by finding the worms or larvae in stool or sputum
2. Maintain fluid and electrolyte balance
3. Strict contact precautions (private room and bathroom, gown, gloves)
4. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
    a. If evacuation will be delayed, consider antihelminth treatment with mebendazole (Vermox®)
    b. Patient may be agitated if worms were visualized or patient has sensation of moving worms. Consider anxiolytic such as diazepam (Valium®)

XXVIII. GASTROINTESTINAL PARASITES, HOOKWORM- *Widespread in the tropics; transmitted by wearing open footwear while walking on contaminated soil or wading in contaminated water. Worms invade directly through the skin.*

A. Signs and Symptoms

1. Itching at site of penetration
2. May see the larval track underneath the skin (cutaneous larva migrans)
3. 8-21 days later- hookworms invade pulmonary tissue, climb the pulmonary tree and are swallowed
4. May be asymptomatic or have mild cough
5. Intestinal infection demonstrates nausea, vomiting, diarrhea, increased flatulence, and abdominal pain that is worse postprandially (after eating)
6. Chronic infection causes nutritional deficiencies and anemia

B. Treatment

1. Diagnosis is made based on seeing larval tracks in the skin or coughed/vomited worms visualized
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Albendazole (Albenza®) single dose is preferred treatment; mebendazole (Vermox®) is an alternative
    b. Iron supplements (Ferrous Sulfate®) if anemic

C. Prevention

1. Do not allow personnel deployed to warm, humid areas to go barefoot or wear open shoes
2. Do not allow deployed personnel to recreationally wade in fresh water

XXIX. GASTRITIS- *Inflammation of the gastric mucosa caused by infections, NSAID use, alcohol, and stress. Gastritis can be classified as wither erosive or non-erosive, acute or chronic.*

A. Signs and Symptoms

1. <u>Erosive Gastritis</u>

a. Caused by NSAIDs, alcohol stress, viral infection or trauma
b. May complain of heartburn, dyspepsia, nausea or vomiting
c. May have hematemesis or melena
d. Symptoms begin within 2-5 days of the inciting event
e. Bleeding is mild or moderate

2. Non-erosive gastritis
   a. Mainly the result of H. pylori infection
   b. May complain of mild dyspepsia or vague abdominal symptoms, nausea or epigastric pain

B. Treatment

1. If bleeding is present
   a. **IMMEDIATE ACTION**
      i. Initiate IV access and administer fluid bolus if indicated
      ii. Stop all medications, herbals, and supplements
      iii. If alcohol is possible etiology, prepare for possible seizures and Delirium Tremens (see "Neurological Protocols- Acute Alcohol Withdrawal")
   b. **CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION**
      a) Consider IV acid suppression therapy, such as rinitidine (Zantac®) or famotidine (Pepcid®)
      b) Keep patient NPO unless otherwise directed by Preceptor
2. If pain is less severe and bleeding is absent
   a. Determine the cause and correct it
   b. Administer PO acid suppressants such as rinitidine (Zantac®) or famotidine (Pepcid®)
   c. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
      i. Consider evacuation for H. Pylori serum test or urea breath test
      ii. If positive, treat for peptic ulcer disease (PUD)
   d. Reevaluate in a few days to make sure symptoms have resolved

XXX. DYSPEPSIA ("HEARTBURN")- *The sensation of epigastric discomfort caused by peptic ulcer disease, gastroesophageal reflux, or drugs (NSAIDs and some antibiotics). May be caused by pyloric dysfunction, motility problems, gastritis or malignancy.*

A. Signs and Symptoms

1. Indigestion, gassiness, postprandial (after eating) fullness, gnawing and burning behind sternum
2. If patient complains of chest pain, see "Emergency Protocols, Acute Coronary Syndrome"

B. Treatment

1. For severe discomfort
   a. Give "GI cocktail" of 30 mL Maalox®/Mylanta® + 10 mL viscous lidocaine + 10 mL Donnatal® mixed together and drunk
   b. If no improvement in 1 hr, CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION
2. For mild discomfort

a. Treat symptoms with an H2 blocker such as cimetidine (Tagamet®) or ranitidine (Zantac®), or proton pump inhibitor such as omeprazole (Prilosec®)
b. Test for fecal occult blood and arrange for routine medical evacuation

XXXI. HEMORRHOIDS- *Dilated veins in the tissue in and around the anus.*

A. Signs and Symptoms

1. Chief complaint of blood with bowel movement or with wiping, pruritus, or palpable mass around the anus
2. Risk factors include prolonged sitting, pregnancy, repetitive straining (ie weightlifters), and chronic constipation
3. Mass will be visible and/or palpable on anal exam
4. Consider anoscopy for better visualization

B. Treatment

1. Treat risk factors, if known
2. Increase fiber content of diet to reduce discomfort of bowel movements
3. Fiber will also help when constipation is a contributing factor
4. Sitz baths- submerge the anal area in 2-3 inches of warm (not hot) water perform BID-TID will markedly improve discomfort and itching
5. Hydrocortisone cream applied topically or via suppository twice daily for maximum of 7 days will also reduce itching, but may irritate and damage the anal mucosa
6. Witch hazel will cool area and decrease discomfort

CLINICAL NOTE: Patients may present with acute severe anal pain and palpable hard mass near the anus. This is a thrombosed hemorrhoid. Although very uncomfortable, these patients can be managed with rest and the above treatments plus Acetaminophen and/or NSAIDs PLUS topical lidocaine or witch hazel with probable resolution of symptoms within 5-7 days. If these measures are insufficient, CONTACT PHYSICIAN PRECEPTOR-CONSULTATION for possible surgical correction.

CLINICAL NOTE: All patients with rectal bleeding should be evaluated by endoscopy to make sure the patient doesn't have a more serious disease such as cancer, ulcerative colitis, or Crohn's Disease. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION for all patients with rectal bleeding, even if source of bleeding is clearly identifiable.

XXXII. HEPATITIS (VIRAL)- See "Febrile Illness Protocols, Hepatitis"

XXXIII. INGUINAL AND FEMORAL HERNIAS- *An internal defect in muscle or fascia that allows abdominal contents to move where they don't belong.* ***Umbilical hernias*** *are when abdominal contents protrude through the anterior abdominal wall, creating a visible outpouching near the umbilicus (belly button) with increased abdominal pressure.* ***Direct inguinal*** *hernias can be felt externally along the inguinal creasewith increased abdominal*

pressure. ***Indirect inguinal hernias*** *occur when abdominal contents slide down through the inguinal canal and may be noticeable inside the scrotum; they can be palpated within the inguinal canal.* ***Femoral hernias*** *occur more often in women and are when abdominal contents slide along the anterior/medial thigh.*

A. Signs and Symptoms

1. Sensation of fullness, dull ache, or discomfort that is more pronounced when patient is lifting, coughing, straining, or increasing abdominal pressure
2. Discomfort and pressure typically improve when the lifting or straining is stopped (ie sitting or resting) due to the intestinal contents moving back into the abdominal cavity
3. Severe pain or pain and pressure that do not resolve spontaneously should raise suspicion that the hernia is "incarcerated", which means it is trapped and unable to move back into the abdominal cavity spontaneously
4. While patient is standing facing the seated examiner, have patient cough, strain, or Valsalva and observe and palpate for visible bulge. Complete "hernia check" in males for indirect inguinal hernias

B. Treatment

1. For uncomplicated, spontaneously reducing hernias, patient will require surgery for repair
2. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
   a. Limited duty profile to restrict lifting and gear to no more than 10 lbs
   b. Rest, ice, NSAIDs, and stretching exercises such as yoga will all help reduce symptoms
3. If incarcerated hernia is suspected
   a. Apply gentle direct pressure on visible bulge, manipulating it back toward abdominal cavity
   b. If this is insufficient, place patient flat with head down and feet up (Trendelenburg position)
   c. Administer 5-10 mg diazepam (Valium®) IM and attempt to reduce hernia
   d. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

ACTION ALERT: Observe closely for signs of impending or actual strangulation such as pain and tenderness to palpation in the region of the hernial swelling, discoloration of tissue over swelling site, nausea and vomiting, low grade fever and malaise, or absence of bowel movements or bowel sounds.

XXXIV. MOTION SICKNESS- *A syndrome caused by repetitive vertical and horizontal acceleration and deceleration.*

A. Signs and Symptoms

1. Nausea, vomiting, vague abdominal discomfort
2. Preceded by salivation, pallor, sweating and sleepiness
3. May have headache, fatigue, weakness and dizziness

B. Treatment

1. Patients prone to motion sickness should premedicate with scopolamine hydrobromide (Scopace®)
2. An antihistamine such as diphenhydramine hydrochloride (Benadryl®) taken 1 hour before anticipated movement is also effective

3. Manage vomiting by administration of promethazine (Phenergan®) IM, PO, or PR as long as patient did not premedicate with an antihistamine
4. Avoid rear-facing seating on aircraft and seat patients over the wings to minimize motion

XXXV. PANCREATITIS- *Inflammation of the pancreas. Pancreatitis may be acute or chronic. Alcoholism and biliary tract disease accounts for over 80% of cases of acute pancreatitis. The remaining 20% are attributed to drugs (hormones and diuretics) associated with some disease processes.*

A. Signs and Symptoms

1. Severe epigastric abdominal pain radiating straight through to the back
2. Pain may radiate to the abdomen
3. Pain described as severe, steady, boring and without relief persisting for hours or days and often worsened by ingesting a heavy meal
4. Patient appears restless, anxious or agitated
5. Bending or leaning forward alleviates pain
6. Pain is aggravated by coughing, movement or deep breathing
7. Nausea and vomiting also occur
8. Abdomen may appear distended
9. Tachycardia- pulse may be over 100 BPM, transient hypertension and postural hypotension are common
10. Epigastric abdominal tenderness always occurs
11. Bowel sounds may be hypoactive

B. Treatment

1. **IMMEDIATE ACTION**
   a. Keep patient completely NPO- no pills, fluids, etc.
   b. Establish IV access with Ringer's Lactate unless otherwise directed by preceptor
   c. Monitor I&Os and aggressively hydrate unless ascites, edema or pulmonary findings are present
   d. Pain management with meperidine (Demerol®) 50-100mg IV, especially during transport (morphine IV may be used as an alternative)
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXXVI. PEPTIC ULCER DISEASE ("PUD")- *An erosion in a segment of the GI mucosa, typically in the stomach or the first 5 cm of the duodenum. Nearly all ulcers are caused by Helicobacter pylori or NSAID use.*

A. Signs and Symptoms

1. Burning epigastric pain relieved by food or antacids
2. Pain described as burning or gnawing and usually chronic or recurrent
3. May have postprandial epigastric discomfort, early satiety, or nausea
4. PUD complications include hemorrhage, gastric obstruction and perforation

B. Treatment

1. Therapy is often started without definitive diagnosis
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Consider evacuation to test for H. pylori

b. If H. pylori is present, treatment consists of "triple therapy" with omeprazole (Prilosec®) or rabeprazole (Aciphex®) PLUS amoxicillin (Amoxil®) PLUS clindamycin (Cleocin®)

3. Discontinue tobacco and alcohol use
4. Gastric acidity can be treated by use of proton pump inhibitors such as omeprazole (Prilosec®) or rabeprazole (Aciphex®), H2 blockers such as ranitidine (Zantac®) or cimetidine (Tagamet®) and/or antacids such as aluminum hydroxide or calcium carbonate (Tums®, Rolaids®)

XXXVII. PERITONITIS- *Inflammation of the peritoneal cavity. Peritonitis can be either primary or secondary. Primary peritonitis occurs following an infection in the blood stream. Secondary peritonitis (most common) usually occurs as a result of perforations of the GI tract through a variety of pathology and trauma.*

A. Signs and Symptoms

1. History of trauma or systemic disease
2. Sudden onset abdominal pain can either be localized or diffuse depending on the cause of the peritonitis
3. Fever, large volume ascites with abdominal distention, absent bowel sounds and altered mental status occurs in over 50% of patients
4. Vomiting and high fever >103°F not uncommon

B. Treatment

1. **IMMEDIATE ACTION**
   a. Initiate IV and provide fluid and electrolyte replacement as needed
   b. Keep patient NPO
   c. Insert NG tube
   d. Monitor I&Os
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. If evacuation will be delayed, consider ceftriaxone (Rocephin®) IV
   b. Repeat vitals at least hourly until evacuation

CLINICAL NOTE: If treatment is not accomplished promptly, multisystem failure occurs rapidly. Loss of fluids in the peritoneum leads to severe dehydration and electrolyte imbalances. Emergency laparotomy may be indicated. Definitive treatment involves treating the underlying disease.

XXXVIII. ULCERATIVE COLITIS AND CROHN'S DISEASE (INFLAMMATORY BOWEL DISEASES)- *Chronic recurring episodes of inflammation of the lining of the bowel that usually starts when patient is in their 20's or 30's. Symptoms often start as bright red blood in bowel movements and in underwear.*

A. Signs and Symptoms

1. Patient has symptoms indistinguishable from infectious causes of diarrhea to include frequent loose stools, cramping abdominal pain especially before passage of stool, low grade fever, and abdominal pain
2. History is suggestive of an inflammatory bowel disease over an infectious condition if patient has had similar episodes of bloody stool that resolved spontaneously, if bleeding occurs as bright spots in the underwear in

addition to stools and/or wiping, if exposure to infectious causes are unlikely, and if nausea is not a presenting symptom

3. Blood loss with inflammatory bowel conditions can be severe; patient may be pale, have weight loss, have severe fatigue, etc.

B. Treatment

1. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Mandatory evacuation of all patients with blood in stool, even if patient appears stable
    b. No medications indicated unless directed by preceptor

Chapter 9: Genitourinary/Gynecological Protocols

I. **CANDIDIASIS (CANDIDAL VULVOVAGINITIS)**- *an overgrowth of normal vaginal yeast flora that may occur spontaneously or after taking broad-spectrum antibiotics. Yeast infections may affect men as well as women.*

A. Signs and Symptoms

1. Vulvar pruritus
2. Labial soreness, irritation, and/or redness
3. If discharge is present, it may be increased typical ("normal") discharge or thick and "cottage cheese-like" discharge without odor
4. May have dyspareunia (pain with intercourse) or dysuria (pain with urination)
5. Bright red "beefy" erythema around the labia and/or vulva and possibly excoriations
6. Vaginal vault may have increased or "cottage cheese" discharge
7. Cervical motion tenderness should be absent
8. Lab: Swab taken from vaginal vault and mounted on a slide with KOH shows branching yeast hyphae

B. Treatment

1. Prescribe oral fluconazole (Diflucan®) single dose or topical clotrimazole (Lotrimin®) or miconazole (Monistat®)
2. Instruct patient to avoid tight-fitting clothing, pantyliners, and pantyhose
3. If patient is pregnant or has recurrent episodes (>2 episodes in 3 months or 3 or more episodes per year), CONTACT PHYSICIAN PRECEPTOR-CONSULTATION
4. For males with candidal infection around scrotum, cleanse genitalia with soap and water and apply nystatin (Mycostatin®) cream twice daily

CLINICAL NOTE: Although popularly recommended, the oral or vaginal use of active yogurt cultures has no demonstrated benefit.

ACTION ALERT: Persistent Candidiasis may be a result of diabetes or HIV. Consider causes of immunocompromise in all recurrent cases.

II. **CHANCROID**- *A Sexually Transmitted Infection (STI) caused by Haemophilus ducreyi (gram negative bacillus).*

A. Signs and Symptoms

1. Incubation period up to 1 month, but typically 4-10 days after exposure
2. Initially manifests as 1-2cm, painful, soft, ragged ulcer with necrotic base and surrounding erythema on penis, anus or genitalia
3. Progresses to multiple, localized lesions and painful, tender, unilateral lymphadenopathy
4. Lymph nodes may liquefy and rupture
5. Fever, chills and malaise are usually present with lymph node involvement

6. Lab: Check STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep) and bacterial and viral cultures of the wound; check HCG in females

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
2. Consider Azithromycin (Zithromax®) PO, erythromycin (E-Mycin®) PO or ceftriaxone (Rocephin®) IM
3. Perform needle aspiration or I&D on any swollen, tender, fluctuant lymph nodes
4. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

III. **CHLAMYDIA (NONGONOCCOCAL URETHRITIS OR "NGU")**- *An acute or chronic inflammation of the urethra not caused by gonorrhea. NGU is usually caused by Chlamydia trachomatis but may be caused by mycoplasma or other bacteria.*

A. Signs and Symptoms

1. Urethral discharge most often seen in the morning
2. Itching, dysuria, urgency and frequency, or may be asymptomatic
3. Irritated meatus with visible discharge
4. Males may have prostatitis or epididymitis
5. Prolonged infection in females may lead to infertility/sterility or pelvic inflammatory disease (see "Febrile Illness Protocols, Pelvic Inflammatory Disease")
6. Lab: Check STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep; check HCG in females

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Consider single dose azithromycin (Zithromax®)
   b. Alternatives are doxycycline (Vibromycin®), erythromycin (E-Mycin®) or levofloxacin (Levaquin®)
2. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

IV. **ECTOPIC PREGNANCY**- *A fertilized egg that failed to implant in the uterus. May be located in the fallopian tube or in the abdominal cavity. Fetus can not survive due to poor placental support, but can grow into second trimester and pose a life-threatening risk to the mother as a result of internal damage and hemorrhage.*

A. Signs and Symptoms

1. Symptoms typically develop 6-8 weeks after last menstrual period
2. Abdominal pain, rebound tenderness
3. Amenorrhea and vaginal bleeding
4. May mimic normal pregnancy such as frequent urination, nausea and fatigue
5. Unilateral lower abdominal/pelvic pain with cervical motion tenderness
6. Lab: Check urine HCG for pregnancy

B. Treatment

1. **IMMEDIATE ACTION**
   a. Keep patient NPO

b. Initiate large bore IV and bolus NS if indicated
c. Monitor closely for changing vitals signs suggestive of shock and/or escalating pain which both indicate impending rupture

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

V. **EPIDIDYMITIS-** *An acute inflammation of the epididymis caused by bacteria (gonorrhea or chlamydia), acute physical strain (lifting or prolonged sitting) or trauma (motorcycles or bicycles).*

A. Signs and Symptoms

1. Pain in the scrotum or testicle radiating along the spermatic cord or to the flank
2. Dysuria, frequency, urgency, or difficulty starting stream if bacterial infection suspected (Absent if caused by strain/trauma)
3. Prostate tenderness, epididymal tenderness
4. Lab: Macroscopic UA will show WBCs, RBCs and bacteria if caused by infection (Urine may be normal or have RBC's if trauma or physical strain is the cause)

B. Treatment

1. Place patient on bed rest if pain prevents activity
2. Provide scrotal support (cotton-lined athletic supporter may be used)
3. Administer analgesics for relief of pain, acetaminophen (Tylenol®), ibuprofen (Motrin®), PRN, or acetaminophen with codeine (Tylenol #3®), as needed
4. Warm Sitz baths may offer some relief (See "Gastrointestinal Protocols, Hemorrhoids")
5. Restrict patient's physical activity (no lifting, pushing, or pulling greater than 5 pounds; no running)
6. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION
   a. Consider Ciprofloxacin (Cipro®)
   b. If patient is less than 35 years old or has new sexual partner, test for chlamydia and gonorrhea
   c. Discuss with preceptor and treat empirically for chlamydia and gonorrhea if lab not available
   d. Alternate medications include doxycycline (Vibramycin®) or trimethroprim-sulfamethoxasole (Bactrim®)

ACTION ALERT: If unable to rule out testicular torsion or if patient appears septic, **IMMEDIATE EVACUATION** is warranted.

VI. **GENITAL HERPES-** *Sexually transmitted infection caused by Herpes Simplex Virus (HSV).*

A. Signs and Symptoms

1. Usually incubation period is 2-12 days post exposure, but may have a longer latent period
2. Burning and stinging in area before lesions occur
3. Lesions appear as grouped clear vesicles over an erythematous base
4. Lesions crust and heal in 1-2 weeks

5. May be accompanied by fever, headache, malaise, dysuria
6. Lab: Check STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep) and bacterial and viral cultures of the wound; check HCG in females

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Consider acyclovir (Zovirax®) or valcyclovir (Valtrex®) to reduce symptoms and duration
   b. If patient has been having symptoms for 3 days or more, no treatment required
2. Examine and treat sexual partners. RPR/VDRL initially and at 3 months
3. Observe for secondary infection
4. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

VII. GENITAL WARTS (CONDYLOMA ACUMINATA)- *A sexually transmitted infection caused by the human papilloma virus (HPV) that causes growths on the skin in the genital region. Patient frequently minimizes them or dismisses them as "normal" because they are nontender and grow slowly.*

A. Signs and Symptoms

1. Incubation period 2-18 months after exposure
2. Single or multiple, soft, fleshy skin growths in the genital and perianal region
3. May be small or layered and cauliflower-like
4. Pruritus may be present
5. Lab: Rule out condyloma lata of syphilis. RPR is indicated. Check STI panel (HIV, Hepatitis panel, GC, KOH and wet prep) and bacterial and viral cultures of any open wounds; check HCG in females if indicated

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Consider evacuation if severe and limiting ability to perform duties
   b. Consider cryotherapy, podofilox (Condylox®) or imiquimod (Aldara®) topical treatments
2. Examine and treat sexual partners
3. RPR/VDRL initially and at 3 months
4. Follow-up every 2 weeks. Refer for further evaluation if no improvement
5. Observe for secondary infection
6. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

VIII. GLOMERULONEPHRITIS, ACUTE- *Glomerulonephritis is an acute kidney injury involving inflammation of the filtering unit of the nephron. In young, Active Duty population, the disorder is most often caused by an untreated streptococcal infection.*

A. Signs and Symptoms

1. Hypertension, edema, and decreased urine output
2. Red to brown colored urine (hematuria)
3. Malaise, anorexia and headache

4. Lab: Macroscopic UA shows RBCs, protein, and possibly WBCs. Sediment often visible in sample. If a Basic Metabolic Panel (CHEM 7) is available, creatinine will most-likely be elevated

B. Treatment

1. **IMMEDIATE ACTION**
   a. Patient is at risk of permanent kidney failure
   b. Measure all I&O's; goal output is >30cc/hr
   c. Establish IV access but do not administer fluid bolus unless directed by Preceptor
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. If evacuation will be delayed, discuss food and fluid restrictions
   b. Consider diuretics such as furosemide (Lasix®) if patient is developing volume overload affecting respiration
   c. Monitor vitals at least every 30 minutes

IX. **GONORRHEA**- *A sexually transmitted infection caused by Neisseria gonorrheae (gram negative diplococcus).*

A. Signs and Symptoms

1. Incubation period 2-8 days
2. Common cause of pelvic pain and sterility in females
3. May be asymptomatic in either sex
4. Men—burning on urination, serous or yellow, milky penile discharge; may have tender epididymis or prostate
5. Women—Vaginal itching, thick mucoid discharge; may present as painful urination, urinary frequency and urgency; inflamed cervix and increased vaginal discharge
6. Rectal infection can occur from anal sex or directly spread from vagina
7. Lab: Urethral/cervical swab testing and urine testing is available. Draw STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep); check HCG in females

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Consider ceftriaxone (Rocephin®) IM or cefixime (Suprax®) PO
   b. If allergic, consider high dose azithromycin
   c. Consider empirically treating for Chlamydia, since infections are clinically indistinguishable and co-infection is common
2. Contact and treat any partners suspected of exposure
3. All positive gonorrhea screens must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

ACTION ALERT: If laboratory is not available or patient is a flight risk and clinical suspicion for gonorrhea is high, treat patient in clinic for both gonorrhea and Chlamydia.

CLINICAL NOTE: Atypical sites of primary gonorrhea infection include pharynx, eyes, septicemia, arthritis, meningitis or pneumonia.

X. **HIV INFECTION-** See "Febrile Illness Protocols, Human Immunodeficiency Virus"

XI. **HYDROCELE-** *A fluid-filled sac near or surrounding a testicle that results in swelling of the scrotum. Can arise spontaneously or be related to trauma or infection.*

A. Signs and Symptoms

1. Painless swelling of the scrotum
2. Mass can be transilluminated (bright glowing fluid-filled sac) with otoscope
3. May be painful if associated with epididymitis
4. Lab: Macroscopic UA if infection/STI is suspected

B. Treatment

1. Stable, painless hydrocele does not require evacuation
2. Treat any coexisting genital infections
3. Increased scrotal support (cotton-lined athletic support) may reduce pull on swollen area
4. Persistent, symptomatic hydrocele requires surgery
5. If hydrocele is tense, it may compromise the circulation; IMMEDIATE EVACUATION is required
6. If there is an associated hernia, testicular mass or epididymitis, CONSULT PHYSICIAN PRECEPTOR- EVACUATION for evaluation and treatment

XII. **MENSTRUAL IRREGULARITY/OLIGOMENORRHEA-** *Infrequent or missed periods may be caused by a myriad of conditions in the healthy active duty female population.*

A. General Considerations/Differentials

1. Pregnancy—Check urine HCG; All patients with missed periods are presumed pregnant until proven otherwise
2. Ectopic pregnancy
3. Contraceptive use (especially Depo-Provera®, IUD, Implanon®, or Seasonale®)
4. Anorexia nervosa—Irregular menses, bradycardia, hypotension, arrhythmias, dry, cracked skin (especially around corners of the mouth), brittle hair and hair loss
5. Female athlete triad—Irregular menses, disordered eating, and stress fractures; most commonly found in gymnastics, ballet, figure skating and runners
6. Malabsorption syndromes—Celiac disease, ulcerative colitis, or Crohn's disease
7. Bulimia nervosa
8. Thyroid Disease

B. Treatment

1. Determine underlying cause
2. Determine weight gain or loss history through chart review if possible
3. Take actual weight and monitor regularly
4. Oligomenorrhea is common in Active Duty females, especially with increased exercise; however, oligomenorrhea and amenorrhea are suggestive of

inadequate nutrition or caloric intake necessary to support the level of activity involved

5. For all cases, provide basic dietary counseling
    a. Eat at least three servings of protein daily
    b. Increase iron, calcium and fiber intake
    c. Take a daily multivitamin
6. Reevaluate patient's weight, dietary history, and menstrual history in 1 month

XIII. **NEPHROTIC SYNDROME-** *Kidney damage resulting in leaking large amounts of protein from the blood into the urine. Defined as the triad of high protein in the urine, low albumin in the blood, and edema. Causes include allergy, diabetes, NSAID use, heroin abuse, HIV, obesity, lupus, hepatitis B/C, malignancy, and many more.*

A. Signs and Symptoms

1. Peripheral edema due to loss of protein and low colloid osmotic pressure
2. Dyspnea from pulmonary edema, pleural effusions, and ascites
3. Lab: Macroscopic UA positive for protein. If chemistry analysis (iSTAT) is available, the comprehensive metabolic panel (CHEM 14) will show low albumin, low protein and high cholesterol/LDL/HDL

B. Treatment

1. **IMMEDIATE ACTION**
    a. Determine underlying cause, if possible
    b. Strict I&O's
    c. Establish IV access; do not bolus patient unless directed to do so
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XIV. **ORCHITIS-** *An inflammation of one or both testicles most commonly associated with the virus that causes mumps, but may also be caused by bacterial infections or STIs.*

A. Signs and Symptoms

1. Mild to severe pain and/or swelling in one or both testicles
2. Sitting may cause discomfort
3. Nausea, fever and discharge from the penis may also be present
4. Testicle(s) will be diffusely tender with possible erythema and edema in or around the testicles
5. Lab: obtain macroscopic UA and STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep)

B. Treatment

1. **IMMEDIATE ACTION**
    a. Place patient on bed rest
    b. Elevate and support scrotum with cotton-lined athletic supporter for symptomatic relief
    c. Apply cold/hot compresses (cold compresses if due to mumps/warm compresses if due to bacterial infection)
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Must rule out testicular torsion
    b. Administer analgesics for pain; aspirin or acetaminophen with codeine (Tylenol #3®) if needed

c. If bacterial orchitis is suspected, consider starting empiric antibiotics to cover sexually transmitted infections

XV. **PELVIC INFLAMMATORY DISEASE (PID)-** See "Febrile Illness Protocols, Pelvic Inflammatory Disease"

XVI. **PREGNANCY**- *Defined as the period fom conception to birth when females carry developing offspring within the body.*

A. Signs and Symptoms

1. Fatigue, nausea, and increased frequency of urination
2. Associated with amenorrhea and breast tenderness
3. May have bloating, lightheadedness, food cravings or aversions, mood changes, low back pain and heartburn
4. Lab: Urine HCG. If positive, or lower abdominal tenderness is present, evaluate for ectopic pregnancy

CLINICAL NOTE: Patients may be unreliable historians (either intentionally deceptive or in denial about the reality of their condition). May mistake cramping and spotting associated with early pregnancy for menses, and fail to recognize a missed cycle. Consider sexual trauma in active duty women.

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION
2. Limited duty profile to avoid exposures to toxins and radiation
3. May continue to PT during first trimester, but if unsure of trimester, restrict PT and heavy lifting

XVII. **PROSTATITIS, ACUTE-** See "Febrile Illness Protocols, Prostatitis"

XVIII. **PYELONEPHRITIS, ACUTE-** See "Febrile Illness Protocols, Pyelonephritis"

XIX. **RENAL FAILURE, ACUTE-** *A sudden decrease in kidney function, resulting in an inability to maintain acid-base, fluid and electrolyte balance and ability to excrete nitrogenous wastes. In the young, healthy active duty population is frequently caused by severe dehydration leading to rhabdomyolysis, but many other causes are possible such as medications, supplements, alcohol, severe infection and abnormal eating and drinking habits.*

A. Signs and Symptoms

1. Decreased or absent urine output
2. Nausea, vomiting and malaise
3. Altered mental status
4. Abnormal blood pressure; in general, if patient is hypotensive the kidney damage is a result; if patient is hypertensive, the kidney malfunction is likely the cause
5. May have cardiac abnormalities (murmurs, gallops, rubs, or arrhythmias)
6. Peripheral edema

7. Lab: If basic metabolic panel is available (CHEM 7), will have elevated BUN and creatinine

B. Treatment

1. **IMMEDIATE ACTION**
    a. Bed rest, strict monitoring of intake and output
    b. Establish 2 large bore IV's; if dehydration is presumed cause then administer 2 Liters normal saline bolused followed by two more liters over 1-2 hours; do not administer Lactated Ringer's or any other types of fluid unless directed by a Preceptor
    c. Reevaluate vitals and reassess for peripheral edema and abnormal lung sounds and/or shortness of breath every 10 minutes while running fluid boluses; if any of these abnormalities are found immediately discontinue fluids but maintain IV access
    d. If patient appears to be going into cardiac arrest or shock, immediately administer 1 ampule of sodium bicarbonate
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Discuss fluid management plan until patient reaches higher level of care
    b. Administer medications only under direction of Preceptor
3. Monitor vital signs every 15 minutes until evacuated; monitor strict I&O's
4. Evaluate for possible etiologies with thorough history to include diet, exercise, supplements/medications, urinary history, surgical history, etc.

CLINICAL NOTE: Definitive treatment is based on correcting the underlying cause of the patient's renal failure.

XX. **RHABDOMYOLYSIS-** *Severe muscle injury such as crush injury, extreme physical exertion or heat stress that causes release of muscle enzymes and myoglobin. Myoglobin is toxic to kidney tubules and can cause acute renal failure.*

A. Signs and Symptoms

1. Muscle cramping and pain
2. Dark, red-brown urine
3. History of trauma or extreme physical exertion
4. Severe muscle injury can cause release of calcium which may cause seizures
5. Lab: Macroscopic UA appears bloody, but microscopic UA is negative for RBC's. Serum creatine kinase (CK) if available, will be extremely elevated. If basic metabolic panel (CHEM 7) is available, the serum creatinine and BUN will identify the severity of renal injury

B. Treatment

1. **IMMEDIATE ACTION**
    a. Establish 2 large bore IV's; Administer 2 Liters normal saline bolused followed by two more liters over 1 hour do not administer Lactated Ringer's or any other types of fluid unless directed by a Preceptor
    b. Treat shock (See "Emergency Protocols, Shock") If patient appears to be going into cardiac arrest or shock, immediately administer 1-2 ampules of sodium bicarbonate

c. Treat hyper/hypothermia (See "Environmental Injury Protocols, Hyperthermia" or "Environmental Injury Protocols, Hypothermia")
d. Reevaluate vitals and reassess for peripheral edema, abnormal lung sounds and/or shortness of breath every 10 minutes while administering fluid boluses; if any of these abnormalities are found immediately discontinue fluids but maintain IV access

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Recheck BUN/Creatinine every hour to monitor kidney failure status; may also get abnormalities in potassium, phosphorus, bicarbonate, and calcium
4. Bed rest, moitor vitals and respiratory status every 15 minutes, strict monitoring of intake and output

XXI. **SYPHILIS-** *A multi-staged Sexually Transmitted Infection caused by Treponema Pallidum. Frequently missed as a diagnosis.*

A. Signs and Symptoms

1. Incubation period ranges from 10-90 days with an average of 3-4 weeks
2. Primary Syphilis
   a. Painless papule that develops into an ulcer with clear base and firm margins
   b. Regional, painless, lymphadenopathy
   c. Painless ulcer usually located in genitalia, but may be in the pharynx
   d. Ulcer heals spontaneously without treatment
3. Secondary Syphilis
   a. Nonpruritic maculopapular rash that includes the palms and soles
   b. Large gray/white lesions on mucous membranes (condyloma lata)
   c. Fever, headache, anorexia, sore throat and weight loss
   d. Large, firm, rubbery lymphadenopathy
   e. "Moth eaten" hair loss
   f. Possible uveitis, meningitis and hepatitis
4. Late (Tertiary) Syphilis
   a. May occur any time after secondary syphilis and can involve any tissue in the body
   b. Most common sites for infection are skin, mucous membranes, skeletal system, eyes, respiratory system, GI, cardiovascular system and nervous system (neurosyphilis)
   c. Symptoms vary depending on the system involved
5. Lab: Check STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep); check HCG in females

B. Treatment

1. Confirm RPR. Quantitative reagent tests (VDRL/RPR) are performed at 1, 3, 6, and 12 months or until not reactive
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. Administer penicillin IM (Bicillin®) or PO (Pen-Vee K®)
   b. Alternates include doxycycline (Vibramycin®) or azithromycin (Zithromax®) PO
3. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer

4. Instruct the patient to refrain from sexual intercourse until conclusion of treatment, including treatment of all sexual partners

CLINICAL NOTE: If laboratory is not available or patient is a flight risk and clinical suspicion is high for Syphilis, treat patient in clinic for both chancroid and syphilis.

XXII. TESTICULAR MASS- *Solid mass on the testicle that must be identified. Testicular cancer is the most common cancer occurring in males aged 15-44 years.*

A. Signs and Symptoms

1. Painless enlargement of a testis or unilateral mass on the testis
2. Mass does not transilluminate, but may be accompanied by a hydrocele
3. Masses are assumed to be cancerous until proven otherwise

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION
   a. Further evaluation is required, but urgency will be determined by situation
   b. Transport to a medical facility with capability for biopsy and surgical intervention

XXIII. TESTICULAR TORSION- *A condition where the spermatic cord twists, cutting off the blood supply to the involved testicle.*

ACTION ALERT: If surgery occurs within 6 hours, the majority of torsed testicles can be saved. For every hour that passes, the viability of the testicle decreases and by 24 hours, most have infarcted and can not be saved.

A. Signs and Symptoms

1. Acute onset of unilateral, severe diffuse testicular pain and tenderness
2. "High-riding" testicle (testicle is pulled up into the inguinal canal)
3. Swelling in the scrotal sac
4. Light headedness, nausea and vomiting
5. May present as severe lower abdominal pain

CLINICAL NOTE: Signs and symptoms of testicular torsion are similar to infectious epididymitis and must be differentiated.

B. Treatment

1. **IMMEDIATE ACTION**
   a. Place patient in supine position/position of comfort and on bed rest
   b. Apply ice bag to the scrotum
   c. Administer analgesics for pain, Meperidine (Demerol®) IV PRN
   d. If surgical treatment is going to be delayed, attempt to reduce torsion manually (rotate as below as seen from patient's feet)
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

CLINICAL NOTES: Good pain management will be needed in order to attempt reduction. Consider administering Diazepam (Valium®) IV or IM in addition to narcotics to aid in relaxation of the spermatic cord. Be prepared to provide respiratory support when using combined narcotics and benzodiazepines. To reduce a right-sided torsion - gently rotate testicle counterclockwise one turn. To reduce a left-sided torsion - gently rotate testicle clockwise one turn. (Whether right or left torsion, if pain is increased by attempt at reduction, rotate in opposite direction). Testicular torsion is a true urological emergency. Immediate surgical evaluation is required. If pain has subsided (indicating a torsion/spontaneous de-torsion), evacuation and immediate urological consultation is still required.

XXIV. TOXIC SHOCK SYNDROME- See "Febrile Illness Protocols, Toxic Shock Syndrome"

XXV. TRICHOMONIASIS- *A sexually transmitted infection caused by the protozoa Trichomonas vaginalis.*

A. Signs and Symptoms

1. Purulent, malodorous, thin discharge associated with vulvar burning, itching, dysuria, urinary frequency and dyspareunia
2. Erythema of the vulva and vaginal mucosa
3. Greenish-yellow frothy discharge may be seen
4. Punctate hemorrhages may be visible on the vaginal walls or cervix (aka "strawberry cervix")
5. Males are frequently asymptomatic carriers
6. Lab: Swab taken of vaginal discharge and mounted on a slide with saline shows swimming organisms (do not confuse with sperm, which may be present up to 5 days after intercourse)
7. Check STI panel (HIV, RPR, Hepatitis panel, GC, KOH and wet prep); check HCG in females

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR– MEDICATION
   a. Administer single dose of metronidazole (Flagyl®) or tinidazole (Tindamax®) for patient and known sexual partners. Males continue treatment for 7 days. Repeat infections should be treated for 7 days
   b. Identify and treat all possible partners and refer patients to public health

ACTION ALERT: Warn patient about "Antabuse® effect" while taking Metronidazole (Flagyl®). Mixing alcohol with this medication can cause profound vomiting. Alcohol should not be consumed for at least 48 hours after completion of treatment

2. All STIs must be reported through military public health. Assist the public health office with partner contact/tracking and treatment as directed by the HMTF public health officer
3. Patients should avoid intercourse until 7 days after patient and partners have completed antibiotic therapy

XXVI. URINARY TRACT CALCULI ("KIDNEY STONES")- *Crystals forming in the urinary system dislodge and move into the ureter causing obstruction and severe pain. Stones are caused by super-saturation of the urine with specific salts.*

A. Signs and Symptoms

1. Severe unilateral flank pain
2. Difficulty staying still and staying quiet due to pain
3. Pain associated with nausea and vomiting
4. Episodic pain, radiating anteriorly to the abdomen. As the stone progresses down the ureter, pain may refer to the ipsilateral (same side) testicle or labia
5. Urinary urgency and frequency
6. Lab: Macroscopic UA showing RBCs and may show crystals; stones containing calcium may be visible on X-Ray ordered as "KUB" for "kidneys, ureters, and bladder"
7. Conclusive diagnosis is usually made with CT scan

B. Treatment

1. **IMMEDIATE ACTION**
   a. Monitor and record vital signs every 10 minutes while patient appears unstable
   b. Ensure adequate hydration. Start IV hydration if patient unable to tolerate PO fluids
   c. Provide analgesia: Acetaminophen (Tylenol®), ibuprofen (Motrin®) or Tylenol with codeine (Tylenol #3®)
   d. Provide treatment for nausea: Promethazine (Phenergan®) 25mg orally every 6 hours as needed
   e. If pain symptoms are mild and patient is tolerating oral intake patient may be monitored as an outpatient
   f. Strain all urine in an effort to recover stone for laboratory analysis
   g. Activity as tolerated
   h. Lab: Check Macroscopic UA; Check urine HCG if female
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. If patient is not tolerating oral intake, or is dehydrated, arrange for medical evacuation
   b. Preceptor may choose to force large volumes of fluids and will advise which fluids to use; otherwise Lactated Ringer's, normal, or half normal saline may be used to ensure hydration at routine daily rate
   c. Monitor I&Os
   d. If unable to control pain with oral medications consider meperidine (Demerol®) or morphine IV
   e. If fever or signs of infection, initiate antimicrobial therapy. Consider trimethoprim/sulfamethoxazole (Bactrim®), or gentamycin (Garamycin®)

CLINICAL NOTE: Suspect pyelonephritis in any patient with symptoms of a UTI accompanied with fever, flank pain, costovertebral angle tenderness or nausea/vomiting.

CLINICAL NOTE: Many STIs also present with voiding symptoms as a chief complaint. Perform thorough sexual history and **consider** a pelvic exam in any female patient with dysuria. Consider prostate exam in male patients with dysuria.

XXVII. **URINARY TRACT INFECTION/CYSTITIS**- *A bacterial infection of the lower urinary tract (below the level of the kidneys) to include the bladder and urethra. "Cystitis" specifically refers to irritation of the bladder. Note: painful urination can have other causes, but only urinary infection is addressed here.*

A. Signs and Symptoms

1. Dysuria, pyuria, urinary urgency and increased frequency
2. Sensation of incomplete void; nocturia
3. Absence of CVA tenderness (if present, see "Acute Febrile Illness Protocols, Pyelonephritis")
4. Pyelonephritis and cystitis can both be present at the same time (upper and lower tract disease)
5. Lab: Macroscopic UA positive for WBCs, leukocyte esterase, bacteria and possibly RBCs

B. Treatment

1. Increase PO fluid intake; patient should complete a full-bladder void every 1-2 hours
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION
    a. Consider trimethoprim-sulfamethoxazole ("TMP-SMX", Bactrim DS®, Septra®) or nitrofurantoin (Macrobid®) to treat infection
    b. Consider phenazopyridine (Pyridium®) for severe dysuria
        i. Warn patient that Pyridium® can stain all body fluids orange, including tears
        ii. patient should not wear contacts while taking this medication due to potential staining effects
    c. Cranberry juice is popularly recommended, but this has only been shown to improve symptoms if it is the pure juice concentrate with no added sugar (only available at food specialty stores) and should not be recommended by the IDMT

XXVIII. **BACTERIAL VAGINOSIS**- *An overgrowth of normal vaginal anaerobic (gram negative rod) bacteria due to a decrease in vaginal lactobacilli.*

A. Signs and Symptoms

1. "Fishy smelling", off-white, thin discharge most noticeable after coitus
2. Risk factors include multiple or new sexual partners (although the condition is not an STI), douching, cigarette smoking and sexual contact with other females
3. BV increases risk of acquiring STIs due to change in vaginal pH
4. Normal labia and vulva; vault with thin, homogenous "fishy smelling" grey discharge
5. Lab: Swab taken from vaginal wall cells (not discharge) and mounted on a slide with saline shows "clue cells" (vaginal wall cells studded with bacteria)

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR – MEDICATION
    a. Consider metronidazole (Flagyl®) PO or PV, clindamycin (Cleocin®) PO or PV
    b. If symptoms recur, treat sexual partners concurrently

c. Instruct patient to avoid tight-fitting clothing, pantyliners, and pantyhose
d. If symptoms fail to resolve with antibiotics, consider diagnosis of trichomonas
e. If symptoms recur more than once in a 6 month period, refer to HMTF for further evaluation

CLINICAL NOTE: Although popularly recommended, the oral or vaginal use of live yogurt strains has no demonstrated benefit.

XXIX. VARICOCELE- *A dilation of the blood vessels of the spermatic cord supplying a testis.*

A. Signs and Symptoms

1. Pulling or aching unilaterally in the groin. Worse after prolonged standing
2. Patient may complain of "feeling something" inside the scrotal sac
3. Examination of the scrotal sac reveals a tortuous, dilated blood vessel commonly described as "bag of worms"

B. Treatment

1. No treatment necessary unless pain is severe
2. Avoid "boxers" undergarments
3. May wear athletic support to decrease pulling sensation

Chapter 10: Endocrine Protocols

XXIV. DIABETES MELLITUS- *Impaired insulin secretion or sensitivitiy leading to hyperglycemia. There are two main types of diabetes mellitus, type 1 and type 2. Type one is referred to as juvenile onset and is characterized by the autoimmune destruction of beta islet cells in the pancreas leading to insulin deficiency. Type 2, or adult-onset diabetes describes a condition where insulin secretion is present, but cells are resistant to insulin effects.*

A. Signs and Symptoms

1. Hyperglycemia (normal blood glucose is 60-100mg/dl; patients in diabetic ketoacidosis may have glucose readings over 500mg/dl)
2. Urinary frequency, polyuria, polyphagia, and polydipsia that progress to orthostatic hypotension and dehydration with weight loss
3. As dehydration worsens, weakness, fatigue and mental status changes occur
4. Type 2 diabetics may have symptoms of hyperglycemia, but more commonly are asymptomatic and the condition is discovered on routine physical exam

B. Treatment

1. **IMMEDIATE ACTION**
   a. Initiate IV and administer saline bolus if indicated
   b. Lab: check fingerstick glucose; check urine for glucose and ketones; check iSTAT Basic Metabolic Panel (CHEM 7) if available
   c. Strict I&O's
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXV. DIABETIC KETOACIDIOSIS (DKA)- *Acute complication of diabetes characterized by hyperglycemia, hyperketonemia and metabolic acidosis.*

A. Signs and Symptoms

1. Nausea, Vomiting, abdominal pain
2. Hyperglycemia
3. pH <7.3 (iSTAT)
4. All signs and symptoms of DM
5. Hypotension/tachycardia
6. Kussmaul's respirations (air hunger, gasping breaths)

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access and administer infusion of 1-3L normal saline rapidly
   b. Check vital signs every 15 minutes while resuscitating
   c. Check blood glucose every 60 minutes while resuscitating
   d. After initial bolus, administer additional 3 L of NS over 1st 5 hours
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXVI. HYPOGLYCEMIA- *Low blood glucose (<60mg/dL) that causes CNS dysfunction.*

A. Signs and Symptoms

1. Sweating, nausea, warmth, anxiety, palpitations, tremulousness, hunger and paresthesias
2. Headache, blurred vision, confusion, seizures and coma
3. BG<60mg/dL

B. Treatment

1. **IMMEDIATE ACTION**
    a. Administer oral glucose, juice, soda, or candy if conscious
    b. If patient is unable to drink or eat, administer glucagon 1mg, IM. OR
    c. 50-100 mL IV bolus of 50% glucose
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXVII. **HYPOTHYROIDISM-** *A condition in which the thyroid gland does not make enough thyroid hormone, affecting patient's homeostasis and metabolism. Treatment is based on correcting the underlying cause and/or replacing inadequate thyroid hormone.*

A. Signs and Symptoms

1. Weight gain, fatigue, cold intolerance, arthralgias, myalgias, depression, menorrhagia, amenorrhea, hoarseness
2. Fingers or toes changing color and cramping when exposed to the cold (Raynaud's syndrome)
3. Physical findings of dry skin, brittle hair, bradycardia, delayed deep tendon reflexes, thin/brittle nails, thinning hair (especially the outer half of eyebrows), peripheral edema, puffy face and eyelids, skin pallor or jaundice
4. Palpably enlarged thyroid (goiter) may be present
5. Note: All patients being evaluated for depression should have their thyroid function assessed prior to starting antidepressant therapy

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION
    a. Patient requires lab evaluation of thyroid stimulating hormone
    b. Also needs to be evaluated for possible coexisting conditions

CLINICAL NOTE: Life threatening hypothyroidism, called **myxedema crisis**, can occur. These patients have impaired cognition, confusion, or coma and peripheral edema. As with any cause of altered mental status, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION.**

XXVIII. **HYPERTHYROIDISM-** *Caused by the thyroid gland secreting too much T3 and T4 hormone, affecting a patient's homeostasis and metabolism.*

A. Signs and Symptoms

1. Jitteriness, anxiety, palpitations, weakness, fatigue, weight loss, heat intolerance, loose stools, sweating nervousness, menstrual irregularities
2. Physical findings of warm moist skin, tachycardia, lid lag (whites are visible above iris as patient looks down), bulging eyes (exophthalmos), tremors
3. May have palpably enlarged thyroid

B. Treatment

1. CONTACT PHYISICIAN PRECEPTOR- CONSULTATION AND ROUTINE EVACUATION
2. Treatment typically involve treatments to suppress or destroy the thyroid gland or surgical removal
3. After destructive treatment, patient will require lifelong daily oral replacement of thyroid hormone

Chapter 11: Integumentary Protocols

CLINICAL NOTE: The management of integumentary disorders will basically be the same in isolated environments as in other spheres. A good patient history is vital to determining cause and accurate diagnosis.

ACTION ALERT: General eruptions often accompany or precede an anaphylactic reaction. A Timely yet complete evaluation of the patent is essential. Be prepared to take emergent actions for managing an anaphylactic episode. If patient becomes short of breath or displays other signs of anaphylaxis, refer to "Emergency Protocols, Anaphylaxis"

CLINICAL NOTE: Due to the nature of integumentary problems, they often lend themselves to telemedicine or emailing of digital photographs. Always ensure patient privacy and ensure that telemedicine consultation procedures are reviewed/approved by the host MTF.

I. **ACNE-** *A common inflammatory skin disease of unknown etiology with onset in puberty or delayed until 3rd or 4th decade of life. Factors include plugging of follicles, retention of sebum, overgrowth of Propionibacterium acnes and staphylococcal bacteria, irritation by accumulated fatty acids, and inflammatory reaction to extrafollicular sebum.*

A. Signs and Symptoms

1. Soreness or pain from open or closed comedones (hallmark of the disease)
2. Papules, pustules, cysts or nodules
3. May occur over face, neck, upper chest, back or shoulders

B. Treatment

1. Do not pick or manipulate lesions, it will result in scarring
2. Use only mild soaps (Ivory® or Cetaphil®, for example) on acne-prone area
3. Do not use "scrubbing" creams, "microbeads" or any other abrasive action on the acne
4. Do not wash face more than twice daily (including showering)
5. Do not use astringent washes on face unless they are part of prescribed regimen
6. If skin drying or flaking occurs, add hypo-allergenic facial moisturizer developed for acne prone skin
7. Apply sunscreen every day before leaving the house; use only mild hypo-allergenic preparations
8. Medications used for acne are based on the severity of the condition
    a. Comedonal Acne: Topical retinoids, start with tretinoin 0.25% cream (Retin-A®, Avita®)) topically applying a pea-sized amount over clean skin twice per week and slowly increasing until using nightly. If patient is unable to tolerate, may use adapalene gel 0.1% (Differin®) topically nightly. If no response to retinoid alone, may add Benzoyl peroxide 2.5%-5% daily
    b. If pustules, papules and/or nodules present and/or if acne is prominent chest and back, CONTACT PRECEPTOR- MEDICATION

early in the course of treatment for addition of topical and/or oral antibiotics

CLINICAL NOTE: Results of acne therapy may not be apparent for 6-12 weeks after initiating new treatment regimen. The condition often worsens for 1-4 weeks before improving. Every time changes are made, wait at least 8 weeks before altering therapy again. Patient must be counseled that developing a successful treatment takes several visits and several months and must keep their expectations realistic.

II. CORNS- *Corns are thickened outer layers of skin caused by friction and pressure (usually on the feet), and include a painful hyperkeratotic core.*

A. Signs and Symtpoms

1. Thickened skin at a pressure point; if patient has worn ill-fitting shoes to the clinic, may see an area of redness around the thickened skin
2. Painful growth or "corn" deep to the outermost skin layer
3. Overlying skin is smooth, unlike warts which have a bumpy, cauliflower-like texture; skin lines will be preserved

B. Treatment

1. Debulk the corn with scalpel by keeping blade parallel to skin surface and gently scraping away successive layers of thickened skin (if black spots are visible in the inner layers of the lesion, consider diagnosis of warts)
2. Apply salacylic acid plaster 40%, cut to the size of the lesion
3. Leave plaster in place for 48-72 hours; keep plaster dry
4. Remove plaster, pare down whitened skin, and repeat
5. Patient may pare own skin and replace plaster every 24 hours until lesion has resolved

C. Prevention

1. Avoid tight or ill-fitting footwear!
2. May use moleskin cut in ring to pad the areas around the lesion to disperse the pressure applied to the corn
3. Keep feet clean and dry

III. ECZEMA (ATOPIC DERMATITIS)- *"Dry skin". Starts as itchy vesicles and progresses to dry flaky eruption on the face, neck, trunk, and flexor surfaces of wrists, hands, and antecubital and popliteal fossae.*

A. Signs and Symptoms

1. Pruritic, rough, red plaques with thick flaky scale
2. Commonly affects the flexor surfaces of knees and elbows
3. If chronic, the skin is dry, leathery, and lichenified; in dark-skinned individuals, the affected area may be hypopigmented
4. During acute flares, may have widespread redness with weeping plaques
5. Severe cases also include cracking, fissuring and bleeding around plaques

B. Treatment

1. Patient education is key! Medications are of limited value, patient must control the illness with their own behavior

2. Start skin rehabilitation with bathing routine—immediately after showering or bathing, towel-dry skin and while it is still warm, apply copious amounts of Vaseline® to all affected areas
3. If bathing at night, recommend dedicating a pair of pajamas to wear after ointment treatments
4. Do not bathe more than once daily and handwashing should be minimized
5. Avoid exfoliating products such as washcloths and scrub brushes
6. Discontinue all deodorant or antimicrobial soaps; use only mild body saps such as Keri Soap® or Ivory®
7. Keep a portable tube of ointment in pocket; every time hands are washed (after using the restroom, after washing dishes, etc) immediately apply ointment to towel-dried hands. May also carry handkerchief to wipe off excess
8. If groin, trunk, or buttocks are affected, recommend cotton underwear
9. Avoid nylon or synthetic clothing
10. Use only mild laundry soaps without softeners or bleaches
11. For severe cracking of hands, recommend washing hands, towel drying, copious ointment, and then placing hands inside plastic gloves and placing gloved hands inside mittens or wool socks. The warmth and occlusion will enable the Vaseline® to deeply penetrate and moisturize the skin, but patient should be warned that ointment will get runny and potentially make a mess
12. Counsel patients that lotions are water-based and will ultimately dehydrate the skin
13. Creams are more penetrating and effective than lotions and ointments are more penetrating and effective than creams; therefore ointments are treatment of choice
14. For pruritic (itching) lesions, apply small amount of topical corticosteroid cream or ointment such as hydrocortisone 1% or triamcinolone 0.1% twice daily; counsel patient to slowly taper off steroid cream but continue aggressive Vaseline® treatment

CLINICAL NOTE: Topical corticosteroids must be used with caution. Use only as long as needed and taper down. Only low potency steroids should be used on the face. Avoid steroids in folds of the skin and around the eyes.

15. For acute, weeping lesions apply Burow's solution (Domeboro®, Campho-Phenique®) solution or colloidal oatmeal dressings for 10-30 minutes 2-4 times per day; these treatments will soothe the skin and they act as an astringent to speed healing
16. Fingernails should be kept short to prevent excoriations and secondary infection
17. If patient has open breaks in the skin that will effect ability to perform duties (healthcare workers, food service workers, etc), consider duty profile
18. Secondary infections should be treated with systemic antibiotics such as dicloxacillin (Dynapen®)

CLINICAL NOTE: Smallpox Vaccine is usually contraindicated in patients with eczema/atopic dermatitis. Consult your physician preceptor if smallpox vaccine is indicated in such an individual.

IV. **CELLULITIS-** *Diffuse spreading infection of the dermis and subcutaneous tissue.*

A. Signs and Symptoms

1. Begins as small tender erythematous, edematous patch
2. Patch expands over several hours to days, with increased swelling, erythema and pain
3. Erythematous "streaks" away from lesion and/or regional lymphadenopathy are possible
4. As lesion spreads patient may develop fever, chills, malaise

CLINICAL NOTE: Cellulitis is common on the lower legs, especially in patients that have chronic lower leg swelling due to venous insufficiency. Infection usually caused by Staphylococcal and Streptococcal infections (organisms that normally live harmlessly on the skin but can invade when the skin is weakened).

B. Treatment

1. If patient is hypotensive, start a large bore IV of NS or Ringer's Lactate and consider diagnosis of septic shock (See "Emergency Protocols- Septic Shock")
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
   a. For mild infections: consider dicloxacillin (Dynapen®) or cephalexin (Keflex®)
   b. If "spider bite" appearance or possible MRSA-consider trimethoprim-sulfamethoxazole (Bactrim®) or clindamycin (Cleocin®)
   c. For severe infections, consider IV antibiotics

ACTION ALERT: If systemic symptoms such as fever, lymphadenopathy, erythematous streaking from the wound, wound is progressing rapidly, or patient appears ill, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION.** If hypotension or other signs of sepsis are present, see "Emergency Protocols- Septic Shock", start large bore IV, discuss IV antibiotics with Physician Preceptor, and arrange immediate evacuation.

V. **CHICKENPOX-** *A pox virus that produces characteristic rash largely eliminated in the United States due to vaccination. Spreads through contact with the fluid from direct contact with vesicles or from aerosolized nasal secretions from an infected individual.*

A. Signs and Symptoms

1. Incubation period up to two weeks
2. Prodrome of fever, malaise, pharyngitis, and loss of appetite occurring 24 hours before the appearance of the rash
3. Pruritic lesions start as macules, then progress to vesicles, that turn into pustules that crust
4. Vesicles have a very characteristic "dew drop on rosepetal" appearance with the vesicle in the center of the macule
5. Lesions typically are in different stages on the face, trunk, and extremities

B. Treatment

1. If discomfort from itching is severe, administer oral antihistamines such as hydroxyzine hydrochloride (Atarax®) or diphenhydramine (Benadryl®)

2. Acetaminophen (Tylenol®) for fever
3. Do not give aspirin-containing analgesics, since these have been linked to potentially fatal Reye's Syndrome
4. Cut fingernails very close to avoid potential for bacterial superinfection of lesions
5. Consider treatment with oral acyclovir (Zovirax®) within 24 hours of rash onset

VI. **CONTACT DERMATITIS-** *Acute or chronic dermatitis that results from direct skin contact with chemicals or allergens.*

A. Signs and Symptoms

1. History of exposure to allergen or irritating substance such as poison ivy/oak/sumac, topical medications, hair care products, nickel in jewelry, fragrances in perfumes or cosmetics, deodorants, preservatives, latex, rubber, adhesive tape, or detergents
2. Identifying the cause is key to prevent recurrence; history should include changes in soaps and detergents, location of lesions, occupation, and hobbies
3. Eruption of intensely pruritic erythematous popular or vesicular rash on the skin that was contacted; usually occurs hours after the exposure
4. Vesicles and bullae may rupture and cause oozing of serous fluid
5. Lesions due to direct contact with plant resins frequently occur on the hands, forearms, or face in linear lesions from brushing contact with stems or leaves
6. Plant resins on the hands may be transferred to other skin sites directly or by transferring the resin to fomites (touching towels, sheets, etc)
7. Patients may believe that oozing lesions are "spreading", or causing new lesions to occur, though this usually does not happen. Instead, patients are probably touching a contaminated fomite with clean skin, causing new lesions

B. Treatment

1. Remove all contaminated fomites and offending agents, if still present
2. Wash the contaminated area of skin thoroughly with mild detergent within 30 minutes of exposure
3. Immediately remove any contaminated clothing and wash thoroughly
4. If the affected area is small (less than 5% of body surface area), prescribe a topical corticosteroid such as Triamcinolone 0.1% cream (Kenalog® cream) applied sparingly to affected area
5. If wheals or itching are prominent, administer oral antihistamines such as hydroxyzine hydrochloride (Atarax®) or diphenhydramine (Benadryl®)
6. If topical agents and antihistamines and are not effective or a larger area of skin is involved, CONTACT PHYSICIAN PRECEPTOR- MEDICATION and consider oral steroids such as prednisone (Deltasone®)

CLINICAL NOTE: Warn patient about drowsiness when first taking Atarax® or Benadryl® and complete profile restricting from driving/hazardous duty if patient intends to take Atarax® during waking hours.

VII. **DERMATITIS MEDICAMENTOSA-** *Skin rash caused by medication.*

A. Signs and Symptoms

1. Vary based on medication; may be fine maculopapular, urticarial, or exfoliative
2. Usually occurs in a central or a sun-exposed distribution
3. Simple- occurs during 2nd week of therapy, no constitutional signs
4. Complex- occurs during 3rd week of therapy, has constitutional signs

B. Treatment

1. Supportive therapy; treatment based on symptoms
2. Discontinue offending drug
3. Consider diphenhydramine (Benadryl®) or hydroxyzine (Atarax®) for urticarial lesions
4. If lesions are blistering and extensive, CONSULT PRECEPTOR- EVACUATION because patient may require inpatient burn unit care

CLINICAL NOTE: Medicamentosa is an adverse reaction to a medication but is not an "allergic reaction". Avoid re-exposure. Common medications include NSAIDs, sulfa drugs, narcotics, and neuropsychiatric medications.

VIII. **FOLLICULITIS-** *A superficial bacterial skin infection of the hair follicles that does not involve the deeper tissues. Folliculitis is caused by follicle inflammation and production of purulent material. Folliculitis is a superficial skin infection, whereas abscesses and furuncles extend into the deeper layers of the skin.*

A. Signs and Symptoms

1. Clusters of multiple small, raised, pruritic or burning erythematous papules less than 5 mm in diameter
2. Pustules may be present in the centers of the lesions
3. Folliculitis is often observed in areas that are hairy or frequently shaved
4. Differentiated from lesions of pseudofolliculitis, which are pustules caused by loose ends of shaved hair curling back to puncture and irritate the skin
5. Pseudomonas is one causative agent—patients with folliculitis caused by pseudomonas may have sharply demarcated areas of erythema following exposure to inadequately chlorinated hot tubs, whirlpools, or swimming pools

B. Treatment

1. Warm compresses TID will usually resolve lesions regardless of causative organism
2. Avoid shaving in affected areas
3. Make sure all water sources that may be possible sources of pseudomonas are adequately treated
4. Avoid tanning booths, local pedicure providers, local saunas and swimming at local watering holes
5. For severe cases, CONTACT PHYSICIAN PRECEPTOR to consider topical antibiotics such as erythromycin (E-Mycin®)

IX. **HERPES SIMPLEX VIRUS-** Viral vesicular/ulcerative lesion of orolabial and genital areas caused by herpesvirus (a member of the poxvirus family).

A. Signs and Symptoms

1. Burning and stinging of orolabial or genital area followed by eruption of small grouped vesicles on an erythematous base
2. Vesicles usually rupture and heal within 1 week
3. May have local lymphadenopathy

B. Treatment

1. For orolabial herpes, oral medications usually are not required
2. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Genital herpes- For first clinical episode acyclovir (Zovirax®) or valacyclovir (Valtrex®) x 7-10 days; Recurrent genital herpes- valacyclovir (Valtrex®) twice daily x 3 days, or acyclovir (Zovirax®) 5 times daily x 5 days
    b. Orolabial herpes- For first clinical episode can give acyclovir (Zovirax®) or valacyclovir (Valtrex®) x 7-10 days

CLINICAL NOTE: Compliance is better with valacyclovir (Valtrex®) because fewer doses are required per day. To be most effective treatment needs to be started at first onset of symptoms (during burning or stinging, before lesions even appear). Topically applied medicines are not very effective. Encourage patient to use sunscreen, which may prevent sun-induced recurrences.

X. HERPES ZOSTER- *Acute viral vesicular eruption caused by reactivation of latent chickenpox infection. Usually occurs when patient is stressed, pregnant, or immunocompromised.*

A. Signs and Symptoms

1. Pain along a dermatome followed by unilateral grouped vesicles along the same dermatome 3-7 days later
2. Most commonly on face and trunk
3. Vesicles are exquisitely tender, even to light touch such as sheets or clothing
4. Vesicles may rupture and ooze clear fluid
5. Up to 20 lesions may be found outside the affected dermatome
6. May have regional lymphadenopathy

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Use systemic antivirals in patients >age 50 or with moderate to severe pain or rash
    b. Goal of antivirals is not to cure the outbreak; the goal is to reduce the severity of nerve damage and chronic pain (postherpetic neuralgia)
    c. Acyclovir (Zovirax®) or valacyclovir (Valtrex®) x 7 days
    d. Consider oral steroids such as prednisone for 3 weeks with taper
    e. Treat pain with appropriate analgesic for level of pain (may require narcotics); preceptor may prefer to treat with neuropathic pain medications (tricyclic antidepressants or gabapenting, for example)
    f. If lesions resolve and patient still has pain in the area, patient probably has postherpetic neuralgia; CONTACT PHYSICIAN PRECEPTOR- MEDICATION for appropriate pain control
2. Symptoms usually lasts 2-3 weeks and does not recur

3. May consider Calamine® lotion topically
4. If lesions are on forehead, eyelid, or anterior scalp, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION** for Ophthalmology consult
5. Prevention-Vaccine (age 60 and up)

CLINICAL NOTE: This is a recurring disorder, and the antivirals work better the earlier the prescription is started. For these reasons it is advisable to give the patient a 5 day supply of acyclovir (Zovirax®) or valacyclovir (Valtrex®) after symptoms resolve to be initiated at the first sign of recurrence. This is particularly important if the patient is traveling to a remote location.

XI. **IMPETIGO**- *A superficial skin infection caused by broken skin in the presence of colonizing gram positive cocci such as Staphylococcus or Streptococcus.*

A. Signs and Symptoms

1. Clusters of vesicles, bullae, or pustules that rupture and develop thick, adherent, honey-colored granular crusts over the lesion
2. Lesions are typically fluid-filled and surrounded by erythema
3. Vesicular fluid typically starts as clear yellow and becomes darker and more turbid as lesions approach rupturing
4. Risk factors include warm humid conditions with overcrowding and poor hygiene

B. Treatment

1. If lesions are limited and without bullae, consider mupirocin (Bactroban®) topically
2. If lesions are bullous or more widespread, consider oral erythromycin (E-Mycin®)
3. Do not delay treatment since lesions can progress into cellulitis, lymphangitis, furunculosis, hyper/hypopigmentation, and scarring
4. Emphasize handwashing due to easy transmission from contaminated hands
5. If lesions develop abscesses, I&D (incise and drain) the lesion and consider clindamycin (Cleocin®) or trimethoprim-sulfamethoxazole (Bactrim®) due to possible superinfection with MRSA

CLINICAL NOTE: Even though impetigo skin disease is relatively minor, patients are at risk for poststreptococcal glomerulonephritis leading to kidney failure, and also rheumatic fever leading to fevers, cardiac problems, blindness, and death. Have high suspicion for these conditions in any patient that presents with malaise following impetigo infection.

XII. **INGROWN TOENAIL**- *Curving in or impingement of a nail border into its adjacent nail fold, causing pain and swelling. Area may become infected.*

A. Signs and Symptoms

1. Causes include tight shoes, abnormal gait, bulbous toe shape, and cutting nails too short or too rounded
2. Pain occurs at the corner of the nail fold or along the lateral margin
3. Exam shows swollen, tender posterior or lateral nail fold that may also be draining or collecting purulent fluid and/or granulation tissue

B. Treatment

1. Treatment is based on the extent of the disease; attempt nonsurgical methods first (choose one or more of the below treatments):
    a. Warm water soaks 10-20 minutes with or without Epsom salts three times a day Instruct patient to push the lateral nail fold gently away from nail plate after each soak. Perform first treatment in clinic
    b. Insert small amount of cotton removed from a cotton ball or cotton-tipped applicator (Q-Tip®) or dental floss between the ingrown nail and the nail fold. If this is too painful, place cotton under the unaffected portion of the distal nail to help prop up the ingrowing corner
    c. If pustule is present, consider incision and drainage (see "Abscess" for procedure description)
2. Most ingrown nails will resolve with the above treatments and patience

CLINICAL NOTE: For severe nails refractory to the measures above, consider partial or full excision of the nail. Oral antibiotics typically not indicated, even if pus is present. Perform local I&D instead.

XIII. LEISHMANIASIS- *Skin or systemic infection caused by Leishmania, a protozoan group of parasites transmitted by sandfly bites. Commonly seen in Iraq, Afghanistan, and other arid desert regions.*

A. Signs and Symptoms

1. Cutaneous**:** Single or multiple cutaneous swellings that start as small papules and develop into nonulcerated dry plaques or large encrusted ulcers
    a. Ulcers are well demarcated raised and indurated margins
    b. Lesions are painless unless secondarily infected
    c. Localized lymphadenopathy is common, but systemic symptoms are uncommon
    d. Healing occurs spontaneously in months to a few years with significant scarring
2. Mucocutaneous (espundia): Nasal congestion followed by ulceration of mouth, lips, palate, pharynx and larynx.
    a. Occurs months or years after resolved cutaneous lesions
    b. Fatality >90% if left untreated, primarily from hemorrhage and superinfection with other organisms
3. Visceral (kala azar): Nonulcerating nodule at site of bite may precede systemic symptoms but is usually not apparent
    a. Fever, chills, sweats, weakness, anorexia, weight loss, cough and diarrhea
    b. Splenomegaly, +/- Liver enlargement and lymphadenopathy
    c. Hyperpigmentation of skin
    d. Skin lesions, petechiae, gingival bleeding, jaundice, edema and ascites
    e. Progression causes wasting and malnutrition and death due to secondary infections

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if leishmaniasis is even remotely possible
2. Evaluation requires skin biopsies and specialist consultation with Dermatologist and/or Infectious Disease specialist; treatment includes IV medications

CLINICAL NOTE: Prevention is KEY! Use insect repellants, fine insect netting, long sleeves and pants. Continuously revaluate disease control measures where leishmaniasis is endemic. Consider large-scale prevention measures such as destruction of animal reservoir, mass treatment for personnel in disease-prevalent areas, pesticide spraying in dwellings, etc. Counsel personnel to avoid contact with dogs and other domesticated animals in high-risk areas.

XIV. **LYMPHANGITIS**- *Inflammation of the lymphatic channels or regional lymph nodes caused by infection with bacteria, mycobacteria, viruses, fungi, or parasites. Often accompanied by lymphadenitis, or painful swollen regional lymph nodes.*

A. Signs and Symptoms

1. Throbbing pain in area of cellulitis with red streak in skin extending toward regional lymph nodes
2. Local lymphadenopathy
3. Malaise, anorexia, sweating, chills, and fever may develop rapidly
4. Lab: CBC-Shows elevated WBC count; get blood cultures if available

B. Treatment

1. Use moist heat compresses, elevation, and immobilize infected area
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION
   a. Patient may require IV antibiotics and inpatient admission
   b. Consider coverage for MRSA such as trimethoprim-sulfamethoxazole (Bactrim®) or clindamycin (Cleocin®)

ACTION ALERT: REASONS TO CONTACT PRECEPTOR- IMMEDIATE CONSULTATION:

1. Masses with associated soft tissue infection will likely require systemic antibiotics
2. Multiple new lesions associated with systemic signs and symptoms will likely require systemic antibiotics
3. Non healing ulcers of more than a few weeks duration will likely require specialist evaluation and/or biopsy

XV. **LYME DISEASE**- See "Febrile Illness Protocols, Lyme Disease"

XVI. **MOLLUSCUM CONTAGEOSUM**- *Papular skin condition of groin, inner thighs, and lower abdomen caused by a poxvirus.*

A. Signs and Symptoms

1. Single or multiple small, smooth, umbilicated papules on penis, pubis, inner thighs, face, or lower abdomen
2. Lesions have characteristic dome shape with central ubilication (looks like a papule with a belly button)

3. Lesions may have surrounding erythema, scale, and pruritus
4. Transmitted by skin-to-skin contact in wrestlers or during intimacy
5. Wrestlers may have lesions on defensive areas such as forearms and thighs
6. Can continuously auto-inoculate (patient spreads it to other areas themselves) through direct contact of lesions with nearby areas of moistened or sweaty skin
7. Expose the whole patient and identify/treat all lesions

B. Treatment

1. Two treatment options:
   a. Apply liquid nitrogen to each lesion for 5-10 seconds being careful not to contact any of the surrounding normal skin
   b. Using a curettage instrument, cauterize each lesion being careful not to contact any of the surrounding normal skin
2. Regardless of treatment method chosen, be sure to treat all lesions to avoid autoinoculation

ACTION ALERT: If there is any evidence of secondary or systemic infection, CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE CONSULTATION.

XVII. PEDICULOSIS (LICE) INCLUDING CAPITIS/CORPORIS/PUBIS- *Parasitic skin infection caused by insects living on hairy areas of head or body.*

A. Signs and Symptoms

1. May be asymptomatic or have intense pruritus
2. Careful examination includes gentle combing wet hair with a fine-toothed comb near the crown and observing for insects
3. Entire head or area should be combed twice
4. Nits (larvae) fluoresce pale blue with Wood's lamp
5. Body lice infestations not likely to be found by examining the skin—instead check in seams of clothing
6. Insects can survive on fomites such as headphones, towels, and bedding for days or longer

B. Treatment

1. **Pediculosis Capitis**
   a. Permethrin 1% (Nix®) cream, apply to head/scalp and leave on for 10 minutes, then rinse off
   b. Comb out nits
   c. Eyelashes treated with petrolatum jelly (Vaseline®) twice a day x 8 days and then nits plucked off
   d. Dispose of infested hats and headgear
2. **Pediculosis Corporis**
   a. Dispose of infested clothing
   b. If nits are seen on hair shafts treat with Permethrin cream 5% (Nix®) leave on for 8 hours, then rinse off
3. **Pediculosis Pubis**
   a. Apply Permethrin 1% cream (Nix®) to affected area for 10 minutes and then rinse off
   b. Sexual contacts should also be treated

4. For all forms of pediculosis, wash all towels and bedding in boiling water and dry with high heat (see below)

CLINICAL NOTE: Do not allow patient to wash using other shampoos or conditioners prior to treatment because they may reduce the efficacy of medications.

CLINICAL NOTE: Wash and dry all clothes and bedding at high temperatures (ie boiling water and high heat dryer). If it is not possible to wash and dry with high heat, then store all bedding and clothes in a plastic bag for two weeks and this should be long enough to starve all lice and nits. Examine all household contacts and treat as necessary.

ACTION ALERT: If signs or symptoms of systemic infection are present contact physician preceptor.

XVIII. PSEUDOFOLLICULITIS BARBAE- *Inflammation around a hair follicle where a curly hair shaft folds over and re-enters the skin.*

A. Signs and Symptoms

1. Inflammation around the embedded hairs results in firm, skin colored or erythematous papules that become irritated from shaving
2. Differentiated from acne by noting that lesions are adjacent to hair follicle, not arising from the follicle
3. Pustules develop as disease progresses

B. Treatment

1. Grow beard (requires profile to allow growth of 1/8")
2. Instruct patient to clip/shave in direction of hair growth, not against it
3. Encourage patient to use warm cloths before shaving to open skin pores
4. May use buff puff pad in circular motions to release trapped hairs prior to clipping
5. If no improvement, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND MEDICATION about possible treatments with retinoids, topical steroids or topical antibiotics

CLINICAL NOTE: Educating the patient is critical regarding proper release of hair shaft. Advise patient not to pluck/tweeze ingrown hairs. Use only sharp razors. May use depilatories (Magic Shave®) but these often cause extensive irritation. Education regarding depilatories: do use depilatory within 36 hours after shaving, use sensitive skin formulation for initial use, and apply only to small skin area for first use to monitor for irritation. Discontinue use if skin irritation occurs.

XIX. PSORIASIS- *A benign, chronic inflammatory skin condition that may also involve the nails and joints.*

A. Signs and Symptoms

1. Silvery scales on bright red, well demarcated plaques with varying degrees of pruritus
2. Usually located on extensor surfaces of elbows, knees, sacral region and scalp and occasionally on palms and soles
3. Examine patient's nails; if nail pitting is present it is suggestive of psoriasis
4. Ask about joint pain, joint stiffness, and complete a full review of systems questioning about systemic disease

B. Treatment

1. If scalp is involved, use coal tar shampoo (Tarsum®) daily
2. Exposing involved skin to sunlight is effective, but be careful to avoid sunburn because skin is already damaged
3. If lesions are present on more than 10% body surface area, if symptoms persist more than a month despite treatment, or if any systemic or joint involvement is present, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION

ACTION ALERT: if joint pain and skin findings are both present at the same time, CONTACT PHYSICIAN PRECEPTOR- EVACUATION. Patient requires workup for psoriatic arthritis and prompt treatment to prevent progression and possibly permanent joint destruction.

CLNICAL NOTE: This disorder carries significant psychosocial impact; patients are often very self-conscious about lesions, especially on the scalp and face. Medications that exacerbate lesions include β-blockers, anti-malarial antibiotics, statins, lithium and oral steroids.

XX. **ROCKY MOUNTAIN SPOTTED FEVER**- See "Febrile Illness Protocols, Rocky Mountain Spotted Fever"

XXI. **SCABIES**- *A skin infection caused by the insect Sarcoptes scabiei. Causes intense pruritus, which comes from deposition of feces and eggs from mites travelling* underneath the skin.

A. Signs and Symptoms

1. Small erythematous nondescript papule that are usually excoriated and tipped with hemorrhagic crusts
2. Small vesicles or pustules may be present
3. "Burrows" may be seen- thin, grey, red, or brown lines several mm long; burrows are frequently obscured by excoriation
4. Lesions are typically located in web spaces, heels of palms, wrist flexors, elbows, axillae, waistline, and groin
5. Patient typically describes intense itching worse at night out of proportion to exam findings
6. Transmission is from skin-to-skin direct contact with infected individual; in young adults transmission is usually through sexual contact

B. Treatment

1. Consider topical permethrin cream (Nix®) or oral ivermectin (Stromectol®)
2. Itching typically persists several weeks after eradication of infection; nighttime itching can be controlled by 1st generation antihistamines such as diphenhydramine (Benadryl®) or hydroxyzine (Atarax®); daytime itching

can be controlled by 2nd generation antihistamines such as loratadine (Claritin®) or cetirizine (Zyrtec®), but do not give more than one antihistamine

3. Localized itching can be treated with topical hydrocortisone cream (Westcort®)
4. Mites cannot survive more than 3 days away from skin; wash all clothing and bedding that has been used in the past 3 days by machine washing in hot water and drying in hot dryer. If that is not possible, fabrics may also be stored in a tightly sealed plastic bag for 14 days and then washed by whatever means available

CLINICAL NOTE: Excoriated lesions are at high risk for superinfection including cellulitis, impetigo, etc. Suspect superinfection if lesions have surrounding erythema, edema, or are weeping (scabies lesions are typically dry with excoriations that have bleeding and scabbing). Also a high risk of spread to household; treat all family members/contacts.

XXII. SEBORRHEIC DERMATITIS- *Dry scaling skin with underlying erythema involving the head, scalp, or face.*

A. Signs and Symptoms

1. Dry scales or oily yellowish scales commonly on the scalp, face, nasolabial folds, between the eyes, and on the eyebrows and eyelid margins
2. Can also occur on the chest, back, umbilicus and body folds
3. Has varying degrees of pruritus (itchiness)

B. Treatment- based on location

1. Scalp
   a. Treat with shampoos containing zinc (Denorex®) or selenium (Head and Shoulders®, Selsun Blue®) used daily
   b. Two days a week, treat with ketoconazole shampoo (Nizoral®) left in for at least 5 minutes instead of the zinc or selenium
   c. Rotating shampoos is often helpful
2. Face
   a. Hydrocortisone 1% used intermittently (small amount every other day and avoid area around eyes)
   b. If no improvement add ketoconazole 2% cream (Nizoral®) twice daily
3. Eyelids
   a. Gentle cleansing nightly with warm compresses
   b. May also use diluted baby shampoo applied with a cotton swab and rinsed thoroughly
4. Intertriginous areas (skin folds)
   a. Hydrocortisone 1% applied twice daily for 5-7 days
   b. Once improved, can drop down to once weekly for maintenance

XXIII. SKIN ABSCESSES AND FURUNCLES- *"Abscess" describes any collection of pus within the dermis and deeper skin tissues. "Furuncles" are related to hair follicles and are a deep-*

seated pustular infection caused by Staphylococcus aureus involving the entire hair follicle and adjacent subcutaneous tissue.

A. Signs and Symptoms

1. Common on hairy skin exposed to irritation, friction, pressure or moisture
2. Pain, erythema, warm skin surrounding rounded or conical abscess
3. Gradually enlarges, becomes fluctuant and then softens
4. May spontaneously open and drain serosanguinous or pustular fluid
5. Regional lymphadenopathy and streaking away from wound may occur
6. Lab: Culture fluid draining from abscess if possible

B. Treatment

1. Antibiotics will not penetrate a closed abscess; incision and drainage is required
2. Incision and drainage procedure:
    a. Obtain written consent
    b. Clean area with topical chlorhexidine (Hibiclens®) if available or povidone-iodine (Betadine®) preparation and allow to dry
    c. With shallow approach, inject wound just underneath top layer of skin with 1% lidocaine (Xylocaine®) with epinephrine (do not use epinephrine on fingers, toes, ears, nose, or near genitalia or mucous membranes). Insert needle through the ulceration if lesion is open. Inject lidocaine at least 5 mm past furthest edge of erythema in all directions
    d. Cleanse area again with povidone iodine (Betadine®) preparation
    e. Drape area with sterile drape or dressing, if available

CLINICAL NOTE: If using an analgesic that includes epinephrine, the numbing effect will be anywhere that is blanched by the epinephrine in the solution. This makes it easy to see visually where the patient will be numb. Unforutnately, pain control will be sub-optimal in infected areas due to neutralizing effect of pus on lidocaine.

    f. Check anesthetized area with something sharp; patient may feel pressure but should not feel sharp or pricking pain
    g. Use sterile blade to create an incision approximately ½ the diameter of the pustular portion of the lesion
    h. Apply firm pressure to margins of lesion and attempt to express pus and debris
    i. Explore inside of lesion with blunt probe and break up any septations (walls)
    j. Pack lesion with wick material, leaving at least 1 inch of packing outside the skin
    k. Draw line with permanent marker around margin of erythema
    l. Cover open wound and wick with sterile dressing
    m. Follow patient daily for wound management/repacking and to redraw line around the erythema to make sure it is not advancing
    n. May discontinue packing when minimal to no drainage occurs over 24 hour interval and cavity is minimized

3. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
    a. Treat patient with antibiotic that kills MRSA
    b. Consider clindamycin (Cleocin®) or trimethoprim-sulfamethoxazole (Bactrim®)

ACTION ALERT- CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION is required if erythema extends past permanent marker line, there are new erythema streaks spreading away from wound, if patient develops fevers, or if any signs of hemodynamic instability/shock.

CLINICAL NOTE: How to complete informed consent for incision and drainage:

1. Type of procedure: incision and drainage of skin abscess
2. Why procedure is necessary: remove infected debris from wound
3. Risks of procedure: pain, bleeding, reaction to anesthesia, infection may not resolve, risk of repeat or additional procedures, poor cosmetic outcome (scarring)
4. Risk of not doing the procedure: spread of infection in skin or into the blood
5. Type of anesthesia to be used (including strength and with or without epinephrine. Example: "2 mL of 1% lidocaine with epinephrine"
6. Patient's statement of his known allergies and that he/she does not have a known allergy to latex, iodine, lidocaine, or the medications you plan to prescribe. Example: "Patient denies food, drug, and contact allergies" or "Patient denies known allergies to latex, lidocaine, and clindamycin; has received procaine (Novocaine®) for dental procedures without adverse reaction"
7. Your signature/date
8. Patient signature/date
9. Witness signature/date

CLINICAL NOTE: Example procedure note for incision and drainage:
"Written consent obtained and placed in patient chart. Area was cleansed with betadine solution. Abscess was injected in all directions with 2 mL of 1% lidocaine with epinephrine. Area of anesthesia extended 5 mm past visible erythema. Area was prepped and draped in sterile fashion. A 3-mm incision was made directly over pustule. Applied pressure yielded 2 mL of thick debris. Wound was explored with blunt probe and septations were broken. 5 inches of iodoform (Invacare®) gauze was packed into wound, leaving a 1-inch wick. Wound was dressed with sterile dressing. Patient tolerated procedure well. No complications. Estimated blood loss: less than 5 mL."

XXIV. SMALLPOX- *A viral illness caused by variola, a poxvirus. The virus is very hardy and can survive for long periods away from a host. Although it was globally eradicated in nature, it remains a concern since it has the potential to be weaponized.*

A. Signs and Symptoms

1. Incubation period of 10-14 days; followed by sudden onset high fever, severe headache, backache, malaise, and vomiting
2. After 2-4 days, symptoms progress to tiny red spots on the tongue and palate

3. Approximately 24 hours after mucosal rash appears, the oral lesions evolve into papules and vesicles and the skin develops a maculopapular red rash
4. As the rash evolves, areas develop papules, then vesicles, then pustules; the pustules eventually rupture and crust
5. Smallpox lesions typically all evolve at the same rate; this can be differentiated from Chickenpox since in Chickenpox there are usually separate "crops" of lesions and older crusted lesions will be present while new vesicles and pustules arise
6. Lesions can also occur on the palms and soles

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing,circulation, O_2, monitor
   b. Immediate isolation including respiratory and contact precautions
   c. Limit number of persons exposed to the patient
   d. Establish IV access and provide maintenance fluids if patient is unable to tolerate PO
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. There are no FDA approved treatments for smallpox
4. Provide supportive care as indicated

XXV. SUBUNGUAL HEMATOMA- *Blood under fingernail/toenail from trauma to distal phalanx caused by bleeding from nail bed.*

A. Signs and Symptoms

1. Throbbing pain
2. Blue-black discoloration under the nail as the hematoma progresses
3. Pain is from pressure in confined space and pressing against nerve fibers

B. Treatment

1. Pain is relieved immediately for most patients by making a hole in the nail
2. <u>Procedure</u>
   a. Obtain written consent
   b. Restrain digit as needed
   c. Clean the nail with povidone iodine (Betadine®) swabs using sterile technique
   d. Isopropyl alcohol is highly flammable and should be avoided if electrocautery is used
   e. Puncture the nail with a hot metal wire (an electrocautery device, for example)
      i. The heated device is pressed against the nail in the center of the hematoma while avoiding the lunula and its associated nail matrix
      ii. Care should be taken to prevent contact with the nail bed
   f. The hole should be large enough (three to four millimeters) for continued drainage that may occur for 24 to 36 hours after the injury
3. <u>Alternative procedure</u>
   a. Twirl a needle on the nail until blood is expressed through the hole in the nail

b. If a small bore needle (23 gauge) is used, then multiple holes should be placed to reduce the chance of plugging and hematoma re-accumulation
c. If a large bore needle (18 gauge) is used, then a single three to four millimeter hole should suffice

CLINICAL NOTE: Do not put a hole in the nail if you suspect a distal phalanx fracture. If fracture is present, then by putting a hole in the nail you have created an open fracture. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if you suspect any fractures in the hand.

CLINICAL NOTE: Example procedure note for drainage of a subungual hematoma:
"Nail and folds were cleansed with povidone iodine (Betadine®) solution and allowed to dry. Area was draped with sterile dressing. Electrocautery instrument was applied to nail directly over the hematoma and held in place for 4 seconds, creating a 3 mm hole in the nail. Nail spontaneously drained 3 mL of dark blood. Patient felt immediate relief of pain and pressure. Nail was dressed with sterile dressing and finger was wrapped in gauze to prevent additional trauma. Patient tolerated procedure well without complications. Estimated blood loss: less than 5 mL."

5. Patient's statement of his known allergies and that he/she does not have a known allergy to latex or iodine
6. Your signature
7. Patient signature
8. Witness signature

XXVI. TINEA CAPITIS/CORPORIS/PUBIS/PEDIS- *Fungal infection of head, body, groin, or feet.*

A. Signs and Symptoms

1. Tinea Capitis
   a. Three distinctly different forms, "gray patch," "black dot" and "favus"
   b. Most common is black dot, which is an erythematous, scaling patch on the scalp. It slowly enlarges, usually unnoticed until alopecia (hair loss) develops
   c. Hair shafts break even with skin giving black dot appearance
   d. Without treatment tinea capitis may result in permanent alopecia
2. Tinea Corporis ("ring worm")
   a. Ring shaped plaque with advancing scaly border and central clearing
   b. Lesions are usually on exposed areas of body (face or arms)
3. Tinea Cruris ("jock itch")
   a. Erythematous patch in intertriginous areas
   b. Scaly, spreading, and sharply demarcated border with centrally clearing
   c. Marked pruritus on inner thighs and along groin lines
   d. Lesions usually confined to groin and gluteal cleft
4. Tinea Pedis
   a. Itching, burning, stinging of interdigital webs between toes
   b. Scaling and cracking of soles (or palms—in tinea manuum)

c. May have skin breakdown between digits

5. Lab: May be confirmed by KOH prep (hyphae) or culture; KOH may not be positive if skin is macerated in web spaces; spores identified on hair shafts (capitis)

CLINICAL NOTE: Some forms of tinea have luminescent glow when examined with Wood's Lamp. Iridescence confirms the diagnosis, but absence does not rule out fungal infection.

B. Treatment

1. Tinea Capitis
   a. CONTACT PHYSICIAN PRECEPTOR- MEDICATION
      i. Consider oral griseofulvin (Gris-Peg®) treat for 6-12 weeks
      ii. Avoid topical agents due to high incidence of treatment failure
   b. Lesions should be kept clean and dry
2. Tinea Corporis
   a. Miconazole (Monistat®) 2% cream twice daily until cleared plus 1 week, OR
   b. Clotrimazole (Lotrimin®) 1% cream twice a day until cleared plus 1 week OR
   c. Terbinafine (Lamisil®) 1% cream once daily, continue treatment for 1-2 weeks after clearing
   d. If not clearing in 1-2 weeks contact preceptor for further care
3. Tinea Cruris
   a. Counseling includes: use drying powder; wear loose fitting underwear
   b. Miconazole (Monistat®), clotrimazole (Lotrimin®) or terbinafine (Lamisil®) cream as above
   c. High incidence of relapse if also infected with tinea pedis
4. Tinea Pedis
   a. Treat with topical terbinafine powder for two weeks
   b. Change socks multiple times during the day, keep shoes dry and rotate shoes when possible, wear shower shoes in communal showers, use absorbent non-synthetic socks, make sure feet and web spaces are completely dry before putting shoes and socks on, use over-the-counter anti-fungal powders to prevent recurrences
   c. If macerated: use Aluminum acetate (Domeboro®) solution soaks 20 min twice daily and clotrimazole (Lotrimin®) or miconazole

ACTION ALERT: Tinea capitis requires oral antifungal agent. Delay in treatment can cause permanent scarring.

(Monistat®) cream as above

If dry and scaly: For especially thick, scaly soles use Urea 10-20% cream daily to soften skin and use clotrimazole (Lotrimin®) or miconazole (Monistat®) cream as above

XXVII. **TINEA VERSICOLOR**- *Yeast infection usually of the upper trunk that causes hypo- or hyper-pigmentation of the skin.*

A. Signs and Symptoms

1. Usually asymptomatic
2. May have pruritus
3. Velvety tan, pink or white macules that vary in size
4. Area does not tan or appears darker than surrounding skin if affected areas are not exposed to sun
5. No visible scale but scale occurs by lightly scraping lesion with flat blade or fingernail
6. Macules may appear on trunk, upper arms, neck or groin
7. Lab: Large hyphae and thick walled budding spores "spaghetti and meatball" appearance on slide treated with KOH

B. Treatment

1. Apply selenium sulfide (Head and Shoulders®, Selsun Blue®) from neck to waist, leave on for 5-15 min on one day then no treatment for 6 days
2. Repeat selenium sulfide treatment weekly x 1 month, then once monthly for maintenance
3. To prevent recurrence: ketoconazole (Nizoral®) shampoo 1% or 2% lathered on chest or back, left on for 5 min applied weekly

CLINICAL NOTE: Inform patient that the medications treat the raised and scaly portion of the rash and the pigmentation change will take months to fill or fade. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if no improvement with topical medications after 1 month.

ACTION ALERT: If there are corneal ulcerations or vesicles on the nose, immediately CONTACT PHYSICIAN PRECEPTOR- CONSULTATION.

XXVIII. **URTICARIA**- *Rapidly expanding pruritic lesions that erupt within a few minutes of exposure to a trigger and generally resolve if the trigger is removed. May recur anytime the individual is exposed to the trigger again.*

A. Signs and Symptoms

1. Pruritic red swellings that vary in size, enlarge, coalesce, and disappear over the course of several hours
2. May cause angioedema (involvement of deeper subcutaneous tissue—swelling of lips, eyelids, palms, soles, genitalia)
3. If severe reaction, consider prophylactic intubation; refer to "Emergency Medicine Protocols, Anaphylaxis"
4. Common triggers include taking a new medication, eating a new food, insect stings, latex exposure, temperature changes, and some systemic infections

CLINICAL NOTE: Search for a trigger (medications, food, hot showers, exercise, cold exposure, sun, freshly mown grass, etc). Avoid systemic steroids unless instructed to do so by preceptor. Steroids may temporarily improve the condition but it is likely to return with cessation of steroids.

B. Treatment

1. Antihistamines- diphenhydramine (Benadryl®) or hydroxyzine (Atarax®) (sedating) with better demonstrated response, but if patient must avoid sedation consider cetirizine (Zyrtec®) or loratadine (Claritin®)
2. **CONTACT PHYSICIAN PRECEPTOR** if no response initially, may require up to 4 times normal doses of second generation antihistamines to suppress urticaria

XXIX. WARTS- *Verrucous (cauliflower-like) papules on the skin caused by papillomaviruses.*

A. Signs and Symptoms

1. Plantar warts can be tender with pressure and may resemble corns or calluses
2. Some warts itch, especially anal warts
3. Appearance varies from large cauliflower-like papules with multiple finger-like projections to small flat papules. May see black spots within lesion when illuminated with magnification (can use otoscope)
4. All warts have rough bumpy texture

B. Treatment- Multiple options (Choose only one)

1. Liquid nitrogen may be used on any warts, including genital and anal. Freeze for 5 seconds and thaw for 20-45 seconds for 2 cycles every 2-4 weeks. Do not freeze more than 1 mm of surrounding skin. Risks: painful, may scar, cause hypopigmentation in dark skin
2. Salicylic acid may be used for common or plantar warts. Pare down wart and apply 40% salicylic acid (Mediplast®); keep area occluded for 5-6 days, then remove. Repeat until lesion disappears. Safe and effective with almost no side effects. Avoid occluding normal skin as much as possible.
3. If anal or genital warts, if wart involves a nail fold, or if wart involves a mucous membrane, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
4. If no improvement with common or plantar warts after 3 months, refer for possible treatment with podophyllin (Podofilox®), laser, or surgery.

CLINICAL NOTE: Warts can be frustrating to treat because they do not resolve based on duration of treatment. Lesions sometimes resolve on their own spontaneously. Often the warts seem completely resistant but then resolve spontaneously with the same treatments.

C. Prevention

1. Virus is spread into areas of broken skin through contaminated surfaces (such as shower floors)
2. Advise all in unit to wear shower shoes and dry feet thoroughly prior to applying socks and footwear

Chapter 12: Musculoskeletal Protocols

Introduction to Musculoskeletal Injuries

I. **LYME DISEASE**- See "Febrile Illness Protocols, Lyme Disease"

II. **RHABDOMYOLYSIS**- See "Genitourinary Protocols, Rhabdomyolysis"

III. **GOUT-** *Sedimentation of urate crystals into a joint space causing severe pain. Most frequently affects base of 1st toe (1st metatarsophalangeal joint), or the knee. All individuals with gout have hyperurecemia, which causes the sediment to form.*

A. Signs and Symptoms

1. Severe pain, redness, swelling, active and passive immobility at affected joint; pain peaks in 12-24 hours and resolves within days to weeks
2. Tophaceous deposits (horn-like crusted growths coming out of the skin)
3. History of kidney stones made of uric acid
4. Risk factors include chronic beer or hard liquor consumption, starvation, trauma, fatty foods or overindulgent food behaviors
5. High dietary intake of red meat and seafood

B. Treatment

1. **IMMEDIATE ACTION**
   a. Do not give aspirin, it will worsen the condition
   b. Naproxen (Aleve®) or indomethacin (Indocin®) are good first line agents
   c. Joint will be exquisitely tender- avoid manipulation or any pressure on affected joint, even sheets
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

IV. **SUSPECTED FRACTURES-** *Acute trauma or repetitive stress that leads to a disruption in bone. Stress fractures and multi-system trauma are addressed separately. This section will discuss general management of uncomplicated fractures.*

A. Signs and Symptoms

1. Suspicion is based on mechanism of injury plus at least one of the following: bony deformity, excessive edema, limited range of motion at suspected fracture site, crepitus on palpation
2. Exam should include assessment of neurovascular status to include distal pulses, distal capillary refill, and checking sensation, motion, and reflexes distal to fracture site (as applicable)
3. Exam should always compare injury site to non-affected side
4. Exam should include joints above and below injury site (at a minimum)
5. Reassess neurovascular status after every manipulation of the injury site

B. Treatment

1. Based on location of injury
2. Immobilize injury in position found/position of comfort
3. Reassess neurovascular status after immobilization
4. Acetaminophen (Tylenol®) for pain management. Recent evidence suggests NSAID use may delay fracture healing

5. CONTACT PHYSICIAN PRECEPTOR- EVACUATION

ACTION ALERT: New or worsening pain, diminished circulation, cool or dusky digits, bleeding through splint, and/or feelings of pressure or numbness at the fracture site after fracture is immobilized are signs of constriction- either from too tight of a splint or from compartment syndrome. If patient develops any of these symptoms, IMMEDIATELY remove splint and reevaluate. If symptoms resolve, replace splint more loosely. If they do not resolve, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**. Consider emergency fasciotomy (long cuts through the skin and fascia to release pressure) if evacuation is delayed.

ACTION ALERT: ANY fracture that is accompanied by a break in the skin, no matter how slight, is considered an open fracture. Open fractures are very serious due to the potential for life- and limb-threatening infections. If the skin is broken in any way (abrasion, laceration, etc) **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION.**

V. **DISLOCATIONS-** *Traumatic disruption of a joint most commonly in the fingers, shoulders, or patellae but may be seen in other joints during high velocity trauma.*

A. Signs and Symptoms

1. Mechanism of injury describes non-anatomic movement of a joint (finger bent backwards, etc)
2. Suspicion based on mechanism of injury plus at least one of the following: joint deformity, excessive edema, and limited range of motion at suspected dislocation site
3. Exam should include assessment of neurovascular status to include distal pulses, distal capillary refill, and checking sensation, motion, and reflexes distal to dislocation site (as applicable)
4. Exam should compare injury site to non-affected side
5. Exam should include joints above and below injury site (at a minimum)
6. Reassess neurovascular status after every manipulation of the injury site

B. Treatment

1. Based on location of injury
2. Immobilize injury in position found/position of comfort
3. Reassess neurovascular status after immobilization
4. Acetaminophen (Tylenol®) and/or NSAIDs for pain management
5. CONTACT PHYSICIAN PRECEPTOR- EVACUATION

VI. **LIGAMENT SPRAINS AND TEARS-** *Ligaments connect bone to bone. Sprains refer to unnatural "stretching" of a ligament, whereas a tear refers to partial or complete rupture. Dislocations can accompany complete ligament tears since bones are no longer being maintained in anatomic alignment.*

A. Signs and Symptoms

1. Edema and pain along ligament
2. No "endpoint" when ligament is stressed. Bony elements will continue to move in direction of stress. Examples of this include positive anterior

drawer's test when the Anterior Cruciate Ligament has been damaged; fingers can bend laterally at PIP or DIP joints if the lateral ligaments of the finger have been disrupted

B. Treatment

1. Based on location of injury
2. Rest, ice, compress, elevate (RICE therapy)
3. Injury may or may not require immobilization
4. Acetaminophen (Tylenol®) and/or NSAIDs for pain management

VII. **TENDONITIS/MUSCLE STRAIN/TENDON RUPTURE-** *Tendons connect muscle to bone. If a muscle is poorly flexible and experiences a significant stress, the muscle fibers are unable to react quickly enough to absorb the force by stretching. This can result in inflammation where the tendon attaches to bone (tendonitis) or microtears in the muscle (muscle strain). The worst final outcome of this process is tendon rupture, when the tendon separates from the bone to which it is attached.*

A. Signs and Symptoms

1. Patients have pain with movement of the involved muscle (active worse than passive). They may have swelling or pain where the tendon inserts (tendonitis) or in the belly of the muscle (muscle strain)
2. Those with more serious muscle tear or tendon rupture will have a palpable "knot" where the muscle has torn free and bunched up in or near the belly of the muscle
3. Some strained muscles, particularly of the neck and back, will be associated with muscle spasms

B. Treatment

1. NSAIDs or acetaminophen (Tylenol®) for 5-10 days
2. Reduce the activities that cause pain
3. Increase gentle stretching routine tailored for specific muscle group

ACTION ALERT: Tendonitis, muscle strain, and muscle tear are all overuse injuries due to poor flexibility. Prevention is key! As an IDMT, advise fitness leaders and Commanders regarding workout strategies to avoid injury. All group-led workouts should meet the following criteria:

1. Begin with 5 minutes of light calisthenics (jog in place, jumping jacks, etc)
2. Complete at least 5 minutes of full body warmup stretches, holding each stretch for 10-15 seconds without bouncing
3. Complete planned exercise regimen
4. Cool down with slower activity to reduce heart rate followed by ten minutes of whole body deep stretching, holding each stretch for 30 seconds or two rounds of 15 seconds
5. Additionally, adding a "stretch day" into exercise regimens is recommended to increase flexibility. Yoga is effective at increasing both core strength and flexibility

VIII. **BURSITIS-** *Bursae are fluid-filled cushioning sacs located around bony prominences that help protect ligaments and tendons as they slide over bone. Bursitis occurs when the sac*

becomes inflamed. It is frequently the result of repetitive motion with poor body mechanics or repetitive pressure, such as kneeling.

A. Signs and Symptoms

1. Pain with motion and at rest
2. Pain with active movement of the adjacent structures
3. Tenderness when the bursa is palpated
4. Localized edema confined to the bursa may be present
5. If there is erythema or heat associated with the swelling, assume that it is an infected bursa or joint and **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

B. Treatment

1. Rest, ice after activity, and heat prior to stretching or activity are mainstays of treatment
2. NSAIDs for up to 14 days

CLINICAL NOTE: Bursitis will typically resolve with rest in 5-10 days, but it will return if the damaging activity is resumed. It is vital to find and correct the underlying issue to avoid further debilitation.

Neck, Back, and Torso Disorders

IX. **ACUTE NECK PAIN**-*Muscle strain and/or spasms due to position, load-bearing, or whiplash injury. For traumatic neck injury, refer to "Emergency Protocols, Neck Trauma"*

A. Signs and Symptoms

1. Neck pain and stiffness, often limiting the neck range of motion
2. History of holding head in same position for long periods of time, whiplash, or waking up after "sleeping wrong"

B. Treatment

1. Posture modification (don't sit/sleep like that anymore)
2. ROM exercises for 5 minutes every hour
3. Avoid sustained computer work or fine motor work
4. Sleep on back with low pillow under neck and head, thighs raised on pillows
5. Keep spine in neutral, inline position
6. NSAIDs for first 72 hours after injury. Add acetaminophen (Tylenol®) for additional pain relief
7. Cyclobenzaprine (Flexeril®) for muscle spasms and difficulty sleeping due to spasms
8. Upon waking, prior to sleeping, and at least one other time during the day, perform the following exercises:
    a. Warm compresses on neck and upper shoulders for 10 minutes
    b. Slow ROM of neck, holding each position for at least 10 seconds
    c. Shoulder rolls and rows without weights
9. AVOID immobilization devices (soft neck collars, etc) as long as acute fracture or ruptured disc is not a possibility
10. If symptoms do not resolve within 2 weeks, CONTACT PHYSICIAN PRECEPTOR- EVACUATION

X. **COSTOCHONDRITIS-** *A musculoskeletal source of acute anterior chest or thoracic pain caused by irritation of the cartilage that attaches ribs to the sternum or from microtears in the muscles between ribs from forceful exhaling (coughing, retching, etc).*

A. Signs and Symptoms

1. Able to reproduce the pain by putting pressure on the region
2. Still must complete thorough history and physical to rule out cardiogenic causes (see "Emergency Protocols, Cardiac")

CLINICAL NOTE: No EKG/cardiac testing is required if the patient meets ALL of the the following diagnostic criteria:
1. Patient/examiner can reproduce the pain through movement, palpation or by deep inspiration
2. Symptoms are NOT associated with cyanosis, edema, shortness of breath, flushing or pallor
3. Vital signs are within normal limits

B. Treatment

1. Limit activities involving load bearing, lifting, weight training, running, jumping or any activity causing pain
2. Allow 4-6 weeks for resolution
3. NSAIDs for pain; consider ketorolac (Toradol®) injection upon presentation
4. Consider cyclobenzaprine (Flexeril®) if pain is associated with muscle spasm
5. Tell patient to avoid splinting (holding arm tight to chest or taking shallow breaths to prevent pain)
6. Encourage deep breathing exercises to avoid atelectasis and ensuing pneumonia

XI. **SUSPECTED RIB FRACTURE-** *Fracture of a rib resulting from blunt trauma to the rib cage. Evaluate patient for emergent conditions as described in "Emergency Protocols, Trauma".*

A. Signs and Symptoms

1. Pain in ribs or thorax following trauma
2. Patient will likely be self-splinting the affected rib with their arm to prevent lung expansion

B. Treatment

1. Evaluate patient at 6, 12 and 24 hours after injury to make sure no cardiopulmonary structures have been compromised
2. Limit activities involving load bearing, lifting, running, jumping or anything that causes pain
3. NSAIDS for pain; consider ketorolac (Toradol®) injection upon presentation
4. If muscle spasms are present, consider cyclobenzaprine (Flexeril®)
5. Avoid taping or splinting
6. Encourage patient to complete deep breathing exercises (use incentive spirometry, if available) to avoid atelectasis and ensuing pneumonia

XII. **LOW BACK PAIN ("LUMBAGO")**- *Pain in the lumbar region of the back, typically following a physical stressor (such as lifting) or prolonged immobility.*

A. Signs and Symptoms

1. Recent heavy lifting or increase in strenuous activity
2. Back pain, pressure or spasm, unilaterally or bilaterally in the lumbar region
3. May have shooting or stabbing pains radiating below the knee (sciatica)
4. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear

ACTION ALERT: Low back pain with bowel or bladder incontinence, numbness of the inner thighs, numbness of the genitalia, and new onset of sexual dysfunction are all predictors of paralysis. If any of these symptoms are present, place the patient in the position of comfort, **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION.** See "Emergency Protocols, Trauma" for more information.

ACTION ALERT: If any of the following are newly present associated with back pain, further evaluation is required. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION:

1. Focal neurological deficits such as weakness or numbness that can't be improved by changes of position
2. Weight loss
3. Fevers
4. High velocity trauma

B. Treatment

1. Limit activities that include running, jumping and load bearing
2. Avoid bed rest and immobility
3. Most patients are pain free within 6 weeks regardless of treatment
4. Encourage activity as tolerated, especially walking and gentle stretching
5. NSAIDs and/or acetaminophen (Tylenol®) for acute pain; consider ketorolac (Toradol®) injection at initial presentation for pain control
6. Avoid narcotics for pain control
7. Consider cyclobenzaprine (Flexeril®) for muscle spasms

C. Prevention

1. Emphasize appropriate workout structure with calisthenics, warm up stretches, exercise, and cool down stretches (See "Action Alert, Tendonitis")
2. Add a "stretch day" into weekly exercise routine with 20-60 minutes of focused stretching
3. Core strengthening exercises- incorporate TRX® suspension training, Crossfit®, kettlebell exercises, stability ball exercises, and/or Pilates® to weekly exercise program (see Action Alert below)

ACTION ALERT: Core strengthening is a critical and frequently missed component of a healthy lifestyle. Advise workout leaders and Commanders that weekly exercise routine should include frequent core training elements. Some options include:

1. TRX® suspension training exercises (or other suspension system)
2. Stability ball exercises
3. "Planks" and any of their variations
4. Pilates®
5. Balance boards/wobble boards
6. Bosu® balance board
7. Pull-ups/chin-ups
8. Ab wheel (make sure it is being used correctly- improper use will cause back strain)
9. Rowing machine

Some common home workout regimens that emphasize good core strengthening principles:

10. Crossfit®
11. Insanity®
12. P90X®
13. The Firm®
14. Zumba®

Shoulder Disorders

XIII. **ROTATOR CUFF STRAIN/TEAR**- *Throwing injury or trauma resulting in strain or tear of one or more of the rotator cuff muscles.*

A. Signs and Symptoms

1. History of injury followed by pain and weakness
2. Pain over the lateral deltoid and exacerbated by overhead movements and while sleeping
3. Weakness in any direction of shoulder movement, but most commonly with external rotation
4. Unable to complete adduction (patient's arms are overhead and adduction is attempted; when the adducting arm reaches 90°, it abruptly falls to the patient's side = "drop arm" test)
5. Suspect coexisting labral tear if patient also complains of catching sensation, instability, and crepitus

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION for any labral symptoms (catching, instability, and/or crepitus) or any shoulder weakness worse than 4/5
2. Suspect rotator cuff strain if patient is able to perform all motions and has shoulder strength of 4/5 or greater in all directions
3. Treat with stretching and strengthening exercises, NSAIDs and activity restriction

ACTION ALERT: IMMEDIATE EVACUATION is indicated if there are any changes in sensation, any decreased elbow, wrist, or hand strength, or any decreased circulation (diminished pulses, delayed capillary refill, skin pallor, etc).

XIV. **BRACHIAL PLEXUS INJURY-** *Known as "burners" or "stingers", injury caused by trauma to the neck or shoulder with pain radiating down affected upper extremity.*

A. Signs and Symptoms

1. Frequently caused during participation in contact or high velocity sports such as football, wrestling, or gymnastics or by shoulder-carried load bearing equipment such as ruck sack, shoulder-slung weapons, dive gear, etc.
2. History of trauma to the neck or shoulder region
3. Radiating pain/paresthesias down arm circumferentially (all the way around the arm and not in a dermatomal pattern)
4. Injury can be followed by numbness, tingling or weakness
5. May have scapular winging and shoulder muscle atrophy if episodes have been recurrent
6. Focal neural deficits depending on which nerves are affected

B. Treatment

1. Evaluate for cervical/head trauma (See "Emergency Protocols, Trauma")
2. Limit activities that include lifting/contact until symptoms are resolved
3. Reevaluate patient daily until all symptoms are resolved
4. NSAIDS for pain
5. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION

CLINICAL NOTE: Counsel patient regarding importance of safety equipment during high-risk activities. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION** is indicated if there is cervical pathology, limited ROM of the c-spine is present, or for any neurovascular compromise that does not resolve within the first few minutes after injury.

XV. **CLAVICULAR FRACTURE-** *Fracture of the clavicle (collarbone) is usually caused by falling onto the lateral shoulder, but may be caused indirectly from an anterior blow to the shoulder, seatbelt injury, or fall on an outstretched hand. See "Introduction to Musculoskeletal Injuries, Suspected Fracture" at the beginning of this chapter for additional information.*

A. Signs and Symptoms

1. Pain in clavicle with arm movement
2. Anterior shoulder/clavicle swelling
3. History of clavicular snapping or cracking sensation at time of injury
4. Deformity, point tenderness, edema and ecchymosis at the fracture site
5. May have associated hemothorax, pneumothorax or rib fracture
6. May be accompanied by brachial plexus injury or glenoid fracture

B. Treatment

1. Treat any additional complications from the trauma (See "Emergency Medicine Protocols, Trauma" for more information)
2. Immobilize extremity using a sling and swath
3. Reassess neurovascular function and vital signs regularly

4. CONTACT PHYSICIAN PRECEPTOR-MEDICATION EVACUATION
    a. Acetaminophen (Tylenol®) initially for pain
    b. Avoid NSAIDs unless directed by your preceptor- recent studies show NSAIDs may impair fracture healing
    c. Consider narcotics for severe pain

ACTION ALERT: IMMEDIATE EVACUATION is indicated if glenohumoral fracture is suspected, if there is any neurovascular compromise, severe deformity is present, lung involvement is suspected, or if any break in the skin has occurred.

XVI. ACROMIOCLAVICULAR (AC) SEPARATION– "*Separated Shoulder" due to direct trauma or fall on shoulder. Often unable to differentiate from clavicular fracture without x-rays.*

A. Signs and Symptoms

1. History of prior shoulder injury increases risk
2. Numbness, tingling or weakness in the affected limb
3. Tenderness and swelling on the AC joint with or without deformity
4. Inability to cross arm across body to touch opposite shoulder or complete shoulder exam ranges of motion due to pain

B. Treatment

1. Immobilize extremity in position of comfort using a sling and swathe
2. Rest and ice therapy as tolerated
3. Reassess neurovascular function and vital signs after each movement of arm/shoulder and every 30 minutes while at rest
4. CONTACT PHYSICIAN PRECEPTOR- EVACUATION

ACTION ALERT: IMMEDIATE EVACUATION is indicated if there are any changes in sensation, any decreased elbow, wrist, or hand strength, any decreased circulation (diminished pulses, delayed capillary refill, skin pallor, etc), severe deformity is present, lung involvement is suspected, or any break in the skin has occurred.

XVII. GLENOHUMERAL DISLOCATION- *Displacement of the humeral head anterior to the glenoid fossa. Posterior dislocations are rare. Dislocation is frequently caused by blunt trauma when the shoulder is abducted, externally rotated and arm is extended. Inferior dislocations can occur when patient has fallen from a height and tries to grab something to break the fall.*

A. Signs and Symptoms

1. History of trauma
2. Pain, numbness, and tingling distal to injury
3. Deformity and ecchymosis
4. Anterior dislocations cause the arm to be held in a position of slight abduction and external rotation at rest
5. Do not force arm against patient's side
6. Patient will avoid all active and passive movements

B. Treatment

1. Immobilize extremity in position of comfort
2. CONTACT PHYSICIAN PRECEPTOR- EVACUATION

a. Consider IV narcotic meditations such as morphine or meperidine (Demerol®)
b. If evacuation will be delayed, consider IV benzodiazepine such as diazepam (Valium®)
c. If both medications are administered, be prepared to support ventilations due to the combined respiratory depression effect of narcotics and benzodiazepines

3. Reduction should not be attempted by the IDMT prior to x-rays unless evacuation will be prolonged AND neurovascular compromise is present
   a. Reduction attempt must have specific consent of preceptor and patient
   b. NEVER attempt to reduce a dislocated joint if there is also a suspicion for fracture

ACTION ALERT: IMMEDIATE EVACUATION is indicated if glenohumoral fracture in suspected, any neurovascular compromise is present, or any signs of shock.

XVIII. ADHESIVE CAPSULITIS- *"Frozen Shoulder" is the reversible contraction of the shoulder joint capsule after immobilization or disuse.*

A. Signs and Symptoms

1. Loss of ROM following days to weeks of disuse
2. May have history of minor shoulder injury days or weeks prior, then stopped moving arm or kept arm continuously in a sling
3. Stiffness/immobility more than pain
4. >50% reduction of BOTH active and passive ROM in all directions, but usually worse with abduction and external rotation

B. Treatment

1. Active and passive movement of the joint is the mainstay of therapy
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION

CLINICAL NOTE: Frozen shoulder can be differentiated from other rotator cuff problems because frozen shoulder will have limited active AND passive ranges of motion. Rotator cuff strains and tears will have limited active range of motion but markedly improved passive range of motion.

XIX. BICEPS TENDONITIS AND BICEPS TENDON RUPTURE- *Injuries to the biceps muscle or tendon is caused by sudden extension at the elbow when the biceps is fully contracted and loaded (doing a bicep curl and allowing the weight to snap the arm back straight). Likelihood of injury increases with age.*

A. Signs and Symptoms

1. Anterior shoulder pain with radiation over the biceps muscle
2. Biceps Tendon Rupture--History of trauma and feeling a "pop", followed by sharp pain, ecchymosis, and swelling ("Popeye" deformity)
3. Tendonitis- Absence of trauma, pain increases with time; history of repetitive motions

4. May have history of other shoulder problems prior to development of symptoms
5. Visible/palpable deformity of the biceps muscle
6. Palpation of the biceps groove of the humerus and resisted supination will cause pain

B. Treatment

1. NSAIDS for up to ten days and rest
2. If biceps tendon rupture is suspected, CONTACT PHYSICIAN PRECEPTOR-CONSULTATION AND EVACUATION
3. Ice and compression may reduce hematoma formation and deformity

Elbow Disorders

XX. **EPICONDYLITIS (MEDIAL OR LATERAL)** - *Repetitive motion injury caused by repetitive hand gripping ("tennis elbow"), handshake motions, or repetitive use of screwdrivers. Frequently develops in carpenters, construction workers, and dentists as well as racquet sport enthusiasts and golfers.*

A. Signs and Symptoms

1. History of repetitive motion tasks or the above risk factors
2. Pain and tenderness with palpation of the affected epicondyle and associated muscles of the forearm
3. Pain with resisted wrist flexion (medial epicondylitis)
4. Pain with resisted wrist extension (lateral epicondylitis)
5. Decreased grip strength

B. Treatment

1. Limit activities that cause pain
2. NSAIDs- use topical diclofenac (Solaraze®), or if not available, use oral diclofenac (Voltaren®). If neither is available, may use other NSAIDs
3. Prescribe exercises that increase forearm strength
4. If no relief after six weeks with the above therapies, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION for specific exercises and additional therapies such as counter bracing

ACTION ALERT: IMMEDIATE EVACUATION is indicated if there are any changes in sensation, any decreased elbow, wrist, or hand strength, or any decreased circulation (diminished pulses, delayed capillary refill, skin pallor, etc) that does not resolve with change in position or decreased physician.

XXI. **ELBOW DISLOCATION AND ELBOW FRACTURE**- *Caused by direct trauma to the elbow or fall on outstretched limb. Over half of adults with elbow dislocations have associated fractures.*

A. Signs and Symptoms

1. History of trauma to extremity or sudden "pop" while throwing
2. Inability to complete full range of motion
3. May have gross deformity, discoloration

4. Elbow dislocation and fracture are unlikely if full active range of motion is possible (able to fully flex and extend even if painful)
5. Some active joint movement is usually still possible

B. Treatment

1. Splint in position of comfort
2. Reassess neurovascular status after splinting
3. Avoid NSAIDs unless directed by your preceptor- recent studies show NSAIDs may impair fracture healing
4. CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND EVACUATION
5. Reassess neurovascular status after any movement of limb or patient and every 30 minutes

Wrist and Hand Disorders

XXII. CARPAL TUNNEL SYNDROME- *A condition of median nerve entrapment as it passes through the carpal tunnel. The carpal tunnel is located on the ventral side of the angle of the wrist.*

A. Signs and Symptoms

1. Numbness or tingling in the thumb and first fingers

CLINICAL NOTE: If pain is in the 4th and 5th digits, consider diagnosis of "cubital tunnel syndrome", which affects the ulnar nerve.

2. History of repetitive motion work during the day, such as typing or using tools
3. Symptoms worsen at night and relieved by wringing hands, running warm water over hands or shaking the hand out
4. May have pain with repetitive motions or sustained positions
5. Pain may radiate up forearm
6. May complain of reduced grip strength or decreased dexterity

CLINICAL NOTE: If no history of trauma, have patient hold hands above head for 60 seconds. If numbness or tingling occur in all the digits sparing the 5th, this is suggestive of carpal tunnel syndrome.

B. Treatment

1. Identify nerve entrapment in the extremity by checking sharp, dull and light touch sensitivity in all the fingers
2. CONTACT PHYSICIAN PRECEPTOR- EVACUATION if patient displays sensory changes or muscle atrophy
3. Splint wrist every night and during the day if daytime symptoms are present
4. Recovery may take weeks or months

5. If symptoms at work are limiting duty, consider profile for 2-4 weeks avoiding activities causing pain. Do not give profile for "no pushups"- patient may do modified pushups on knuckles or using grip bars

CLINICAL NOTE: There are no documented benefits from using NSAIDs (or other analgesics) in the treatment of carpal tunnel syndrome.

XXIII. SCAPHOID "SNUFFBOX" FRACTURE- *A fracture of the scaphoid bone, the carpal at the base of the thumb. Usually caused by a fall on an outstretched hand.*

A. Signs and Symptoms

1. History of trauma
2. Pain localized to radial side of wrist at base of thumb ("anatomic snuffbox")
3. May present with continuing or worsening pain several weeks after initial injury
4. Pain with radial deviation
5. Minimal swelling and full ROM
6. Tenderness elicited when palpating the scaphoid with the wrist in ulnar deviation and slight flexion
7. X-rays are often negative for 1-2 weeks post injury

B. Treatment

1. Thumb spica splint or cast
2. Consider acetaminophen (Tylenol®) for pain
3. RICE (rest, ice, compression, elevation) for three days after initial injury
4. Limit activities involving pressure on the wrist
5. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION

XXIV. DORSAL GANGLION- *A cystic swelling overlying a joint or tendon sheath most commonly in the wrists or fingers.*

A. Signs and Symptoms

1. Spontaneously occurring, painful mass on dorsal wrist
2. Lesion fluctuant, mobile, demarcated, and transilluminates well

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
2. Consider limiting activities if pain is severe
3. Medications are not generally helpful

XXV. DEQUERVAIN'S TENOSYNOVITIS- *Inflammation of the tendons that extend and/or cross the thumb across the palm most commonly caused by repetitive gripping and grasping.*

A. Signs and Symptoms

1. Thumb movements cause catching
2. No trauma history
3. Swelling at the base of thumb along anatomic snuffbox

CLINICAL NOTE: Perform Finkelstein's maneuver--Have the patient close fingers over the thumb and ulnar-deviate the wrist. This test is diagnostic, causes extreme pain and may cause the patient to lift shoulder to prevent examiner from stretching the tendons.

B. Treatment

1. Restrict grasping and gripping motions
2. Ice/NSAID therapy
3. Thumb spica cast or splint
4. If symptoms persist for >6 weeks, consult physician preceptor and consider evacuation for further evaluation and possible steroid injection

XXVI. BOXER'S FRACTURE- *A fracture of the 4th and/or 5th metacarpal necks, usually of the dominant hand, caused by punching a solid object with a closed fist.*

A. Signs and Symptoms

1. Dorsal hand swelling, bony tenderness over the affected metacarpal bone and loss of contour of the affected knuckle (appears sunken in or depressed)
2. Tenderness and ecchymosis of the palmar bony metacarpal surfaces
3. Inability to straighten fingers or form a fist well
4. May have inability to straighten distal affected finger
5. Thoroughly evaluate for any traumatic breaks in the skin of the hand AT ALL. If the individual has sustained a "fight bite" the fracture is classified as an open fracture even if the lesions are minor

CLINICAL NOTE: Have patient flex at the MCP joints while keeping the DIPs and PIPs in extension so the fingers form a 90° angle with the palm. All fingernails should be in the same horizontal plane.

B. Treatment

1. Thoroughly wash any open lesions and decontaminate wounds as much as possible
2. Splint in position of comfort
3. Reassess neurovascular status after splinting
4. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
5. Consider antibiotics if wound is open and evacuation will be delayed

ACTION ALERT: Patients with open wounds associated with a fracture (no matter how minor) and/or patients with decreased sensation should be evaluated by a surgeon as soon as possible.

Hip/Pelvis/Upper Leg Disorders

XXVII. ILIOTIBIAL BAND SYNDROME- *The iliotibial (IT) band originates at the lateral hip and inserts on the lateral knee at the lateral femoral epicondyle. A tight muscle at its origin causes pain and inflammation along the IT band.*

A. Signs and Symptoms

1. Dull aching or burning pain at the lateral femoral condyle radiating up the lateral thigh toward the hip

2. Risk factors include excessive running mileage, old running shoes, or running on uneven terrain
3. Focal tenderness with or without palpable clicking at the lateral femoral condyle
4. Remainder of knee exam is usually normal and pain free

B. Treatment

1. Limit activities involving running and jumping
2. NSAIDs no more than 2 weeks
3. Stretching is key! Prescribe stretching exercises of the lateral thigh and strengthening of the medial thigh
4. Rehab exercises should be completed three times daily until pain free (may take several months)

XXVIII. TROCHANTERIC BURSITIS- *Inflammation of the lubricating sac on lateral trochanter of the proximal femur (on the point of the hip) caused by abnormalities in gait or repetitive hip flexion (climbing stairs).*

A. Signs and Symptoms

1. Severe, lateral hip pain and pain over the outer thigh
2. May cause difficulty walking and aggravated by direct pressure (such as laying on that side)
3. May have leg length discrepancy, fallen or high foot arches
4. Shoes may have unequal tread wear
5. Hip exam many be normal except for point tenderness over the greater trochanter

B. Treatment

1. Correct any misalignments
2. Limit activities that cause pain
3. Apply heat for 15-20 minutes followed by stretches of the IT band and lateral hip three times daily for 2 weeks
4. Consider ibuprofen (Motrin®) or naproxen (Aleve®, Naprosyn®) for no more than 2 weeks

XXIX. PELVIS FRACTURE- *Disruptions of the pelvic ring, acetabulum or sacrum caused by high-energy blunt trauma to the pelvic girdle, such as vehicular and motorcycle accidents.*

A. Signs and Symptoms

1. History of trauma
2. Abnormal positioning of lower extremity/extremities
3. Pelvic instability (gentle bilateral compression at the iliac crests- do not rock)- if any instability or crepitus is discovered avoid any further manipulation
4. Bowel or bladder incontinence
5. Vaginal or rectal bleeding, hematuria
6. Numbness or weakness distal to injury
7. Bruising of the buttocks, flanks, genitalia, lower abdomen, inner thighs, skin folds
8. Blood at the penile meatus

9. Rectal exam is required (both males and females) to check sphincter tone and palpate for bony fragments of the pelvic ring; use double glove technique to help protect finger from fracture fragments
10. High incidence of associated organ injuries and severe bleeding

B. Treatment

1. **IMMEDIATE ACTION**
   a. Stabilize life threatening injuries (see "Emergency Protocols, Trauma")
   b. Initiate airway management, IV and O_2
   c. Wrap pelvis in a folded sheet, covering the lateral hips and buttocks, tighten until lower extremities are brought back into anatomic alignment, and secure with towel clamps
   d. Sheet wrap should improve comfort- do not over-compress
   e. Do not give NSAIDs or aspirin; avoid ketorolac (Toradol®)
2. **CONTACT PHYSICIAN PRECEPTOR- MEDICATION AND IMMEDIATE EVACUATION**
   a. Consider morphine or meperidine (Demerol®) for pain control
   b. Consider IV cefuroxime (Ceftin®) and tetanus prophylaxis

ACTION ALERT: Ominous signs include rectal/vaginal/penile bleeding, urinary incontinence, and open fractures.

XXX. **FEMUR FRACTURE-** *A fracture of the femur, usually from a high-energy trauma such as a motor vehicle accident. Femur fractures can be life-threatening and very difficult to treat due to the strength of the surrounding musculature.*

A. Signs and Symptoms

1. History of high-energy trauma (if trauma seems minor for extent of injury, suspect tumor or congenital defect)
2. Extreme pain, deformity, swelling and obviously shortened and/or rotated thigh
3. Extensive soft tissue injury, hemorrhage and shock are common

B. Treatment

1. **IMMEDIATE ACTION**
   a. Stabilize life-threatening injuries (see "Emergency protocols, Trauma")
   b. Initiate airway management, IV and O_2
   c. Splint the limb in the position found unless neurovascular compromise has occurred
   d. If sensation is altered or evidence of bleeding, reduce the fracture with a mechanical traction device or splint
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider morphine or meperidine (Demerol®) for pain control
   b. Consider IV cefuroxime (Ceftin®) and tetanus prophylaxis for open fracture
3. If evacuation is delayed, monitor for development of compartment syndrome and consider emergency fasciotomy

Knee/Lower Leg Disorders

XXXI. **CALF STRAIN/RUPTURE-** *A pull or tear in the gastrocnemius muscle.*

A. Signs and Symptoms

1. Sudden onset of posterior calf pain following abrupt push-off (ie beginning a sprint or leaping) without adequate warm up
2. Pain or inability to tiptoe walk on affected side
3. Swelling or palpable "knot" in the gastrocnemius muscle (rupture)
4. Calf circumference may be greater than the opposite side

B. Treatment

1. Limit activities causing pain
2. NSAIDs for no more than 2 weeks
3. Stretching and strengthening exercises tailored for calf strain
4. Exercises should be completed three times daily until pain free

C. Prevention

1. Stretching is key! Injury occurs due to inadequate compliance of muscle
2. See text box "Musculoskeletal Injuries, Tendonitis"

XXXII. **DEEP VENOUS THROMBOSIS (DVT)-** *Blood clot that forms in a large vein of an upper or lower extremity. Blood clot can break off and travel to distant sites such as the lungs, known as "pulmonary embolism."*

A. Signs and Symptoms

1. Risk factors include history of immobilization (such as airplane ride or recent lower extremity injury), bed rest, recent surgery, smoking, obesity, prior episode of DVT, use of oral contraceptives and pregnancy/postpartum status
2. Swelling, pain and erythema of the affected leg
3. Palpable, painful, warm cord in the lower extremity
4. Asymmetrical calf circumference

B. Treatment

1. **IMMEDIATE ACTION**
   a. Confine patient to bedrest
   b. Establish IV access, may saline lock or KVO
   c. Avoid aggressive palpation of limb or manipulation of limb
   d. Monitor patient for signs and symptoms of pulmonary emboli
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXXIII. **PREPATELLAR BURSITIS–** *Inflammation of the bursal sac located between the patella and the skin.*

A. Signs and Symptoms

1. Swelling anterior to the kneecap after kneeling or falling on the knee; swelling is very superficial and located between patella and skin
2. Pain is present without motion (while knee is at rest)
3. Tenderness specifically localized to area of swelling/trauma
4. Knee ROM and strength are not limited unless due to swelling

CLINICAL NOTE: If erythema and warmth are present, consider systemic cause of bursitis such as gout or infection.

B. Treatment

1. If fluid collection is superficial and margins are easy to identify, consider therapeutic aspiration procedure
2. Wrap knee after procedure (or instead of procedure) with compression dressing for 3 days to help reduce swelling
3. May use NSAIDs for up to two weeks
4. RICE (rest, ice, compression, elevation) therapy three times daily for 3 days
5. On fourth day, patient can begin quad-strengthening exercises, discontinue compression dressing, and use knee sleeve if desired for comfort
6. Limit squatting and kneeling exercises for 4 weeks
7. Advise patient to use proper PPE to include knee pads to avoid re-injury

CLINICAL NOTE: How to complete informed consent for therapeutic aspiration of prepatellar bursitis:

1. Type of procedure: therapeutic aspiration of prepatellar bursitis
2. Why procedure is necessary: remove painful collection of fluid from anterior knee
3. Risks of procedure: pain, bleeding, infection, risk of repeat or additional procedures, poor cosmetic outcome (scarring)
4. Risk of not doing the procedure: continuing pain and delayed healing
5. Type of anesthesia to be used: N/A
6. Patient's statement of his known allergies and that he/she does not have a known allergy to latex or iodine or the medications you plan to prescribe. Example: "Patient denies food, drug, and contact allergies" or "Patient denies known allergies to latex, lidocaine (Xylocaine®), and ibuprofen (Motrin®); has taken OTC Motrin® without adverse reaction"
7. Your signature/date
8. Patient signature/date
9. Witness signature/date

CLINICAL NOTE: Example procedure note for therapeutic aspiration of prepatellar bursitis: "Written consent obtained and placed in patient chart. Area was cleansed with betadine solution. Area was prepped and draped in sterile fashion. A 5 mL syringe with 18 gauge needle was inserted into the bursal sac using a shallow approach. The needle was rotated 180° so the bevel was facing the patella bone. Gentle aspiration yielded 4 mL of clear fluid. Knee was wrapped with compression dressing and patient advised to avoid all squatting, jumping, running, and kneeling exercises for 3 days. Patient tolerated procedure well. No complications. Estimated blood loss: less than 1 mL."

If fluid collected is turbid, murky, or thick, CONTACT PHYSICIAN PRECEPTOR-CONSULTATION AND MEDICATION. Consider culture of fluid if possible and antibiotics due to probable infectious cause.

XXXIV. **PATELLOFEMORAL SYNDROME**- *Overuse injury causing anterior knee pain. Occurs due to abrupt increase in strenuous exercise without appropriate conditioning.*

A. Signs and Symptoms

1. History of recent increase in exercise, especially running
2. BMI over 25 or increased load bearing
3. Patellar tenderness, crepitus and popping
4. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (see "Musculoskeletal Protocols, Flat Foot")
5. May have a normal exam

B. Treatment

1. Modify activities for 4 weeks and only limit those activities that cause pain
   a. If patient is limping, limit all running, jumping, marching
   b. If patient always has pain with running that occurs during run, limit running but keep other lower body exercises
   c. If patient has pain with stairs, limit only squats, step-ups, lunges, and running on hills and uneven terrain. Track running should still continue
   d. If patient has pain intermittently with running or pain occurs after running, limit sprints, long distances, and running on hills and uneven terrain. Patient may continue light running workouts
2. Ibuprofen (Motrin®) or naproxen (Aleve®, Naprosyn®) for inflammation not to exceed 2 weeks
3. Quadriceps strengthening exercises starting with isometric tightening of the quadriceps and slowly adding more strenuous exercises such as wall squats and lunges

C. Prevention

1. Progressive, realistic training is key
2. Start new exercise regimen at 70% of current ability level and progress up to 100% challenge after the first two weeks
3. Multiple patients in same unit with overuse injuries such as tendonitis and patellofemoral syndrome warrant evaluation of the unit exercise regimen and intervention by the IDMT

XXXV. **PATELLAR TENDONITIS "JUMPER'S KNEE"**- *Caused by chronic inflammation of the patellar tendon due to running and jumping.*

A. Signs and Symptoms

1. Pain inferior to the patellar pole worse with jumping
2. Aggravated by going up or down stairs and prolonged sitting
3. Tenderness on the inferior pole of the patella or patellar tendon
4. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (See "Musculoskeletal Protocols, Flat foot")
5. May have normal exam

B. Treatment

1. Modify activities to avoid those causing pain (see "Patellofemoral syndrome, above)
2. Ibuprofen (Motrin®) or naproxen (Aleve®, Naprosyn®) for inflammation not to exceed 2 weeks
3. Stretching and strengthening exercises tailored for patellar tendonitis

4. Quadriceps strengthening exercises starting with isometric tightening of the quadriceps and slowly adding more strenuous exercises such as wall squats and lunges

C. Prevention

1. Progressive, realistic training is key
2. Start new exercise regimen at 70% of current ability level and progress up to 100% challenge after the first two weeks
3. Multiple patients in same unit with overuse injuries such as tendonitis and patellofemoral syndrome warrant evaluation of the unit exercise regimen and intervention by the IDMT

XXXVI. SHIN SPLINTS/TIBIAL STRESS FRACTURE- *Overuse injuries of the lower leg due to prolonged walking, running, jumping or marching. "Shin splints" generally refer to muscular derangements and "stress fractures" to bony derangements, but clinical evaluation is usually not able to discern the difference. Both conditions occur as gradual progression of activity-related pain resulting from poor body mechanics. Treatment is initially the same, but if patient fails to respond, stress fracture should be considered.*

A. Signs and Symptoms

1. History of overuse, increase in exercise regimen, or change in regimen
2. May have poor footwear or history of running on irregular terrain
3. May have history of knee, shin, ankle or foot injuries
4. Females experiencing irregular/absent menses are at higher risk for stress fractures
5. Shin Splints—generalized anterior lower leg pain; tenderness while hopping on the affected leg; pain escalates through the run but usually able to continue
6. Stress Fracture—localized, bony surface pain, cannot complete 10 hops on the affected leg. Pain begins within the first steps of running
7. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear

CLINICAL NOTE: Patients experiencing foot numbness, tingling or foot drop/motor deficits should be evaluated for exercise-induced compartment syndrome (see below).

B. Treatment

1. Treat underlying cause (purchase new shoes/boots, add inserts if flat-footed, etc)
2. Ice 10 minutes three times daily (avoid applying ice directly to the skin)
3. Limit activities involving running/jumping
4. Stretching and strengthening rehabilitation exercises to include heel raises, toe raises and squats (as long as activities are painless)
5. Reevaluate after 2 weeks/introduce additional rehabilitation exercises
6. If pain persists after six weeks, CONTACT PHYSICIAN PRECEPTOR-CONSULTATION

CLINICAL NOTE: If tenderness extends into the joint space/tibial plateau or to the medial malleolus, CONTACT PHYSICIAN PRECEPTOR- EVACUATION. Fractures in these areas are high risk for non-union and require further evaluation.

XXXVII. EXERCISE INDUCED COMPARTMENT SYNDROME- *Increased tissue pressure in a closed muscle compartment that compromises local circulation and neuromuscular function. Occurs when individuals with congenitally tight compartments increase blood flow to lower extremities (by running or marching, for example), which increases pressure within the compartment.*

A. Signs and Symptoms

1. Pain begins after a specific distance or time into the exercise
2. Pain does not resolve as soon as activity ceases, but does resolve with rest 10-60 minutes after stopping the activity
3. Pain during exercise described as severe and debilitating
4. May have numbness/tingling distal to pain and may complain of foot drop or foot "slapping" when pain is severe
5. Exam at rest is usually normal, but may have associated swelling/tenderness along the affected compartment
6. Check peripheral pulses and capillary refill
7. Pain/tenderness often out of proportion to exam findings
8. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (See "Musculoskeletal Protocols, Flat foot")

B. Treatment

1. Treat underlying cause (purchase new shoes/boots, add inserts if flat-footed, etc)
2. Limit activities involving impact activities for 4 weeks
3. Condition is chronic and may require permanent profile or surgery

ACTION ALERT: IMMEDIATE EVACUATION is indicated if numbness, tingling or foot drop persists for greater than 1 hour after activity is stopped, if extremity is cold to touch, or if pulses are absent. Emergent fasciotomy may be required.

XXXVIII. MENISCAL TEAR- *Disruption in the horseshoe-shaped cushion (meniscus) between the femur and the tibia.*

A. Signs and Symptoms

1. History of injury involving weight bearing while twisting
2. Clicking, locking, popping and catching are common
3. Weakness or sudden "giving out" of the knee is common
4. Tenderness with palpation along the joint line

B. Treatment

1. RICE (rest, ice, compression, elevation) therapy for three days
2. Crutches for stability if needed (patient should still be able to do some weight-bearing)
3. Pain control for first 5 days: NSAIDs with/without acetaminophen (Tylenol®)
4. For severe pain, CONTACT PHYSICIAN PRECEPTOR- MEDICATION and consider acetaminophen/oxycodone (Percocet®) or acetaminophen/ hydrocodone (Vicodin®)
5. Avoid bracing unless necessary for pain relief or severe instability

6. Rehabilitation exercises should begin as soon as movement is tolerated; may start isometric quadriceps exercises immediately
7. Reassess patient weekly for six weeks, adding additional rehabilitation exercises each visit

ACTION ALERT: IMMEDIATE EVACUATION is indicated if there is complete inability to bear weight, evidence of deep bruising (hemarthrosis) is present, joint is locked or unable to move, or there is suspected ACL/PCL/MCL/LCL involvement. Routine evacuation is indicated if there is no clinical improvement or if swelling is still substantial after 3 weeks.

XXXIX. MCL/PCL/LCL/ACL SPRAINS AND TEARS- *Trauma causes disruption of one or more of the ligaments of the knee that connects the femur to the tibia. Often accompanied by a meniscal tear.*

A. Signs and Symptoms

1. History of trauma, not overuse
    a. MCL- Caused by sudden change in direction and speed or by direct blow to the lateral knee (ie low football tackle)
    b. PCL- Direct trauma to the upper anterior tibia when the knee is flexed (ie sliding into a base); injury often perceived as minor and patient continues activity but later may report swelling, deep joint pain, and sensation of instability
    c. LCL- Less common; caused by blow to the medial knee (ie being kicked in the medial knee by an oncoming soccer player)
    d. ACL- Direct trauma or hyperextension while foot is planted. May be complicated by meniscal injury; athlete typically falls down grabbing the knee because pain is immediate and limiting after injury; may report sensation that the knee shifted or a pop inside the knee

CLINICAL NOTE: For all ligamental injuries, exam will be of limited benefit if begun more than 30 minutes after injury due to muscle spasm, swelling and pain. Observe gait as the patient enters the room. Patient must have relaxed quadriceps and hamstrings to be able to examine thoroughly.

1. Examination
    a. MCL- swelling unlikely unless meniscus or ACL are involved; decreased flexion and extension compared to the opposite side; no decreased passive ROM; pain/laxity with valgus stress test
    b. PCL- Pain or no endpoint with posterior drawer test
    c. LCL- Laxity with varus stress test; gross instability suggests PCL and meniscus disruption
    d. ACL- Swelling from hemarthrosis; sensation of instability, difficulty or inability to walk; no endpoint with anterior drawer test

B. **Treatment**

1. RICE (rest, ice, compression, elevation) therapy for three days
2. Crutches for stability if needed (patient should still be able to do some weight-bearing)

3. Pain control for first 5 days: NSAIDs with/without acetaminophen (Tylenol®)
4. For severe pain, CONTACT PHYSICIAN PRECEPTOR- MEDICATION and consider acetaminophen/oxycodone (Percocet®) or acetaminophen/ hydrocodone (Vicodin®)
5. Avoid bracing unless necessary for pain relief or severe instability
6. Rehabilitation exercises should begin as soon as movement is tolerated; may start isometric quadriceps exercises immediately
7. Reassess patient weekly for six weeks, adding additional rehabilitation exercises each visit

ACTION ALERT: IMMEDIATE EVACUATION should be considered for all ligamental injuries and is required if there is complete inability to bear weight, evidence of deep bruising (hemarthrosis) is present, joint is locked or unable to move, or there is suspected ACL/PCL/MCL/LCL involvement. Routine evacuation is indicated if there is no clinical improvement or if swelling is still substantial after 3 weeks.

XL. **ANSERINE BURSITIS**– *Inflammation of the anserine bursa, located on the medial knee several cm below the joint line. Usually cased by poor body mechanics.*

A. Signs and Symptoms

1. Medial knee pain, especially at night
2. Pain specifically located in the anserine bursa (can point with one finger)
3. If pain is diffuse, consider medial collateral ligament or bony abnormality
4. Tenderness of the anserine bursa with direct palpation
5. Minor palpable, localized swelling
6. Medial collateral (valgus) stress test does not aggravate pain
7. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (See "Musculoskeletal Protocols, Flat foot")

B. Treatment

1. Apply ice to affected area 2-3 times per day
2. Avoid applying ice directly to the skin
3. Sleep with pillow between legs at night to avoid direct pressure on the bursa
4. Modify activities to exclude those involving squats and repetitive knee bending
5. Stretching and strengthening exercises tailored for anserine bursitis

CLINICAL NOTE: Failure to correct underlying gait abnormality will lead to further episodes.

Ankle and Foot Disorders

XLI. **ACHILLES TENDONITIS AND ACHILLES TENDON RUPTURE**- *Damage to the Achilles tendon due to excess strain, often due to running, jumping, or sudden acceleration or deceleration. Most common when tendon becomes damaged due to trauma or inelastic due to poor stretching regimen for the posterior legs.*

A. Signs and Symptoms

1. Pain or stiffness a few centimeters above the posterior calcaneus
2. History of increased training regimen or activity after sedentary period
3. Whereas ankle sprains tend to occur when landing, Achilles tendonitis and rupture tend to occur with push-off
4. Sensation of being struck in the heel or hearing a "pop" followed by severe pain
5. Risk factors include poor flexibility, cold weather training, misalignment, poor running mechanics, increased speed training or hill work, inappropriate footwear, and leg length discrepancy
6. Unable to bear weight
7. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (See "Musculoskeletal Protocols, Flat foot")

CLINICAL NOTE: Thompson's Test- examine the Achilles tendon while patient is lying prone with feet hanging off the table. Squeeze the belly of the gastrocnemius, which should plantar-flex the foot. No movement is diagnostic of an Achilles tendon rupture.

B. Treatment

1. Achilles rupture is based on Thompson's test and inability to plantar-flex; not all Achilles ruptures are painful
2. If Achilles tendon is ruptured, CONTACT PHYSICIAN PRECEPTOR-EVACUATION
3. If no rupture is suspected:
   a. Limit activities causing pain
   b. NSAIDs; consider ibuprofen (Motrin®) or naproxen (Aleve®, Naprosyn®) for 7-10 days only
   c. RICE (rest, ice, compression, elevation) therapy for 3 days
   d. Stretching and strengthening exercises tailored for Achilles tendonitis

XLII. ANKLE SPRAIN/FRACTURE - *Ankle sprains are a disruption of the stabilization ligaments of the ankle. Ankle fractures occur when the distal tibia or fibula are fractured.*

A. Signs and Symptoms

1. Severe sprain or fracture
   a. Pain, swelling, tenderness and ecchymosis
   b. Mild to moderate joint instability (lack of endpoint with passive range of motion or anterior ankle drawer test)
   c. May not be able to bear weight
2. For milder sprains, patients should be able to partially bear weight

CLINICAL NOTE: Ottawa Ankle Rules define which patients require x-rays. Routine medical evacuation is indicated for x-rays are indicated if any of the following conditions are met:

1. Unable to bear weight at the time of incident
2. Unable to bear weight at the time of examination
3. Pain with palpation of the inferior or posterior edge of the medial malleolus
4. Pain with palpation of the inferior or posterior edge of the lateral malleolus

B. Treatment

1. RICE (rest, ice, compression, elevation) therapy for first three days after injury
2. Crutches for stability
3. Continuous NSAIDs for pain control and anti-inflammatory effect for no more than 10 days
4. ROM exercises to reduce risk of "frozen ankle". Exercises include circles, plantar flexion/dorsiflexion and should begin as soon as pain permits and continue until patient returns to full activity
5. Splints and braces allow for improvement in weight bearing and offer good protection against injury; patient should remove brace three to five times daily to complete ROM exercises
6. Tight-laced combat boots work well if commercial splints/braces are not available
7. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION** if neurovascular impairment exists, or if fracture with gross deformity has occurred. Splint extremity in position of comfort for transportation unless otherwise directed by preceptor
8. If ankle has been sprained before, double the rehabilitation time

XLIII. PLANTAR FASCIITIS- *Inflammation where the plantar fascia attaches to the base of the calcaneus.*

A. Signs and Symptoms

1. Severe pain with first steps of the morning with symptoms improving with walking
2. Risk factors include obesity, frequent high-heel wear, prolonged standing or jumping, flat feet or high arches, old footwear, and running on hard surfaces
3. Patients usually have tight hamstrings which exacerbate the pressure on the plantar fascia insertion
4. Check for leg length discrepancy, fallen or high arches and shoes for unequal tread wear

B. Treatment

1. Limit activities causing pain, especially running and jumping
2. Suggest frozen bottle massage to bottom of foot BID
3. Correct problems with gait or footwear
4. NSAIDs for no more than 14 days
5. Exercises include dorsiflexion of the foot and hamstring stretches before getting out of bed in the morning and after sitting for prolonged periods
6. Discourage walking bare footed or flat-soled shoe wear; patients should have cushioned footwear even when in the shower
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if symptoms do not improve after 4 weeks of conservative treatment

XLIV. STRESS FRACTURE OF THE FOOT- *"Marching fracture" is an overuse injury of the metatarsal shaft caused by repetitive stress to the forefoot from running, jumping or marching.*

A. Signs and Symptoms

1. Gradually worsening forefoot pain that initially occurs with a specific activity, but progresses to a poorly-defined forefoot pain induced by minimal activity
2. Point tenderness along the metatarsal shaft
3. Risk factors include tight gastrocnemius, poorly fitting shoes, osteopenia associated with amenorrhea, obesity, or increased load bearing, increase in training volume or change in training surfaces
4. Check for leg length discrepancy, fallen or high arches, and check shoes for unequal tread wear (See "Musculoskeletal Protocols, Flat foot")
5. Females with disordered eating and/or amenorrhea are at increased risk for stress fractures

CLINICAL NOTE: if fracture is suspected, hold each toe in line with its metatarsal and load the toe by pushing it in towards the metatarsal. If the patient has soft tissue injury alone, this maneuver should not produce pain, but if a bony abnormality is present, the maneuver will cause pain.

B. Treatment

1. Replace old or worn out running shoes and boots (life of a running shoe is six months or 500 miles, whichever happens first)
2. Identify and correct the underlying risk factors
3. Limit activities causing pain for 6-8 weeks
4. Crutches if pain occurs with walking
5. Consider acetaminophen (Tylenol®) for pain control

CLINICAL NOTE: Stress fractures are difficult to observe on x-ray and findings may not be visible until 2-6 weeks after the fracture occurs. Physical examination is more sensitive. Untreated stress fractures can progress to complete fractures of the metatarsal.

XLV. PES PLANUS ("FLAT FEET")– *Congenital or traumatic ("fallen arches") loss of the arch of the foot.*

A. Signs and Symptoms

1. Feet display little or no anatomic arch
2. Have patient face you and stand at attention; individuals with fallen arches will appear as if their medial ankles are close to the ground or the lateral foot is rolled out

CLINICAL NOTE: Have barefoot patient face away and rise up on tiptoes. If there is no arching visible in the midfoot, **CONTACT PHYSICIAN PRECEPTOR** for further evaluation. Alternatively, have the patient wet their feet on a damp towel, then step on to a piece of paper. The midfoot width in contact with the paper should be approximately ½ the width of the total midfoot.

B. Treatment

1. If no foot, ankle, knee or back pain, no treatment necessary

2. If patient complains of lower-body musculoskeletal issues, counsel patient regarding stretching of the gastrocnemius and foot
3. Recommend appropriate footwear- "motion control" style running shoes prevent overpronation and associated lower extremity injury
4. Consider arch supports

XLVI. TARSOMETATARSAL FRACTURE ("LISFRANC" FRACTURE)- *A fracture-dislocation of the metatarsals from the tarsals caused by forceful pronation of the foot.*

A. Signs and Symptoms

1. History of traumatic pronation of the foot such as stepping awkwardly in a hole
2. Pain with direct palpation of the tarsometatarsal joints
3. Dorsal and plantar ecchymosis in the midfoot
4. Pain with passive pronation of the transverse tarsal joint

B. Treatment

1. Crutches; no weight bearing
2. RICE therapy
3. Acetaminophen (Tylenol®) for pain
4. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION if Lisfranc fracture is suspected based on mechanism of injury and physical exam
5. Lisfranc fractures have a high rate of poor healing and severe morbidity

XLVII. LEG LENGTH DISCREPANCY– *Congenital condition frequently missed on initial military physicals that can lead to permanent damage of the feet, knees, hips, and back if not corrected. Frequently a contributing factor to many conditions listed above.*

A. Signs and Symptoms

1. Pattern of back and lower extremity overuse injuries
2. Uneven wear on shoes
3. History or exam suggestive of scoliosis
4. Have patient stand at attention facing away from you and observe level of iliac crests for unevenness
5. Have patient lay supine and measure from anterior superior iliac spine along medial knee to tip of medial malleolus; 0.5-1.0 cm difference is normal

B. Treatment

1. For discrepancies up to 2 cm, use heel cup or wedge in shorter leg's shoe
2. Discrepancies greater than 2 cm require podiatrist evaluation upon return to home station

Chapter 13: Mental Health/Substance Abuse Protocols

CLINICAL NOTE: The management of most psychiatric conditions will be determined by mental health professionals, not the IDMT. Administer all drugs; DO NOT give them to the patient for self-administration.

Mood Disorders

I. **DEPRESSION-** *Depression is extremely common with an incidence of up to 30% of military primary care patients having some depressive symptoms. Depression may be the expression of genetic or developmental factors or psychological stresses. Rates of depression are especially high in combatants who have sustained relatively minor trauma such as mild concussion/traumatic brain injury. Military personnel frequently avoid medical attention for depression due to fear of career repercussions.*

A. Signs and Symptoms

1. Mood varies from mild sadness to intense feelings of guilt, worthlessness and hopelessness
2. Difficulty in thinking, inability to concentrate, rumination and lack of decisiveness
3. Loss of interest, diminished involvement in work and recreation
4. Somatic complaints such as headache, sleep problems, fatigue, appetite change and decreased sexual drive
5. Patients appear fatigued, frequent crying, sighing, poor hygiene, missed appointments or tardiness
6. Severe depression is characterized by anxiety, social withdrawal, and suicidal ideations
7. Ask about mania episodes (see "Bipolar Disorder") to rule out manic-depression

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
   a. If you have ANY suspicion patient may cause harm, immediately assign nonmedical attendant to one-to-one direct observation
   b. Do not allow patient out of sight (even in the bathroom) and do not allow them access to anything potentially harmful
   c. Rule out any organic defect (check thyroid, examine for evidence of alcohol or substance abuse)
3. Establish probable diagnosis based on the history and physical examination findings and CONTACT PHYSICIAN PRECEPTOR based on "9-symptom questionnaire" guidance below

4. If the patient is severely agitated then subdue with Haloperidol (Haldol®) 10 mg oral concentrate or 5 mg IM; CONTACT PHYSICIAN PRECEPTOR-MEDICATION prior to use if possible

CLINICAL NOTE: Screen all potentially depressed patients with "9-Symptom Questionnaire". Rate each question on a scale of 0-3, with 0 meaning not at all, 1 means several days, 2 means more than half the days, and 3 means almost every day:

1. Little interest or pleasure in doing things
2. Feeling down, depressed or hopeless
3. Trouble sleeping or sleeping too much
4. Feeling tired or little energy
5. Poor appetite or overeating
6. Feeling bad about self
7. Trouble concentrating on things
8. Moving or speaking slowly or being restless
9. Thoughts that you would be better off dead
10. Scores greater than 15, CONTACT PHYSICIAN PRECEPTOR- EVACUATION for further evaluation
11. Scores between 4 and 15, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION

5. If evacuation is required
   a. Advise patient's Command of the need for further evaluation; do not suggest diagnosis- diagnosis will be determined by mental health provider
   b. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations
6. If evacuation is not required
   a. Have daily/weekly talks with patient; encourage patient to speak freely
   b. Schedule adequate time, don not appear rushed or hurried
   c. Increase quantity and intensity of exercise, be outdoors during hours of light as much as possible, increase water intake, regulate sleep schedule, start a multivitamin
   d. Avoid all substances that could alter mood, including herbs, supplements, caffeine, and especially tobacco and alcohol

ACTION ALERT: If you believe patient may be a threat to themselves or others, patient may require physical and/or chemical restraints (sedation). Frequently reassess patient for risk of suicide or homicide. Even if a patient is not an immediate risk to self or others, depressed individuals can cause inadvertent harm to self or others through inattention and neglect. See "Suicidal Gestures/Attempts".

II. **ADJUSTMENT DISORDER WITH DEPRESSED MOOD-** *Occurs within 3 months of the stressor and causes significant impairment in social or occupational functioning. Care is focused on rapid return to functionality and focusing on present/future rather than the past.*

A. Signs and Symptoms

1. May occur in relation to identifiable stressors or adverse life situations such as death, divorce, or loss of an established role such as being a care giver
2. Anger associated with loss and intense guilt
3. Sadness, irritability, lack of concentration, anxiety, and worry
4. Somatic complaints are common (headache, stomach/bowel problems, etc)
5. Ask about mania symptoms (see "Bipolar Disorders")

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
   a. If you have ANY suspicion patient may cause harm, immediately assign nonmedical attendant to one-to-one direct observation
   b. Do not allow patient out of sight (even in the bathroom) and do not allow patient access to anything potentially harmful
3. Rule out any organic causes (check thyroid, examine for evidence of alcohol or substance abuse)
4. Establish probable diagnosis based on the history and physical examination findings and CONTACT PHYSICIAN PRECEPTOR based on "9-symptom questionnaire" guidance
5. Treatment emphasis is on keeping patient engaged in daily activities and planning for the future
6. Encourage relaxation techniques (journaling, creative outlets, sports, yoga, meditation)
7. Have daily/weekly talks with patient; encourage patient to speak freely
8. Scheule adequate time, don not appear rushed or hurried
9. Increase quantity and intensity of exercise, be outdoors during hours of light as much as possible, increase water intake, regulate sleep schedule, start a multivitamin
10. Avoid all substances that could alter mood, including herbs, supplements, caffeine, and especially tobacco and alcohol

III. BIPOLAR DISORDER ("MANIC DEPRESSION")- *Episodic mood shifts between mania, major depression, hypomania and mixed mood states. The ability of bipolar disorder to mimic aspects of many other major mental health disorders and its high comorbidity with substance abuse can make the initial diagnosis of bipolar difficult.*

A. Signs and Symptoms-Manic episode

1. Extravagance, elation, hyperactivity, over-involvement in life activities, increased irritability, hypersexuality, flight of ideas, easy distractibility and little need for sleep
2. Pressured speech, hyperverbosity
3. History of overspending, quitting jobs, quick (short-lived) marriages and sexual promiscuity
4. May include gross delusions, severe paranoia, and auditory hallucinations
5. Acute onset lasting several days to months
6. Manic episodes typically shorter than depressive episodes

B. Signs and Symptoms- Depressive Episode

1. See "Depression" and "Adjustment Disorder" above

2. Bipolar patients have high incidence of attempted and completed suicide and high incidence of substance abuse
3. Must ask about mania symptoms for any patient with depressive symptoms to avoid missing diagnosis of bipolar disorder

C. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
    a. If you have ANY suspicion patient may cause harm, immediately assign nonmedical attendant to one-to-one direct observation
    b. Do not allow patient out of sight (even in the bathroom) and do not allow patient access to anything potentially harmful
3. Rule out any organic causes (check thyroid, examine for evidence of alcohol or substance abuse)
4. Establish probable diagnosis based on the history and physical examination findings and CONTACT PHYSICIAN PRECEPTOR based on "9-symptom questionnaire" guidance
5. Treatment emphasis is on keeping patient engaged in daily activities and planning for the future
6. Encourage relaxation techniques (journaling, creative outlets, sports, yoga, meditation)
7. Have daily/weekly talks with patient; encourage patient to speak freely
8. Scheule adequate time, don not appear rushed or hurried
9. Increase quantity and intensity of exercise, be outdoors during hours of light as much as possible, increase water intake, regulate sleep schedule, start a multivitamin
10. Avoid all substances that could alter mood, including herbs, supplements, caffeine, and especially tobacco and alcohol

IV. **HOMESICKNESS-** *Distress and functional impairment caused by actual or anticipation of separation from loved ones and familiar situations.*

A. Signs and Symptoms

1. Recurrent, intrusive thoughts of home that impact work or daily functioning
2. Risk factors include little or no prior experience away from home, presently insecure home environment, young age, negative attitude towards separation, no input into the decision to be separated
3. May have behavior issues such as anger, fighting, irrational behavior, tardiness, insubordination
4. May have somatic complaints such as headaches, upset stomach, fatigue

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
    a. If you have ANY suspicion patient may cause harm, immediately assign nonmedical attendant to one-to-one direct observation
    b. Do not allow patient out of sight (even in the bathroom) and do not allow patient access to anything potentially harmful

3. Treatment is focused on engaging in their new environment and establishing new social support network
4. Initially limit contact with home, especially phone calls/video chats due to their emotional nature
5. Increase structure and scheduled activities
6. Rule out any organic causes (check thyroid, examine for evidence of alcohol or substance abuse)
7. Treatment emphasis is on keeping patient engaged in daily activities and planning for the future
8. Encourage relaxation techniques (journaling, creative outlets, sports, yoga, meditation)
9. Have daily/weekly talks with patient; encourage patient to speak freely
10. Scheule adequate time, don not appear rushed or hurried
11. Increase quantity and intensity of exercise, be outdoors during hours of light as much as possible, increase water intake, regulate sleep schedule, start a multivitamin
12. Avoid all substances that could alter mood, including herbs, supplements, caffeine, and especially tobacco and alcohol

Anxiety and Dissociative Disorders

V. **GENERALIZED ANXIETY DISORDER-** *Most common anxiety disorder; initial manifestations appear at age 20-35 and is slightly higher in women. The two primary components are psychological and somatic. Consider differential diagnosis of illegal drug and steroid use.*

A. Signs and Symptoms

1. Symptoms of anxiety are present more days than not and are present for at least 6 months
2. Apprehension, insomnia, worry, irritability, difficulty concentrating
3. Somatic complaints that include cardiac (chest pain, palpitations, etc), GI (nausea, vomiting, gas, diarrhea, constipation, etc) and neurologic (headaches, numbness, tingling, weakness, pseudoseizures, etc)
4. Ask about supplement use- many weight loss preparations and muscle building preparations induce anxiety

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
3. Rule out any organic defect (check thyroid, examine for evidence of alcohol or substance abuse)
4. Establish probable diagnosis based on the history and physical examination and CONTACT PHYSICIAN PRECEPTOR- EVACUATION. Patient will require further workup before diagnosis of anxiety can be made
5. If the patient is severely agitated then subdue with diazepam (Valium®) 2 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. Advise patient's Command of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider

7. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

VI. **PANIC DISORDER-** *Short lived, recurrent, unpredictable episodes of intense anxiety accompanied by severe physical symptoms.*

A. Signs and Symptoms

1. Agoraphobia (irrational fear of public or open places or embarrassing situations where escape will not be feasible), palpitations, paresthesias (pricking or tingling feeling, such as "limb going to sleep"), nausea, dyspnea, headaches, choking, bloating, tachycardia, dizziness, claustrophobia (fear of small spaces without escape), or "smothered" feeling
2. Feeling of impending doom
3. Differentiating panic disorder from physical disorders (such as myocardial infarction) can often be made by the patient's description of the "psychic pain" that precipitates episodes
4. Symptoms resolve completely in less than an hour
5. Ask about supplement use- many weight loss preparations and muscle building preparations induce anxiety

CLINICAL NOTE: Panic disorder tends to be familial with onset usually under age 25. Female to male ratio is 2:1. Patients with recurrent panic disorder become demoralized, hypochondriacal (afraid that they have severe or fatal medical conditions), agoraphobic (fear of public situations from which they can't easily escape) and depressed. Suicide attempts and major depression are often associated with panic disorder.

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
3. Rule out any organic causes (check thyroid, examine for evidence of alcohol or substance abuse)
4. Establish probable diagnosis based on the history and physical examination and CONTACT PHYSICIAN PRECEPTOR- EVACUATION, patient will require further evaluation before a diagnosis of panic disorder can be made
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
7. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

VII. **OBSESSIVE-COMPULSIVE DISORDER ("OCD")-** *A mental condition with predominant features of obsessions (constant recurrent intrusive thoughts) and compulsions (intense need to perform repetitive actions such as door locking, handwashing, ordering of objects, etc.)*

A. Signs and Symptoms

1. Irrational ideas or impulses persistently intrude into awareness
2. Actions recognized as absurd and are resisted, but anxiety is relieved only by ritualistic performance of the action or by deliberate contemplation of the intruding thought or idea
3. Compulsions are time consuming, cause significant distress, and produce shame
4. Patients are usually predictable, orderly, conscientious and intelligent
5. May be associated with tics, Tourette syndrome or eating disorders

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Examine for signs of alcohol or substance abuse
3. Assess patient's risk for harm to self or others
4. Establish probable diagnosis based on the history and physical examination
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. CONTACT PHYSICIAN PRECEPTOR- EVACUATION; patient will require further evaluation before a diagnosis of obsessive-compulsive disorder can be made
7. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
8. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

VIII. ANOREXIA NERVOSA AND BULIMIA NERVOSA- *Obsession with weight loss leads to recurrent intrusive thoughts about food, weight loss, exercise, and ritualistic eating behaviors focused on secrecy and denial. Although disorders manifest themselves with regards to food and eating behaviors, source of disorders are an attempt to gain control over their environment.*

A. Signs and Symptoms- Anorexia Nervosa

1. Refusal to maintain body weight at or above a minimally normal weight for age and height
2. Intense fear of gaining weight or becoming fat despite being underweight
3. Distorted perception of body weight and shape
4. Amenorrhea

B. Signs and Symptoms- Bulimia Nervosa

1. Recurrent episodes of binge eating and inappropriate compensatory behavior to prevent weight gain (laxative abuse, induced emesis, excessive exercise)
2. Binge/purge cycles average at least twice per week for three months

C. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Examine for signs of alcohol or substance abuse
3. Assess patient's risk for harm to self or others
4. Establish probable diagnosis based on the history and physical examination

5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. CONTACT PHYSICIAN PRECEPTOR- EVACUATION; patient will require further evaluation before a diagnosis of eating disorder can be made
7. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider

IX. **PHOBIC DISORDER-** *Clinically significant irrational fear of an object or situation that leads to dysfunctional avoidance of the stimulus.*

A. Signs and Symptoms

1. Excessive fear triggered by specific object or situation such as performance anxiety, bridges, heights, spiders, water, or agorophobia (fear of not being able to get away from a situation, such as when one is in a crowd)
2. Exposure to stimulus or even mention of the stimulus evokes an immediate anxiety response and may be associated with severe panic attacks
3. The individual recognizes the fear is excessive or unreasonable
4. The fear causes significant distress and interference in daily functioning

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
3. Rule out any organic defect (check thyroid, examine for signs of alcohol or substance abuse)
4. Establish probable diagnosis based on the history and physical examination
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. If evacuation is required
        i. Advise patient's commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
        ii. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations
    b. If evacuation is not required
        i. Treatment is based on desensitization to the stimulus that causes the phobia
        ii. Have frequent talks with the patient
        iii. Acknowledge negative reactions

X. **DISSOCIATIVE DISORDER-** *A disunion of mind of which the patient is not aware. Examples include dual personality, dissociative fugue, somnambulism (sleepwalking), selective amnesia, and depersonalization.*

A. Signs and Symptoms

1. Reaction often precipitated by emotional crisis

2. Repression, isolation, limited scope of attention
3. Dissociative symptoms result in reduced anxiety

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
   a. Patients with dissociative symptoms typically do not desire to hurt themselves because their dissociative symptoms take away reduce their anxiety but they may act irrationally, erratically, or destructively
   b. If you have ANY suspicion patient may cause harm to self or others, immediately assign nonmedical attendant to one-to-one direct observation
   c. Do not allow patient out of sight (even in the bathroom) and do not allow the patient access to anything potentially harmful
3. Rule out any organic causes (check thyroid, evaluate for alcohol and substance abuse)
4. Establish probable diagnosis based on the history and physical examination and CONTACT PHYSICIAN PRECEPTOR- EVACUATION
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
7. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

Somataform Disorders (in these disorders physical symptoms are not intentionally induced)

XI. **CONVERSION DISORDER**- *Conversion of psychic conflict into physical sensory or motor symptoms manifesting as a single, continuous or recurrent symptom.*

A. Signs and Symptoms

1. Sudden blindness or paralysis are common presentations
2. Symptoms are not intentional or for personal gain, and can not be explained by a medical condition
3. History of coping through repression and isolation
4. History of serious precipitating emotional event (combat or natural disaster)
5. May occur when others present with same symptoms (ie several members of unit were exposed to bright flash and blinded; patient not exposed but develops sudden onset blindness)
6. Often associated with other psychopathology

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
3. Rule out any organic defect (check thyroid, evaluate for alcohol and substance abuse)

4. Establish probable diagnosis based on the history and physical examination and CONTACT PHYSICIAN PRECEPTOR- EVACUATION; patient will require further evaluation before a diagnosis of conversion disorder can be made
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
7. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

XII. **SOMATIZATION DISORDER**- *Syndrome of physical symptoms without medical basis resulting in high utilization of medical care without meeting diagnostic criteria for specific illnesses.*

A. Signs and Symptoms

1. History of long-standing medical utilization for multiple nonspecific complaints
2. Extensive workups that reveal normal exams and testing
3. Common complaints include headaches, dysmenorrhea, nausea/vomiting, bloating/diarrhea, back pain, symptoms of the sex organs, amnesia, lump in throat, dyspnea, pain in extremities
4. History of multiple symptoms, multiple referrals, and no solid diagnosis
5. Frequently has anxiety, panic, and depression components
6. Disorder is a reflection of maladaptive coping techniques; patient uses symptoms to avoid responsibility or intimacy

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
3. Rule out any organic defect (check thyroid, evaluate for alcohol and substance abuse)
4. Establish probable diagnosis based on the history and physical examination and CONTACT PHYSICIAN PRECEPTOR- EVACUATION
5. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible
6. Advise patient's Commander of evacuation and need for further evaluation; do not suggest diagnosis- diagnosis is determined by mental health provider
7. Establish nonmedical attendant for continuous direct observation of patient, even if patient does not report suicidal or homicidal ideations

Personality Disorders

XIII. **PERSONALITY DISORDERS**- *Recurrent, maladaptive behavior resulting from a long history dating to childhood. Patients with personality disorders generally have low self-esteem and lack of confidence. They exhibit difficulties with interpersonal relationships or social*

functioning and have a tendency to blame others. Most develop anxiety and depression when the maladaptive behavior fails.

CLINICAL NOTE: IDMTs do not diagnose or treat personality disorders. They are mentioned here as a reference to recognize behavior and guide referrals.

A. Signs and Symptoms

1. All personality disorders share common symptoms of frequent mood swings, feeling there is nothing wrong with their behavior, need to be center of attention, feeling the external world is to blame for their behaviors and feelings
2. **Paranoid**- Defensive, oversensitive, secretive, suspicious, hyper-alert, with limited emotional response
3. **Schizoid**- Shy, introverted, withdrawn, avoids close relationships
4. **Obsessive Compulsive**- Perfectionistic, egocentric, indecisive, with rigid thought patters and need for control
5. **Histrionic** (hysterical)- Dependent, immature, seductive, egocentric, vain, emotionally labile
6. **Schizotypal**- Superstitious, magical thinking, socially isolated, suspicious, with limited interpersonal ability, eccentric behaviors and odd speech
7. **Narcissistic**- Exhibitionist, grandiose, preoccupied with power, lacks interest in others, excessive demands for attention (often associated with histrionic behavior)
8. **Avoidant**- Fears rejection, hyper-reacts to rejection and failure, with poor social endeavors and low self esteem
9. **Dependent**- Passive, over-accepting, unable to make decisions, lacks confidence, poor self esteem
10. **Antisocial**- Selfish, callous, promiscuous, impulsive, unable to learn from experience, has legal problems
11. **Borderline**- Impulsive; has unstable and intense interpersonal relationships; is suffused with anger, fear and guilt; lack self control and self-fulfillment; has identity problems and affective instability; is suicidal or self-mutilating; aggressive behavior, feelings of emptiness and occasional psychotic decompensation. This group has a high drug abuse rate and up to 80% of hospitalized borderline patients make a suicide attempt at some time after treatment

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
2. Avoid all medications, especially benzodiazepines such as diazepam (Valium®) unless prescribed by a clinician who has evaluated the patient
3. Antipsychotics such as haloperidol (Haldol®) for chemical restraint may be required for emergent conditions
4. If the patient is severely agitated or experiencing a psychotic break (disconnected from reality) administer haloperidol (Haldol®) 5 mg IM

Acute Stress Conditions

XIV. **POST TRAUMATIC STRESS DISORDER ("PTSD")-** *PTSD is the complex cognitive and behavioral effects of psychological trauma. PTSD syndrome characterized by "re-experiencing" a traumatic event, with decreased responsiveness and avoidance of current events associated with the trauma. PTSD is more common when the event is associated with a physical injury.*

A. Signs and Symptoms

1. Hyper-arousal, intrusive thoughts, nightmares, impulsivity, hyper-alertness, startle reactions, illusions, sleep problems, flashbacks, difficulty concentrating, impaired social functioning and social withdrawal
2. Attempts to avoid thoughts or flashbacks may result in fleeing or combativeness
3. Symptoms may be precipitated or exacerbated by events that are a reminder of the original stress
4. Onset of symptoms is often weeks to months after the initial stressful event
5. Symptoms frequently arise after a long latent period

B. Treatment

1. Perform a neuropsychiatric evaluation of the patient (comprehensive history interview and physical examination)
2. Assess patient's risk for harm to self or others
   a. If you have ANY suspicion patient may cause harm to self or others, immediately assign nonmedical attendant to one-to-one direct observation
   b. Do not allow patient out of sight (even in the bathroom) and do not expose patient to anything potentially lethal
3. Establish history of exposure to a life-threatening event followed by intrusive or avoidance symptoms
4. Rule out any organic defect (check thyroid, rule out current traumatic brain injury, examine for evidence of alcohol or substance abuse)
5. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
6. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to use if possible

CLINICAL NOTE: For diagnosis of PTSD, the following criteria must be met:

1. Severe traumatic event causing intense fear or helplessness (rape, capture, torture, etc)
2. Symptoms in each category of:
   a. Re-experiencing the event in flashbacks
   b. Avoidance of triggers
   c. "Jumpy" or on edge
3. Social or occupational impairment due to symptoms
4. Symptoms and functional impairment lasting more than one month after the traumatic event ends

ACTION ALERT: Therapy should begin as soon as possible after the traumatic event and should be kept brief with quick return to unit and regular activities. Recognizing symptoms and early intervention after a traumatic event is the key to successful treatment.

XV. **COMBAT OPERATIONAL STRESS REACTION ("ACUTE STRESS DISORDER")**- *Traumatic event directly witnessed or indirectly experienced by an individual or group that leads to a maladaptive behavioral response. The Air Force takes these exposures very seriously and treats them with a comprehensive team approach called the "Traumatic Stress Response Team". At a minimum, the team includes mental health providers and technicians, spiritual leaders, and the Airmen and Family Readiness Center. Medical records are not maintained and treatment is confidential unless patient is actively homicidal or suicidal. Treatment focuses on fostering resilience, individual responsibility and self-esteem, and unit cohesion. See AFI 44-153 for more information.*

XVI. **SUICIDAL OR HOMICIDAL GESTURES/ATTEMPTS**- *Individuals become suicidal when they perceive stressors as overwhelming and permanent and they desire to escape the stressors by ending their lives. Individuals become homicidal when they perceive stressors as overwhelming and permanent and desire to hurt those whom they believe are the source of the stressors.*

A. Signs and Symptoms

1. Start assessment by evaluating patient's responses to the 9-symptom questionnaire for severe depression (see below)

CLINICAL NOTE: Screen all potentially depressed patients with "9-Symptom Questionnaire". Rate each question on a scale of 0-3, with 0 meaning not at all, 1 means several days, 2 means more than half the days, and 3 means almost every day:

1. Little interest or pleasure in doing things
2. Feeling down, depressed or hopeless
3. Trouble sleeping or sleeping too much
4. Feeling tired or little energy
5. Poor appetite or overeating
6. Feeling bad about self
7. Trouble concentrating on things
8. Moving or speaking slowly or being restless
9. Thoughts that you would be better off dead

2. If you believe the patient may be at risk for harming self or others, even if they don't report wanting to hurt themselves, assess suicide/homicide risk through the following criteria:
    a. Are they thinking about killing self or others (suicidal/homicidal ideation)? If they were going to hurt somebody or themselves, how would they do it (plan)? Do they wish they could (intent)?

b. Do they have access to a means to commit suicide or homicide and the lethality of those means (weapons, pills, etc)?
c. Do they have psychotic symptoms such as command hallucinations ("the voice in my head is telling me to kill him"), or severe anxiety?
d. Is there any present or history of alcohol or substance abuse?
e. Is there history of prior attempts and how close they came to completion?
f. Is there family history of suicide or recent exposure to suicide?

3. The patient may make statements indicating lack of future thinking ("When I'm gone…")
4. The patient may make statements indicating intense desire to avoid activities of daily living ("I wish I would never wake up….")
5. The patient may have disproportionate anger to relatively minor stressors (shattering a computer monitor when the network system is slow)
6. The patient may take routine frustrations personally and then generalize them into statements of low self-worth ("I never pass the training pretest. I'm worthless to the Air Force")
7. The patient may give away items of high personal value or seem disinterested when something of high personal value is damaged; may destroy important personal artifacts such as pictures of family, etc.

B. Treatment

1. <u>If patient talks of committing suicide or homicide</u>
   a. Initiate 24-hour 1-to-1 direct surveillance with patient in protective custody
   b. Remove all of patient's belongings to a secure location
   c. Prevent access to potentially harmful items
   d. Arrange for immediate evacuation to a medical facility for specialized care
   e. Employ appropriate physical and/or chemical restraint if necessary, until evacuation can occur
   f. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR-MEDICATION prior to use if possible
2. <u>If patient has attempted suicide</u>
   a. Render emergency treatment required to stabilize patient (treat for drug overdose, physical injury, etc. See "Emergency Protocols")
   b. Initiate 24-hour surveillance with patient in protective custody
   c. Employ appropriate physical and/or chemical restraint if necessary, until evacuation can occur
   d. If the patient is severely agitated then subdue with diazepam (Valium®) 5 mg IM; CONTACT PHYSICIAN PRECEPTOR-MEDICATION prior to use if possible
3. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

Substance Abuse

XVII. OPIATE (NARCOTIC) INTOXICATION/OVERDOSE- *Includes many prescription and recreational sedatives including heroin, opium, morphine, hydrocodone, oxycodone,*

Percocet®, Vicodin®, Tylenol #3®, and Tylox®. If overdose of prescribed substance containing acetaminophen (Tylenol®) is suspected, be alert for possible severe liver damage.

A. Signs and Symptoms

1. Dissociation, amnesia, confusion, apathy
2. Bradycardia, severe respiratory depression
3. Fixed, constricted pupils, decreased bowel sounds
4. Hypothermia

B. Treatment

1. **IMMEDIATE ACTION**
   a. Support respirations, be prepared to intubate for deteriorating respiratory function, administer CPR if necessary
   b. Administer oxygen 15 LPM by non-rebreather mask for respiratory distress, may adjust according to patient comfort as needed
   c. Establish IV access
   d. Attach EKG if available and continually monitor; use waveform capnography to monitor respirations if available
   e. Administer naloxone hydrochloride (Narcan®) 0.4 mg IV STAT; then, if patient is having severe respiratory distress or is not breathing, administer Narcan® 0.2 mg IV every 2 to 3 minutes until breathing and alertness are restored or until three doses have been administered NOTE: may give Narcan® at double the dose through endotracheal tube if IV access is not available
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Do not perform gastric lavage or administer activated charcoal unless directed by preceptor

ACTION ALERT: Due to the short half-life of Narcan®, the dose of Narcan® may need to be repeated. If patient responds, monitor closely to determine if additional doses are needed. If there is no improvement in patient's condition after three doses of Narcan®, consider other sources of patient's symptoms.

Narcan® is of no value in treating other intoxicants such as alcohol, barbiturates, cocaine, and LSD.

XVIII. OPIATE WITHDRAWAL SYNDROME- *Unpleasant but rarely life-threatening side effects when chronic opiate users discontinue the drug. Seen in active duty populations primarily in chronic pain patients who are taking daily medications containing narcotics such as Percocet® or Vicodin®.*

A. Signs and Symptoms

1. Restlessness, rhinorrhea, lacrimation, myalgias, arthralgias
2. Incapacitating nausea/vomiting/cramping/diarrhea
3. Pupillary dilation, tachycardia, abnormally elevated or decreased blood pressure (decreased if severe diarrhea/emesis)
4. Increased bowel sounds, yawning, and piloerection (goosebumps)
5. Opiate withdrawal is rarely deadly, although patients will assert that they will die without treatment

B. Treatment

1. May treat diarrhea with loperamide (Immodium®)
2. May treat nausea/vomiting with ondansetron (Zofran®)
3. If patient is severely agitated or violent, may give diazepam (Valium®) 10 mg PO or 5 mg IM; may repeat 5 mg PO every 8 hours PRN

C. Prevention

1. As an IDMT, never prescribe narcotics for chronic pain
2. Patients taking as few as 1-2 daily pills containing narcotics can quickly develop tolerance and withdrawal symptoms if withheld

CLINICAL NOTE: Opioids have little direct effect on the heart; they usually cause death by severe respiratory depression. If adequate ventilator support is provided, the patient will likely recover, even from massive overdose.

As evidenced by their contribution to multiple high-profile celebrity deaths (Michael Jackson, Heath Ledger, and Elvis Presley to name a few), prescription narcotics and sedatives are dangerous and potentially deadly. As an IDMT, avoid prescribing more than 5 narcotic-containing pills at a time. Reevaluate patient's pain level each time before refilling.

XIX. BARBITURATES AND TRANQUILIZER INTOXICATION/OVERDOSE- *Often prescribed as sleeping pills, muscle relaxers, and anti-anxiety medications. Includes phenobarbital (Donnatal®), lorazepam (Ativan®), diazepam (Valium®), alprazolam (Xanax®). Street drugs generally classified as "downers".*

A. Signs and Symptoms

1. Sleep, stupor, sedation, coma
2. Respiratory depression
3. Short-term memory impairment, inability to form new memories
4. Ataxia, impaired psychomotor skills
5. New onset sleepwalking or complex behaviors such as driving when the patient is not fully awake and after which the patient has no memory

B. Treatment

1. **IMMEDIATE ACTION**
   a. Support respirations, be prepared to intubate for deteriorating respiratory function, administer CPR if necessary
   b. Administer oxygen 15 LPM by non-rebreather mask for respiratory distress, may adjust according to patient comfort as needed
   c. Establish IV access
   d. Attach EKG if available and continually monitor
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider administering flumazenil (Ramazicon®) if available
   b. Do not perform gastric lavage or administer activated charcoal unless directed by preceptor

CLINICAL NOTE: Benzodiazepines are commonly prescribed in the military due to high incidence of sleep problems, muscles spasms, and anxiety disorders. They have little direct effect on the heart; they usually cause death by severe respiratory depression. If adequate ventilator support is provided, the patient will likely recover, even from massive overdose.

Celebrity deaths associated with tranquilizer, barbiturate, or benzodiazepine overdose include Marilyn Monroe, Judy Garland, Jimi Hendrix, and Anna Nicole Smith, to name a few.

XX. **BARBITURATE/BENZODIAZEPINE WITHDRAWAL SYNDROME**- *Abrupt discontinuation of prescription or recreational sedative-hypnotics produces life-threatening withdrawal syndrome.*

A. Signs and Symptoms

1. Life threatening
2. Tremors, anxiety, psychoses, dysphoria, and seizures
3. May have complete break with reality
4. History of chronic medication or drug ingestion; identify substance taken, if possible

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access
   b. Attach EKG if available and continually monitor
   c. Cover patient's eyes with cool damp cloth to minimize visual stimuli (this helps to stop hallucinations)
   d. Reassure patient frequently
   e. Treatment of barbiturate withdrawal syndrome is administration of a long acting benzodiazepine such as diazepam (Valium®). CONTACT PHYSICIAN PRECEPTOR- MEDICATION prior to administration
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXI. **STIMULANT DRUG INTOXICATION/OVERDOSE**- *Legal stimulants include caffeine, weight loss products, nicotine, energy drinks, and medications used to treat Attention-Deficit Hyperactivity Disorder such as dextroamphetamine (Adderall®). Illegal stimulants include cocaine, crystal meth, crack, crank, and ecstasy.*

A. Signs and Symptoms

1. Agitation, tachycardia, hyperthermia
2. Headache, psychoses, seizures, focal neurologic symptoms
3. Patient likely to become violent; agitation and violence can be abrupt and unpredictable
4. Death by cardiac arrhythmias and multi-organ system failure
5. If drug inhaled by snorting, may have ulcers, holes or burns in nasal septum
6. Consider acute alcohol withdrawal syndrome as a differential diagnosis

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access; maintain adequate oral/IV intake to prevent dehydration; cool patient to core temperature below 102°F
   b. Attach EKG if available and continually monitor

c. If the patient becomes violent, administer 10 mg diazepam (Valium®) IV (only use IM if IV not feasible)
d. Repeat diazepam (Valium®) 5 mg IV or IM every 10 minutes after initial dose until appropriate response is reached; be prepared for respiratory arrest
e. Use physical restraints and keep patient in protective custody
f. Severe agitation is a predictor for cardiac arrest- review ACLS guidelines and be prepared for cardiac failure
g. Cover patient's eyes with cool damp cloth to minimize visual stimuli (this helps to stop hallucinations)

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXII. HALLUCINOGEN INTOXICATION/OVERDOSE/FLASHBACK- *Dextromethorphan, a cough suppressant commonly found in over-the-counter cold medicines (Day-Quil®, Ny-Quil®, Robitussin®, Pediacare®, Triaminic®, and Theraflu®, etc) is readily available and becoming a common recreational hallucinogen. Illegal hallucinogens include "magic mushrooms", PCP, LSD, special K, and ecstasy. Ketamine is an anesthetic used in veterinary medicine and on the battlefield instead of sedatives; enables soldier to dissociate from pain of severe injury so that they can participate in their own rescue.*

A. Signs and Symptoms

1. In general, blending of senses such as "hearing colors", feelings of expansiveness, out of body experiences, distortion of time, euphoria
2. Unwanted effects include fear, panic, frightening imagery, intense dread
3. Frank psychoses may last for days

B. Treatment

1. **IMMEDIATE ACTION**
   a. Establish IV access; maintain adequate oral/IV intake to prevent dehydration
   b. Attach EKG if available and continually monitor
   c. Sedation generally not necessary; if the patient becomes violent, use physical restraints and protective custody
   d. Cover patient's eyes with cool damp cloth to minimize visual stimuli (this helps to stop hallucinations)
   e. Reassure patient frequently
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

XXIII. MARIJUANA INTOXICATION- *A controversial herb that can be smoked to produce generalized euphoria and relaxation. It is legal in some states for medicinal use to gain weight, especially in cancer patients.*

A. Signs and Symptoms

1. Euphoria, more vivid sensations, distortion of time
2. Impaired cognition and impaired reaction time last much longer than the "high"
3. Tachycardia, increased blood pressure, increased respiratory rate, conjunctival injection (red eyes), dry mouth, increased appetite ("munchies")

B. Treatment

1. Maintain adequate oral/IV intake to prevent dehydration
2. If the patient becomes violent, use physical restraints and protective custody

3. If patient becomes severely agitated, may use diazepam (Valium®) 5 mg IV or IM
4. CONTACT PHYSICIAN PRECEPTOR- EVACUATION

XXIV. **ALCOHOL INTOXICATION AND ALCOHOL WITHDRAWAL**- See "Neurologic Disorders, Alcohol Intoxication" and "Neurologic Disorders, Alcohol Withdrawal"

Chapter 14: Poisonings, Bites, and Stings Protocols

Marine Life

I. **GENERAL CONSIDERATIONS-** *Many marine life envenomations can cause paralysis, including respiratory paralysis. Be prepared to provide long-term ventilator support from the beginning of the incident through the entire evacuation. These patients can be "locked in", meaning they are fully aware but unable to communicate, breathe, or move. If full cardiorespiratory support is provided, these patients can often have a complete recovery.*

A. Signs and Symptoms

1. Pain, local skin reaction, and edema spreading from bite/sting site are common to all marine envenomations
2. Rapid progression to shortness of breath, paralysis, shock, and respiratory arrest

B. Treatment

1. Basic Life Support, especially respiratory support is the most critical action following marine bites and stings
2. Marine wounds are prone to infection and poor wound healing; consider antibiotic prophylaxis
3. Treatment of some envenomations and stings call for hot water immersion, which destroys some of the toxins
    a. Make sure patient is hemodynamically stable before starting an immersion attempt
    b. Immerse the extremity in water as hot as will be tolerated or at (45°C/113°F); patient may have lost sensation in affected extremity so be sure to monitor water temperature closely to avoid scald burns
    c. Keep extremity immersed for 30-90 minutes; refresh the hot water as needed to keep it hot

CLINICAL NOTE: "Venomous" means that a creature can attack using its toxin by biting, stabbing, or stinging. "Poisonous" means the creature must be touched, bitten, or eaten for its toxin to take effect.

II. **COELENTERATES/CNIDARIA-** *These include jellyfish, box jellyfish, Man o' War, soft corals, and anemones. These creatures are common in both warm and cold brackish salt water, with about a million envenomations occurring around the Chesapeake Bay and Florida Gulf Coast annually. Australian box jellyfish and the Portuguese Man o' War are especially fatal.*

CLINICAL NOTE: Suspect possible Cnideria envenomation in any unresponsive patient found in surf/seawater. The Box Jellyfish (Chrionex fleckeri) can cause death in a matter of minutes, and it is the only Cnideria for which there is an antivenin (Australian species only). If Chrionex envenomation is suspected, and antivenin is available, give the antivenin immediately.

A. **Signs and Symptoms**
   1. Immediate burning, itching, throbbing pain at the site of the sting
   2. "Tentacle prints" - linear, papular eruptions that develop into raised lines, surrounded by red, inflamed dermis
   3. Weakness, nausea, headache, muscle pain, and muscle spasms
   4. Lacrimation, nasal discharge, increased perspiration, changes in pulse rate

B. **Treatment**
   1. **IMMEDIATE ACTION**
      a. Provide life support: airway, breathing, O_2, circulation
      b. The respiratory system is often the most severely affected; death can sometimes be prevented by basic respiratory support
      c. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
      d. Remove unfired nematocysts and rinse with salt water

ACTION ALERT: Cnidarian victims often have unfired nematocysts on their skin. Avoid skin to skin contact: Wear gloves/use universal precautions. Brush off unfired nematocysts with a plastic object such as a credit card; applying shaving cream if available and scraping with credit card enhances removal. Use salt water for rinsing injury site after brushing off nematocysts. DO NOT use cool fresh water to rinse off nematocysts because this will cause them to fire.

      e. After all tentacles have been removed, rinse or soak wound in acetic acid (vinegar)
      f. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
      g. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
      h. Limit patient's movements and carry them when possible; envenomations are hastened by muscular effort pushing more of the poison towards the central circulation
      i. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
   2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
      a. Consider pressure bandage (requires preceptor approval)
      b. Consider topical lidocaine (Xylocaine®) or benzocaine (Oragel®, Anbesol®) applied directly to wound area

ACTION ALERT: How to apply pressure immobilization to an extremity:

1. Apply 6-8 cm gauze pad with 2-3 cm thickness directly to bite or sting
2. Hold in place using 15-18 cm wide bandage
3. Tighten enough to occlude venous and lymphatic vessels only- pulse should still be present distal to bandage site
4. Immobilize and elevate extremity above level of the heart

3. For minor Cnidarian wounds without systemic symptoms, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Consider topical hydrocortisone 1% cream bid (do not use with oral steroids)
    b. If patient does not respond to topical treatment, consider prednisone (Deltasone®)60 mg PO daily (requires physician approval)
    c. Acetaminophen (Tylenol®) and/or ibuprofen (Motrin®) for pain
    d. Administer tetanus immunization if last immunization is unknown or if it was greater than 5 years ago
    e. Be prepared to treat delayed reactions
    f. Limited duty profile or bed rest

CLINICAL NOTE: Patients often experience fever, fatigue, malaise, joint pain, and swelling intermittently for 1-2 months after Cnidarian sting. This is termed "delayed reaction."

C. Prevention

1. Know the toxic marine life in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Monitor local warnings regarding movements/migrations of sea creatures; consider banning marine recreation during periods of increased creature presence
3. Wearing nylons while swimming or wading can prevent tentacles from attaching to the legs
4. Wear thick-soled shoes while swimming or wading to prevent injury to the feet

III. CONE SHELLS, SNAILS, OCTOPUS, AND NUDIBRANCH (SEA SLUGS)- *Most ocean species of snails, octopi, and slugs are not venomous but many are poisonous. Sea creatures with bright blue coloring tend to be venomous, including the blue ringed octopus and the blue sea slug. Cone shell snails are found in many tropical and reef environments; their shells are highly prized due to their smooth contour and lattice-like or polka-dotted decorations.*

A. Signs and Symptoms

1. Localized severe pain, swelling, redness, and numbness, and paralysis of affected limb
2. Wounds cause significant tissue damage and necrosis
3. Pain is sometimes so severe victim may request amputation
4. Rapid progression to shock, paralysis, and respiratory failure
5. Although patient is unable to communicate or move, their sensation, pain, and awareness remains intact ("locked in")

B. Treatment

1. **IMMEDIATE ACTION**
    a. Provide life support: airway, breathing, O_2, circulation
    b. The respiratory system is often the most severely affected; death can sometimes be prevented by basic respiratory support
    c. If prolonged cardiopulmonary support can be administered, patients sometimes can experience complete recovery as the poison is metabolized—however this can take up to 24 hours
    d. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
    e. Rinse with salt water or normal saline
    f. Attempt to find and remove any foreign bodies (radula)
    g. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
    h. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
    i. Limit patient's movements and carry them when possible; envenomations are hastened by muscular effort pushing more of the poison towards the central circulation
    j. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Consider pressure bandage (requires preceptor approval) – see text box above
    b. Consider topical lidocaine (Xylocaine®) or benzocaine (Oragel®, Anbesol®) applied directly to wound area
    c. If pain is severe, consider IV morphine, however this will worsen respiratory depression
    d. Administer tetanus prophylaxis if last immunization is unknown or greater than 5 years

C. Prevention

1. Know the toxic marine life in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Wear thick-soled shoes while swimming or wading to prevent injury to the feet
3. Counsel all personnel to avoid handling venomous creatures; make sure all personnel are familiar with local threats

IV. **VENOMOUS FISH STINGS-** *From least to most potent, these envenomations include the lionfish, scorpionfish, and stonefish. Lionfish tend to live in more tropical and subtropical areas, while scorpionfish and stonefish are more temperate. Scorpionfish and stonefish are both bottom-dwellers and most envenomations come from waders inadvertently stepping on them.*

A. Signs and Symptoms

1. Excruciating localized pain and swelling, peaking in intensity in <90 minutes and gradually diminishing over hours to days

2. Syncope, weakness, nausea, and anxiety are common, followed by hypotension, respiratory collapse, and possibly death
3. Peripheral vasodilation, vomiting, diarrhea, sweating, muscle cramps
4. Lymphangitis (inguinal or axillary swollen tender lymph nodes proximal to envenomation site)
5. Puncture wounds on the soles of the feet (stonefish and scorpionfish) or on the hands (lionfish) surrounded by blue, cyanotic tissue
6. Severe envenomations can cause vesicle formation followed rapidly by tissue sloughing and excruciating pain

B. Treatment

1. **IMMEDIATE ACTION**
   a. Provide life support: airway, breathing, O_2, circulation; as with other marine envenomations, prolonged cardiopulmonary support may be required
   b. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
   c. Rinse with salt water or normal saline
   d. Attempt to find and remove any retained sheaths and spines
   e. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
   f. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
   g. Immerse affected limb in hot water- see "General Principles" at the beginning of this chapter
   h. Limit patient's movements and carry them when possible; envenomations are hastened by muscular effort pushing more of the poison towards the central circulation
   i. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider topical lidocaine (Xylocaine®) or benzocaine (Oragel®, Anbesol®) applied directly to wound area
   b. If pain is severe, consider IV morphine, however this will worsen respiratory depression
   c. Administer tetanus prophylaxis if last immunization is unknown or greater than 5 years

C. Prevention

1. Know the toxic marine life in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Wear thick-soled shoes while swimming or wading to prevent injury to the feet
3. Counsel all personnel to avoid handling venomous creatures; make sure all personnel are familiar with local threats

V. **SEA SNAKES-** See Hydrophiidae in Snakes section of this chapter, under "Coral Snakes".

VI. **SEA URCHINS, CORALS, AND STINGRAYS-** *Although not commonly fatal, sea urchins, corals, and stingray wounds cause extensive local damage and are prone to infection. Sea urchins are small spiny creatures that live in coral reefs and at the bottom of the ocean. Corals form extensive rigid reefs that can inflict jagged knife-like wounds and abrasions with minimal contact. Stingrays live on the ocean floor in warm shallow waters; human injuries commonly occur when a wader steps on the stingray. Since a stingray tail is barbed, wounds from stingrays are typically deep puncture surrounded by jagged tissue damage.*

A. Signs and Symptoms

1. Pain, bleeding, burning, stinging, swelling, and itching at the injury site
2. Wounds are prone to poor healing; secondary infection is common
3. May have urticaria, vesicles, and fever develop within a few hours after injury- these are reactions to the poison and not evidence of infection
4. Systemic symptoms are less common but include nausea, vomiting, muscle cramping and malaise
5. Death from corals and sea urchins is rare, but stingray envenomations can progress to hypotension, respiratory failure, seizures, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Provide life support if needed: airway, breathing, O_2, circulation; as with other marine envenomations, prolonged cardiopulmonary support may be required
   b. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
   c. Rinse wound with salt water or normal saline
   d. Attempt to find and remove any retained sheaths and spines
   e. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
   f. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
   g. Immerse affected limb in hot water- see "General Principles" at the beginning of this chapter
   h. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider topical lidocaine (Xylocaine®) or benzocaine (Oragel®, Anbesol®) applied directly to wound area
   b. If pain is severe, consider IV morphine, however this will worsen respiratory depression
   c. Administer tetanus prophylaxis if last immunization is unknown or greater than 5 years
3. For minor wounds without systemic symptoms, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Do not suture wound closed; may draw edges loosely together with Steri-Strips® or loose suture if approved by preceptor
   b. Significant discoloration around wound after 48 hours indicates probable retained foreign body that requires surgical debridement

c. Acetaminophen (Tylenol®) for pain
d. Consider antibiotics for prevention of secondary infection; discuss with preceptor
e. Administer tetanus immunization if last immunization is unknown or if it was greater than 5 years ago
f. Be prepared to treat delayed reactions
g. Limited duty profile or bed rest

C. Prevention

1. Know the toxic marine life in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Wear wetsuits and gloves when swimming or diving around reefs
3. Wear thick-soled shoes while swimming or wading to prevent injury to the feet (will not protect against stingrays or sea urchins)
4. Shuffle feet when wading in waters inhabited by stingrays
5. Counsel all personnel to avoid handling venomous creatures; make sure all personnel are familiar with local threats

Arachnids and Insects

VII. **BLACK WIDOW SPIDER BITE-** *Black widows are found throughout the US and around the world, especially around human habitation near wooded areas. Bites most commonly occur when an unsuspecting human disturbs a female's nest and threatens her egg sac. The spiders grow up to 1.5 inches long, is glossy black or dark brown, with bright red or orange "hourglass" on the underside of the abdomen. Black widow webs are typically irregular mesh-type in dark corners, often in clutter around houses, woodpiles, and garages. If the spider was not witnessed, always consider other possible diagnoses, especially focal infection with skin bacteria.*

A. Signs and Symptoms

1. Positive identification of spider or history of high risk activity such as sitting on outdoor furniture, working in garage or outdoors clearing debris, gardening, etc
2. Immediate sharp, stinging pain at bite site; wound is a blanched circular patch with central puncture wound or pair of wounds and blanched region has ring of surrounding erythema
3. In approximately an hour of bite, affected limb develops unilateral diaphoresis with muscle pain and cramping
4. Can progress into systemic symptoms of diaphoresis (sweating), restlessness, anxiety, dizziness, nausea, and vomiting, and generalized peripheral vasodilation
5. Abdominal cramping and muscular rigidity involving major muscle groups can be severe and can be indistinguishable from acute abdomen, however abdominal sounds will normally still be present with widow bite and absent in cases of acute abdomen
6. Severe envenomations may have respiratory distress/tachypnea, salivation, and hypertension- mimicking myocardial infarction, tetanus, or rabies

B. Treatment

1. **IMMEDIATE ACTION**
   a. Provide Basic Life Support if needed

b. Obtain IV access with normal saline at maintenance rate
c. Clean bite site with copious soap and water
d. Warm baths may provide some relief for cramping and muscle spasms
e. For severe allergic reaction, see "Emergency Protocols- Anaphylaxis"

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Administer acetaminophen (Tylenol®) PO for pain; if pain is severe, consider acetaminophen/narcotic combination such as Vicodin® or Percocet®
   b. For severe muscle contractions/spasms, consider IV benzodiazepine such as diazepam (Valium®) or lorazepam (Ativan®)
   c. Administer tetanus toxoid if last immunization is unknown or >5 years ago; consider human tetanus immunoglobulin

VIII. **BROWN RECLUSE SPIDER BITE-** *Also known as the "fiddleback" due to its distinctive brown violin shape on its cephalothorax. They are found primarily in the South, West, and Midwest United States as well as in the Mediterranean and South America. They live primarily in homes in dark quiet areas such as attics, cupboards, behind bookshelves, and in basements.*

A. Signs and Symptoms

1. Initially painless bite that increases over the next several hours
2. Bite site begins to swell with local erythema, small puncture wounds may be visible
3. Within a few hours of bite, lesion develops "Bull's Eye" appearance with central pallor and surrounding erythema and ecchymosis
4. Lesion may resolve over several days or become dusky and insensate, rupture, and leave a central ulceration; these lesions can develop significant tissue necrosis
5. Systemic reactions are rare but may include nausea, vomiting, fevers and myalgias (indistinguishable from sepsis)

B. Treatment

1. **IMMEDIATE ACTION**
   a. Obtain IV access on unaffected limb with normal saline at maintenance rate if changes in sensation, systemic symptoms, severe pain, or any signs of anaphylaxis
   b. Ice the bite site taking care to protect tissue since patient may have altered sensation
   c. Immobilize extremity and elevate (possibly for several days until lesion starts to resolve or until necrosis starts)
2. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. **IMMEDIATE EVACUATION** for any patient with severe or symptoms, signs of necrosis, or unconfirmed diagnosis
   b. Administer tetanus toxoid if last immunization is unknown or >5 years ago; consider human tetanus immunoglobulin
   c. Consider antibiotic prophylaxis such as dicloxacillin (Dynapen®) or clindamycin (Cleocin®)

IX. **BEES, WASPS, HORNETS, AND YELLOW JACKETS-** *These are easily recognizable flying insects that predominantly sting when threatened. Sugar, scented flowers, and sweat*

attract bees. Meat, sugar, and fruit attract yellow jackets, which are typically found at picnic sites and garbage bins. Wasps and hornets make paper nests that are frequently attached to gutters, eaves, and the sides of buildings. All of these insects can secrete pheromones that incite other members of the hive to become aggressive and attack any creatures that threaten a nest- this behavior can lead to massive swarming attacks.

A. Signs and Symptoms

1. Uncomplicated local reaction
    a. Immediate pain and burning at the site of the sting
    b. Red, swollen wheal up to 5 cm develops quickly and should resolve over several hours
2. Large Local Reaction
    a. Exaggerated erythema and swelling (typically 10 cm) develops over 1-2 days and resolves over 5-10 days
    b. Swelling crossing a joint limits movement and can be debilitating
3. Anaphylaxis
    a. Skin symptoms: generalized urticarial, flushing of the skin, swelling of the face and around the eyes
    b. Respiratory symptoms: hoarse voice, wheezing, swelling of the pharynx and epiglottis, severe shortness of breath
    c. Cardiovascular symptoms: lightheadedness, hypotension, shock, death

B. Treatment

1. Bee stingers continue to inject venom for several seconds after the sting- it is therefore important to try to remove the stinger as fast as possible; after 10 seconds removal is less critical because all of the venom is injected
2. Use cold compresses on sting site and elevate extreme
3. Control pain with NSAIDs
4. Control itching and swelling with cetirizine (Zyrtec®) or diphenhydramine (Benadryl®) PLUS high potency topical corticosteroid such as clobetasol (Temovate®) 0.05% ointment applied to sting every 4 hours until itching subsides
5. Administer tetanus toxoid if last immunization is unknown or >5 years ago; consider human tetanus immunoglobulin
6. CONTACT PHYSICIAN PRECEPTOR- MEDICATION; may administer prednisone 60 mg one-time dose or with taper over 3-5 days for large local reactions
7. For systemic symptoms, see "Emergency Protocols- Anaphylaxis"

C. Prevention

1. Keep trash bins cleaned regularly to reduce attractive residues
2. If wasp nests must be sprayed, do so at night when insects are less active
3. Prescreen all deployers for history of anaphylaxis and make sure they bring multiple sets of epinephrine autoinjectors (EpiPen®)

X. **FIRE ANT STINGS**- *Fire ants are found throughout the US with greatest density in the Southern states. They grow up to 1 cm in length and are usually reddish brown or reddish black in color. They are aggressive and attack with minimal provocation; like other Hymenoptera they release pheromones that can incite a swarming attack. Fire ants*

attach with strong jaws and can deliver multiple stings within seconds. If a bite is felt, immediately crush the ant to prevent more stings.

A. Signs and Symptoms

1. Intense burning at the site of the sting changing to itching after 20-30 minutes
2. Red swollen wheal at sting site
3. Within 24 hours a sterile pustule (not infected with bacteria) develops at the bite site
4. Since lesions are itchy and unsightly, patient will be tempted to pick and scratch at pustules; this should be avoided because it increases risk of infection and scarring
5. Large local reactions can occur (as with other Hymenoptera stings) as well as anaphylaxis reactions

B. Treatment

1. Use cold compresses on sting site and elevate extremity
2. Control pain with NSAIDs
3. Control itching and swelling with cetirizine (Zyrtec®) or diphenhydramine (Benadryl®) PLUS high potency topical corticosteroid such as clobetasol (Temovate®) 0.05% ointment applied to sting every 4 hours until itching subsides
4. CONTACT PHYSICIAN PRECEPTOR- MEDICATION; may administer prednisone (Deltasone®) 60 mg one-time dose or with taper over 3-5 days for large local reactions
5. For systemic symptoms, see "Emergency Protocols- Anaphylaxis"
6. Administer tetanus toxoid if last immunization is unknown or >5 years ago; consider human tetanus immunoglobulin

C. Prevention

1. Ants can not bite through fabric or nylon; therefore wear closed-toe shoes, socks, (not sandals or flip-flops), long sleeves, and long pants when working outdoors
2. Avoid placing campsites on or near ant mounds
3. Insect repellants are NOT effective against fire ants

XI. CENTIPEDES, MILLIPEDES, AND CATERPILLARS- *Rarely deadly, these arthropods can cause substantial irritation. Centipedes live under rocks or fallen logs and will bite when disturbed. Millipedes have a chemical arsenal of poisons including cyanide; some tropical species of millipedes can shoot their poisons over a distance of two feet or more. Millipede reactions resemble 1st or 2nd degree burns. Caterpillars can implant tiny hairs that release venom into the skin.*

A. Signs and Symptoms

1. Pain at bite site, especially with larger specimens
2. Local tissue redness and swelling at bite site, often with pruritus
3. Swollen, painful lymph nodes
4. Systemic symptoms are less common but can include headache, palpitations, nausea/vomiting, and anxiety or anaphylaxis
5. Millipede wounds typically start as brown-stained skin that progresses into blisters and sloughing of the skin

6. Observe patients for a minimum of four hours since delayed systemic symptoms are common

B. Treatment

1. For caterpillars and millipedes- copiously irrigate the area with sterile water and dry without touching the skin (use fan or hairdryer)
2. Retained caterpillar hairs can be removed by applying scotch tape or duct tape and pulling away. If this is ineffective, use fingernail polish or rubber cement, allow it to dry, and peel away
3. Cold compresses are the historical mainstay of therapy, but recent evidence suggests that hot water immersion may actually be more soothing for centipede bites
4. For centipede bites, patients receive prompt relief if bite site is injected with local anesthetic such as lidocaine 1% (Xylocaine®); CONTACT PHYSICIAN PRECEPTOR- CONSULTATION for approval
5. For systemic symptoms, see "Emergency Protocols- Anaphylaxis"
6. Administer tetanus toxoid if last immunization is unknown or >5 years ago; consider human tetanus immunoglobulin

C. Prevention

1. Know the toxic arthropods in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Wear proper clothing, gloves, and footwear when working outdoors
3. Counsel all personnel to avoid handling venomous creatures; make sure all personnel are familiar with local threats

XII. SCORPIONS- *Found on all continents except Antarctica, scorpions dwell in desert, arid, and tropical environments. These arthropods grab prey with their pincers and whip their tails overhead to sting with a neurotoxic poison, with symptoms peaking within 5 hours. Scorpions fluoresce under black light or Wood's lamp, making them easy to locate in the dark.*

A. Signs and Symptoms

1. Immediate pain at the sting site, but unlike hymenoptera stings erythema and immediate edema ("wheel and flare") do not occur
2. Cranial nerve dysfunction includes hyper-salivation, involuntary conjugate roving eye movements or sharp chaotic nystagmus
3. Skeletal muscle dysfunction includes restlessness, twitching, jerking movements, and writhing movements of the trunk that may resemble seizure- however unlike seizure patient remains alert during movements
4. Systemic symptoms include tachycardia, vomiting, hyperthermia to 104°F, diaphoresis, hypertension, dyspnea, priapism, pancreatitis, delirium, shock, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Provide life support if needed: airway, breathing, O_2, circulation; as with other envenomations, prolonged cardiopulmonary support may be required
   b. If bronchospasm is present give albuterol (Ventolin®)2 puffs inhaled orally with spacer every 4 hours
   c. Cleanse wound with soap and water

d. Place patient on bed rest and immobilize affected limb; apply cold packs to sting site
e. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
f. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
g. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Take scorpion with patient for identification at treatment facility
   b. Administer tetanus prophylaxis if last immunization is unknown or greater than 5 years
   c. NSAIDs for pain; if this is insufficient, fentanyl (Duragesic®) patch/IM/IV/lollipop is the preferred narcotic since it won't interfere with administration of antivenom
3. For minor wounds without systemic symptoms observe for a minimum of four hours to make sure no systemic symptoms occur, then CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
   a. Consider antibiotics for prevention of secondary infection; discuss with preceptor
   b. Administer tetanus immunization if last immunization is unknown or if it was greater than 5 years ago
   c. Be prepared to treat delayed reactions
   d. Limited duty profile or bed rest

C. Prevention

1. Examine all clothes and shoes prior to putting them on
2. Shake out bedrolls before entering, especially if they are on the ground
3. Shake out all camping gear and equipment prior to using
4. Reduce brush and debris in campsite to reduce habitat
5. Spraying insecticides around base of tent won't kill the scorpions but will reduce their food supply

Lizard Bites

XIII. **GILA MONSTER AND BEADED LIZARD-** *Found in the Southwestern United States and Mexico, these lizards are becoming endangered species due to human encroachment on their habitat. Gila monsters are stout black lizards with orange or yellow marbling or banding patterns and grow approximately 1.5 feet in length. Beaded lizards are also stout, grow up to 3 feet in length, are brown with bright white spots and live in scrubland, tropical forests, and pine forests.*

A. Signs and Symptoms

1. Bites usually occur from attempting to interact with the lizard; lizard clamps down with strong jaws and is difficult to remove
2. Chewing action instills poison in the wound
3. Severe pain, burning, and progressive edema from the bite site

4. Systemic symptoms include generalized weakness, nausea/vomiting, shortness of breath, profuse sweating, dizziness, and numbness/tingling
5. Exam findings include tachycardia, regional lymphadenopathy, hypotension, significant local tissue trauma, retained teeth, and cyanosis/ecchymosis at the bite site

B. Treatment

1. **IMMEDIATE ACTION**
   a. If lizard is still attached, remove by submerging in running hot water or pry jaws loose- be careful not to injure victim or allow rescuer to be bitten
   b. Keep patient calm and immobile; DO NOT allow patient to walk or run
   c. Immobilize bitten area and DO NOT manipulate limb; remove obvious retained foreign bodies and apply direct pressure to control bleeding
   d. Provide life support if needed: airway, breathing, O_2, circulation; as with other envenomations, prolonged cardiopulmonary support may be required
   e. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
   f. Cleanse wound with soap and water
   g. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
   h. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
   i. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Do not give NSAIDs or aspirin for pain because these may increase bleeding from wound
   b. May give acetaminophen (Tylenol®)
   c. May consider acetaminophen with codeine (Tylenol #3®) with preceptor approval, however this will enhance respiratory depression

C. Prevention

1. Know the toxic reptiles in your area, availability of antitoxin, and creature-specific treatments prior to deployment
2. Wear long pants and boots when hiking or working outdoors
3. Counsel all personnel to avoid handling venomous creatures; make sure all personnel are familiar with local threats
4. Watch venomous lizards only from a distance; these reptiles are shy and will try to hide. Do not attempt to poke or manipulate them to "get a better look"

Snake Bites

CLINICAL NOTE: Severed snake heads can still bite up to 60 minutes after decapitation. Venom in fangs remains active for days. Avoid handling severed heads unless necessary for patient care.

XIV. VIPERS (INCLUDING VENOMOUS VIPERS, COTTONMOUTHS, COPPERHEADS, AND RATTLESNAKES)- *These snakes are found worldwide in warm and temperate areas; in the United States, cottonmouth, copperhead, and rattlesnakes are primarily found in the warmer regions of the Southern and Western States. Their venom is primarily hemotoxic.*

A. Signs and Symptoms

1. Fang marks at bite site (presence of fang marks does not necessarily indicate delivery of venom)
2. Erythema, edema, warmth, and ecchymosis at the bite site and adjacent tissues within 60 minutes
3. Edema progresses up proximal extremity within hours
4. Systemic symptoms include nausea/vomiting, diaphoresis, chills, oral paresthesias, unusual taste in mouth, and fasciculations
5. Late systemic effects include bleeding, rhabdomyolysis (severe muscle cramping and pain), kidney failure, tachycardia, tachypnea, shock, altered mental status, seizures, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Bring snake/snake parts with patient for positive identification if it will not put anyone else at risk; avoid direct handling of snake heads
   b. Keep patient calm and immobile; DO NOT allow patient to walk or run
   c. Immobilize bitten area and DO NOT manipulate limb and DO NOT apply suction devices or constricting bands unless directed by preceptor, and DO NOT cut or incise the bite area
   d. Provide life support if needed: airway, breathing, O_2, circulation; as with other envenomations, prolonged cardiopulmonary support may be required
   e. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
   f. Cleanse wound with soap and water
   g. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
   h. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
   i. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Do not give NSAIDs or aspirin for pain because these may increase bleeding from wound
   b. May give acetaminophen (Tylenol®)
   c. May consider acetaminophen with codeine (Tylenol #3®) with preceptor approval, however this will enhance respiratory depression
   d. Swelling may become so severe it prevents blood flow to the distal limb- if evacuation is delayed be prepared to consider fasciotomy

XV. CORAL SNAKES, SEA SNAKES, BOOMSLANGS, BIRD SNAKES, AUSTRALIAN DEATH ADDERS AND TAIPANS- *These snakes are found worldwide in a variety of habitats. Their venom is primarily neurotoxic.*

A. Signs and Symptoms

1. Mild to moderate pain and swelling with weakness of the bitten extremity occurring within hours
2. Unlike viper bites, there is minimal local tissue damage
3. Cranial nerve palsies/deficits, double vision, sagging eyelids, difficulty swallowing, and hyper-salivation occurs within hours
4. Dizziness, nausea/vomiting, abdominal pain may occur
5. Late systemic effects include systemic weakness, drowsiness, euphoria progressing to muscle flaccidity, respiratory distress, and respiratory failure

B. Treatment

1. **IMMEDIATE ACTION**
   a. Bring snake/snake parts with patient for positive identification if it will not put anyone else at risk; avoid direct handling of snake heads
   b. Keep patient calm and immobile; DO NOT allow patient to walk or run
   c. Immobilize bitten area and DO NOT manipulate limb, DO NOT apply suction devices, and DO NOT cut or incise the bite area
   d. Provide life support if needed: airway, breathing, O_2, circulation; as with other envenomations, prolonged cardiopulmonary support may be required
   e. If bronchospasm is present give albuterol (Ventolin®) 2 puffs inhaled orally with spacer every 4 hours
   f. Cleanse wound with soap and water
   g. Obtain IV access early in unaffected limb, even if it seems unnecessary at the time; give normal saline at maintenance rate unless hypotensive
   h. If patient is hypotensive give crystalloid such as Lactated Ringer's solution 1-3 liters then maintenance at 125 cc/hour
   i. Monitor airway for anaphylaxis-type reaction (See "Emergency Medicine Protocols, Anaphylaxis")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. May give acetaminophen (Tylenol®)
   b. May consider acetaminophen with codeine (Tylenol #3®) with preceptor approval, however this will enhance respiratory depression
   c. If evacuation will be delayed, consider pressure immobilization within 30 minutes of bite
   d. Perform neurologic assessments of cranial nerves and strength every 15-30 minutes in order to monitor progressive neurotoxicity

Miscellaneous Exposures

XVI. **POISONOUS PLANTS**- See "Integumentary Protocols, Contact Dermatitis"

XVII. **ANIMAL BITES AND RABIES**- *Rabies is a potentially fatal virus that is transmitted when the patient is bitten by an infected dog, raccoon, bat, skunk, fox, or rodent. The virus travels along nerves to the brain and spinal cord; once clinical manifestations occur it will universally progress to neurologic deficits, encephalitis, and death. Assume any animal that bites has rabies unless it is a pet and its vaccination status is well-documented.*

A. Signs and Symptoms of Rabies

1. Incubation period- lasts days to months after bite; bites close to the head will have shorter incubation periods than bites on the extremities
2. Prodromal period- flu-like symptoms lasting up to a week that include nausea, malaise, anorexia, sore throat, headache, irritability, and low grade fever; percussion of muscular areas will cause mounding up of tissue at the percussion site
3. Encephalitic rabies- fear of water, painful pharyngeal spasms upon encountering a draft of air, excessive salivation, fluctuating consciousness, cardiac arrhythmias lasting 2-7 days
4. Paralytic rabies- less common than encephalitic rabies, it resembles Guillain-Barré syndrome with weakness progressing to flaccid paralysis and respiratory failure; it is different than Guillain-Barré in that it is accompanied by persistent fever, intact sensation, and bladder dysfunction
5. Coma- both encephalitic and paralytic rabies progress to coma, flaccid paralysis, and respiratory and vascular collapse

B. Treatment

1. Apply direct pressure to control bleeding
2. Cleanse superficial wounds with soap and water; if tissue damage is extensive inject local anesthesia into wound prior to cleansing and clean with gently pressurized sterile water; remove any devitalized tissue prior to closure
3. The IDMT should not close bite wounds from cats or humans, puncture wounds, bites > 12 hours old, bites associated with crush injury, or bites on the hands or feet due to increased risk of deep infections
4. Survival from rabies is dependent on being vaccinated prior to the appearance of symptoms; therefore it is critical that patient be evaluated for rabies whenever an animal bite from a potential carrier occurs
5. Every effort should be made to capture/kill the animal so that it can be evaluated for presence of rabies virus
6. In the absence of the animal, the patient may be required to undergo lengthy and painful treatments
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION AND EVACUATION for further evaluation and management

Chapter 15: Environmental Injury Protocols

Heat Stress Injuries

I. **HEAT SYNCOPE-** *Occurs if patient stands still immediately after exertion in a hot environment. While working or exerting, the muscles help to pump blood up to the brain. By stopping suddenly, peripheral vasodilation and venous pooling are too substantial for the heart to compensate and the patient loses consciousness briefly.*

A. Signs and Symptoms

1. Strenuous exertion in a hot environment followed by standing still
2. Brief loss of consciousness with rapid recovery as soon as the patient is supine

3. Normal core body temperature

B. Treatment

1. Move patient to a cool environment
2. Provide life support if needed: airway, breathing, O_2, circulation, monitor
3. Patient should immediately regain consciousness
4. Administer oral rehydration with salty fluids or rehydrate while eating salty foods
5. If patient does not respond to oral treatments, if they are nauseous, or if they are orthostatic, obtain IV access and give normal saline 1-2 L fluid bolus
6. Keep patient supine until asymptomatic

C. Prevention

1. Educate troops regarding appropriate cool down exercises after strenuous workout
2. After severe exertion ensure troops slow down gradually so that heart rate and blood distribution can normalize over several minutes
3. Once heart rate normalizes and skin is no longer pink, recommend 3-5 minutes of stretching exercises, holding each stretch for a minimum of 30 seconds per side (See "Musculoskeletal Protocols")
4. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; discontinue use in hot conditions

II. **HEAT CRAMPS-** *Muscular cramps and spasms that occur during or after activity in a hot environment that occurs due to loss of sodium chloride. Individuals that have been sweating and aggressively hydrating with water replaces fluid but not salt balance and places them at greater risk.*

A. Signs and Symptoms

1. Abrupt onset of cramping in large muscle groups such as thighs, calves, abdomen
2. Painful muscle spasms and profuse sweating; core temperature normal or elevated, but not above 104°F
3. May progress to "heat tetany" with severe muscle spasms, hyperventilation, cramping of the upper extremities, and perioral numbness

B. Treatment

1. Symptoms can continue to progress even after activity has ceased; therefore be prepared to offer life support measures
2. Move patient to a cool environment
3. Provide life support if needed: airway, breathing, O_2, circulation, monitor
4. Administer oral rehydration with salty fluids or rehydrate while eating salty foods
5. If patient does not respond to oral treatments or if they are nauseous, obtain IV access and give normal saline 1-2 L fluid bolus
6. Work muscles with passive stretching techniques and massage
7. If patient is hyperventilating, use a partial rebreather mask delivering less than 5 liters per minute of oxygen; this will enable the patient to get adequate amounts of both carbon dioxide and oxygen and will reduce the hyperventilation

C. Prevention

1. Ensure adequate work/rest cycles are followed and that workers in hot environments have a cool place available where they can recover
2. Ensure appropriate hydration is taking place; workers should alternate water and salty/sugary fluids (such as Gatorade®), mix Gatorade® ½ and ½ with water, or have salty snacks such as chips, nuts, or crackers available on breaks
3. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; recommend discontinuing use in hot conditions

III. HEAT EXHAUSTION- *Fatigue due to loss of fluids and salts.*

A. Signs and Symptoms

1. Dizziness, confusion, headache, fatigue, weakness
2. May have mild confusion that returns to normal within 30 minutes of cooling
3. Nausea, vomiting, tachycardia, profuse sweating, orthostatic hypotension
4. Ashy gray, pale, cool and clammy skin, elevated body temperature, thirst
5. Symptoms can rapidly progress to heat stroke; if any neurologic symptoms are present other than mild confusion, move to stroke protocols

B. Treatment

1. Symptoms can continue to progress even after activity has ceased; therefore be prepared to offer life support measures
2. Move patient to a cool environment
3. Provide life support if needed: airway, breathing, O_2, circulation, monitor
4. Administer oral rehydration with salty fluids or rehydrate while eating salty foods if patient can tolerate
5. If patient does not respond to oral treatments or if they are nauseous, obtain IV access and give normal saline 1-2 L fluid bolus
6. Reassess vitals every 15 minutes until stable
7. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
    a. Patient may require evacuation if recovery is slow
    b. No medications indicated

C. Prevention

1. Ensure adequate work/rest cycles are followed and that workers in hot environments have a cool place available where they can recover
2. If work/rest cycles can not be followed, recommend close medical monitoring (high ops tempo environments where the mission takes priority over health). The IDMT should closely observe all individuals involved in the activity
3. Ensure appropriate hydration is taking place; workers should alternate water and salty/sugary fluids (such as Gatorade®), mix Gatorade® ½ and ½ with water, or have salty snacks such as nuts, chips, or crackers available on breaks
4. Ensure that all troops that are new to a hot environment acclimatize to activity over several weeks
5. Ensure that all troops rehydrate fully after exertion- repeated heat stresses without full recovery of fluid status will markedly increase potential for heat injury

6. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; recommend discontinuing use in hot conditions

IV. **HEAT STROKE-** *A true medical emergency, heat stroke is caused by elevation in the core body temperature that overwhelms the central nervous system and leads to a failure of thermoregulation. Patients have a high morbidity and mortality rate even when symptoms are rapidly recognized and treated.*

A. Signs and Symptoms

1. Elevated core body temperature, usually over 105°F, rectal temperature is most accurate
2. Flushing, tachypnea, pulmonary crackles, altered mental status
3. Skin may be diaphoretic or dry, depending on how long it took symptoms to develop
4. May spontaneously bleed or bruise due to clotting mechanism failure
5. Rapidly progresses to seizures, tachycardia, tachypnea, hypotension, shock, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Move patient to a cool environment
   b. Provide immediate life support: airway, breathing, O_2, circulation, monitor
   c. Obtain IV access and give normal saline 1-2 L fluid bolus
   d. Initiate rapid cooling measures (any combination) until core temperature is below 102°F
      i. Spray unclothed patient with cool mist and use fans to rapidly evaporate water
      ii. Immerse patient in ice water (don't perform on patients over age 65)
      iii. Use ice packs on pulse points in axillae, neck, and groin
      iv. Use room temperature to slightly cool IV fluids
   e. Reassess vitals every 15 minutes including rectal core temperature checks during and after rapid cooling
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

C. Prevention

1. Ensure adequate work/rest cycles are followed and that workers in hot environments have a cool place available where they can recover
2. If work/rest cycles can not be followed, recommend close medical monitoring (high ops tempo environments where the mission takes priority over health). The IDMT should closely observe all individuals involved in the activity if possible
3. Ensure appropriate hydration is taking place; workers should alternate water and salty/sugary fluids (such as Gatorade®), mix Gatorade® ½ and ½ with water, or have salty snacks such as chips, nuts, or crackers available on breaks
4. Ensure that all new troops to a hot environment acclimatize to activity over several weeks

5. Ensure that all troops rehydrate fully after exertion- repeated heat stresses without full recovery of fluid status will markedly increase potential for heat injury
6. Personnel with a prior history of heat injury are at greater risk for developing subsequent heat related illness
7. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; recommend discontinuing use in hot conditions

Cold Stress Injuries

V. **TRENCH FOOT AND IMMERSION FOOT/SYNDROME-** *"Trench foot" occurs when wearing wet boots and socks for a prolonged period of time. Immersion foot and Immersion syndrome come from prolonged immersion of the lower extremities in water or mud. Both occur when ambient temperatures are above freezing. Nerve damage occurs when environment causes skin breakdown and tight footwear combined with immobility causes vasoconstriction.*

A. Signs and Symptoms

1. Prehyperemic phase (before rewarming)
   a. Skin is white, yellow or mottled, edema is present
   b. Complete absence of sensation, leading to "walking on air" feeling
   c. Exam findings include sluggish capillary refill, loss of sensation including light touch, sharp/dull, and proprioception, and abnormalities in gait
2. Hyperemic phase (Starts within a few hours of rewarming)
   a. Extremity become hot, red, painful, very swollen, and covered with petichiae
   b. Severe pain with any heat, movement, or touch, including pressure from sheets
   c. Vibratory sense usually absent, proprioception usually present
3. Vaso-spastic phase (several days after rewarming)
   a. Area becomes pale or cyanotic when elevated and deep red or purple and with tense edema when not elevated
   b. Diminished pulses, vesicles/bullae, edema, hemorrhagic blisters, skin sloughing and area may develop gangrene

B. Treatment

1. **IMMEDIATE ACTION**
   a. Dry patient and keep core temperature warm (treat coexistent hypothermia) while keeping extremities cool; avoid massaging or rewetting
   b. Recommended method is keeping extremities elevated and exposed to a cool fan; this treatment should be continued until pain and swelling has resolved (may take days)

C. Prevention

1. Ensure proper fitting boots prior to deployment
2. Change socks 2-3 times per day in sweaty or wet environments

3. Recommend boots be removed completely for a minimum of 8 hours out of every 24; feet should be completely dry and warm and preferably exposed to air during those periods

VI. **FROSTBITE-** *Crystallization of water in the distal tissues due to exposure to subfreezing temperatures; most commonly affects the ears, nose, cheeks, chin, fingers, and toes. Risk factors include smoking, exhaustion, dehydration, alcohol use, and exposure to vibration.*

A. Signs and Symptoms

1. Patient reports cold, numbness, and clumsiness of affected area
2. Skin is without sensation, grayish yellow, and hard or waxy to the touch
3. First degree frostbite- superficial, central area of pallor and anesthesia surrounded by edema
4. Second degree frostbite- large blisters containing clear fluid surrounded by erythema and edema; blisters may form black eschar but they slough off and reveal healthy tissue; no permanent tissue loss
5. Third degree frostbite- deep forming black eschar, blisters are smaller; hemorrhagic bullae present
6. Fourth degree frostbite- extends deep into muscle and bone and causes complete tissue necrosis and mummification

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, O_2, circulation, treat hypothermia
   b. Do not allow frozen tissue to thaw if there is any chance of re-freezing before evacuation is complete
   c. Remove all wet or constrictive clothing and replace with dry, loose fitting garments
   d. Avoid allowing patient to walk on frostbitten extremities; if walking is necessary do not allow any rewarming to take place
   e. Cover frozen tissue with large dry bulky dressing and avoid manipulation or pressure on frozen part
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. If evacuation will be delayed, consider active rewarming with the approval of preceptor; do not rewarm if there is any potential for tissue to refreeze
      i. Rewarming is painful; consider morphine or meperidine (Demerol®) IV during rewarming process
      ii. Submerge injured part in warm water bath (99-102°) ideally without letting injured part touch sides or bottom of container
      iii. Continue to replenish warm water until skin is pliable, painful, and erythematous at most distal part of injury
      iv. Allow injury to air-dry; DO NOT towel dry as this may increase pain and tissue injury
      v. Keep blisters intact and protect from trauma; avoid manipulation or pressure on rewarmed part; do not allow blankets to apply pressure to rewarmed part
      vi. After rewarming, elevated extremity, put patient on bed rest, and keep injured part open to the air as much as possible

C. Prevention

1. Dress to maintain whole body warmth; use mittens instead of gloves as much as possible; make sure vulnerable areas such as face, neck, nose and corneas are covered; clothes should be adequately ventilated
2. Avoid perspiring in cold weather; activity should be slowed to prevent sweating
3. Keep toenails and fingernails trimmed
4. Avoid tobacco and alcohol in extreme environments, even in camp
5. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; recommend discontinuing use in extreme conditions
6. Make sure tetanus is up to date prior to deployment

VII. **HYPOTHERMIA-** *Condition from exposure to prolonged cold or being immersed in cold water which results in core body temperature dropping below 95°F.*

A. Signs and Symptoms

1. Mild- core temperature 90-95°. Tachypnea, tachycardia, ataxia, dysarthria, impaired judgment, shivering, urinating
2. Moderate- core temperature 82-90°. Slowing of pulse and breathing, loss of shivering, decreased level of consciousness, arrhythmias
3. Severe- core temperature below 82°. Pulmonary edema, coma, hypotension, bradycardia, arrhythmias, asystole

CLINICAL NOTE: Beware of vital signs inconsistent with temperature. Tachycardia or hyperventilation with low core temperature suggests hypoglycemia, hypovolemia, or overdose/poisoning.

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, O_2, circulation, monitor
   b. Remove wet/cold clothing and cover with warm/dry blankets
   c. Moderate and severely hypothermic patients are often hypovolemic; initiate IV and administer warm saline or D5W 500 mL-2 Liters
   d. Avoid administering Lactated Ringer's solution due to potential complications with potassium and lactate contents
   e. If above measures are insufficient, use warm packs to chest, neck, armpits and groin
   f. "Not dead till you're warm and dead"- if CPR becomes necessary, follow ACLS protocols including up to three rounds of shocks and medications; if patient doesn't respond, continue CPR until core temperature is above 90°F then renew attempts with ACLS shocks and medications per ACLS protocols
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

CLINICAL NOTE: Move carefully; rough handling may precipitate ventricular fibrillation. Keep patient in horizontal position to avoid aggravating hypotension and circulatory collapse. Remember, resuscitation efforts for hypothermia victims are not considered failed until the patient core temperature is above 95° F and there is no response.

C. Prevention

1. Dress to maintain whole body warmth; use mittens instead of gloves as much as possible; make sure vulnerable areas such as face, neck, nose and corneas are covered; clothes should be adequately ventilated
2. Avoid being wet in cool or temperate environments
3. Avoid perspiring in cold weather; activity should be slowed to prevent sweating
4. Ensure troops maintain adequate hydration and nutrition
5. Avoid tobacco and alcohol in extreme environments, even in camp
6. Many weight loss and muscle building supplements decrease the body's ability to compensate for environmental stressors; recommend discontinuing use in extreme conditions

Burns

ACTION ALERT: The following patients should be referred to a Burn Center:

1. Any deep partial thickness or full thickness burns of face, eyes, ears, hands, feet, genitalia, perineum, or major joints
2. Full thickness burns of more than 5% BSA in any age group
3. Partial or full thickness burns greater than 20% BSA in all age groups
4. Burns involving more than 10% BSA (partial thickness and full thickness) in patients older than 50 or younger than 10

Additional Burn Center referrals include:

1. Electrical burns (especially lightning)
2. Inhalation injuries with burns
3. Any circumferential burns
4. Concurrent burn and trauma
5. Pediatric burns
6. Victims with pre-existing medical problems
7. Chemical burns threatening disfigurement or loss of function

CLINICAL NOTE: As a quick rule of thumb, use the size of a patient's palm to measure 1% body surface area (BSA).

For larger burns, use the "Rule of Nines"- each of the following areas represents 9% of total body surface area:

1. Anterior of one leg
2. Posterior of one leg
3. One arm
4. Chest
5. Abdomen
6. Upper back
7. Lower back
8. Head

Genitalia = 1% total body surface area

VIII. **THERMAL AND RADIATION BURNS**- *Burns were previously categorized as 1st, 2nd, or 3rd degree, but now are classified by depth of injury to more accurately reflect which burns will need surgical intervention. When analyzing the burn, consider length of exposure, temperature of environment, thickness of skin at burn area, inhalation of gases at time of burn, what burned material was made of (plastics, chemicals, etc), patient age, and patient comorbidities such as diabetes or coronary artery disease.*

A. Signs and Symptoms

1. Superficial- dry, red, painful burn; blanches with pressure; peels after several days

2. Superficial partial thickness- blisters, moist, red, weeping, blanches with pressure, painful to temperature and air
3. Deep partial thickness- blisters, variable colors from cheesy white to red, waxy dry areas, does not blanch with pressure, sensation intact to pressure only
4. Full thickness- Waxy white to leathery grey to charred and black, dry and inelastic, sensation absent or intact to deep pressure only
5. Fourth degree- Extends into fascia and/or muscle, sensation absent or intact to deep pressure only

B. Treatment

1. **IMMEDIATE ACTION**
   a. Ensure safety of patient and rescue team; evacuate away from disaster if indicated; carefully remove any burning material and jewelry to avoid further injury
   b. Airway, breathing, O_2, circultion, monitor NOTE: pulse oximetry is NOT reliable in thermal burn patients due to possibility of carbon monoxide poisoning
   c. Treat any life threatening trauma (See "Emergency Protocols-Trauma")
   d. Immediately cool wounds with sterile water or saline-soaked gauze to prevent further tissue damage
   e. Continually reassess respiratory rate and depth; be prepared for abrupt airway closure necessitating intubation and/or cricothyroidotomy (will be more likely in patients with facial burns, neck burns, cough, wheeze, hoarseness, and/or charred material in sputum or when blowing nose)
   f. Obtain 2 large-bore IV's placed in unburned areas (if possible); give Lactated Ringer's or other crystalloid according to Parkland Burn Formula
   g. Administer single dose of morphine up to 10 mg IV if patient's pain is incapacitating
   h. Administer single dose of lorazepam (Ativan®, Temesta®) up to 4 mg IV if patient is severely anxious (do not mix in same IV line)
   i. Be prepared for severe respiratory depression requiring mechanical ventilation if both medications are administered; onset of respiratory depression may occur an hour or more after medication administration

ACTION ALERT: How to use the Parkland Burn Formula (using pounds):

To calculate total fluids that need to be administered over first 24 hours:

1.8 × patient's weight in pounds × percentage of body burned with 2nd or 3rd degree burns

Administer half the fluid in the first 8 hours, and half the fluid over the next 16 hours

EXAMPLE: 200 lb male with 2nd degree burns covering chest and abdomen (9% + 9% = 18%)

1.8 × 200 × 18 = 6480 mL in first 24 hours

Fluid quantity for first 8 hours = ½ (6480) = 3240 mL administered in first 8 hours
Fluid rate for first 8 hours = 3240 mL ÷ 8 hrs = 405 mL/hr or about 1 L every 2 ½ hours

Fluid quantity for remaining 16 hours = ½(6480) = 3240 mL administered over 16 hours
Fluid rate for remaining 16 hours = 3240 mL ÷ 16 hrs = 202.5 mL/hr or about 1 L every 5 hours

CLINICAL NOTE: Fluid under-resuscitation and over-resuscitation are equally dangerous. Not providing enough fluids leads to decreased organ perfusion, kidney failure, multi-organ system failure, and death. Giving too many fluids leads to acute respiratory distress syndrome, compartment syndrome, multi-organ system failure, and death. It is therefore CRUCIAL to calculate the correct amount of fluids to give (See "How to use the Parkland Burn Formula" above). Following the formula will usually suffice; monitor urinary output and blood pressure to ensure adequate perfusion. If patient is unable to urinate within 4 hours of starting IV therapy, or at any point if patient is orthostatic, CONTACT PHYSICIAN PRECEPTOR- CONSULTATION to consider fluid bolus or increase in IV fluid rate.

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
    a. Confirm fluid resuscitation requirements
    b. Notify if morphine and/or lorazepam were administered
    c. Burn patients may require extremely large doses of morphine; consider administering an additional 5 mg morphine every 10 minutes until pain is no longer incapacitating (Requires preceptor approval)
    d. Burn patients may be extremely anxious; consider administering additional 2 mg lorazepam every 20 minutes until anxiety is no longer incapacitating (requires preceptor approval)
    e. Monitor for airway compromise and reassess vitals every 10 minutes for at least two hours or until patient is stabilized (whichever is longer)
    f. Consider nasogastric tube and urinary catheter placement

g. Patients are at high risk for hypothermia; consider using warmed fluids and monitor core temperature with each reassessment

3. Debride and dress wounds
   a. Use a sterile dry sheet or dressing material if transport to burn center can occur in under an hour
   b. Apply silver sulfadiazine (Silvadene®) to all partial or full thickness burns that do not involve the face or genitalia
   c. Apply Polysporin® ointment to all partial or full thickness burns of the face or genitalia
   d. Dress all partial and full thickness burns with Xeroform Gauze®
4. Administer tetanus toxoid if last immunization is unknown or > 5 years; consider tetanus immunoglobulin

IX. **CHEMICAL BURNS**- *Treatment of chemical burns is dependent on the type of chemical involved.*

A. General Considerations

1. Remove all clothing, jewelry, gear, etc that may be potentially contaminated
2. Identify substance, if possible
3. Instruct non-medical assistant to locate MSDS data sheet and follow substance-specific treatment guidelines
4. Brush away all dry particles and powders
5. Unless specified below, immediately irrigate with copious amounts of high volume, low pressure warm water
6. Irrigate all contaminated skin surfaces continuously for a minimum of 20 minutes, even if "burning" sensation has resolved
7. Irrigate all eye contaminations a minimum of 30 minutes; alkali exposures will require several hours of irrigation

B. Decontamination Based on Type of Chemical

1. Dry lime or soda ash- brush away; don't get wet while particles are still present; water will increase corrosive potential and extent of injury
2. Phenol- absorb with sponges soaked in 50% polyethylene glycol; if polyethylene glycol is not available, use copious low-pressure irrigation
3. Lithium, Magnesium, and Sodium metals- Explosive when contacted with water; immerse particles in oil
4. Sulfuric acid- heats up when combined with water producing full-thickness burns; treat with soap or lime to neutralize prior to irrigation
5. White Phosphorus- burns in contact with air and can produce gas from patient contamination; brush dry particles away and then gently cleanse with solution of soap and water (phosphorus pentoxide is the off-gas. When hydrolyzed,(exposed to mucous membrane) it takes on the oxygen from the water and becomes phosphuric acid…making the off-gas extremely dangerous)

C. Treatment

1. **IMMEDIATE ACTION**
   a. Ensure safety of patient and rescue team; evacuate away from disaster if indicated; wear appropriate PPE based on contaminant to avoid injury during rescue effort

b. See "Environmental injury protocols, thermal burns" for burn treatment guidelines

D. Prevention

1. Be familiar with hazardous chemicals and their treatments in each work area
2. Ensure up-to-date MSDS safety sheets are immediately available in each work area where hazardous chemicals are used
3. Ensure toxin-specific treatments are available in work area where hazardous chemicals are used (example: polyethylene glycol in work areas using phenols)

X. **ELECTRICAL INJURIES**- *Severity of wound determined by type and amount of current exposure, duration of exposure, and whether or not current crossed vital organs. Injury patterns involve three aspects: direct effect of electricity on the functioning of the body, burn injuries associated with exposure, and mechanical trauma to the body due to muscle contractions (like the injuries seen in seizure patients) or falls.*

A. Signs and Symptoms

1. History of witnessed electrical injury or suspicion based on location patient is found is sufficient to consider electrical injury
2. Expose patient; observe for any entrance and exit wounds
3. Size of electrical wounds DOES NOT indicate severity of injury; small wounds often indicate deep burns and severe electrical effects
4. Assume multiple organ involvement
5. Assume head and neck trauma from seizure-like activity; assume major trauma from falls or being thrown by high voltage
6. Fixed, dilated pupils are a result of paralysis of the ocular muscles and NOT necessarily an indication of severe brain injury

B. Treatment

1. **IMMEDIATE ACTION**
   a. Ensure safety of patient and rescue team; evacuate away from disaster if indicated
   b. For multiple electrocution casualties, treat THOSE REQUIRING CPR FIRST (normally these patients would be considered "expectant" or "deceased" using standard mass casualty triage)
   c. Airway, breathing, O_2, circulation, monitor
   d. Be prepared for prolonged administration of CPR; patients are more likely to recover from electrical causes of cardiopulmonary arrest than other causes
   e. Treat any life threatening trauma (See "Emergency Protocols-Trauma")
   f. Immediately cool wounds with sterile water or saline-soaked gauze to prevent further tissue damage
   g. Continually reassess respiratory rate and depth; be prepared for abrupt airway closure necessitating intubation and/or cricothyroidotomy (will be more likely in patients with facial burns, neck burns, cough, wheeze, hoarseness, and/or charred material in sputum or when blowing nose)
   h. Obtain 2 large-bore IV's placed in unburned areas (if possible)

i. Victims of electrical burns are similar to patients with severe crush injuries and often need massive fluid resuscitation; DO NOT use Parkland Burn Formula® since this is likely to grossly underestimate needed fluids; DO NOT use any fluids containing potassium (such as Lactated Ringer's) since this will increase potential for cardiac arrhythmias and renal failure
j. Administer 2 Liters normal saline if available or may use normal saline; bolus the 2 Liters over one hour
k. Decrease fluid rate to 500 mL per hour and switch fluids to D5 ½ NS or as directed by preceptor after first hour
l. Administer single dose of morphine up to 10 mg IV if patient's pain is incapacitating
m. Administer single dose of lorazepam (Ativan®, Temesta®) up to 4 mg IV if patient is severely anxious (do not mix in same IV line)
n. Be prepared for severe respiratory depression requiring mechanical ventilation if both medications are administered; onset of respiratory depression may occur an hour or more after medication administration

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Confirm fluid resuscitation requirements
   b. Insert indwelling urinary catheter to measure urinary output; urine output must be 100 mL per hour or greater in the first two hours of resuscitation; if it is not; discontinue IV fluids
   c. Notify if morphine and/or lorazepam (Ativan®, Temesta®) were administered
   d. Monitor for airway compromise and reassess vitals every 10 minutes for at least two hours or until patient is stabilized (whichever is longer)
   e. Consider nasogastric tube placement
   f. Patients are at risk for hypothermia; consider using warmed fluids and monitor core temperature with each reassessment
3. Administer tetanus toxoid if last immunization is unknown or > 5 years; consider tetanus immunoglobulin

C. Prevention

1. Work with safety monitor; perform regular shop visits and correct any electrical safety hazards
2. Ensure all safety equipment is well maintained (no holes in electrical gloves, etc)

Altitude Illnesses

CLINICAL NOTE: The onset of altitude illness can occur as low as 6,000 feet. Twenty percent of cases occur when ascending to greater than 8,200 feet in less than one day. Ascents should be slowed to less than 2,000 feet per day when moving over 8,000 feet.

XI. **ACUTE MOUNTAIN SICKNESS (AMS)-** *Common syndrome caused by decreased oxygen availability to the tissues due to higher altitudes and compounded by inadequate acclimatization. NOTE: acute mountain sickness is the early stage of high altitude cerebral edema (HACE). Prompt recognition and treatment of acute mountain sickness is necessary to avoid potentially fatal progression to HACE.*

A. Signs and Symptoms

1. Risk factors include flying directly to higher altitude (such as ski resort cities), heavy exertion before acclimatization is complete, rapid ascent greater than 2,000 feet per day above 8,000 feet, and residence at low altitude
2. Headache within first 24 hours of reaching altitude
3. Fatigue, nausea, dyspnea, sleep disturbances, dizziness, mild shortness of breath with exertion
4. Check for pulmonary crackles and complete a full neurologic exam

B. Treatment

1. Mild cases should resolve spontaneously within 48 hours of reaching altitude
2. Symptoms should be easily reversed with administration of supplemental oxygen; if oxygenation does not improve symptoms, consider alternate diagnosis
3. Limit exercise or activity for 48 hours to aid with acclimatization
4. Patient must be monitored closely for progression to HACE over subsequent 48 hours
5. Headaches may be treated with aspirin or ibuprofen (Motrin®)
6. Nausea and vomiting may be treated with ondansetron (Zofran®)
7. Consider descending to lower altitude
8. CONTACT PHYSICIAN PRECEPTOR- CONSULTATION
9. Consider dexamethasone (Decadron®) or acetazolamide (Diamox®) per preceptor
10. Consider staying at higher altitude during the day and sleeping at lower altitude

C. Prevention

1. Proper mission planning should include gradual ascent and time to acclimatize at each change in elevation
2. Monitor new arrivals over first 48 hours
3. Proper hydration
4. Nutritious diet
5. Avoid vigorous exertion in first few days at new altitude
6. Alcohol, tobacco, sleeping medications, and many weight loss and muscle-building supplements interfere with the body's ability to respond to altitude stress; therefore all of these should be discontinued prior to journey to altitude

XII. **HIGH ALTITUDE CEREBRAL EDEMA (HACE)-** *Encephalopathy (brain swelling) that causes neurologic dysfunction and can be fatal. Symptoms are usually preceded by symptoms of acute mountain sickness. Prompt recognition and treatment is critical.*

A. Signs and Symptoms

1. Severe headache, irritability, ataxic (stumbling) gait, drowsiness, confusion, hallucinations, and coma progressing to death

2. Severity of illness often overlooked; can be mistaken for fatigue or exhaustion; patients will often minimize the severity of their symptoms and go somewhere alone and quiet
3. Stumbling gait is usually first symptom
4. Impaired coordination with finger-nose and heel-shin testing is common

B. Treatment

1. **IMMEDIATE ACTION**
   a. Prompt recognition is critical while patient is still ambulatory! Immediate descent is the key to survival
   b. Recommend bed rest, but if patient is in isolated environment and ambulatory, patient may assist in their own evacuation in order to make descent more rapid
   c. Airway, breathing, O_2, circulation, monitor; O_2 administered at 2-4 LPM via nasal cannula is usually sufficient and can be lifesaving as long as HAPE is not present
   d. If not ambulatory, obtain IV access and administer normal saline at maintenance rate
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Hyperbaric therapy if immediate descent is not possible

ACTION ALERT: At first sign of ataxia or mental status change, descent should be started. Do not delay descent while considering alternative diagnoses or attempting to reach preceptor.

C. Prevention

1. Proper mission planning should include gradual ascent and time to acclimatize at each change in elevation
2. Monitor new arrivals over first 48 hours
3. Proper hydration
4. Nutritious diet
5. Avoid vigorous exertion in first few days at new altitude
6. Alcohol, tobacco, sleeping medications, and many weight loss and muscle-building supplements interfere with the body's ability to respond to altitude stress; therefore all of these should be discontinued prior to journey to altitude

XIII. HIGH ALTITUDE PULMONARY EDEMA (HAPE)- *The abnormal collection of fluid in the lung due to a breakdown in the lung's blood-gas barrier from exposure to decreased atmospheric oxygen levels.*

A. Signs and Symptoms

1. Onset is usually subtle with mild nonproductive cough, mild shortness of breath, and difficulty walking uphill occurring within first week of being at new altitude (similar to all persons trying to acclimatize)
2. Progresses to dyspnea at rest, which is how the IDMT can tell when interventions are necessary; pulse oximetry readings typically 10 points below expected value for altitude; patients typically look better than expected

3. Approximately 50% of cases have signs and symptoms of acute mountain sickness
4. Tachycardia, tachypnea, low grade fever, and crackles initially in the right middle lobe (listen at right lateral mid-chest) and progresses to all lobes
5. Untreated, can progress to cough productive of pink/frothy sputum, frank hemoptysis, cyanosis, neurologic dysfunction, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Bed rest, remove from cold environment
   b. Prompt recognition is critical! Patient's initial symptoms will be subtle
   c. Airway, breathing, O_2, circulation, monitor; provide enough oxygen to maintain O_2 saturation greater than 90% by pulse oximetry
   d. Have patient exhale through pursed lips on expiration to increase expiratory pressure in lungs
   e. If patient is unconscious, consider EPAP (expiratory positive airway pressure) or CPAP (continuous positive airway pressure) if equipment available to do so
   f. If not ambulatory, obtain IV access and administer normal saline at maintenance rate
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Hyperbaric therapy if immediate descent is not possible

Barotrauma

XIV. **BAROTRAUMA-** *Barotrauma is tissue injury resulting from pressure changes relative to gas-containing structures of the body. Barotrauma can affect the lungs, ears, sinuses, GI tract and tooth fillings. Most barotrauma-related injuries resolve spontaneously and require only symptomatic, outpatient treatment.*

A. Signs and Symptoms

1. Ear fullness and pain during descent
2. Patient may complain of tinnitus, which may be debilitating
3. Middle ear hemorrhage and TM rupture is possible
4. Decreased TM mobility
5. Positive vestibular tests (Romberg, nystagmus) if oval window is affected

B. Treatment

1. Have patient attempt autoinflation by swallowing, yawning or Valsalva maneuver
2. Administer systemic decongestants such as pseudoephedrine (Sudafed®)
3. Administer topical decongestants such as oxymetazoline hydrochloride (Afrin®) 2 sprays into each nostril, BID no more than 3 days
4. Administer analgesics for pain such as acetaminophen (Tylenol®) or ibuprofen (Motrin®)

XV. **BAROTRAOUMA, SINUS-** *Barotrauma is tissue injury resulting from pressure changes relative to gas-containing structures of the body. Barotrauma can affect the lungs, ears,*

sinuses, GI tract and tooth fillings. Most barotrauma-related injuries resolve spontaneously and require only symptomatic, outpatient treatment.

A. Signs and Symptoms

1. Most often affects the frontal sinuses
2. Mild pressure, pain, and tenderness
3. Hemorrhage, epistaxis

B. Treatment

1. Have patient attempt auto-inflation by swallowing, yawning or Valsalva maneuver
2. Administer systemic decongestants such as pseudoephedrine (Sudafed®)
3. Administer topical decongestants such as oxymetazoline hydrochloride (Afrin®) 2 sprays into each nostril, BID no more than 3 days
4. Administer analgesics for pain such as acetaminophen (Tylenol®) or ibuprofen (Motrin®)

XVI. BAROTRAUMA, DENTAL- *Barotrauma is tissue injury resulting from pressure changes relative to gas-containing structures of the body. Barotrauma can affect the lungs, ears, sinuses, GI tract and tooth fillings. Most barotrauma-related injuries resolve spontaneously and require only symptomatic, outpatient treatment.*

A. Signs and Symptoms

1. Occurs when a patient with infected tooth or recently repaired tooth is subjected to pressure changes
2. Tenderness with percussion (tooth rake)

B. Treatment- See "Dental Protocols, Barotrauma- Dental"

XVII. BAROTRAUMA, GASTROINTESTINAL- *Barotrauma is tissue injury resulting from pressure changes relative to gas-containing structures of the body. Barotrauma can affect the lungs, ears, sinuses, GI tract and tooth fillings. Most barotrauma-related injuries resolve spontaneously and require only symptomatic, outpatient treatment.*

A. Signs and Symptoms

1. Caused by swallowing air during a dive
2. Cramps, abdominal fullness, pain, belching and flatulence
3. Visceral organ rupture is characterized by intense pain, positive rebound tenderness, and guarding

B. Treatment

1. **IMMEDIATE ACTION**
   a. Aggressive fluid resuscitation
   b. Do not give anything PO—no oral medications, fluids, etc.
   c. Keep patient in position of comfort and minimize abdominal exams/manipulation
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Initiate broad spectrum antibiotics IV per preceptor instructions and availability
   b. Immediate surgical consultation

XVIII. BAROTRAUMA, PULMONARY- *Barotrauma is tissue injury resulting from pressure changes relative to gas-containing structures of the body. Barotrauma can affect the*

lungs, ears, sinuses, GI tract and tooth fillings. Most barotrauma-related injuries resolve spontaneously and require only symptomatic, outpatient treatment.

A. Signs and Symptoms

1. Caused by expanding gas during dive ascent or rapid decompression at altitude
2. Alveolar rupture/pneumothroax causing dyspnea, chest pain and decreased breath sounds
3. Pneumomediastinum—Sensation of fullness, neck pain, pleuritic chest pain, dyspnea, coughing and hoarseness with subcutaneous emphysema ("Rice Krispies®" under skin)
4. Positive Hamman's sign (mediastinal crunch or rasping that occurs with the heartbeat)
5. May result in arterial gas embolism

B. Treatment

1. **IMMEDIATE ACTION**
   a. 100% O_2 via non-rebreather mask
   b. Patients displaying neurologic signs or those with pulmonary/GI complaints should be immediately evacuated to a recompression chamber for treatment
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Patients displaying neurologic signs or those with pulmonary/GI complaints should be immediately evacuated to a recompression chamber
   b. If recompression will be delayed, consider other compression options such as repeat dive
3. Altitude restriction during evacuation should be considered

Chapter 16: Occupational Exposure Protocols

I. **LEAD POISONING**- *Leaded paint was commonly used in paint products distributed in the U.S. before 1960 and is still used in developing countries. Exposure occurs during remodeling projects or accidental ingestion of chipped paint. Some ceramic glazes also contain lead and pose an ingestion hazard when used serving food or drink (particularly when used with acidic food products like fruit juice, tomato products and wine from leaded crystal). In manufacturing, lead is used in glass making, bronzing, welding, brass manufacturing, and leaded gasolines. Military occupations that may involve exposure to lead include plumbers, electricians, mechanics, shipbuilders, printers, police officers, steel workers, and firing range instructors.*

A. Signs and Symptoms

1. Acute poisoning- abdominal pain, constipation, tremors, mood changes, joint pain, muscle aches, headache, anorexia, difficulty concentrating, decreased libido, short-term memory loss
2. Chronic exposure- Personality changes (anxiety and hostility), headache, myalgias, fatigue, irritability, insomnia, anorexia, difficulty concentrating, and short-term memory loss, decreased sperm count
3. Chronic low-level exposure- cataracts, hearing loss, tooth loss, kidney cancer
4. Patients develop anemia because lead interferes with the body's production of hemoglobin
5. Diagnosis is made by history of exposure and increased lead levels in the blood.

B. Treatment

1. **IMMEDIATE ACTION (Acute poisoning)**
   a. Eliminate source of contamination
   b. Check others in work area for similar symptoms
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Perform gastric lavage and irrigation of the digestive tract using polyethylene glycol if directed by preceptor
   b. Lab: CBC, lead levels, CHEM 7, peripheral smear
   c. Patients with lead exposure require heavy metal chelating drugs; this therapy should not be administered by the IDMT
3. **PREVENTION**
   a. At risk populations should be screened for lead annually
   b. Tobacco products can contain low levels of lead; recommend smoking cessation
   c. Personnel should use appropriate PPE while at work and change clothing and footwear prior to returning home
   d. Personnel should shower prior to sleep
   e. Personnel should not be allowed to eat or smoke in or near their work area; all personnel must wash hands prior to eating or smoking to reduce chance of ingestion

II. **CARBON MONOXIDE POISONING (CO POISONING)**- *Carbon monoxide is primarily released as a byproduct of incomplete combustion. Carbon Monoxide is odorless, tasteless, and has twice the affinity for hemoglobin than oxygen. This characteristic poses a serious hazard to exposed individuals because it tightly binds the hemoglobin and thus prevents*

hemoglobin from carrying oxygen to the tissues. Sources include acetylene gas, automobile exhaust, fossil fuel combustion, furnace gas, illuminating gas and marsh gas. Exposures most commonly occur when motorized machines, electrical generators, or gas heaters are used in closed environments such as vehicles in a parking garage or warehouse, kerosene heaters indoors, or charcoal grills in the garage.

A. Signs and Symptoms

1. Variable depending on length of exposure and concentration of the inhaled gas
2. Headache, exertional shortness of breath, dizziness, nausea/vomiting, drowsiness, dry mouth
3. Bright red/rosy cheeks and lips, bounding pulse, hypertension, tachypnea, dyspnea
4. Often mistaken for viral syndrome- key differences are drowsiness and pupil dilation
5. Progresses to confusion, syncope, visual disturbances, muscle fasciculations (twitching), seizures and coma (Death occurs when carboxyhemoglobin blood levels are >60%)

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, 100% O_2 via tight fitting aviator's mask, monitor
   b. Hyperbaric O_2 if available
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**

C. Prevention

1. Smokers have up to 20% carboxyhemoglobin levels at rest; recommend smoking cessation
2. Check CO levels in shops or living areas where CO is suspect and ventilation is poor
3. Install CO detectors in living/working areas

III. **BENZENE**- *Benzenes include toluene, xylene, benzol, and toluol. These toxins are common in household items such as plastic adhesives and glue. They are also found in industrial fluids such as JP-8, Avgas, etc. These compounds are also frequently abused by "huffing" due to the euphoric effects at lower doses.*

A. Signs and Symptoms

1. Acute and chronic exposure (ingestion, inhalation, or absorption) can all cause dizziness, weakness, headache, euphoria, nausea and vomiting, ventricular arrhythmias, ataxia, blurred vision, paralysis and seizures
2. Acute exposure often presents with fever up to 104°F
3. If route was inhalational, may have evidence of pulmonary injury such as coughing, choking, gagging, tachypnea, dypsnea, cyanosis, wheezing, nasal flaring, or grunting respirations
4. Chronic, long term exposures cause aplastic anemia, hypokalemia, leukemia and CNS depression

B. Treatment

1. **IMMEDIATE ACTION**
   a. Decontaminate patient with water and prevent aspiration of vomitus
   b. Airway, breathing, circulation, O_2, monitor

ACTION ALERT: Ventricular fibrillation can occur quickly due to hypersensitive myocardium. **EPINEPHERINE SHOULD BE AVOIDED** unless directed by your Preceptor.

c. Initiate IV with normal saline at maintenance rate; bolus as necessary
d. If seizures occur, administer lorazepam (Ativan®) 4 mg IV/IM or diazepam (Valium®) 5 mg IV
e. Keep patient NPO

2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. For severe anemia, consider walking blood bank/transfusion
   b. Replace K (give "banana bag" – 10 mL liquid multivitamins + 100 mg thimine + 1 mg folic acid mixed in 1 Liter normal saline) if available

C. Prevention

1. Personnel should use appropriate PPE while at work and change clothing and footwear prior to returning home
2. Personnel should shower prior to sleep
3. Personnel should not be allowed to eat or smoke in or near their work area; all personnel must wash hands prior to eating or smoking to reduce chance of ingestion

IV. **HYDROCARBONS (PETROLIUM DISTILLATES)-** *Poisoning results from ingestion or inhalation that can result in aspiration pneumonitis. Hydrocarbons include asphalt, benzenes, gasoline, kerosene, lubricating oils, paint thinner and tar. Ingestion or inhalation occurs most often while attempting to siphon a liquid hydrocarbon from one container to another. These compounds are also frequently abused by "huffing" due to the euphoric effects at lower doses.*

A. Signs and Symptoms

1. Initially, patients cough, choke, and/or vomit
2. Patients complain of epigastric burning sensation, diarrhea and pneumonia if aspiration has occurred
3. Aspiration pneumonitis causes hypoxia and respiratory distress
4. For vapor inhalation, patients complain of chest pain/burning, headache, nausea, weakness, CNS depression, confusion, dyspnea, tachypnea and rales

B. Treatment

1. CONTACT PHYSICIAN PRECEPTOR- EVACUATION
2. Gastric lavage is usually not necessary for accidental ingestion
3. Provide supportive care for pulmonary edema, O_2 and respiratory support as necessary

C. Prevention

1. Ensure proper equipment for siphoning is present where needed
2. Personnel should use appropriate PPE while at work and change clothing and footwear prior to returning home
3. Personnel should shower prior to sleep
4. Personnel should not be allowed to eat or smoke in or near their work area; all personnel must wash hands prior to eating or smoking to reduce chance of ingestion

V. **ORGANOPHOSPHATE POISONING-** *Organophosphates include many pesticides, insecticides, herbicides and chemical weapons (nerve agents and defoliants). In addition to exposures to weaponized nerve agents, deployed personnel may also be exposed due to unregulated use/distribution of these products in under developed countries. Organophosphates cause extreme stimulation of secretions and smooth muscle. Poisoning can occur through inhalation, absorption or ingestion. NOTE: This section covers organophosphate poisoning in general and does not review standing orders/instructions for response to casualty exposure from chemical weapons.*

A. Signs and Symptoms

1. Excessive salivation, lacrimation (tearing), sweating, and rhinorrhea
2. Nausea, abdominal cramping, diarrhea, urination, vomiting
3. Increased pulmonary secretions, frothing from the mouth
4. Blurred vision, miosis, slurred speech, confusion, headache
5. Difficulty breathing, seizures, coma, and death

B. Treatment

1. **IMMEDIATE ACTION**
   a. Airway, breathing, O_2, monitor, consider endotracheal intubation
   b. Remove clothing, brush off dry chemical and flush with copious amounts of water
   c. Use M-291 kit, 0.5% hypochlorite solution, or soap and water to decontaminate skin
   d. Initiate large-bore IV, bolus 1 Liter normal saline or Lactated Ringer's
   e. Administer atropine 5 mg IV/IM for adults; repeat every 5 minutes until excessive secretions stop; may require HUNDREDS of mg of atropine over several days
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
   a. Consider pralidoxime (2-PAM®) 1-2g IV over 15-30 minutes if directed by Preceptor
   b. Consider diazepam (Valium®) 5 mg IV if seizures occur or if directed by Preceptor

C. Prevention

1. Wear PPE when handling/disposing of known organophosphates
2. Detectors include:
   a. M256A1 chemical agent detector kit NSN 6665-01-133-4964
   b. Improved Chemical Agent Alarm (ICAM) NSN 6665-01-199-4153
   c. M-8 paper NSN 6665-00-050-8529
   d. M-9 paper NSN 6665-01-049-8982
   e. M-272 chemical agent water testing kit NSN 6665-01-134-0885
3. Healthcare workers should wear appropriate PPE when decontaminating patients
4. Personnel should use appropriate PPE while at work and change clothing and footwear prior to returning home
5. Personnel should shower prior to sleep
6. Personnel should not be allowed to eat or smoke in or near their work area; all personnel must wash hands prior to eating or smoking to reduce chance of ingestion

VI. **RADIATION SYNDROME**- *Radiation is the process in which energy is emitted from one source and absorbed by another. Radiation is classified as either ionizing or non-ionizing. Ionization is the removal of electrons from an atom. Ionizing radiation is any high energy radiation or any particle having the potential energy to remove electrons from another atom. Examples of ionized radiation include alpha, beta, photon, neutron and x radiation. Ionizing radiation is measured in rads, Becquerel (Bq), grays (Gy), and Sieverts (Sv)*

Low energy radiation [non-ionizing, or radio frequency radiation (RFR)] does not have enough energy to produce ions in other substances, but it does have enough energy to cause vibration, heat, and coagulation in protein. Examples of non-ionizing radiation include radio waves, microwaves, infrared waves and ultraviolet waves. RF radiation is measured in MHz. Typical injuries from non-ioninzing radiation sources include skin and ocular burns.

A. Signs and Symptoms

1. Hematopoietic Syndrome
    a. Exposure of 100-500 rad
    b. Nausea, vomiting and fatigue
    c. Leukopenia (50% drop in 24 hours =lethal dose)
    d. Can progress to death within a few weeks to months
2. Gastrointestinal Syndrome
    a. 600-1200 rad
    b. Nausea, fever, vomiting, fatigue, diarrhea and abdominal cramps
    c. Diarrhea may be bloody
    d. Cardiovascular collapse, kidney failure and shock from fluid shifts
    e. Can progress to death within 2 weeks
3. Neurovascular Syndrome
    a. >1500 rad
    b. Nausea, Vomiting, neurologic deficits within 1 hour
    c. Diarrhea, ARDS, and prostration within 4-5 hours
    d. Death within 2 days
4. Chronic exposure
    a. Thyroid cancer, leukemia, growth retardation, cataracts
    b. Infertility, fetal abnormalities

B. Treatment

1. **IMMEDIATE ACTION**
    a. Airway, breathing, O_2, circulation, monitor
    b. Decontaminate if alpha/beta particulates are suspect
    c. Treat all traumatic injuries (See "Emergency Medicine Protocols, Trauma")
2. **CONTACT PHYSICIAN PRECEPTOR- IMMEDIATE EVACUATION**
3. Provide supportive care as necessary

C. Prevention

1. Do not become a casualty. Ensure radiation-emitting equipment is powered down
2. Wear proper PPE when required
3. Operate within safe working distances as defined by the Technical Order appropriate for the weapon system

4. The IDMT should familiarize his/her self with the hazards associated with the assigned weapon system. Antenna arrays, radar, x-ray equipment, aircraft antennas, and anti-armor munitions (not to mention nuclear weapons) all emit radiation capable of injury or death. Understanding the occupational risks in the workplace is the first step in appropriate patient care
5. See HMTF BEE for site-specific information on workplace hazards

Chapter 17: Basic Laboratory Information

I. **MACROSCOPIC URINE ANALYSIS (URINE DIPSTICK)** - *A rapid and non-invasive test that assesses the genitourinary system, hydration status, and metabolic abnormalities.*

A. Specimen Handling

1. Sample should be fresh, well mixed, and room temperature for examination
2. Unless otherwise specified, patient should be instructed in proper cleansing technique and collect clean midstream urine
3. May use same urine sample for urine HCG sampling, but may not use with gonorrhea/chlamydia testing since G/C sample should be from first portion of void and patient should not cleanse beforehand

B. Color (normal: pale yellow, clear)

1. Red/brown- In young, healthy active duty population usually indicates blood. Red urine may also come from ingestion of food dyes, beets, or fava beans. IDMT should consider whether blood is coming from a leaky kidney or from trauma to the urinary system somewhere between the kidney and the urethra. Common causes include kidney stones, rhabdomyolysis, and pyelonephritis. Common drugs that may cause this color change include metronidazole (Flagyl®) and nitrofurantoin (Macrobid®)
2. Orange- May be related to taking bladder analgesic phenazopyridine (Pyridium®) used in treating urinary tract infections. May indicate metabolic conditions such as diabetes or hypothyroidism. May be related to ingestion of B-vitamins, beta-kerotenes (carrots), blackberries, beets, or rhubarb
3. Green- may indicate food dyes, drinking mouthwash, eating asparagus
4. Black- ingestion of stimulant laxatives such as senna (in Ex-Lax®), melanoma in the bladder

C. pH- (normal: 5.5-6.5; range 4.5-8.0)

1. Alkali (high pH)- may indicate low carb or vegetarian diet, aspirin overdose, systemic alkylosis, or urinary tract infections by most common bacteria
2. Acid (low pH) may indicate high protein diet or starvation, diarrhea illnesses, diabetes, or systemic acidosis

D. Specific Gravity (normal: 1.002-1.030g/mL)

a. Low specific gravity- excessive water intake, kidney damage
b. High specific gravity- dehydration, metabolic abnormalities

E. Leukocytes/white blood cells/WBC (normal: none)

1. Urine should be sterile (without bacteria or white blood cells)
2. Presence of white blood cells usually indicative of contaminated sample (poor clean catch technique)
3. White blood cells may also be present in patients with urinary tract infection, but their presence is not diagnostic
4. False positive caused by trichomonas or dyes that stain urine red

F. Leukocyte Esterase (normal: none)

1. Comes from damaged or destroyed white blood cells
2. Indicates possible urinary tract infection

G. Nitrites (normal: none)

1. Come from presence of pathologic bacteria converting dietary nitrates to nitrites; especially in presence of leukocyte esterase, is diagnostic for urinary tract infection
2. False negative can occur if urine has not been in bladder long enough for bacteria to cause conversion (patient provides sample too soon after last void)

H. Urobilinogen (normal: 0.2-1.0 mg/dL)

1. Elevated in some liver diseases, hemolytic anemia, malaria, and mononucleosis
2. Low in some liver diseases

I. Blood (normal: none)

1. Measures intact hemoglobin only. Positive result caused by trauma, infection, inflammation, kidney stones, vascular red blood cell damage (intravascular hemolysis)
2. Red, bloody appearance of urine in the absence of +blood on urine dipstick indicates probable presence of myoglobin; occurs in rhabdomyolysis, crush injuries, and electrocution

J. Bilirubin (normal: none)

1. Elevated in liver disease, and different forms of hepatitis or gall bladder disease. May also be caused by pancreatitis.
2. False positives seen in patients taking high doses of vitamin C

K. Protein (normal: none)

1. Kidney damage such as pyelonephritis or nephrotic syndrome destroys the filter ability of the kidney and allows proteins from the blood to leak into the urine
2. May also be caused by electrical injury, gout, or low blood potassium
3. Drugs that may cause elevations in urine protein include aminoglycoside, penicillin and sulfa antibiotics and NSAIDs

L. Ketones (normal: none)

1. May indicate high fat/high protein diet, starvation, alcoholism, or metabolic diseases such as diabetes or hyperthyroidism
2. Drugs that cause ketones in the urine include antibiotics used for treatment of TB such as isoniazid

M. Glucose (normal: none)

1. Most associated with diabetes, but may be associated with other metabolic abnormalities of the adrenal gland or thyroid gland or diseases of the liver
2. May be seen in pregnancy (suggestive of diabetes of pregnancy)
3. May be caused by multiple classes of antibiotics and steroids
4. False positives seen in patients taking high doses of vitamin C and when urine has high specific gravity and/or ketones

II. BASIC METABOLIC PANEL ("BMP", "CHEM 7")- *A blood test that gives basic information about the current status of the kidneys, blood sugar, electrolyte balance, and acid/base balance. Abnormalities and combinations of abnormal results can help explain abnormalities found on physical exam or in vital signs.*

A. Sodium (normal 135-145 mEq/dL)

1. <u>In the body</u>
   a. Positive ion (cation), 85% found in blood and lymph

b. Indicates volume status of the patient
c. Regulated by kidney and adrenal gland via aldosterone
d. Used in nerve impulse transmission and water balance regulation

2. Interpretation
   a. Increased in: dehydration, steroids, contraceptives, licorice, overhydrating with IV normal saline
   b. Decreased in: excessive sweating/sweating and replacing water without replacing salt, blood pressure meds, heart failure
   c. Sodium falls about 1.6 mmol/dL for every 100 mg/dL increase in glucose

B. Potassium (normal: 3.5-5.3 mEq/dL)

1. In the Body
   a. Positive ion (cation) regulated by renal excretion
   b. Critical for nerve impulse transmission and muscle contraction
2. Interpretation
   a. Increased in: tissue damage, rhabdomyolysis, dehydration, dietary intake, blood pressure medicines
   b. Decreased in: low dietary intake, prolonged vomiting or diarrhea, hyperglycemia
   c. Often artificially elevated due to poor handling of sample causing lysis of red blood cells (sometimes from prolonged fist clench during draw, or tourniquet on too long during draw)

C. Chloride (normal: 98-107 mEq/L)

1. In the Body
   a. Negative ion (anion) maintains normal acid-base balance
   b. Increased levels cause acidosis, decreased levels cause alkylosis
2. Interpretation
   a. Increased in: renal failure, nephrotic syndrome, dehydration, overtreatment with IV normal saline, salicylate (aspirin) toxicity
   b. Decreased in: vomiting, diarrhea, overtreatment with blood pressure medications, excessive sweating/sweating and replacing water without replacing salt, laxative overuse, burns

D. Carbon Dioxide/Bicarbonate (CO_2/HCO_3^-) (normal: 21-28 mEq/L)

1. In the Body
   a. Carbon dioxide is a waste product from oxygen consumption by tissues, but in the body is found primarily in the form of bicarbonate ion
   b. Levels are primarily regulated by the kidneys by urinating out bicarbonate, but also by the lungs through respiration
2. Interpretation
   a. Increased in: severe vomiting, hypoventilation
   b. Decreased in: chronic diarrhea, starvation, diabetic ketoacidosis, shock

E. Blood Urea Nitrogen (BUN) (normal: 10-20 mg/dL)

1. In the Body
   a. Urea is formed in the liver from ammonia, which is a byproduct of protein metabolism

b. BUN is urea moving from liver where proteins were broken down to the kidneys, where urea is excreted
c. Measures both liver's ability to break down protein and kidney's ability to secrete and filter

2. Interpretation
a. Elevated in: impaired renal function (shock, dehydration, etc), urinary tract obstruction, large protein breakdown such as supplement users; starvation due to muscle breakdown
b. Liver failure (can't break down protein), malabsorption/malnutrition

F. Creatinine (normal: 0.4-1.5 mg/dL)

1. In the Body
a. Breakdown product of skeletal muscle contraction
b. Is produced at a constant rate based on muscle mass of individual
c. Easily excreted by kidneys
d. Primarily reflects filter capability of the kidney; if produced at a constant rate, should be filtered at a constant rate unless filter is broken

2. Interpretation
a. Increased in: kidney disease, pyelonephritis, rhabdomyolysis, urinary tract obstruction, massively increased muscle mass
b. Decreased in: decreased muscle mass, debilitation

G. Glucose (normal; 65-110 mg/dL)

1. In the Body
a. Comes from carbohydrate digestion and conversion of glycogen to glucose by liver
b. Controlled by pancreatic hormones glucagon (raises blood sugar by increasing glycogen to glucose) and insulin (drops blood sugar by enabling sugar to cross from bloodstream into cells)

2. Interpretation
a. Increased in: Severe stress (infection, burns, surgery), pancreatitis, steroid therapy or abuse, diabetes mellitus
b. Decreased in: insulin overdose, starvation, hypothyroidism

III. COMPLETE BLOOD COUNT ("CBC")- *Measures the numbers of red blood cells, white blood cells, and platelets in the blood as well as some indicators of the health of these cells.*

A. Red Blood Cell Count ("RBC") (normal: males 4.6-6.2, females 4.2-5.4)

1. In the Body
a. Total number of red blood cells in an mL of blood
b. Formed in bone marrow and have a lifespan of 120 days
c. Females tend to have lower counts than males partly due to female menstruation
d. Counts 10% or more below normal are called "anemia"

2. Interpretation
a. Increased in severe diarrhea, burns
b. Decreased in: anemia, hemorrhage, blood transfusion with incompatible blood, severe reactions to drugs

B. White Blood Cell Count ("WBC", "Leukocytes") (normal: 5,000-10,000)

1. In the Body

a. Fights infections throughout the body
b. Lymphocytes primarily attack viruses
c. Neutrophils primarily attack bacteria
d. Eosinophils attack parasites and are present in allergies

2. Interpretation
   a. Increased in: infection, trauma, stress, hemorrhage, cancers, dehydration, steroid use
   b. Decreased in: drug toxicity (chemotherapy), overwhelming infection, dietary deficiency such as B12 or iron, enlarged spleen (spleen pulls WBCs out of blood stream)

C. Hemoglobin ("Hgb") (normal: males 13.5-18 g/dL, females 12-16 g/dL)

1. In the Body
   a. The oxygen and carbon dioxide carrying component of a red blood cell
   b. The oxygen carrying capacity is determined by the hemoglobin concentration, rather than the RBC count
2. Interpretation
   a. Increased in severe diarrhea, burns
   b. Decreased in: anemia, hemorrhage, blood transfusion with incompatible blood, severe reactions to drugs

D. Hematocrit ("Hct") (normal: males 40-54%, females 38-47%)

1. In the Body
   a. The percentage of total blood volume that is made up by the RBC's
   b. Normally about 3 times the hemoglobin concentration
2. Interpretation
   a. Increased in severe diarrhea, burns
   b. Decreased in: anemia, hemorrhage, blood transfusion with incompatible blood, severe reactions to drugs

E. Platelets ("thrombocyte count") (normal: 150,000-400,000)

1. In the Body
   a. Essential for blood clotting
   b. Survive 7-10 days
   c. 75% usually in blood stream, 25% in liver and spleen
2. Interpretation
   a. Increased in: some cancers, rheumatoid arthritis, iron deficiency anemia
   b. Decreased in: enlarged spleen, hemorrhage, hemolytic anemia, infection

IV. TESTING FOR SEXUALLY TRANSMITTED INFECTIONS

- This section reviews the common laboratory tests used to screen for different genital infections. Note that many genital infections, including Molluscum contageosum and genital warts, have no laboratory tests associated with them. Many infections, such as chancroid and herpes, can only be tested if active lesions are present. Patients often come in requesting sexually transmitted infection checks "for everything" after unprotected sexual encounters. It is CRITICAL to educate them that although there are lab tests available for some infections, there are many more infections than those that have suitable screening tests. Clinic visits for infection checks are NOT a substitute for condom use or abstinence.

A. Human Immunodeficiency Virus (HIV) (blood)

1. Blood test
2. Screening test is usually rapid HIV or HIV enzyme immunoassay
3. If patient is having upper respiratory symptoms, fever, or other signs of HIV seroconversion, order HIV RNA test instead (check with local lab)

B. Gonorrhea (urine or probe swab)

1. Multiple collection methods possible (test includes chlamydia)
    a. Can be tested on "dirty void" of first morning urine sample- have patient collect the very first urine produced and don't cleanse beforehand, rather than the more common collection method of midstream clean catch
    b. Can be tested on dirty voids later in the day, but higher likelihood of false negative test
    c. Can be tested by urethral swab or endocervical swab; but higher likelihood of false negative test if patient has voided recently
2. If ordering urinalysis for the same patient, be aware of logistical complications to collect both specimens
    a. Performing swab test before urine analysis may introduce blood into the urine sample
    b. Performing urinalysis before gonorrhea swab may wash away infected sample
    c. Performing both urinalysis and gonorrhea on urine requires patient void dirty urine into one cup and clean midstream urine into another cup; has high risk of getting samples mislabeled even if performed correctly

C. Chlamydia (urine or probe/swab)

1. Multiple collection methods possible (test includes gonorrhea)
    a. Can be tested on "dirty void" of first morning urine sample- have patient collect the very first urine produced and don't cleanse beforehand, rather than the more common collection method of midstream clean catch
    b. Can be tested on dirty voids later in the day, but higher likelihood of false negative test
    c. Can be tested by urethral swab or endocervical swab; but higher likelihood of false negative test if patient has voided recently
2. If ordering urinalysis for the same patient, be aware of logistical complications to collect both specimens
    a. Performing swab test before urine analysis may introduce blood into the urine sample
    b. Performing urinalysis before chlamydia swab may wash away infected sample
    c. Performing both urinalysis and chlamydia on urine requires patient void dirty urine into one cup and clean midstream urine into another cup; has high risk of getting samples mislabeled even if performed correctly

D. Trichomonas (wet mount of vaginal secretions)

1. For females, performing a wet mount on vaginal secretions may show mobile trichomonads

2. Culture of penile discharge or vaginal discharge is the test of choice, but these tests are not always available and take 7 days to get a result
3. New GC commercial testing kits are being developed that will include trichomonas, but these are not yet available in the US

E. Syphilis (blood)

1. Rapid plasma reagin testing is most commonly used in the military
2. RPR testing is relatively quick and inexpensive, but has high false positive rates; positive tests should automatically be confirmed by treponemal testing; if a positive result comes back to the IDMT, double check to make sure it has already had treponemal testing

F. Herpes (genital lesion culture)

1. Although blood tests do exist, blood testing is primarily used to identify high-risk carriers of herpes (such as pregnant women during routine prenatal visits)
2. At this time, there are no recommended routine screening tests for the asymptomatic patient requesting sexually transmitted infection screening
3. For patients with active lesions, unroof a vesicle and collect fluid on viral culture swab for diagnosis

G. Hepatitis (blood)

1. Hepatitis B and C are transmitted through blood and genital fluids; a screening blood test is available
2. Hepatitis A is primarily a foodborne illness (fecal-oral transmission) but is often included in sexually transmitted infection panels due to other potential for fecal-oral spread

H. Human papillomavirus (cervical swab; "Pap smear")

1. Different serotypes of human papillomavirus are responsible for genital warts and for triggering cervical cancer
2. No good screening test exists for males; no good screening test exists for any of the serotypes associated with genital warts
3. Cervical swab specimens are tested for abnormal cells ("Pap smear") and if abnormal cells are identified, specimen is evaluated for presence of HPV
4. Although HPV infections can progress to invasive cervical cancer, in adolescent patients pap screening is no longer recommended due to the high incidence of spontaneous resolution of the infection; pap screening is now recommended starting at age 21

I. Common Order Sets (every preceptor is different)

1. <u>Asymptomatic male screening for sexually transmitted infection</u>
   a. HIV
   b. RPR
   c. GC
   d. Hepatitis B/C or hepatitis A/B/C
   e. Viral culture if lesion is present
   f. Bacterial culture if lesion is present
2. <u>Asymptomatic female screening for sexually transmitted infection</u>
   a. HIV
   b. RPR
   c. GC
   d. Hepatitis B/C or hepatitis A/B/C

   e. HCG
   f. Pap smear if over age 21
   g. Viral culture if lesion is present
   h. Bacterial culture if lesion is present
3. <u>Male presenting with dysuria or urethral discharge</u>
   a. HIV
   b. RPR
   c. GC
   d. Hepatitis B/C or hepatitis A/B/C
   e. Urinalysis (see notes about collection of GC and urinalysis)
   f. Viral culture if lesion is present
   g. Bacterial culture if lesion is present
4. <u>Female presenting with vaginal discharge</u>
   a. HIV
   b. RPR
   c. GC
   d. Hepatitis B/C or hepatitis A/B/C
   e. HCG
   f. KOH
   g. Wet Prep
   h. Viral culture if lesion is present
   i. Bacterial culture if lesion is present

Attachment 1: GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

Websites

Up to Date; www.uptodate.com, current edition.

MD Consult; www.mdconsult.com, current edition.

Center for Disease Control and Prevention; www.cdc.gov

References

DOD 5210.42-R, *Nuclear Weapon Personnel Reliability Program*

AFPD 41-1, *Health Care Programs and Resources*

AFPD 44-1, *Medical Operations*

AFI 44-103, *The Air Force Independent Duty Medical Technician Program*

AFI 44-158, *The Air Force Independent Duty Medical Technician Medical and Dental Treatment Protocols (obsolete)*

AFI 10-246, *Food and Water Protection Program*

AFI 10-2501, *Air Force Emergency Management (EM) Program Planning and Operations*AFI 24-301, *Vehicle Operations*

AFPD 24-3, *Management, Operation and Use of Transporation Vehicles*

AFI 36-2201, *Air Force Training Program*

AFI 36-2910, *Line of Duty (Misconduct) Determination*

AFI 41-101, *Obtaining Alternative Medical and Dental Care*

AFI 41-209, *Medical Logistics Support*

AFI 41-301, *Worldwide Aeromedical Evacuation System*

AFI 41-115, *Authorized Health Care and Health Care Benefits in the Military Health Services System (MHSS)*

AFI 44-102, *Medical Care Management*

AFI 44-108, *Infection Control Program*

AFI 44-119, *Medical Quality Operations*

AFI 47-101, *Managing Air Force Dental Services*

AFI 48-101, *Aerospace Medicine Enterprise*

AFI 48-102, *Medical Entomology Program*

AFI 48-105, *Surveillance, Prevention, and Control of Diseases and Conditions of Public Health or Military Significance*

AFJI 48-110, *Immunizations and Chemoprophylaxis*

AFJI 48-131, *Veterinary Health Services*

AFI 48-115, *Tuberculosis Detection and Control Program*

AFI 48-116, *Food Safety Program*

AFI 48-117, *Public Facility Sanitation*

AFI 48-123, *Medical Examination and Standards*

AFI 48-124_IP *Armed Forces Institute of Pathology and Armed Forces Histopathology Centers*

AFI 48-144, *Drinking Water Surveillance Program*

AFI 48-145, *Occupational and Environmental Health Program*

AFI 48-149, *Squadron Medical Elements*

AFJI 48-104, *Quarantine Regulations of the Armed Forces*

AFI 91-204, *Safety Investigation and Reports*

AFMAN 23-110, *USAF Supply Manual*

AFJMAN 24-306, *Manual for the Wheeled Vehicle Driver*

AFMAN 44-144, *Nutritional Medicine*

The AF IDMT Medical and Dental Treatment Protocols

AFMAN 48-125, *Personnel Ionizing Radiation Dosimetry*

AFMAN 48-138, *Sanitary Control and Surveillance of Field Water Supplies*

AFMAN 48-153, *Health Risk Assessment*

AFPAM 91-211, *USAF Guide to Aviation Safety Investigation*

AFPAM 48-151 *Thermal Injury*

AFOSHSTD 48-9, *Electro-Magentic Frequency (EMF) Radiation Occupational Health Program*

AFOSHSTD 48-14, *Swimming Pools, Spas and Hot Tubs, and Bathing Areas*

AFOSHSTD 48-20, *Occupational Noise and Hearing Conservation Program*

AFOSHSTD 48-137, *Respiratory Protection Program*

AFOSHSTD 48-139, *Laser Radiation Protection Program*

4N0X1, Aerospace Medical Service Career Field Education and Training Plan.

NIOSH Publication #78-210, *Pocket Guide to Chemical Hazards, National Institute of Occupational Safety and Health (NIOSH)/Occupational Safety and Health Administration (OSHA).*

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorder*s, Washington, DC: American Psychiatric Association, current edition.

Solomon, Elder P., *Human Anatomy and Physiology*, Ontario, Canada: W.B. Saunders; current edition.

Beers, Mark H., Berkow, Robert, *The Merck Manual of Diagnosis and Therapy,* current edition. Merck Research Laboratories, Whitehouse Station N.J.

Benenson, Abram S., *Control of Communicable Disease Manual*, Washington, DC: American Public Health Association, current edition.

Katzung, Betram G. *Basic and Clinical Pharmacology*, San Francisco: McGraw-Hill, current edition.

Bledsoe, Bryan E., Cherry, Richard A., Porter, Robert S., *Brady Intermediate Emergency Care*, New York: Prentice Hall, current edition.

Comprehensive Drug Abuse Prevention Act of 1970. Available from **http://www.usdoj.gov/deas/agency/csa.htm**

Emergency Care and Transportation of the Sick and Injured, current edition. American Academy of Orthopedic Surgeons, 222 South Prospect Avenue, Park Ridge, IL 60068.

Emergency War Surgery, U.S. Government Printing Office.

Food and Drug Administration Food Code, Food, Drug, and Cosmetic Act. Available from **http://vm.cfsan.fda.gov/~dms/foodcode.html**

DeWitt, Susan C., *Fundamentals Concepts and Skills for Nursing*, Philadelphia: W.B. Saunders, current edition.

Habif, Thomas, *Clinical Dermatology*, current edition, Elsevier, Inc 3251 Riverport Lane, Maryland Heights, MO 63043.

Auerbach, Paul S., *Management of Wilderness and Environmental Emergencies*, St Louis: Mosby, current edition.

Murray, P. R., Rosenthal, K. S.,Kobayashi G. S. and M. A. Pfallar, *Medical Microbiology*, St Louis: Mosby, current edition.

National Association of Emergency Medical Technicians and American College of Surgeons, *Prehospital Trauma Life Support*, St Louis: Mosby, current edition.

Rouzier, Pierre, et. al., *The Sports Medicine Patient Advisor,* SportsMedPress, 518 Bay Road, Amherst, MA 01002.

Sarwark, John, *Essentials of Musculoskeletal Care,* current edition. American Academy of Orthopaedic Surgeons, 6300 North River Road, Rosemont, Illinois, 60018-4262.

All of the following publications are required for all MTF/HMTF IDMT training programs, remote sites:
Only publications annotated with a * are required for MMU/SME/UTC IDMTs:

*Bates, Barbara, Bickley, Lynn S., Hoekelman, Robert A., *A Guide to Physical Examinations and History Takin*g, New York: Lippincott, William & Wilkins,current edition.

* Benenson, Abram S., *Control of Communicable Disease Manual*, Washington, DC: American Public Health Association, current edition.

Mistovich, Joseph J., Hafen, Brent Q., and Keith S. Karren , *Brady's Prehospital Emergency Care*, New Jersey: Prentice Hall, current edition; or other similar reference published by a different company.

*Dorland's Medical Dictionar*y, Philadelphia: W.B. Saunders, current edition.

Grant, Harvey, Murrary, Jr., Robert, Bergeron, David, and Michael F. O'Keefe, *Brady Emergency Care*, New York: Prentice Hall, current edition.

Nettina, *Sandra M., Lippincott Manual of Nursing Practic*e, Washington DC: Lippincott Williams & Wilkins, current edition. Consider Mosby's Online or Mosby's Nursing

*Tierney, Lawrence M., McPhee, Stephen J., and Maxine A Papadakis, *Current Medical Diagnosis and Treatmen*t, San Francisco: McGraw-Hill/Applelon & Lange,current edition.

Cummins, Richard O., *Textbook of Advanced Cardiac Life Suppor*t, American Heart Association, current edition.

*Cowl, Clayton T., *Physician's Drug Handbook*, Minnesota: Lippincott Williams & Wilkins, current edition.

Attachment 2: DRUG FORMULARY

Antihistamine / Decongestant / Expectorant/Antitussive Medications

Generic Name	Trade Name
Benzonatate	Tessalon Perles®
Brompheniramine	Dimetapp® Extentabs
Chlorpheniramine	CTM®
Chlorpheniramine/Pseudoephedrine	Deconamine®
Cetirazine	Zyrtec®
Dimenhydrinate	Dramamine®
Diphenyhydramine Hydrochloride injection	Benadryl®
Diphenyhydramine tablets	Benadryl®
Guaifenesin	Robitussin®
Guaifenesin/Dextromethorphan	Robitussin DM®
Hydroxyzine hydrochloride 10 mg	Atarax®
Loratadine	Claritin®
Oxymetazoline hydrochloride	Afrin®
Phenylephrine/Guaifenesin	Entex® Entex LA®
Pseudoephedrine	Sudafed®

Systemic Anti-infective Agents - All require physician prescription/consultation

Generic Name	Trade Name
Acyclovir "MD"	Zovirax®
Albendazole “MD”	Albenza®
Amoxicillin "MD"	Amoxil®
Ampicillin "MD"	Ampicillin
Amoxicillin clavulanate "MD"	Augmentin®
Azithromycin "MD"	Zithromax®
Cefotaxime for I.V. use "MD"	Claforan®
Cefoxitin "MD"	Mefoxin®
Ceftriaxone "MD"	Rocephin®
Cephalexin monohydrate "MD"	Keflex®
Ciprofloxin "MD"	Cipro®
Clindamycin hydrochloride "MD"	Cleocin®

Dicloxacillin sodium "MD"	Dynapen®
Doxycycline "MD"	Vibramycin®
Erythromycin "MD"	E-Mycin®
Gentamycin sulfate "MD"	Garamycin®
Griseofulvin “MD”	Gris-Peg®
Ivermectin “MD”	Stromectol®
Mebendazole "MD"	Vermox®
Metronidazole "MD"	Flagyl®
Nitrofurantoin "MD"	Macrobid®
Penicillin G benathine "MD"	Bicillin LA®
Penicillin G procaine "MD"	Wycillin®
Penicillin V potassium "MD"	Pen Vee K®
Praziquantel “MD”	Biltricide®
Tetracycline "MD"	Achromycin®
Trimethoprim/sulfamethoxazole "MD"	Bactrim® Septra®
Valacyclovir “MD”	Valtrex®

Topical Anti-infective Agents

Generic Name	**Trade Name**
Bacitracin	Bacitracin® Ointment
Bismouth impregnated petroleum gauze	Xeroform gauze®
Coal tar shampoo	T-gel® shampoo, Tarsum®
Condoms	Condoms
Clindamycin jel	Cleocin® Jel
Clotrimazole	Mycelex G ®, Gyne-Lotrimin®
Chlorhexidine gluconate	Hibiclens®
Iodoform gauze packing strips	N/A
Ketoconazole 2% cream	Nizoral cream®
Ketoconazole shampoo	Nizoral shampoo®
Lindane Shampoo	Kwell®
Miconazole vaginal	Monistat® Cream/ Suppositories
Miconazole Derm cream/lotion	2% cream / lotion
Mupirocin 2% topical	Bactroban®
Permethrin 1% cream	Nix®
Polymyxin B-bacitracin–neomycin ointment	Neosporin®

Pramoxine with Hydrocortisone foam	Proctofoam®
Selenium sulfide	Head and Shoulders®, Selsun Blue®
Silver sulfadiazine	Silvadene®
Terbenifne cream	Lamisil® cream
Tolnaftate	Tinactin®

Analgesic Drugs

Generic Name	**Trade Name**
Acetaminophen	Tylenol® **IDMT MAX 3000 MG/DAY**
Acetaminophen, Aspirin, Caffeine	Excedrin®
Acetaminophen and codeine sulfate "MD"	Tylenol #3® **MAX DISPENSE 5 TABS/DAY**
Aspirin	Bayer®
Clove Oil	Eugenol®
Cyclobenzaprine HCl "MD"	Flexeril®
Ibuprofen	Motrin® **IDMT MAX 2400 MG/DAY**
Indomethacin "MD"	Indocin®
Ketorolac “MD”	Toradol®
Lidocaine anesthetic injection 1%, 2%	Xylocaine®
Lidocaine injection with epinephrine 1%	Xylocaine with Epinepherine®
Lidocaine, viscous	Xylocaine®
Meperidine "MD"	Demerol®
Methocarbamol "MD"	Robaxin®
Morphine sulfate "MD"	Morphine
Naproxen	Naprosyn®
Phenazopyridine	Pyridium®
Propoxyphene "MD"	Darvocet-N®
Salicylic Acid	Mediplast®
Sumitriptan “MD”	Imitrex®
Vitrabond®	Vitrabond®
Zolmitriptan “MD”	Zomig®

Overdose / Poison Antidotes

Generic Name	**Trade Name**
Acetylcysteine	Mucomyst®
Activated charcoal	Charcoaid®
Flumazenil	Flumazenil®

Ipecac Syrup	Ipecac
Magnesium citrate	Mag-Citrate
Naloxone hydrochloride	Narcan®

Topical Preparations

Generic Name	**Trade Name**
Aluminum acetate	Dome Boro Tablets®
Analgesic balm	Ben-Gay®
Anapalene 0.1% jel	Differin®
Anusol-HC	Anusol-HC®
Benzoyl peroxide gel 5%	Desquam-X-5®
Benzoyl peroxide gel/soap	Fostex ®, Neutrogena®
Boric acid/Acetic acid	Burrow's Solution®
Calamine lotion	Calamine lotion®
Chlorahexidine gluconate	Peridex®
Dichlorotetrafluorethane	Skin Refrigerant
Hank's Balanced Salt Solution	Save-A-Tooth®
Hydrocortisone, 1% topical cream	Cort-Aid®
Hydrogen peroxide	Hydrogen peroxide
Glass ionomer restorative	Fuji 9®, IRM®, Vitrabond®
Isopropyl alcohol	Isopropyl alcohol
Orabase	Orabase®
Povidone-Iodine	Betadine®
Silver nitrate applicators	Silver nitrate applicators
Sunscreen	Sunscreen
Tretinoin 0.25% cream "MD"	Retin-A®
Triamcinolone	Kenalog®
Triamcinolone acetonide 0.1% topical	Aristocort®
Trichloroacetic acid solution	Trichloroacetic acid solution
Zinc Oxide	Zinc Oxide

Ophthalmic Ointments and Solutions

Generic Name	**Trade Name**
Acetazolamide "MD"	Diamox®
Ciprofloxacin ophthalmic "MD"	Ciloxan®
Erythromycin ophthalmic "MD"	Ilotycin®
Gentamicin sulfate opthalmic "MD"	Garamycin®

Homatropine ophthalmic 5%	Isopto Homatropine®
Polymyxin B-Bacitracin-Neomycin ophthalmic ointment “MD”	Neosporin®
Polymyxin B-Neomycin-Gramicidin ophthalmic solution "MD"	Neosporin®
Proparacaine Hydrochloride	Ophthaine®/Alcaine®
Scopolamine ophthalmic solution (1/4%) "MD"	Isopto Hyoscine®
Sodium fluorescein applicator strips	BioGlo®, Ocu-Flur 10®
Sulfacetamide "MD"	Sulamyd®
Visine	Visine AC® eye drops

Otic and Nasal Preparations

Generic Name	**Trade Name**
Acetic acid solution	Burrow’s otic®
Anesthetic throat lozenges	Chloraseptic lozenges®
Antipyrene and Benzocaine	Auralgan®
Chlorhexidine gluconate	Peridex®
Cipro Otic® ”MD”	Cetraxal®
Fluticasone “MD”	Flonase®
Mometasone “MD”	Nasonex®
Polymyxin B-Neomycin-Hydrocortisone Otic suspension	Cortisporin®

Bronchodilators / Asthma Medications

Generic Name	**Trade Name**
Albuterol inhaler	Proventil®
Albuterol nebulizer “MD”	N/A
Spacer for use with metered dose inhalers	Aerochamber®, Inspir-Ease®

Systemic Steroid Medications

Generic Name	**Trade Name**
Dexamethasone IM, IV"MD"	Decadron®
Methylprednisolone IV "MD"	Depo-Medrol®, Solu-Medrol®
Prednisolone “MD”	Orapred®, Prelone®
Prednisone tablets "MD"	Deltasone®

Gastrointestinal Drugs

Generic Name	Trade Name
Aluminum-OH 400mg / Magnesium-OH 400mg / Simethicone 40mg	Mylanta II liquid®
Aluminum-OH 200mg / Magnesium OH 200mg / Simethicone 25 mg	Maalox Plus®
Belladonna and Phenobarbital "MD"	Donnatal®
Bisacodyl	Dulcolax®
Bismuth subsalicylate	Pepto-Bismol®
Cimetidine	Tagamet®
Diphenoxylate "MD"	Lomotil®
Docusate sodium	Colace®
Esomeprazole	Nexium®
Hydrobromide Scopolamine	Scopace®
Kaopectate	Kaopectate®
Loperamide	Imodium®
Meclizine hydrochloride	Antivert®
Metaclopramide	Reglan®
Methylcellulose	Citrucel®
Milk of Magnesia	Milk of Magnesia®
Omeprazole	Prilosec®
Ondansatron	Zofran®
Promethazine	Phenergan®
Psyllium	Metamucil®
Rabeprazole	Aciphex®
Ranitidine	Zantac®
Simethicone	Mylicon®
Sucralfate	Carafate®
Trimethobenzamide HCl	Tigan®
Witch hazel	Preparation H®, Tucks®

Cardiac Drugs

Generic Name	Trade Name
Adenosine	Adenocard ®
Amioderone	Corderone®, Pacerone®
Atropine sulfate injectable “MD”	N/A

Calcium Gluconate “MD”	Kalcinate®
D50	D50
Dopamine hydrochloride “MD”	Intropin®
Epinephrine 1:1000 aqueous solution “MD”	Adrenalin®
Ferrous Sulfate	Prenatal vitamin
Furosemide “MD”	Lasix®
Lidocaine IV	Xylocaine®
Nitroglycerin tabs "MD"	0.3 mg Nitrostat®
Vasopressin	Pitressin®

Psychological/Neurological Drugs

Generic Name	**Trade Name**
Diazepam injection "MD"	Valium®
Diazepam tablets "MD"	Valium®
Haloperidol "MD"	Haldol®
Lorazepam "MD"	Ativan®
Nicotine transdermal patches "MD"	Habitrol®

I.V. Solutions

Generic Name	**Abbreviation**
Dextrose 5% in Water	D5W
Dextrose 5% Normal Saline	D5NS
Half Normal Saline	1/2NS
Normal Saline	NS
Lactated Ringer’s	LR
Lactated Ringer’s with 5% Glucose I.V.	D5LR

Immunizations/Miscellaneous

Generic Name	**Trade Name**
Diagnostic strips (blood and urine)	Labstix®
DPT and HIB	Tetramune®
Hepatitis A	N/A
Hepatitis B	Heptavax-B®
Measles, mumps, and rubella	MMR
Meningococcal meningitis	N/A
Oral polio	OPV
Oxygen	Oxygen, USP

Sodium Fluoride "MD"	Sodium Fluoride
Temporary Dental Restorative Material	IRM®, Vitrabond® etc.
Tetanus Diphtheria	T-D
Tetanus immune globulin	Tetanus immune globulin
Tetanus toxoid	Tetanus toxoid
Tuberculin skin test (IPPD)	Tuberculin skin test
Typhoid	Typhim®

"MD" denotes those items that may be dispensed by the IDMT only with physician consultation.

IDMTs are cautioned to remain extremely conservative when dispensing medication and must document specific instructions and the name of the physician prescribing the medication.

Attachment 3: List of Acronyms

AC	acromioclavicular
AC>BC	Air Conduction better than Bone Conduction
ACL	Anterior Cruciate Ligament
ACLS	Advanced Cardiac Life Support
ACS	Acute Coronary Syndrome
AED	Autmated External Defibrillator
AFI	Air Force Instruction
AFMOA	Armed Forces Medical Operations Agency
AFRC	Air Force Reserve Component
AHA	American Heart Association
AMS	Acute Mountain Sickness
ANG	Air National Guard
ANUG	Acute Necrotizing Ulcerative Gingivitis
AP	Anterior/Posterior
ARDS	Acute Respiratory Distress Syndrome
ASA	Aspirin/Acetylsalicylic Acid
ASAP	As Soon As Possible
AVPU	Alert Verbal Pain Unresponsive
BG	Blood Glucose
BID	Twice a day (bis in die)
BLS	Basic Life Support
BMI	Body Mass Index
BP	Blood Pressure
BPA	Blanket Purchase Agreement
BRAT	Bananas, Rice, Applesauce and Toast
BSA	Body Surface Area
BUN	Blood Urea Nitrogen
CA	Cancer
CAD	Coronary Artery Disease
CC	Commander
cc	Cubic Centemeter (one milliliter)
CD	Compact Disk
c-diff	Clostridium dificile
CHEM 14	Complete Metebolic Panel
CHEM 7	Basic Metabolic Panel
CK	Creatinine Kinase
cm	Centemeter
CN	Cranial Nerve
CNS	Central Nervous System
CPR	Cardio Pulmonary Resuscitation
CRAO	Central Retinal Artery Occlusion
CVA	Cerebrovascular Accident/Costovertebral angle
DCAP BTLS	Deformities, Contusions, Abrasions, Punctures, Burns, Tenderness, Lacerations, Swelling
DIP	Distal Interphalangeal (Joint)
dL	Deciliter
DM	Diabetes Mellitus
DVT	Deep Vein Thrombosis
EAC	External Auditory Canal

EBV	Epstein-Barr Virus
ECG	Electrocardiogram
EKG	Electrocardiogram
EOMI	Extra-Occular Movements Intact
ET	Endotracheal
EVAC	Evacuation
FHT	Fetal Heart Tones
ga/GA	Gauge
GCS	Glascow Coma Score
GI	Gastrointestinal
gtts/gtt	Drops/drop (guttae)
h/hr	Hour
HA	Headache
HACE	High Altitude Cerebral Edema
HAPE	High Altitude Pulmonary Edema
HCG	Human Chorionic Gonadotropin
HDCV	Human Diploid Cell Rabies Vaccine
HDL	High Density Lipid
HIV	Human Immunodeficiency Virus
HMTF	Host Medical Treatment Facility
HNP	Herniated Nucleus Pulposis
HPI	History of Previous Illness
HR	Heart Rate
HTN	Hypertension
I &O	Intake and Output
I&D	Incision and Drainage
IAW	In Accordance With
ie	That Is to say (id est)
Ig	Immunoglobulin
IG	Inspector General
IM	Intramuscular
IO	Intraosseous
IOP	Intraoccular Pressure
IPPD	Intradermal Purified Protien Derivative
IRM	Intermediate Restorative Material
IT	Illio-Tibial
IUD	Intrauterine Device
IV	Intravenous
IVP	Intravenous pyelogram; intravenous push
kg	Kilograms
KOH	Potassium Hydroxide
KVO	Keep Vein Open
L	Liters
LAD	Lymphadenopathy
lbs	Pounds
LCL	Lateral Collateral Ligament
LDL	Low Density Lipoprotein
LLQ	Left Lower Quadrant
LMA	Laryngeal Mask Airway
LMP	Last Menstrual Period
LNMP	Last Normal Menstrual Period
LPM	Liters Per Minute

LR	Lactated Ringers
LUQ	Left Upper Quadrant
MAST	Military Anti Shock Trousers
mcg	Micrograms
MCL	Medial Collateral Ligament
MCP	Metacarpophalangeal
MD	Medical Doctor
MDI	Metered Dose Inhaler
mg	Milligrams
MI	Myocardial Infarction
Min	Minutes
mL	mililiters
mmHg	Millimeters of Mercury
MOI	Mechanism Of Injury
MRSA	Methicillin-Resistant Staphylococcus Aureus
NG	Naso-gastric
NS	Normal Saline
NSAID	Non-Steroidal Anti-Inflammatory Drug
OCD	Obsessive Compulsive Disorder
OD	Right Eye (Oculus Dexter)
OPA	Oro-Pharyngeal Airway
OPQRST	Onset, Provocation, Quality, Radiation, Severity and Timing
OS	Left Eye (Oculus Sinister)
OTC	Over the Counter
OU	Both Eyes (Oculus Uterque)
oz	Ounce
PCL	Posterior Cruciate Ligament
PCN	Penicillin
PEEP	Positive End Expiratory Pressure
PEF	Peek Expiratory Flow
PERRLA	Pupils Equal Round Reactive to Light and Accomodation
PHS	Public Health Service
PID	Pelvic Inflammatory Disease
PIP	Proximal Interphalyngeal (Joint)
PMH	Past Medical History
PMI	Point of Maximal Impulse
PO	Orally (per os)
PRN	As Required (pro re nata)
psi	Pounds Per Square Inch
PSVT	Paroxysmal Supraventricular Tachycardia
PT	Physical Training/Therapy; Prothrombin time
PTL	Physical Training Leader
PTSD	Post-Traumatic Stress Disorder
PUD	Peptic Ulcer Disease
PV	Per Vagina
q/Q	Every (quaque)
QHS	Every night/Hours of Sleep
QID	Four Times per Day (quater in die)
RBC	Red Blood Cell
RICE	Rest, Ice, Compression, Elevation
RIG	Rabies Immunoglobulin
RLQ	Right Lower Quadrant

RMCL	Right Medial Collateral Ligament
ROM	Range of Motion
ROSC	Return Of Spontaneous Circulation
RPR	Rapid Plasma Reagin test
RSV	Respiratory Syncytial Virus
RUQ	Right Upper Quadrant
rx	Prescription/Prescribe
SAMPLE	Signs, Symptoms, Allergies, Medications, Previous History, Last Meal and Events Leading up to the injury or illness
SAO2	Percentage of oxygen saturated arterial blood
sec	Seconds
SOAP	Subjective, Objective, Assessment, Plan
SOB	Shortness of Breath
SPO2	Percentage of oxygen saturation in the blood measured by pulse oximetry
STAT	Immediately (statim)
STI/STD	Sexually Transmitted Infection/Disease
sub-Q	Subcutaneous
TB	Tuberculosis
tbsp	Tablespoon (15 mililiters)
TID	Three times a day (ter in die)
TKO	To Keep Open
TM	Tympanic Membrane
tsp	Teaspoon (5 mililiters)
tx	Treatment
UA	Urinalysis
uL	Microliters
URI	Upper Respiratory Infection
UTI	Urinary Tract Infection
UV	Ultra Violet
VA	Visual Acuity
VDRL	Venerial Disease Research Lab
WBC	White Blood Cell
WP	White Phosphorus
YO	Years Old

Attachment 4: PATIENT ENCOUNTER NOTETAKER

NAME_____UNIT _____OCCUPATION _____CELL #_____
GRAD DATE _____ PRP? Y/N FLY STATUS? Y/N DEPLOYMENT RELATED? Y/N
DAY/NIGHT SHIFT AGE_____SEX: M / F RACE _____STATUS/BRANCH _____ TOS _____
C/C/FOR HOW LONG _____
DDx _____

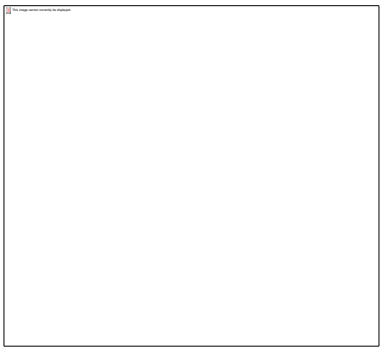
HPI (tell a story + OPQRST and OPQRST for other + symptoms; for severity include comparison and limitations)
_____
_____
_____
_____
_____
_____
_____
_____
_____
_____
_____
PMHX _____
PSHX/YEAR/COMPLICATIONS _____
ALLERGIES MEDICATIONS/SEASONAL/CONTACT_____
REACTION_____
MEDS_____
OTC/HERBS/SUPPLEMENTS_____
GYN HX LMP_____ Last 3 regular? Y/N Possibly pregnant? Y/N birth control _____
Total # of Pregnancies _____Living ___Miscarriages ___ Abortions _____ Abnormal Pap Y/N___
Sexually active Y/N # of partners in past year_____ Hx of STDs Y/N _____ Protection Y/N
SOC HX: TOB (type, amount, how long) _____Alcohol: ___servings/day or week C A G E
Caffeine intake _____ servings/day or week Typical Diet?_____ Changed by condition Y/N
Sleep/Rest_____hrs/day Changed by condition? Y/N How? _____
Exercise ___hrs/wk Strength____Cardio_____ Changed by condition? Y/N How? _____
Pertinent/High Risk Hobbies_____
Recent Travel (deployment) _____ Exposures_____ Uses PPE Y/N
Immunizations UTD? Per record or patient?

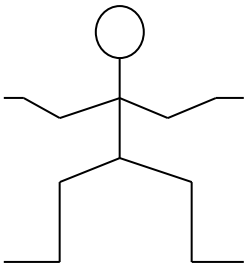
FAMILY HISTORY	HTN	STROKE	CANCER	CARDIAC	ASTHMA	SUDDEN DEATH	DIABETES	OTHER		ALIVE	AGE
MOTHER											
FATHER											
SIS/BRO											
PGF											
PGM											
MGF											
MGM											

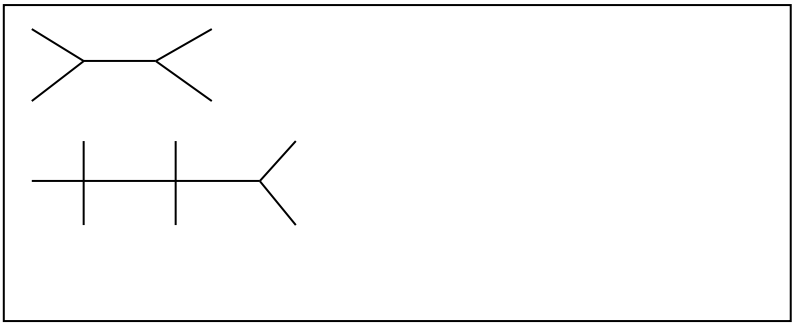
Additional negative symptoms

GENERAL Fever, fatigue, recent wt change **HEENT** HA, dizzy, vision/hearing change, ear pain, sore throat
NECK Stiffness, pain, swollen glands **RESP** Cough, wheeze, shortness of breath **CARDIO** Chest pain, palpitations
GI: N/V, diarrhea, constipation, abdominal pain, stool changes **GU:** Last urine output, discharge, urine changes
NEURO: Numbness, tingling, change in strength **SKIN**: Rashes **MUSCULOSKELETAL**: Back pain, other joint pain
PSYCH anxiety, depression, suicidal/homicidal ideations, stress, Able to handle? Y/N Want to see life skills? Y/N

O: General Impression: Well developed? Well nourished? No acute Distress? A&OX3? Appears _____
BP ___/___ L/R P _____ R _____ T ____F/C Ht _____ cm Wt _____ lbs O2 Sat _____ on _____
Visual Acuity OS 20/____ OU 20/____ OD 20/____ Uncorrected
OS 20/____ OU 20/____ OD 20/____ Corrected
Orthostatic: Laying BP___/____ P____ Sitting: BP___/_____ P_____ Standing: BP___/____P_____
Peak Flow1._____2._____3._____ Predicted_____ % Predicted _____
Head_____
Ears _____
Eyes _____
Nose_____
Throat _____
Neck/Lymph _____
Cardio _____
Resp _____
Abd_____
GU_____
Spine _____
Shoulder_____
Elbow/wrist_____
Hip _____
Knee _____
Hand/Foot _____
Skin _____
Neuro _____







Repeat VS Time_____ BP ___/____P____R____T____ Time_____ BP___/___P____R____T____
Time/Intervention _____
Time/Intervention _____
Time/Intervention _____
Time/Intervention _____
A: Dx or final DDx's _____
Additional Dx's _____
P: Med 1 (sig + instructions) _____
Med 2 _____
Med 3 _____
Tx/Education_____
_____
_____
_____
_____
F/U _____
Worse _____
Better _____
Well _____
P2 _____

Attachment 5: HISTORY AND PHYSICAL EXAM GUIDE

SUBJECTIVE

PATIENT IDENTIFICATION- Age, sex, race, occupation, and/or status.

CHIEF COMPLAINT (CC)- Use patient's own words if possible, include duration of this episode.

HPI

Provocative/Palliative? (What makes it better/worse?)
Quality? (Sharp, dull, ache, throbbing, etc.?)
Region/Radiation? (Where does it hurt and does it radiate?)
Severity/Strength? (1 to 10 compared to what? and/or limitations?)
Time/Treatment? Started when? Had it before? When? Duration?
Treatment (self/MD Rx'd)? Medications, dose, frequency, duration? Tx/medication giving relief?

HPI with injuries- What was the patient doing prior to the injury? What was the patient doing at the time of the injury? What did the patient do and what was he/she capable of doing immediately after the injury? How did the patient land? Mechanism of injury? What can the patient do now compared to at the time of injury?

ASSOCIATED SYMPTOMS- OPQRST all positive responses and include them in the HPI.

OTHER PROBLEMS- Active on going problems being treated, or being followed without treatment, not related to the "Chief Complaint".

PMH- (Past Medical History) Any significant medical history (i.e. childhood illnesses, lost time from school/work, surgeries, hospitalizations, injuries, blood transfusions, etc.). Be thorough and concise.

EXAMPLE: Appendectomy age 8, May 1962, ruptured, received transfusion, hospitalized 6 weeks, no complications, no sequelae.

CURRENT MEDICATIONS- Include meds frequently over looked, aspirin, birth control pills, vitamins, etc.

IMMUNIZATIONS- Types, current/noncurrent.

ALLERGIES- Drugs? Insects? Food? Other? Address each separately; include manifestation of symptoms, duration, previous treatment, etc.

HABITS- Alcohol? CAGE +/-? Tobacco? Caffeine? Sleep? Diet? Other? Address each separately; include pertinent history, amount, etc.

HOBBIES- What type and the tools, chemicals, etc. used for the hobby?

RELIGIOUS PREFERENCE- Include if it impacts care to be given.

LIVING CONDITIONS- Dorm, house, apartments, alone, with others, etc.?

EXPOSURES

People; family, coworkers, roommates, etc.
Things; environmental, occupational, etc.

FAMILY HISTORY- Mother, MGM, MGF, Father, PGM, PGF, Siblings? If living include age and state of health. If deceased include age at death, cause of death. Also, include any history of cancer, diabetes, hypertension, thyroid, heart, lung or kidney disease and mental disorders prior to age 55. If adopted, state "Adopted, Family History unknown."

ROS- (Review of Systems) Include negative responses in the organ system being evaluated and 1-3 questions from each organ system geared towards identifying life threatening illnesses

OBJECTIVE

VITAL SIGNS: B/P, Temp, Pulse, Respirations, O_2 saturation if indicated, visual acuity, height, weight

GENERAL IMPRESSION: Well developed? Well nourished? Oriented to person, place, and time? No acute distress? Appears _____. (Fatigued, restless, uncomfortable, etc)

Head: Normocephalic with even hair distribution, no masses or swelling. Mastoid nontender.

TMJ: TMJ nontender no crepitus; Full range of motion without pain, popping, or locking.

Eyes

Visual acuity- right, left, bilateral, uncorrected AND corrected.
Visual fields by confrontation = to examiner.
PERRLA, EOMI without nystagmus.
Eyes, Eyelids, Eyebrows- symmetrical without edema.
Conjunctiva- moist, not injected, no cobblestoning.
Sclera- no icterus or hemorrhage.
Lacrimal apparatus-without edema or regurgitation.
Cornea and Lens- without opacities.
Funduscopic- Disc margins sharp, no hemorrhage or exudate. No photophobia.

Ears

Auricles (or pinna)- symmetrical, nontender.
External ear canals (EAC)- no inflammation, discharge, or obstruction. (cerumen present?)

TM's- pearly gray and translucent, handle of malleus, umbo, pars tensa, pars flaccida , cone of light anterior/inferior, TM mobile with Valsalva.
Able to distinguish two syllable whispered words.
Weber- equal bilaterally (or without lateralization).
Rinne- AC>BC bilaterally.

Nose

Nose is midline without lesions, tenderness, or deformity.
Septum- midline.
Nares- patent.
Turbinates- pink and moist; no polyps, edema, or discharge.
Sinuses- nontender with palpation and percussion.
Negative head tilt.

Throat

Lips and buccal mucosa- pink and moist without lesions
Tongue- midline no masses
Teeth- intact and nontender to teeth rake, no appliances.
Gums- no gingivitis
Tonsils- Grade __, without exudates.

Neck

Supple. Nontender spinous processes and paravertebral muscles. No stepoffs.
Full active range of motion and strength 5/5 with flexion, extension, rotation, and lateral bending. No pain with movements.
No lymphadenopathy in occipital, pre/post auricular, submental, submandibular, tonsillar, anterior /posterior cervical, or supra/infraclavicular locations.
Trachea- midline.
Thyroid- nontender no masses.

Respiratory

Chest symmetrical, no bruising, retractions, crepitus, or tenderness.
Respiratory excursion symmetrical.
Tactile fremitus = bilaterally in all fields.
Resonant to percussion.
Diaphragmatic excursion __cm bilaterally.
Breath sounds clear bilaterally in all fields. No wheezes, rales, or ronchi.
No bronchophony, egophony, or whispered pectoriloquy.

Cardiovascular

Skin warm and dry. Capillary refill less than 2 seconds.
No lifts or heaves. PMI not displaced.
No pitting edema (where checked).
Peripheral pulses- list where checked and use scale of measurement- 0 = absent, 1+ = weak, 2+ = strong/normal 3+ = bounding. EXAMPLE: "Dorsalis pedis pulse 2+ bilaterally."
Rate and rhythm regular (RRR), without murmurs.

No bruits (specify where checked).

Breasts

Skin- smooth, no masses or dimpling.
Nipples- symmetrical without discharge.
Areola- symmetrical without scaling or discharge.
Axillary Lymph nodes- no lymphadenopathy in central, lateral, pectoral, or subscapular lymph nodes bilaterally.

Abdomen

Flat without discolorations or distension.
Bowel sounds- active in all 4 quadrants.
Liver span- __cm, RMCL.
Spleen- no change in typani at Traube's space with deep inspiration.
Bladder and stomach- tympanic
Kidneys- nontender. No CVA tenderness.
Nontender in all four quadrants and epigastrum. No rebound tenderness.
Obturator and Psoas- negative.
Aortic pulse- __cm diameter.

Genitalia (Male)

Penis- circumcised, no lesions, ulcerations, or discharge.
Scrotum- no edema or indirect hernias.
Testes- firm, ovoid, equal size, without masses or tenderness.
Epididymis- superior and posterior; without masses or tenderness.
Spermatic cord- without masses or tenderness.
Inguinal canal- no direct hernias or lymphadenopathy.

Genitalia (Female)

External Genitalia- normal hair distribution, no lesions.
Vagina- mucosa pink, moist, without discharge.
Cervix- pink, no lesions or discharge. Os transverse. No cervical motion tenderness.
Uterus- nontender.
Ovaries- nontender (or nonpalpable).

Rectal

Perianal region- without hemorrhoids.
Sphincter tone- intact.
Rectal wall- smooth, no masses or fissures. No stool in vault.
Prostate- consistency, size, shape; no masses or tenderness. Median sulcus midline.
Guaiac- negative.

Neurological

CN-I- identified and discriminated odors of __and __, bilaterally.
CN-II- visual acuity. Visual fields by confrontation = to examiner. Funduscopic.
CN-III, IV, VI- PERRLA, EOMI, no nystagmus.

CN-V- temporal and masseter muscles- symmetrically strong, facial sensation intact to light touch, sharp/dull.
CN-VII- facial movements (eyebrow raise, lids close tight, smile, show teeth, puff out cheeks) symmetrical, no weakness
CN-VIII- Weber = bilatearally, Rinne AC>BC, able to distinguish 2 syllable whispered words.
CN-IX, X- swallowing and Gag reflex intact. Uvula rises midline.
CN-XI- head movements and shoulder shrug symmetrically strong, against resistance.
CN-XII- tongue midline, no fasciculations.
Motor- no atrophy, weakness or tremors. Strength 5/5 (and where checked).
Cerebellar- point to point and rapid alternating movements (RAM)- speed and rhythym fluent.
Negative Romberg.
Gaits- heel-to-toe, heel walk, toe walk, and normal gait- no limp, hesitation or loss of balance.
Sensory- bilaterally intact to; light touch, pin prick, two-point discrimination, hot/cold, proprioception, and stereognosis. Identify where checked. Upper extremities? Feet?
DTR's- (0-4+, 2+ = normal) (draw diagram with plus signs). No ankle clonus.
Plantar reflex- toes curl.

NOTE: If documented elsewhere, tests do not need to be redocumented in this section.

Shoulders

No edema, deformity, or ecchymoses.
No tenderness or crepitus in Acromioclavicular (AC), sternoclavicular (SC) or glenohumoral joints.
Full active ROM with 5/5 strength with flexion, extension, internal/external rotation, abduction, and adduction. No pain with movement or against resistance.

Elbows

No edema, deformity, ecchymosis, tenderness, or crepitus.
Full active ROM with flexion, extension, supination, and pronation. Strength 5/5 with flexion and extension. No pain with movement or resistance.

Wrists

No edema, deformity, ecchymosis, tenderness, or crepitus.
Full active ROM with flexion, extension, radial and ulnar deviation. Strength 5/5 with flexion and extension. No pain with movements or against resistance.

Hands/fingers

No edema, deformity, ecchymosis, tenderness, or crepitus.
Full active ROM with flexion, extension, abduction, and adduction of the digits.
Grip strength, thumb opposition, and finger abduction equal bilaterally.
VASCULAR AND NEURO CHECK: check capillary refill, peripheral pulses and light touch distal to injury.

Spine

Spine without deformities. Shoulders and iliac heights equal bilaterally.
No tenderness of the spinous processes or paravertebral muscles. No stepoffs or spasms.
Full active range of motion and strength 5/5 with flexion, extension, rotation, and lateral bending. No pain with movements.

Straight leg raise- to ____^{o}, without tenderness or sciatica bilaterally.

Hips

Levels of iliac crests equal bilaterally. No deformities or ecchymosis. Iliac crests and greater trochanters nontender bilaterally.
Full passive range of motion and 5/5 strength with flexion, abduction, adduction, and internal/external rotation. No pain with movements or against resistance.

Knees

No swelling, deformity, or ecchymosis.
Patella, joint space, tibial tuberosity, and patellar tendon nontender.
MCL and LCL without laxity with collateral stress tests.
ACL and PCL without laxity with Anterior and posterior drawer's tests.
Full active ROM and 5/5 strength with flexion and extension. No pain with movement or against resistance.

Ankles

No swelling, deformity, or ecchymosis.
Achilles tendon, medial, and lateral malleoli without tenderness or crepitus.
Full active ROM with dorsiflexion, plantarflexion, inversion, and eversion. Strenth 5/5 planterflexion and dorsiflexion. No pain with movements or against resistance.

Feet/Toes

No swelling, deformity, ecchymosis, skin abnormalities, or tenderness.
Passive inversion and eversion of the transverse tarsal joint without pain or crepitus.
Full active ROM and strength 5/5 with toe flexion and extension.
VASCULAR AND NEURO CHECK: check capillary refill, peripheral pulses and light touch distal to injury.

LAB
X-RAYS
INTERVENTIONS

ASSESSMENT

If you have more than one diagnosis, list each separately and number them in accordance to priority.

PLAN

(1) Rx (include meds, strength, dose, frequency, ROA, #days, #Rxd, refills)
(2) Therapy, crutches, handouts, exercises, etc.
(3) Administrative support- sick slip, profile, medication sheet.

(4) Patient education/instructions- DOs/DONTs.
(5) Expectation of wellness; Follow up instructions.
(6) P2- Preventative counseling that applies to this patient but is not the purpose of the visit

ADDITIONAL INFORMATION

COLORS:

Erythematic/Erythematous- used for red skin and TMs.
Injected- used for red conjunctiva and TMs (engorged vessels).
Icteric- (jaundiced) skin, sclera.

MASSES AND LESIONS DESCRIPTION GUIDE:

Masses- include location, size, shape, consistency, tenderness, mobility.
Lesions- include location, distribution, grouping, arrangement, color, type (macule, papule, vesicle, etc), size, shape, texture, exudate, bleeding, drainage.

GUIDE TO LOCATION AND DIRECTION WORDS:

Anatomic position- standing, facing forward, arms at sides, palms facing forward
Anterior- on the front side of the body when in the anatomic position
Posterior- on the back side of the body when in the anatomic position
Right/Left Lateral- towards the right/left side
Transverse/horizontal plane-Describing a cut through the body that is parallel to the ground
Ipsilateral- same side
Contralateral- opposite side
Superficial- towards the surface
Deep- away from the surface
Superior- towards the head
Inferior- towards the feet
Dorsal- towards the spine, top of foot, or back of hand
Palmar/volar- towards the palm of the hand
Plantar- towards the sole of the foot
Proximal- towards the center of the body
Distal- away from the center of the body
Midclavicular line- runs vertically down the anterior side of the body passing through clavicle's midpoint
Midaxillary line- runs vertically down the lateral side of the body passing through the apex of the axilla
Midpupillary line- runs vertically down the face through the midpoint of the pupil

Attachment 6: Materials Needed for Pelvic Exam

Materials in Room

Stand, tray, desk, or place to put instruments
Exam table with lithotomy stirrups
If not available, place inverted bedpan under supine patients' buttocks, knees bent
Place Chux pad under where patient's hips will be
Light source
Sheet to cover patient

Equipment on Tray

Place Chux pad down to protect surface of tray/table
Speculum (metal or plastic; white handle plastic = small, green handle plastic = medium)
Lubricant
Fox swabs (large cotton-tipped applicators)
PPE (examiner's gloves, consider face mask with shield)

Specimen Collection

For Gonnorhea/Chlamydia: GC collection kit
-Take sample from inside cervix
For Herpes external lesions: Viral culture media and viral cotton-tipped applicator
-Unroof external vesicles and collect vesicular fluid
For trichamonas and bacterial vaginosis: "Wet Prep"
-Use wooden cotton-tipped applicatior along vaginal wall
-Place specimen in ordinary blood collection tube
-Use saline bullet to fill collection tube halfway with normal saline
For yeast: "KOH Prep"
-Use wooden cotton-tipped applicator along vaginal wall
-Place specimen in ordinary blood collection tube
-Use saline bullet to fill collection tube halfway with normal saline
-Add several drops of KOH solution
For Pap Smear
-Thin prep container/solution
-Use flat spatula around outside of cervix, larger side towards os, swish in solution
-Use cytobrush inside cervix, may cause trace spotting, swish in solution

Attachment 7: Biological Agents with Potential for Weaponization

VII. ***CATEGORY A AGENTS-*** *These organisms are hearty with good survivability in multiple environments. They can be easily released or spread person to person, likely to cause high degree of social disruption, and require special public health preparedness.*

- **A.** Veriola Major (Smallpox)
- **B.** Bacillus anthracis (anthrax)
- **C.** Yersinia pestis (plague)
- **D.** Clostridium botulinum toxin (botulism)
- **E.** Francisella tularensis (tularemia)
- **F.** Filoviruses (Ebola, Marburg)
- **G.** Arenaviruses (Lassa, Junin, etc)

VIII. ***CATEGORY B AGENTS-*** *These agents are moderately easy to spread but generate less morbidity and mortality than category A agents.*

- **A.** Coxiella burnetti (Q fever)
- **B.** Brucella species (brucellosis)
- **C.** Burkholderia mallei (glanders)
- **D.** Burkholderia pseudomallei (meliodosis)
- **E.** Chlamydia psittaci (psittacosis)
- **F.** Rickettsia prowazekii (typhus fever)
- **G.** Alphaviruses (eastern equine encephalitis, western equine encephalitis, Venezuelan equine encephalitis)
- **H.** Ricin toxin (found in castor beans)
- **I.** Clostridium perfringens toxin
- **J.** Staphlococcus enterotoxin B
- **K.** Salmonella species
- **L.** Shigella dysenteriae (shigellosis)
- **M.** Escherichia coli O157:H7
- **N.** Vibrio cholera
- **O.** Cryptosporidium parvum

IX. ***CATEGORY C AGENTS-*** *Agents that could be weaponized due to easy availability and significant morbidity and mortalty.*

- **A.** Nipah virus
- **B.** Hantaviruses
- **C.** Tickborne hemorrhagic fever viruses
- **D.** Tickborne encephalitis viruses
- **E.** Yellow fever
- **F.** Multiple drug resistant tuberculosis

Attachment 8: Chemicals with Potential for Weaponization

I. ***NERVE AGENTS-*** *Organophosphates that are colorless, odorless, tasteless, and nonirritating to the skin. They cause severe secretions (tears, runny nose, drooling), stimulation of smooth muscle (diarrhea, urination, vomiting), and inhibition of skeletal muscle, which paralyzes the diaphragm and causes respiratory failure. They are used commercially as insecticides.*

A. Sarin
B. VX
C. Tabun
D. Soman

II. ***TOXIC ASPHYXIANTS-*** *These agents are colorless; cyanide smells like almonds. They cause respiratory depression from severe lactic acidosis. Venous blood turns red because hemoglobin is unable to deposit oxygen in the tissues. They are used commercially to fumigate ships and buildings and to clean metals.*

A. Cyanide
B. Arsine (arsenic trihydride)

III. ***PULMONARY IRRITANTS****- Common industrial chemicals and household cleaners that cause direct pulmonary tissue injury and pulmonary edema. Irritation also occurs to the skin and eyes.*

A. Phosgene
B. Chlorine
C. Ammonia
D. Diphosgene

IV. ***BLISTER AGENTS (VESICANTS)-*** *These agents interfere with DNA synthesis and are carcinogenic, in addition to being directly toxic. Symptoms include blistering of skin and lungs and eye burns. Sulfur mustard and Lewisite have no industrial use, but the other agents on the list are used for cancer therapeutics.*

A. Sulfur mustard
B. Lewisite
C. Nitrogen mustard (mechlorethamine)
D. Cyclophosphamide
E. Alkyl sulfonate (busulfan)

V. **HALLUCINOGENS-** *Renders soldiers unable to fight due to hallucinations*

A. Quinuclinyl benzilate (QNB)
B. LSD

ATTACHMENT 9: Head-to-Toe Physical Exam

Exam		*Verbal Indicator*	Skill		Comments
			S	U	
Introduction & General Assessment					
1.	Indicate wash hands and dons PPE as indicated prior to exams	*Verbalize step*			
2.	Introduces self as IDMT student	*"Good morning, I am Sgt/Airman ___, I am an IDMT student performing a physical exam for you today"*			
Head, Face, and Scalp					
3.	Inspect head/hair/scalp	*"normocephalic with even hair distribution; no masses or swelling"*			
4.	Palpate head/hair/scalp	*"hair texture is normal without infestations; scalp has no scaling or lesions"* *"mastoid is nontender and without discoloration"*			
5.	Inspect, palpate and perform ROM of TMJ (Open, close, side to side, forward, back)	*"TMJ not tender and without swelling; no pain, popping or locking with ROM"*			
6.	CN V (Clench teeth-temporalis/masseter)	*"CN V intact"*			
7.	CN VII (Raise eyebrows, frown, close eyes and attempt to open with resistance, show teeth, smile and puff out cheeks)	*"CN VII intact"*			
Eyes					
8.	Verbalize visual acuity	*"Visual acuity checked"*			
9.	Inspect eyes, brows, lashes and lids	*"no exophthalmos, ptosis, periorbital discoloration, edema, or skin lesions"*			
10.	Inspect conjunctiva and sclera	*"no injection, icterus, lesions, edema or foreign bodies"*			
11.	Inspect and palpate lacrimal duct	*"no swelling and no regurgitation"*			
12.	Evaluate visual fields by confrontation	*"visual fields are full and equal to examiner"*			
13.	Check extraocular movement and lid lag	*"EOMIs equal without nystagmus, no lid lag"*			
14.	Assess for pupillary reaction to convergence and accommodation (near/far)	*"eyes converge equally and pupils constrict with near focus"*			
15.	Evaluate position and alignment of eyes and cover/uncover	*"no strabismus observed"*			
16.	Inspect iris and pupillary reaction to light (direct and consensual)	*"pupils are equal, round, and reactive to light and accomodation bilaterally"*			
17.	Anterior chamber depth	*"No crescent shadow"*			
18.	Inspect for red reflex	*"red reflex present"*			
19.	Inspect clarity of cornea, anterior chamber, lens, and vitreous	*"cornea, anterior chamber, lens, and vitreous are clear without opacities, haze or encroachments"*			
20.	Inspect optic disc	*"disc color is _____, margins are sharp, and cup-to-disc ratio is 1:2"*			
21.	Inspect retinal vessels	*"AV ratio is 2:3 without nicking or spasms"*			
22.	Inspect fundus	*"there are no hemorrhages or exudates"*			
23.	Inspect macula (have pt look into light)	*"no lesions on macula"*			

24.	Ophthalmoscopic exam technique				
Ears					
25.	Inspect auricles and Tragus	*"no lesions or discoloration"*			
26.	Palpate Auricles and Tragus	*"nontender bilaterally"*			
27.	Inspect external auditory canals	*"no swelling, redness, or obstruction"*			
28.	Inspect general appearance of TMs	*"TMs are pearly gray and translucent; light reflex is ant-inf. quadrant; pars tensa, pars flaccida, umbo, handle of malleus visualized and undistorted"*			
29.	Inspect for fluid in middle ear	*"no fluid behind TM"*			
30.	Inspect mobility of TMs	*"TM's mobile"*			
31.	Otoscope exam technique				
32.	Whisper test	*"CN VIII intact"*			
33.	Perform Weber test	*"Weber is equal bilaterally'*			
34.	Perform Rinne test	*"AC > BC bilaterally"*			
Nose & Sinuses					
35.	Inspect/palpate nose	*"nose is midline and without lesions or tenderness"*			
36.	Check patency of nares	*"nasal passageways are patent"*			
37.	Discrimination of odors	*"CN I intact"*			
38.	Inspect septum, turbinates, and floor of nasal cavity	*"mucosa is pink and moist without lesions; septum is without perforations; superior, middle, inferior turbinates without polyps, edema, exudates or discharge"*			
39.	Technique of nasal speculum				
40.	Palpate and percuss paranasal sinuses	*"no maxillary or frontal tenderness"*			
Mouth & Throat					
41.	Inspect lips, oropharyngeal mucosa, gingiva	*"lips are pink, mucosa moist without lesions, swelling or discoloration"*			
42.	Inspect tonsils and uvula	*"Tonsils are [grade size] and equal with midline uvula; no exudates or lesions"*			
43.	Inspect tongue surface	*"no discoloration or lesions"*			
44.	Inspect Stensen's and Wharton's Ducts	*"no stones or obstructions"*			
45.	Perform bimanual exam (Palpate lips, tongue, gingival and buccal mucosa)	*"no edema, masses or lesion"*			
46.	CN XII	*"Tongue is midline, CN XII intact"*			
47.	Inspect and percuss teeth	*"teeth are in good repair and nontender"*			
48.	CN IX/X	*"Uvula rises with phonation and gag reflex is present"*			
Neck					
49.	Inspect skin of neck	*"no lesions, swelling, discoloration and musculature is symmetrical"*			
50.	Inspect for trachea position	*"trachea is midline"*			
51.	Palpate trachea for mobility	*"trachea is mobile"*			
52.	Palpate lymph nodes of neck and head (occipital, pre and postauricular, submental, submandibular, tonsillar, superficial cervical, pre and posterior cervical, supra and infraclavicular)	*"no lymphadenopathy"*			
53.	Inspect and palpate thyroid gland	*"thyroid is normal size, shape, and consistency without nodules or tenderness"*			
54.	Test active ROM of neck (flexion, extension, lateral bending, rotation)	*"full active ROM without pain"*			
55.	Strength testing (rotation and shoulder shrug)	*"CN XI intact"*			

Lungs & Chest					
56.	Inspect chest	*"no deformities; AP < lateral diameter"*			
	Posterior chest				
57.	Inspect posterior thorax	*"posterior chest is symmetric without lesion or cyanosis"*			
58.	Palpate chest wall expansion	*"symmetric expansion"*			
59.	Palpate chest wall	*"no masses, crepitus, or tenderness"*			
60.	Palpate for tactile fremitus	*"equal in all fields"*			
61.	Percuss lung fields	*"resonant with no areas of dullness"*			
62.	Percuss for diaphramatic excursion	*"___cm with inspiration"*			
63.	Auscultate lung fields	*"equal breath sounds bilaterally with no adventitious sounds"*			
64.	Voice Sounds (All 3 of the following: Egophony, Bronchophony, Whispered Pectoriloquy)	*"No egophony, Bronchophony or whispered pectoriloquy"*			
65.	Percuss for CVA tenderness	*"no CVA tenderness"*			
	Anterior chest				
66.	Inspect anterior thorax	*"anterior chest is symmetric without lesion or cyanosis"*			
67.	Palpate chest wall expansion	*"symmetric expansion"*			
68.	Palpate chest wall	*"no masses, crepitus, or tenderness"*			
69.	Palpate for tactile fremitus	*"equal in all fields"*			
70.	Percuss lung fields	*"resonant with no areas of dullness"*			
71.	Auscultate symmetric lung fields	*"equal breath sounds bilaterally with no adventitious sounds"*			
72.	Voice Sounds (All 3 of the following: Egophony, Bronchophony, Whispered Pectoriloquy)	*"No egophony, bronchophony or whispered pectoriloquy"*			
Cardiovascular					
	Continue in seated position				
73.	Auscultate with diaphragm at 4 valvular locations and left 3rd ICS	*Verbalize areas as you proceed*			
74.	Auscultate with bell at 4 valvular locations and left 3rd ICS	*Verbalize areas as you proceed*			
75.	Auscultate for S2 split	*"I am listening for S2 split with inspiration"*			
76.	Auscultate for aortic regurgitation (left sternal border, leaning forward, exhale holding breath)	*"I am listening for an early diastolic murmur of aortic regurgitation"*			
77.	Palpate, then auscultate carotid arteries with bell	*"no carotid bruits"*			
	Move patient to supine position with HOB @ 30 degrees with tangential light				
78.	Inspect precordium at 4 valvular areas and locate PMI	*" no lifts or heaves; PMI is not displaced"*			
79.	Palpate precordium for thrills	*"No vibrations/thrills"*			
80.	Auscultate with diaphragm at 4 valvular locations and left 3rd ICS	*Verbalize areas as you proceed*			
81.	Auscultate with bell at 4 valvular locations and left 3rd ICS	*Verbalize areas as you proceed*			
82.	Auscultate for gallops (Listening with bell at left lateral recumbent position"	*"I am listening for S3 and S4 with bell"*			
83.	Verbalize quality	*"rate and rhythm are regular without murmurs, gallops, or rubs"*			
84.	Auscultate Aorta, Renal, Iliac, Femoral arteries for bruits (May be done with abdominal exam)	*"no aortic, renal, iliac, or femoral artery bruits"*			
85.	Palpate bilateral peripheral pulses (Brachial, radial, ulnar, popliteal, dorsal pedis	*"pulses are full and equal bilateral"*			

	and posterior tibialis) (May be done during MS exam and femoral may be performed during abdomen exam)				
86.	Inspect nailbeds and estimate cap refill (May be done with MS exam)	*"no peripheral cyanosis or clubbing; cap refill is < 2 seconds"*			
Abdomen					
87.	Inspect abdomen	*"no discoloration, lesions, or scars; contour is ___ with visible aortic pulsations"*			
88.	Auscultate for bowel sounds (4 quadrants and epigastrium)	*"normal active bowel sounds"*			
89.	Palpate femoral pulse and inguinal lymph nodes	*"femoral pulse is full and equal bilaterally; no inguinal lymphadenopathy"*			
90.	Percuss abdomen (four quadrants)	*"no discomfort or large areas of dullness with percussion"*			
91.	Percuss stomach and bladder	*"no dullness to percussion"*			
92.	Estimate liver size with percussion	*"liver size estimated at [cm] at R MCL"*			
93.	Estimate spleen size with percussion	*"No dullness to percussion anterior to midaxillary line and no change in tympany with inspiration at Traubes space"*			
94.	Light palpation of all 4 quadrants with knees flexed	*"no tenderness, guarding, or rigidity with light palpation"*			
95.	Deep palpation of all 4 quadrants with knees flexed	*"no tenderness, guarding, rigidity, or masses with deep palpation"*			
96.	Palpate for rebound tenderness	*Verbalize that palpation will be done remote from area of pain*			
97.	Palpate liver edge	*"liver edge is smooth and nontender"*			
98.	Palpate for spleen enlargement	*"spleen is nontender and not enlarged"*			
99.	Palpate for kidneys	*"kidneys are nonpalpable"*			
100.	Palpate abdominal aorta	*"abdominal aorta is estimated at __ cm wide without lateral expansion of pulsation"*			
101.	Psoas sign	*"No RLQ pain with resisted R hip flexion"*			
102.	Obturator sign (passive flexion/internal rotation of right hip)	*"No RLQ pain with R hip flexion and internal rotation"*			
Upper Extremities					
103.	Inspect soft tissues, bones and joints of the upper extremities	*"shoulders, elbows, wrists, and fingers are symmetrically aligned without scars atrophy, deformities, fasciculations, or tremors"*			
104.	Palpate the soft tissues, bones, joints of shoulder (SC joint, AC joint, Glenohumoral joint, clavicle, scapula, humerus, bicepital groove)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
105.	Assess active ROM of the shoulder (flexion, extension, abduction, adduction, internal rotation, external rotation)	*"full active range of motion without pain"*			
106.	Evaluate muscle strength of shoulder (flexion, extension, abduction, adduction)	*Shoulder strength -/5*			
107.	Palpate the soft tissues, bones, joints of elbow (epicondyles, joint spaces, ulnar nerve, olecrannon, forearm)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
108.	Assess active ROM of the elbow (flexion, extension, supination, pronation)	*"full active range of motion without pain"*			

109.	Evaluate muscle strength of elbow (flexion, extension)	*Elbow strength -/5*			
110.	Palpate the soft tissues, bones, joints of wrist (joint spaces, radial and ulnar styloid, carpal bones, scaphoid)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
111.	Assess active ROM of the wrist (extension, flexion, abduction, adduction)	*"full active range of motion without pain"*			
112.	Evaluate muscle strength wrist (flexion, extension)	*Wrist strength -/5*			
113.	Palpate the soft tissues, bones, joints of hands/fingers (joint spaces, tendon sheaths, MCP, PIP, DIP, metacarpals and phalanges)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
114.	Assess active ROM of the hand (flexion of all finger joints and extension)	*"full active range of motion without pain"*			
115.	Evaluate muscle strength of hands (grip strength, thumb opposition, finger abduction)	*Hand strength -/5*			
Lower Extremities					
116.	Inspect the soft tissues, bones and joints of the lower extremities	*"hips, knees, ankles and toes are symmetrically aligned without scarring, swelling, atrophy, deformities, fasciculations, or tremors"*			
117.	Palpate the soft tissues, bones and joints of hips (iliac crests, greater trochanter, thigh)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
118.	Assess **passive** ROM of hips (flexion, abduction, adduction, internal rotation, external rotation)	*"full passive range of motion without pain"*			
119.	Evaluate muscle strength of hips (flexion, extension, abduction, adduction)	*"Hip strength _/5"*			
120.	Straight leg raise (with foot in dorsiflexion)	*"No radicular pain bilaterally"*			
121.	Palpate the soft tissues, bones and joints of knees (patella, joint space, popliteal space, popliteal pulse, tibial tuberosity, quadriceps, patellar and hamstring tendons)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
122.	Assess active ROM of knees (flexion, extension)	*"full active range of motion without pain"*			
123.	Evaluate muscle strength of knees (flexion, extension)	*"knee strength _/5"*			
124.	Collateral stress test	*"MCL/LCL without laxity"*			
125.	Drawers sign	*"Drawer sign negative, ACL/PCL without laxity"*			
126.	Palpate the soft tissues, bones and joints of ankles (tibia-check for tibial edema, calf, Achilles tendon, joint spaces, medial and lateral malleolus, and ATFL)	*"skin is warm, no soft tissue swelling, tenderness, boney enlargements or masses"*			
127.	Assess active ROM of the ankles (dorsiflexion, plantar flexion, inversion, eversion)	*"full active range of motion without pain"*			

128.	Evaluate muscle strength of ankles (dorsiflexion, plantar flexion)	*"ankle strength _/5"*			
129.	Palpate the soft tissues, bones and joints of feet/toes (metatarsals, MTP, DIP, PIP, plantar fascia)	*"no soft tissue swelling, tenderness, boney enlargements or masses"*			
130.	Assess passive ROM of forefoot (inversion, eversion)	*"full passive range of motion of forefoot without pain"*			
131.	Assess active ROM of the toes (flexion, extension)	*"full active range of motion without pain"*			
132.	Evaluate muscle strength of great toe	*"great toe strength _/5"*			
Spine					
133.	Inspect soft tissue and joint alignment	*"no deformity of the spine; scapulae, shoulder and iliac height are symmetrically aligned and there is no atrophy, swelling or discoloration"*			
134.	Palpate spinous processes and paravertebral muscles	*"no step-off deformity or tenderness of spinous processes AND there is no swelling, heat or tenderness of soft tissues"*			
135.	Assess active ROM of the lumbar spine (flexion, extension, right and left lateral bending, right and left rotation)	*"full active range of motion without pain"*			
Neurologic					
	Cranial Nerves (Completed throughout exam)				
	Sensory System				
136.	Test sharp vs. dull tactile stimulation of dermatomes (face, UE, LE, trunk)	*"no deficit with pain stimulation"*			
137.	Test light touch tactile stimulation of dermatomes (face, UE, LE, trunk)	*"no deficit with tactile stimulation"*			
138.	Test vibratory sense	*"no deficit with vibratory stimulation"*			
139.	Test position sense	*"no deficit with proprioception"*			
140.	Test stereognosis	*"correct interpretation"*			
	Motor System (Completed throughout MS exams)				
	Reflexes				
141.	Test Biceps reflex	*"Biceps reflex _/4"*			
142.	Test Triceps reflex	*"Triceps reflex _/4"*			
143.	Test Brachioradialis reflex	*"Brachioradialis reflex _/4"*			
144.	Test Patellar reflex	*"Patellar reflex _/4"*			
145.	Test Achilles reflex	*"Achilles reflex _/4"*			
146.	Test Plantar (Babinski) reflex	*"toes curl down"*			
147.	Test Clonus	*"No clonus"*			
	Cerebellar				
148.	Test heel-to-shin	*"smooth and accurate"*			
149.	Test RAM for upper and lower extremities	*"quick, smooth, rhythmic and symmetrical"*			
150.	Test finger-to-nose	*"smooth and accurate"*			
151.	Assess gaits (NL, H, T, H/T)	*"balanced and coordinated without weakness or hesitation"*			
152.	Deep knee bend	*"balanced and coordinated without weakness or hesitation"*			
153.	Test Romberg	*"maintains balance with eyes closed"*			
154.	Test pronator drift	*"no drift"*			

Attachment 10: Quick Reference Guides

EYE DISORDERS QUICK REFERENCE

Symptoms and Signs	**Glaucoma**	**Internal Hordeolum**	**Allergic Conjunctivitis**	**Bacterial/Viral Conjunctivitis**	**Anterior Uveitis**	**Corneal Abrasion/Ulcer/ Foreign Body**
Eye Pain	With closed-angle	Sensation of foreign body	Itching, burning, no pain	Itching or Burning	Moderate to severe dull ache	Moderate to severe
Vision	Loss of peripheral; blurring	Normal	Normal	Normal	Blurred	Blurred
Photophobia	Possible	Absent	Absent	Absent to moderate	Present	Present
Discharge	Lacrimation	Absent	Watery, may have lacrimal regurgitation	Present (purulent or watery)	May have tearing from pain, no discharge	Lacrimation and purulence
Redness and Inflammation	Diffuse; "steamy cornea"	Localized to inner palpebral conjunctiva	Injected conjunctiva, more pronounced away from the cornea	Entire eye red, more pronounced away from the cornea	Pink to red Flush around Cornea	Entire eye red
Pupil Size	Normal	Normal	Normal	Normal	Constricted	Constricted or Normal
Pupil Shape	Fixed/dilated	Normal	Normal	Normal	May be irregular	Normal
Pupil Reaction to Light	Normal	Normal	Normal	Normal	Diminished or absent	Normal or Diminished
Special Notes	Immediate referral	Small elevation or yellow area at site of affected gland, on conjunctival side of lid	Associated with allergic rhinitis	Rarely a serious condition	Immediate referral	Abnormalities noted with fluorescein; immediate referral if ulcerated

EAR DISORDERS QUICK REFERENCE

Symptoms and Signs	**External Otitis**	**Otitis Media**	**Serous Otitis Media**	**Foreign Body**	**Ruptured or Perforated Eardrum**	**Labyrinthitis**

Pain and Location	Moderate to severe earache, especially with pulling on affected ear	Mild to severe earache, may radiate over same side of face	Usually not Present	Sensation of something moving, burning, or pain in canal	Increasing sharp pain followed by sudden release of pressure; may have "rushing air sounds" after rupture	Usually not present
Associated Symptoms and Signs	Ear canal swollen and crusty with discharge, recent exposure to water, * Consider ruptured TM if discharge present	Nausea, purulent discharge may be present, upper respiratory infection may be present	Vertigo and dizziness, history of upper respiratory infection, or changes in atmospheric pressure	Recent exposure	Tear in tympanic membrane, history of trauma, increased atmospheric pressure or exposure to loud noise(s), purulent discharge (if caused by otitis media	Vertigo, nausea, nystagmus, tinnitus, history of recent upper respiratory infection
Body Temperature	Usually normal Except with severe Infection	Elevated based on severity of infection	Normal	Normal	May be elevated if caused by otitis media	May be Elevated
Eardrum Appearance	Normal, if visible	Dull, reddish-pink, bulging, or ruptured	Retracted, "ground glass" appearance, amber, bubbles & fluid	Normal, if visible	Torn	Normal
Hearing	Normal	Normal to slightly impaired	May be impaired, complaints of muffled sounds	May be Impaired due to obstruction	May be Impaired	May be impaired if infection is present
Require Immediate referral	Fever, tender mastoid	Complete hearing loss, dizziness	Complete hearing loss, dizziness, vertigo	Fever, TM rupture	Hx of head trauma, fever, neurological symptoms	Fever, neurological symptoms

NOSE AND THROAT DISORDERS QUICK REFERENCE

Symptoms and Signs	Allergic Rhinitis	Viral Rhino-sinusitis	Sinusitis	Pharyngitis	Strep Throat	Peritonsillar Abscess	Laryngitis
Pain Status and Location	Usually not noted	Pressure in or around the nose and sinuses	Sharp and throbbing headache over sinus, may radiate to teeth	Mild to severe sore throat	Mild to severe sore throat	Severe sore throat. May have trismus (inability to open mouth), jaw pain, ear pain	Mild sore throat
Associated	Itchy, watery	Low fever,	Sore throat or	Enlarged cervical	Tonsils red	Hard to swallow	Larynx red and

Thick mucoid discharge may be clear, yellow, or

Symptoms and Signs	Cerebral Vascular Accident	Head Injuries	Seizures	Meningitis	Anaphylactic Reaction	Alcohol intoxication/ withdrawal	Unconscious patient
Onset	Sudden or Gradual over a few hours	Sudden	Sudden	Gradual	Sudden	Sudden; related to alcohol intake	Sudden
State of Consciousness	Drowsy to comatose	Varying with severity of trauma	Unconscious during attack, drowsy later, amnesia for the event	Drowsy	Varies from anxious to unconscious	Impaired speech, coordination, movement, tremors	Unconscious
Pain Status	Headache may be present	Headache	Usually not reported	Headache and stiff neck	Headache and occasionally abdominal cramps	Headache, can have seizures	Non-responsive
Associated Symptoms and Signs	Convulsions, vomiting, face may be flushed, one-sided paralysis	Vomiting, vertigo, confusion, contusions or lacerations of the head	Nausea, vomiting, muscle spasms, past history of seizures, look for injuries	Nausea, vomiting, chills, convulsions, positive Kernig's and Brudzinski's signs	Heavy perspiration, vomiting, generalized itching, breathing difficulty. Urticaria	Ataxia, abnormal movements, slurring speech, impaired congnition, and loss of bladder control	Variable
Body Temperature	May be elevated to low-grade fever	Normal	Normal or very high	High-grade fever, variable	Normal	Normal	Variable
Pulse	Slow	May be Slow	Rapid	Rapid and weak	Rapid, weak, and irregular	Variable	Variable
Pupil Size	Normal or dilated and unequal	Constricted, normal or dilated, equal or unequal	Normal	Normal	Normal	Sluggish/ constricted (intox) or dilated	Constricted, normal or dilated, equal or unequal
Respiration	Rapid and shallow	Variable	Slow	Variable	Rapid and shallow	variable	Varied to absent

Signs and Symptoms	Angina Pectoris	Pulmonary Edema	Congestive Heart Failure	Costocondritis	Pneumothorax	Pneumonia (Lobar)	Myocardial Infarction
Pain Status and Location	Moderate to intense, pressure-like pain under the sternum, radiates to left arm and shoulder	Usually absent (if present, suspect other conditions)	Usually absent (if present, suspect other conditions)	Sharp chest pain localized to the cartilage between the ribs and sternum	Sudden, sharp, tearing pain in the side of the chest, may be referred to the shoulder blade. Pain worse with breathing	Sudden, sharp or gradual pain in the side of the chest aggravated by breathing	Sharp, moderate to intense crushing pain under sternum, that may radiate to neck, arms, and jaw
Duration of Pain	Usually 3-30 minutes	Not Applicable	Not Applicable	Hours to days	Hours	Hours to days	Hours to several days
Shortness of Breath	Not Present	Severe breathing difficulties	Present, especially when recumbent	May be present	Severe and increasingly worse with time	Common and often severe	May be present
Shock	Not Present	Not Present	Not Present	Not Present	Occasionally mild shock present	Usually not present	Commonly present (cold sweat, fast pulse, blood pressure down)
Associated Findings	Significant elevation in systolic and diastolic blood pressure during attack-however may also be hypotensive. Associated with strenuous activity	History of heart disease or exposure to toxic fumes, productive cough, nauseas and vomiting. Pitting edema of feet and legs. Moist cough with frothy pink or white sputum	Can accompany any serious illness, history of hypertension, nausea, vomiting, edema of the ankles and feet. Moist cough with frothy pink or white sputum	History of similar attacks, temperature may be slightly elevated	Anxiety; flat, quiet chest, collapsed lung	Chills, fever, productive cough, rales, high WBC	Nausea, vomiting, fever, heavy perspiration. Associated with strenuous activity, history of heart disease
EKG	T Wave changes may or may not be present	Variable	Variable	Normal	Normal	Normal	ST-T wave changes, Q wave changes

Symptoms and Signs	Asthma	Pleurisy (irritation of lung lining)	Pneumothorax	Pneumonia	Pulmonary Embolism and Infarction of the Lung	Bronchitis	ARDS	Pulmonary Edema (non-cardiogenic)
Onset	Sudden/ gradual	Gradual	Sudden	Gradual	Sudden	Gradual	Sudden	Sudden

Associated Symptoms and Signs	History of previous attacks, heavy perspiration. Inspiration is short and gasping. Wheezing on expiration	Productive cough, runny nose, inflamed throat	May have open chest wound, decreased chest motion on the affected side, hyperresonant sound over affected side	Shaking chills, productive cough, cyanosis, flaring nostrils	Flaring nostrils, cyanosis, heavy perspiration, hemoptysis	Productive cough, runny nose, inflamed throat	Variable based on etiology	Irritation of the eyes, nose, throat, trachea and bronchitis if caused by gases or smoke

ABDOMINAL PAIN QUICK REFERENCE

Symptoms and Signs	Appendicitis	Cholecystitis	Gastroenteritis	Hepatitis	Inguinal Hernia	Intestinal Obstruction
Age	Any age	Usually 30-50 years	Any age	Any age	Any age	Any age
Onset	Gradual	Gradual	Sudden	Gradual	Usually sudden, occasionally gradual	Gradual or sudden
Pain and Location	Mild to severe, generalized in mid abdomen early, later localizes in RLQ. Rebound	Upper abdominal discomfort to sharp burning in RUQ. May refer to right shoulder	Moderate to severe in epigastric region, and lower abdominal cramps	Constant, dull, moderate to severe, in RUQ	Intermittent dull ache or “dragging sensation” in the inguinal area	Intermittent, moderate to severe, in any area

ABDOMINAL PAIN QUICK REFERENCE, CONTINUED

Symptoms and Signs	Peptic Ulcer	Pneumonia	Pyelonephritis	Urethritis and Cystitis	Urinary Tract Calculi
Age	Usually 25-50 years old	Any age	Any age	Any age	Usually 35-65 years
Onset	Gradual	Gradual or rapid	Gradual or rapid	Gradual	Sudden
Pain and Location	Intermittent sharp burn in epigastric region Deep tenderness may be present	Sharp pain in lower chest, upper abdomen, aggravated by breathing	Flank pain	Dull aching	Colicky abdominal pain radiating from the groin, scrotum, or labia are classic presenting symptoms

GENITOURINARY QUICK REFERENCE

Symptoms and Signs	**Urinary Tract Calculi**	**Acute Pyelonephritis**	**Prostatitis**	**Testicular Torsion**	**Orchitis**	**Epididymitis**	**Hydrocele**
Pain Status and Location	Colicky abdominal pain radiating to the groin, scrotum, or labia are classic presenting symptoms	Moderate to severe dull ache over the kidneys flank, and lower back unilateral or bilateral	Moderate to severe pain in lower back and perineum	Severe pain in scrotum. Radiates into lower abdomen	Moderate to severe pain in scrotum	Moderate to severe pain in scrotum. Usually hemiscrotal	Normally not present
Associated Symptoms and Signs	Anorexia, nausea and vomiting often accompany abdominal pain or flank pain	Nausea, vomiting, chills history of urinary tract conditions	Frequent and urgent urination, difficulty in starting urine	Nausea, vomiting, often occurs at night or following recent strenuous	Nausea, vomiting, previous case of mumps 7 to 8 days earlier	Nausea, vomiting, past history of urinary tract	Swelling transilluminates in scrotum, usually on one side.

Urinalysis	Urine may contain RBCs, WBCs, bacteria, protein, and crystals	Urine contains white blood cells, red blood cells, and bacteria	May contain white blood cells +/- Hematuria +/- Pyuria	Normal	Normal	Urine may contain white blood cells	Normal

PSYCHIATRIC QUICK REFERENCE GUIDE

SYNDROME	**Behavior**	**Speech**	**Thought Content**	**Perception**	**Affect**	**Orientation & Memory**	**Onset & Duration**	**Physical Findings**
Delirium	Agitation (occasionally quiet) lint-picking	Nonspecific	Variable, delusions	Illusions, hallucinations	Fear, anxiety	Disoriented, memory impaired, clouded consciousness	Acute with fluctuating symptoms	Abnormal vital signs
Dementia	Apathy, apraxia, echopraxia	Echolalia, aphasia.	Variable, few if any delusions	Few if any hallucinations	Lability	Disorientation, memory impairment	Insidious, chronic, usually late life	Poor self care
Schizophrenia	Social	Rambling	Bizarre	Hallucinations	Blunt, flat	Intact	Symptoms	None

HEALTH RECORD	CHRONOLOGICAL RECORD OF MEDICAL CARE
DATE	

PATIENT'S IDENTIFICATION *(Use this space for Mechanical Imprint)*	RECORDS MAINTAINED **AT:**		
	PATIENT'S NAME *(Last, First, Middle initial)*		**SEX**
	RELATIONSHIP TO SPONSOR:	**STATUS**	**RANK/GRADE**
	SPONSOR'S NAME		**ORGANIZATION**
	DEPART./SERVICE AF	SSN/IDENTIFICATION NO.	**DATE OF BIRTH**

CHRONOLOGICAL RECORD OF MEDICAL CARE **STANDARD FORM 600**

(EF)

DATE	